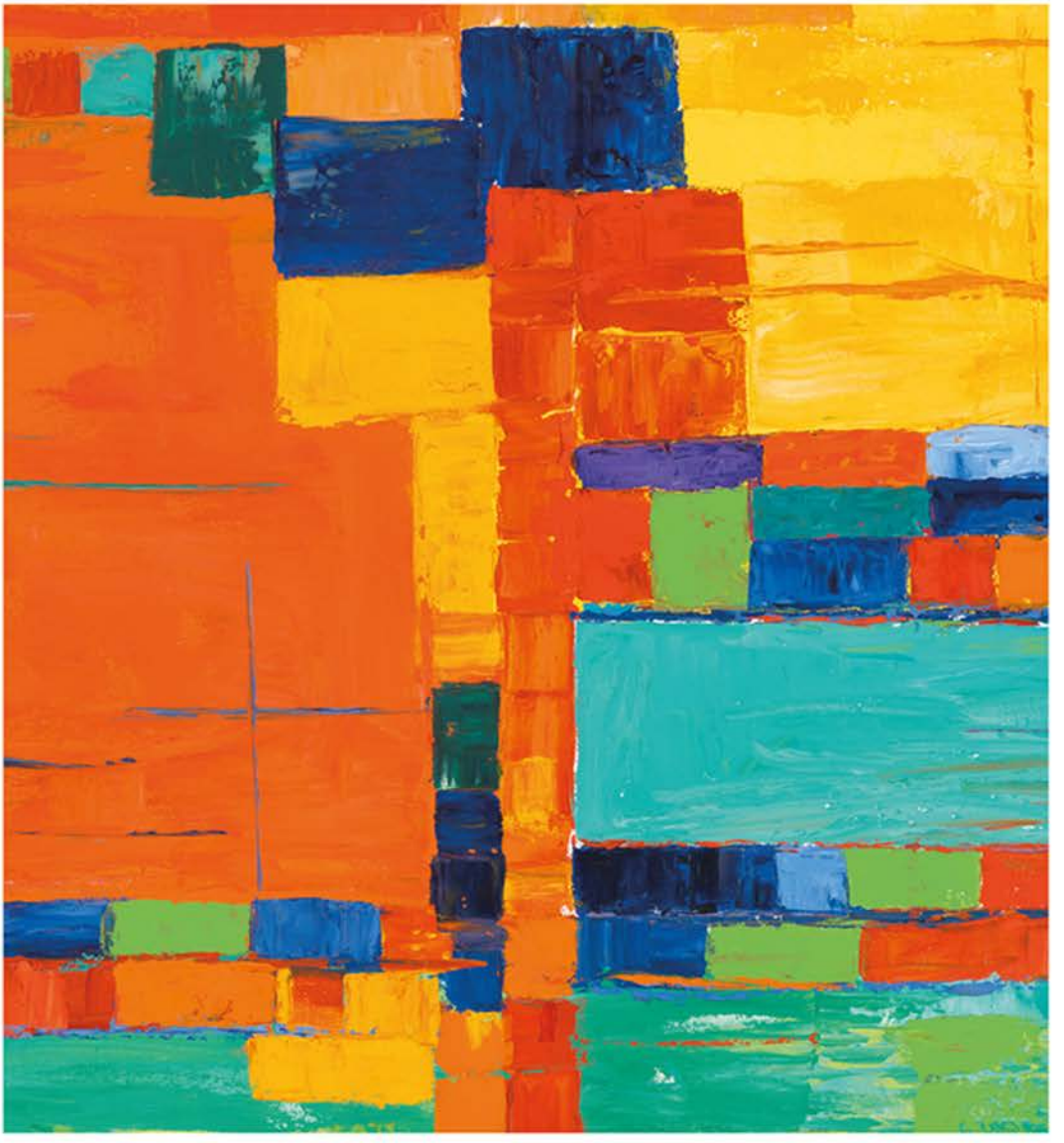
TRACKING COLOR IN CINEMA AND ART

PHILOSOPHY AND AESTHETICS



EDWARD BRANIGAN

ROUTLEDGE

Tracking Color in Cinema and Art

Color is one of cinema's most alluring formal systems, building on a range of artistic traditions that orchestrate visual cues to tell stories, stage ideas, and elicit feelings. But what if color is not—or not only—a formal system, but instead a linguistic effect, emerging from the slipstream of our talk and embodiment in a world? This book develops a compelling framework from which to understand the mobility of color in art and mind, where color impressions are seen through, and even governed by, patterns of ordinary language use, schemata, memories, and narrative.

Edward Branigan draws on the work of Ludwig Wittgenstein and other philosophers who struggle valiantly with problems of color aesthetics, contemporary theories of film and narrative, and art-historical models of analysis. Examples of a variety of media, from American pop art to contemporary European cinema, illustrate a theory based on a spectator's present-time tracking of temporal patterns that are firmly entwined with language use and social intelligence.

Edward Branigan is Professor Emeritus in the Department of Film and Media Studies at the University of California, Santa Barbara. He is the author of *Projecting a Camera: Language-Games in Film Theory*; *Narrative Comprehension and Film*; and *Point of View in the Cinema: A Theory of Narration and Subjectivity in Classical Film*. With Warren Buckland, he is the editor of *The Routledge Encyclopedia of Film Theory*. With Charles Wolfe, he is the general editor of the American Film Institute Film Readers series.


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Tracking Color in Cinema and Art

Philosophy and Aesthetics

Edward Branigan

Routledge
Taylor & Francis Group
NEW YORK AND LONDON

First published 2018
by Routledge
711 Third Avenue, New York, NY 10017

and by Routledge
2 Park Square, Milton Park, Abingdon, Oxon OX14 4RN

Routledge is an imprint of the Taylor & Francis Group, an informa business

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This project was partially supported by funding from the President's Research Fellowships in the Humanities, University of California.

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Library of Congress Cataloging-in-Publication Data
Names: Branigan, Edward, 1945- author.
Title: Tracking color in cinema and art: philosophy and aesthetics/ Edward Branigan.
Description: New York : Routledge, 2017. | Includes bibliographical references.
Identifiers: LCCN 2017018660 | ISBN 9781138230668 (hardback)
Subjects: LCSH: Motion pictures—Philosophy. | Motion pictures—Aesthetics. | Colors in motion pictures. | Color in art.
Classification: LCC PN1995. B7185 2017 | DDC 791.4301—dc23
LC record available at https://lccn.loc.gov/2017018660

ISBN: 978-1-138-23066-8 (hbk)
ISBN: 978-1-315-31750-2 (ebk)

Typeset in Sabon LT Std
by Apex CoVantage, LLC

For Melinda, who looked after and furthered
my memories now lost of many things,
and for our son, Nicholas.


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Preface

> The phenomena of color pose a special puzzle to philosophers characterizing the mind, the world, and the interaction of the two. In various ways, both subjective and objective, both appearance and reality, color has been the subject of wide disagreement.[1]
>
> —Justin Broackes

LANGUAGES

This book examines philosophies of color and their implications for aesthetics and methods of analysis. The book dissects color displays, principally in film, painting, and photography, though a sonnet and short story are also analyzed. Words generate images and images words. Color has no intrinsic meaning; it acquires meaning in human contexts. Many living creatures see colors and shapes, but humans have developed language, culture, and art to create what archaeologists call a "symbolically and conceptually mediated worldview." I am primarily interested not in the symbolic content of what is being represented, but in the various mental and symbolic schemata that constrain and drive what can be said by perceivers about what is represented. It is often asserted that there exists a color "language" as well as a film "language" that mediates a worldview. What needs to be unpacked in these formulations is the meaning of the word "language." My assumption is that how we talk about color (and film) is an indication of how we are able to think about color (and film) and our thinking, in turn, influences what we perceive and expect to see. Thus when we encounter an artwork, we may say that it *speaks* to us. But the *speaking* voice is our own since we are imbued with working vocabularies and structures that draw upon powerful memory systems developed by living in a world.

Color is relational. It is not a substance with independent existence nor is it a quality that is static and simply *possessed* by a thing. Sensing the presence of a hue depends on a series of intertwined relationships: the motivated or unmotivated position and quality of a light source in relation to an object in relation to light from nearby colored objects in relation to a perceiver. A hue is perceived relative to the perceptual, cognitive, emotional, intentional, and memory states of a perceiver. Furthermore, a perceiver's colored mental images are not exclusively internal, but rather they are put in place, and responsive to, community perceptions. It is a mistake to focus exclusively on any one color relation or to imagine an irreversible

perceptual stream traveling from light to object to mind. Color is not a purely objective phenomenon determined by a light meter, but rather the result of many interacting systems.

We must reject the idea that what appears on the screen is possessed by a film and lies somehow *within* the film. To understand color *in* the cinema, we must explore the myriad ways of *thinking about* and *speaking about* what we are seeing within a cognitive, social, and cultural context. Color is given significance and quality in the act of being seen from a place at a time. A light meter does not define what is being seen. Announcing that one perceives "blueness" in an object is already to have formed a mental abstraction about its significance seized from our memories and from a vast, patterned environment—an environment that continuously embeds our mental acts, making us speak and think about what we see, value, and expect. The mind does not seize a color spot and go no further. Seeing the spot is already to have gone further.

Allow me to briefly elaborate. When one recognizes a specific blue hue, the fact of being bluish also constitutes an abstraction because many different things in the world are seen, classified, and said to be "blue." The most recent bluish material – known as YInMn – promises to bring together many new things in new ways. Once one thinks about color in its categorical aspect, a discrete hue acquires mobility and expansiveness (see nos. 6, 9, and 10 in Figure 3.1). The power of color resides in the fact that an object and its parts need *not* be thought to possess merely a single, unique set of qualities, but rather to share its colorfulness more widely among groups of objects.

Language provides the ground for a color spot to move and become significant in consciousness. Michael Gorra writes about the genesis of a book by William Gass, *On Being Blue: A Philosophical Inquiry*:

> In the beginning was the word [blue] not the color, not the shade that the spectrum places alongside green, but rather the sound and all that clumps around it, the fuzz balls of meaning that it's picked up as it has rolled on through time. . . . All words are figurative; no blue is ever just blue.[2]

Put otherwise: what may be said of a color always lies within networks of words. The bare percept of a hue has nothing left in it of life—it says nothing, is blank. In fact, a hue is always of a thing—whether tangible or vaporous or colored as fiction—that connects to other things in a perceiver's thought, being, and life.

COLOR CONSCIOUSNESS

The central premise of my argument about film is that color should be evaluated as a *temporal pattern* that may build from shot to shot and from moment to moment

according to the actions of a perceiver's *phonological rehearsal loop* in working memory. I will argue that the process is one of mental "tracking" as opposed to a perceiver's summoning of semiotic codes from long-term semantic and episodic memories. Semiotics makes use of a priori signs and codes that employ knowledge already "standing" in place. I derive the notion of tracking from P.F. Strawson's work on the mental activity of "reidentification."

I will usually analyze the theoretical framework that a critic has selected, whether explicitly or implicitly, rather than evaluate the adequacy of particular claims made about an artwork. This book will not analyze entire films or paintings. Instead, I will analyze particular moments from nine films, four paintings, and two photographs. I've kept color illustrations to a minimum, relying on the internet to fill gaps. Illustrations may even obscure the point of my commentary. Again, the goal is to demonstrate *general* principles of color analysis along with the competing theoretical schemata employed to justify a specific principle of criticism along with its supporting concepts and rhetoric.

My intention is to examine the nature and sources of a viewer's consciousness of color in the visual arts. Ludwig Wittgenstein's unfinished manuscript, *Remarks on Colour*, is a major point of reference. I also draw on film theory, cognitive science, and philosophy of language, though not exclusively. These disciplines have much to contribute to traditional issues studied in art history. The latter's sophisticated approaches to color, line, depth, implied motion, and pattern may be productively supplemented by considering a perceiver's construction of flow when encountering mobility in film, painting, literature, and, for that matter, when fabricating a coherent *line* through his or her ordinary day. Whether an art object is pictorial or literary, our response to a feeling of movement will be linguistic. Motion and change, like a wandering line in a painting, acts to shape and refine our expectations about meaningful patterns and patterns-to-come. As we shall see, color itself need hardly be static, but can move from shot to shot, word to word, and from one mental state to another to generate a stream of ideas and expectations.

This is not a history text. This book is not a history of color theorizing or a history of color in film or a history of worthy color films or a history of color technology. The book does not historicize the impact of commercially successful industrial practices. This is also not the usual art historical aesthetics book. I do make the requisite gestures to conventional color exposition, but my argument forges new, even startling, pathways, especially concerning color design and film theories. The book aspires to be a roadmap toward acquiring a set of concepts for investigating color in all of its forms and properties. My aim is to provide an overview of our many shortcomings when we try to assess color philosophically. I will also point toward a reconceptualization of film theory, narrative theory, and the role of sensory qualities in film.

FOUR ELEMENTS OF COLOR LANGUAGES

The creation of expectations through text-spectator interaction is a crucial premise of many arguments in this book. Wittgenstein asserts that an expectation, though about the future, is a discursive aspect of how we are *presently using language* to memorialize (know) the present! For Wittgenstein, expectation is defined by, and exists within, practices of language drawn from memory. This is an important point. To state it differently, for Wittgenstein, our sense of the future (expectation) emerges out of a present working language, which has developed out of experience. A person's memory—drawing upon, for example, schemata and cognitive metaphor—is thus not confined to, not locked into, a past "past," but drives a future-oriented present. To analyze what one expects of color, one needs to analyze the kinds of languages that are rooted in culture and bring it about.[3] The labyrinthine and paradoxical cultural stereotypes of specific hues—such as fertile green/sickly green, deathly black/sophisticated black—need to be relocated from what color "must mean" to *one* of the ways color may mean.

The book weaves together four thematic strands concerning language and our consciousness of color in an effort to uncover ways that color may mean:

1. *Color conventions and norms*, including cultural, commercial, institutional, and aesthetic practices (techniques, technologies);
2. *Socially anchored linguistic systems* that are available for expressing our felt consciousness of color, including the manifestation of color through physiology, psychology, and emotion, especially in those situations where color comes to embody important narrative concreta and abstracta;
3. *Mental processes for memorializing and reidentifying color patterns* that may be collectively termed the realm of "memory chromatics," such as the use of root metaphors and folk theories for identifying a resonance of color, which realm is much broader than textual inferences, signs, or cues from a screen or canvas, i.e., broader than adherence to explicit instructions and instead sensitive to activities focused on the implicit, the suppressed, the background, and the interpretive; and,
4. *Narrative interfaces between sensation and thought*, spectacle and rationality, in the life of the mind. What are the functions of sensation in art? Is color only sensation, spectacle merely excess?

Art seems in some way always a spectacle that solicits the emotions and thus subject to the accusation that it is a retreat from rationality, the ordinary, and language. Philosophers have wrestled with a series of dualities whereby sensuous qualities like color have been opposed to cognition. For example, one finds in philosophy

such binaries as emotion vs. reason; sensation vs. import (importance); materiality vs. abstraction; body vs. mind; and flesh vs. spirit. I will propose, however, that these binaries are at best merely provisional points along a sliding scale. I will argue that pressure from top-down, language-like processes continuously operates to recast sensation. Language mediates between sensation and sense, infusing perception with familiar descriptions and goals, problems and projects. Language exists to select, represent, and connect facets of a world with our routines and community activities. Language is firmly grounded in mental schemata, descriptions, narrations, discourses, and interpretations, and hence works to bring together sense and sensation. It makes no sense to say that sensation stands alone. Sensation is what we make of it.

The preceding four strands of color languages sketch a theory of experience in media that is dependent on ordinary verbal discourses about kinesthetic and gestalt sensations as filtered through community norms. Equally relevant are nonconscious, taken-for-granted descriptions of bodily existence as well as beliefs conditioned by the use of social intelligence, folk knowledge, and cognitive schemata, all of which are subject to the perplexing inertia of fixed mental images, e.g., stereotypes and prototypes.[4] The centrality of ordinary language also means that there is no pure sort of "fiction" removed from all daily concerns. Interpretation and discourse bring fictional texts back to earth, so to speak.

I imagine a world of "cities of words." Within each city, there are separately colored structures surrounded by descriptive networks pertaining to each hue. The networks shade off into more generally descriptive semantic fields anchored to culture. This book seeks to chart some of these systems of words—informal, ordinary-language communities that evolve and devolve as they come and go[5]—that key our color consciousness when traveling to cities of our choice.

WHERE COLOR VANISHES

Percy Lubbock explains where words vanish:

> To grasp the shadowy and fantasmal form of a book, to hold it fast, to turn it over and survey it at leisure—that is the effort of a critic of books, and it is perpetually defeated. Nothing, no power, will keep a book steady and motionless before us, so that we may have time to examine its shape and design. As quickly as we read, it melts and shifts in the memory; even at the moment when the last page is turned, a great part of the book, its finer detail, is already vague and doubtful. A little later, after a few days or months, how much is really left of it? A cluster of impressions, some clear points emerging from a mist of uncertainty, this is all we can hope to

> possess, generally speaking, in the name of a book. The experience of reading it has left something behind, and these relics we call by the book's name; but how can they be considered to give us the material for judging and appraising the book?[6]

Lubbock argues that the fleeting quality of streams of words in a book produces "shadowy" forms out of light. I argue that this applies equally to a spectator's experience of fleeting images in a film. In particular, most colors experienced in a film are quickly forgotten, as if never seen. A very few may be condensed, shifted, melted into new form, replaced, revised, or re-crystallized in memory and feeling. How might color be more sharply perceived and acknowledged?

Two main obstacles impede the full recognition of color. The first is that hues are typically seen as merely isolated sensuous spots that slip away rather than as components of emerging patterns across time. A color moves, develops, even changes hue. The second is that hues are usually seen as static symbols or vague bits of atmosphere rather than as dynamic memory elements interacting with complex narrative and thematic designs.

The present book seeks to overcome these obstacles by emphasizing the multiple possibilities of patterns and interactions. No blue is ever just blue. My aim is to bring to attention the contribution that color makes to our enchantment with film and other artworks. To accomplish this aim, one must consider more broadly what color *can do* and this, in turn, requires careful attention to our actions and talk about color.

NOTES

1. Justin Broackes, "Colors" in *The Encyclopedia of Philosophy: Supplement*, ed. by Donald M. Borchert (New York: Simon & Schuster Macmillan, 1996), p. 83.
2. Michael Gorra, "Introduction" in *On Being Blue: A Philosophical Inquiry*, William H. Gass (New York: New York Review Books, 2014), p. x.
3. On the nature of expectation, see Ludwig Wittgenstein, *Philosophical Investigations*, trans. by G.E.M. Anscombe, P.M.S. Hacker, and Joachim Schulte (Oxford: Wiley-Blackwell, 4th ed. 2009), §§ 435–465, 572–586. What we *expect* of color is influenced by a series of judgments that are caught within grammatical fluctuations among criteria and symptoms. Cf. §§ 352–356 and see below Figure 3.1. Chapters 7 and 8 below discuss expectation and types of memory in the guise of the "nearly true" and possible (future) "reidentification." Consider also the predispositions, i.e., expectations, intrinsic to many judgment heuristics as well as the implementation of a "self-fulfilling prophecy," i.e., an expectation fulfilled as if by itself out of the present.
4. More exactly, this book will rely on theories of embodied cognition, situation models, and conceptual metaphor in situating our consciousness of color and our actions in response to it. On embodied cognition generally, see Chapter 7 below, note 24. A bibliography and outline of the importance of situation models may be found in Maarten Coëgnarts, Miklós Kiss, Peter Kravanja,

and Steven Willemsen, "Seeing Yourself in the Past: The Role of Situational (Dis)continuity and Conceptual Metaphor in the Understanding of Complex Cases of Character Perception," *Projections* 10, 1 (Summer 2016), pp. 114–138. They stress, as do I, the importance of *working memory* in a spectator's comprehension of film.

My approach to color might be placed within the method of Coëgnarts et al. by adding a "box" for color in their Figure 1 (p. 122) underneath their "box" for "container," which represents the image schema through which a visual field is conceptualized by a viewer in terms of an inside, a boundary, and an outside, i.e., outside of view. The problem for a theorist is then to identify which specific metaphors and metonymies of embodied perception will function on a local or micro-level of perception to stage a viewer's conceptual inferences about color inside a visual field as well as hypotheses about color lying outside the field. I undertake this task in the present book; see, e.g., Figure 3.1; Chapter 4, section 5; and Chapter 5, section 4.

In Figure 7 (p. 129), Coëgnarts et al. identify five image schemas as the sensory-motor grounding of character perception and time in film. Three of the schemas relate to a character's/spectator's construction of three-dimensional space (front-back, left-right, up-down as related to a person's body); the two others relate to movement or time (source-path-goal) and shape (specific embedded "containers"). To these five, one might add others, including an image schema—call it "radiant mass"—specifically for color (incorporating black and white) in which embodied color perception would amount to a preliminary sketch of the appearance of a bare *materiality* extended in space. Color becomes an enigmatic sign of the *flesh* of an object, hence also a tie to the emotions. In this way, the conventional associations with a given hue are recast as the emotions associated with a specific subset of objects linked to the hue. Thus a hue moving through an artwork may assemble a novel group of objects to which our feelings are drawn and shaped. On color as a sign of materiality, see Chapter 10, section 3, and Appendix, section 2.

5. I owe this phrase to Philipp Schmerheim (private correspondence). On his approach to Stanley Cavell's philosophy and Cavell's *Cities of Words* (p. 298) as well as the resultant implications for media theory, see Philipp Schmerheim, *Skepticism Films: Knowing and Doubting the World in Contemporary Cinema* (New York: Bloomsbury, 2016). I have, of course, appropriated the title of Cavell's book in the text above.
6. Percy Lubbock, *The Craft of Fiction* (London: Jonathan Cape, 1931), pp. 1–2.


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Acknowledgments

This book was written sporadically over a period of eleven years. It derives from teaching one graduate seminar and five undergraduate classes on the philosophy and aesthetics of film color during quarters from Spring 2003 to Spring 2012. It derives also from students teaching me. The course arose from two of my articles: "The Articulation of Color in a Filmic System: *Deux ou trois choses que je sais d'elle*" (1977, reprinted and expanded, 2006) and "Color and Cinema: Problems in the Writing of History" (1979, reprinted and revised, 1985, 1986) (see Works Cited, B2, B5). John Kurten was my invaluable teaching assistant. He contributed regular lectures from a deep and extensive knowledge of color, especially from his original research concerning the technology, business plans, and aesthetic strategies of the Technicolor Motion Picture Corporation. Figure 0.1 is a witty portrayal and summary of the course by a student, Jess Riegel.

Themes from my previous books—involving character subjectivity, point of view, narration, narrative, fiction, camera, language, memory, and film theory—continue to be developed in the present book. These themes were outlined in the final two pages of the preface to my last book, *Projecting a Camera: Language Games in Film Theory*, and in a recent essay, "If-Then-Else: Memory and the Path Not Taken" (Works Cited, B4). Extending the argument of *Projecting a Camera*, it will be found that I treat the sensation of filmic color (like the sensation of a camera's presence) as a visual experience both presenting a sensory quality and *representing* an object as having specific qualities due to a spectator's *projecting* representations from memory onto the object (see no. 9, Figure 3.1). Such mental representations stem from bodily schemata and social interactions that map onto a film image attributing to it and its depicted objects the possession of certain physical and nonphysical properties. Sensed color qualities are thus fictive with respect to actual objects, light sources, and other sensory objects like a screened film since the qualities of a sensation are distinct from the inferred categories and ostensible properties of an object causing the sensation. Thus, despite the visceral qualities of film, the nature of fictiveness makes fiction film especially conceptual and abstract. Clearly, in the following chapters, I will need to elaborate more precisely the nature and function of visual "fictiveness" and its connection to ordinary things.

Mark Johnson states a working assumption of my book succinctly in *The Meaning of the Body: Aesthetics of Human Understanding* (175, Johnson's emphasis; Works Cited, D1):

> As William James claimed more than a century ago, the music of meaning-making is both thought *and* feeling at once, and its notes are the rhythms and tone qualities of our bodily processes.

The word "tone," of course, may also be thought in terms of color; and "rhythms," then, in terms of color patterns moving as music. Thus, for this approach, thought and feeling emerge "at once," in the *same moment* that color becomes conscious as sensation.

The present book was made possible in part by two generous grants: an American Council of Learned Societies Fellowship during 2008–2009 and a President's Research Fellowship in the Humanities from the University of California during 2008–2009.

I thank Mikki Kressbach, Zdenko Mandusic, and Hannah Frank, whose recent death was so untimely, for inviting me to present some of my ideas at the 2014 conference, "The Silver Screen: Theories and Histories of Cinematic Color," at the University of Chicago. Bits and pieces of the book's ideas have appeared in lectures at the Society for Cinema & Media Studies, Stanford University, the University of Bristol, the University of Kansas, and the University of St Andrews.

I am grateful to the following publishers and artists for permission to reprint or reproduce material: Alfred A. Knopf for John Updike's poem, "Perfection Wasted"; American Artists Group for the Dudeen Color Triangle from John Sloan's *Gist of Art: Principles and Practise Expounded in the Classroom and Studio*; Karen Brobeck for her painting on the cover; and the Estate of Roy Lichtenstein for Lichtenstein's painting, *Two Paintings: Dagwood*.

I have once again been exceptionally well served by the professionals at Routledge: Natalie Foster, Senior Publisher; Erica Wetter, Publisher; Felisa Salvago-Keyes; Christina Kowalski; and Mia Moran. I have also benefited from the exceptional project management, copyedit, and composition of the professionals at Apex CoVantage.

Fortunately, Routledge engaged some smart reviewers to read the manuscript of my book. Brian Price, Sarah Street, and Elizabeth Watkins, all of whom have published important work on film color, offered a series of probing comments, corrections, and suggestions, all of which I have taken fully to heart. I also wish to thank an anonymous reader for additional fruitful remarks and analyses. Finally, the manuscript has profited from many valuable contributions from film scholars John Kurten and Jeff Scheible.

Further, I wish to acknowledge James Peterson, an early 1990s enthusiast and scholar of film color; the historian Michael Albright; and Laure Brost, a fellow color conspirator, who has done significant work on filmic color. I have been sustained by my faithful colleague Charles Wolfe and by the ever-essential presence of a dynamic duo whose lessons endure, never fail, David Bordwell and Kristin Thompson. My thoughts are kept current by the filmies: you know who you are.

My deliriously wonderful sons, talented writers and thinkers, Alex, Evan, Liam, and Nicholas remain a main inspiration. Evan worked hard in streamlining

the preface as did the philosopher Philipp Schmerheim. The usual solid support came from my brother and sisters, Will, Suzie, Alison, and Carol; not forgetting the steadfast Roberta, Julie, and Emma. Special thanks to intellectual helpmates Dudley Andrew, Peter Bloom, Warren Buckland, Chris Dzialo, William Germano, Christopher Husted, Allan Langdale, Stephen Mamber, Daniel Morgan, Christian Quendler, Daniel Reynolds, and David Rodowick. A special thanks to the eminent philosopher of color, C.L. Hardin. I am especially grateful for the unwavering encouragement and companionship of Ellen Rabinowich.

My parents are the embodiment of steadfast support. This might have been the third book dedicated to them. I've benefited greatly from the precise, sharp, and wise thinking of my father, Henry Odell, and from the felicity with words of my English-teacher mother, Evelyn Odell, whose eye for the graceful in things pointed the way. Inspiration as a young adult was provided by my mother's dear friend and author, Verna King Gruhlke, and her husband, Carl.

The reference in the dedication is to a type of joint memory known as "transactive," which I shared with Melinda Szaloky and our son, Nicholas Branigan. Melinda offered a series of crucial comments on several sections of chapters. The parts that Melinda tackled, line by line, immeasurably improved the book's arguments and prose. The book would surely have been better had she been able to read more of it.

Thinking over the foregoing acknowledgments as well as previous book acknowledgments that have been written to highlight persons and fortunate events, I am reminded of Stanley Cavell's wise advice in *Little Did I Know: Excerpts from Memory* (443; Works Cited, E3):

> I have found that there is no more choice over the way one writes than over what makes one happy. If that is true then what you need to know is what makes you happy and what you need to do is to write enough to find this state and keep up with its haunts.

Edward Branigan
Bellingham, Washington
June 2017


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Chapter 1 **Introduction and Overview through *Two Paintings: Dagwood***

ABSTRACT

A Lichtenstein painting offers a preview of the method of color analysis presented in this book. Close analysis shows that the painting is better seen as a triptych rather than the diptych indicated by its title. Furthermore, the "hidden" middle part of the painting is analogous to a crucial third and mediating part of the argument of the present book. The first two, explicit parts of Lichtenstein's painting may be visually analogized to the "theory" (left side) and "practice" (right) of color aesthetics. The third part that facilitates an interaction between theory and practice in the painting and in the present book is a zone of *remembered languages* and general schemata (forms) that a perceiver summons to assemble and comprehend aesthetic displays of color.

How might one analyze the mediation of "remembered languages"? Chapter 2 will explore four strands of these languages: (1) norms of color usage; (2) language systems (semantic fields) that speak a felt consciousness of color; (3) mental processes that memorialize and reidentify (track) color, which area of study I refer to as "memory-chromatics"; and, finally, (4) the place of sensation and its supposed excess—spectacle—in the life of the mind. Which color language and associated set of metaphors is a person *most comfortable with*?

There is more to color than its sheen and conventional symbolism. What we choose to see in color is not strictly a personal decision but is shaped by culture, our plans and projects in society, and the way we become comfortable in *talking about* ourselves and our world. It is perhaps strange to think that language shapes a sensation like color, what we think we have seen, i.e., in thinking we see. Color is not so much about the exact sensation at a point in space, but rather about discovering and feeling *patterns* in space. Recognizing one or another pattern is not exclusively objective nor certain, but abstract and contingent; that is, one can point out the elements of a possible pattern, but the *whole* is not there—it comes together in memory.

> Man is always a teller of tales, he lives surrounded by his stories and the stories of others, he sees everything that happens to him through them; and he tries to live his own life as if he were telling a story. . . . Nothing happens while you live. The scenery changes, people come in and go out, that's all. There are no beginnings. . . . That's living. But everything changes when you tell about life.[1]
>
> —Jean-Paul Sartre

PROCESS INTO PATTERN

Think of a diptych: two parts connected, each telling a story out of which may emerge a third story. That's the structure of the present book. What are these parts and the relationship between them that produces a third story? To preview the organization of this book, I will explore an analogy with Roy Lichtenstein's *Two Paintings: Dagwood* (see Figure 1.1). Interestingly, the painting is slyly enigmatic, for in addition to the title's invitation to view it as staging two scenes of interpretation, it may also be seen as a triptych, which suggests that within its placid surface is a third part which, I will argue, acts to blend the "two paintings" into a larger story about the nature of a viewer's confrontation with the expressive. This book, too, will contain a third part that mediates between the theory and practice of color aesthetics.

Though suggestively fluid, the connections among Lichtenstein's panels remain elusive. The left portion of the diptych appears at first to be disorganized while the center portion coalesces into mysterious designs. Do these two sections amount to merely chance, noise, or inert decoration; or else, a depiction of chance, noise, and inert decoration? Not quite. Not when one considers the overall linkages among the dispersed hues and lines. Lichtenstein's process evolves into a pattern. What brings forth an interpretation of this painting is an event that will be prominent throughout this book: the coming into view of *patterns* that map our comprehension.[2]

Dagwood is situated within a severely flattened space where only a shoe suggests his second leg. (Are his feet off the floor or not? Do we see movement or not?) Dagwood is seemingly confined to his own frame, though his yellow belt and the yellow wall behind seem to flow toward a broad enclosing yellow band in the middle section and then toward five scattered yellow streaks and blotches at the left that paradoxically are both blocked by, and visible through, a white wash. One marvels as fresh colors emerge through additive and subtractive combinations at left. Experiments with green at left are apparently transformed into an orderly floor at right. One savors the subtle and minute, steady flow of lightness from the top toward a darker bottom at left and right.

Other connections and patterns among the parts can be found. For example, the diagonal lines of Dagwood's shoes match the diagonals in the middle section. Certain curves of Dagwood's body and clothes mirror curves in the middle and left

sections. Seven slightly curved horizontal lines in the middle may repeat the lines of his hair! This raises a key question: when comparing the various lines of the middle section to the figure of Dagwood, must a viewer decide whether the middle is, or is not, a highly abstracted and fragmented version of Dagwood? That is, might the middle be seen instead as something *in between* decoration and figurative metaphor—as a sort of joint "shadow" cast by literal paint (left) and figurative design (right), something more than literal but less than a metaphor or symbol, something that foreshadows a figure but refuses to rise to the level of ambiguity or hybridity? If so, then this middle portion may exemplify an inchoate or nascent idea *in the process* of being formed into a new pattern at right.

I would like to suggest that this in-between potential of color and line in the middle section is an important stage in our thinking about film (and significance in general), i.e., one must be sensitive to what is *incipient*, to movements that are neither exclusively sensation (left) nor yet idea (right) but that prefigure what may come to be. Sensory data that is located in a middle, unrealized zone and only "nearly true" as figuration (as in Lichtenstein's middle panel) is an important effect in many artworks. It is evidence that color can be seen as more than sensation or figuration, whether insipid or vivid.

Furthermore, I believe that this in-between effect can be revealed by attending closely to certain arrangements in film that escape both decoration and static symbol. Like the activity of a spectator's memory, a film may entertain moods in the process of becoming something other than the indicative or assertive. To watch a film is to struggle within a film's dense, artificial memory where only fragments become visible and audible. At certain times, impressions will be less precise and more tentative, amorphous, multiple, and in process.[3]

THEORY: DRAWING LINES, TAKING SHAPE

Dagwood seems to be emerging from, or else falling back into, paint itself; or rather, the painted, pixelated Dagwood—the pixels are benday dots—is moving from/into the splashed paint at left. (Has he perhaps escaped the black frame line at the far right?) The painted brushstrokes, smears, smudges, spatters, and drips at left—which are represented as having been painted with different kinds of brushes—are painted to suggest the ground (or better, *a* ground) for the startled figure of Dagwood who is reacting to something not seen further off right, something that could have been rendered in a still different form(ing) of paint. Dynamism of color possibilities at left and middle featuring an intense yellow are juxtaposed to dynamism of quite another sort at right as the yellow becomes muted and a new color—red—emerges to characterize Dagwood's energy. It is as if the yellow has transformed into red, or else can be seen as a prior form of the red. The hot red

also offers a stark contrast with the cold blues of the left two-thirds of the painting. The red makes for an unusual contrast with a grayish yellow and two pinks (at left lower right) and with an incipient brown, which deep-down is a very strange color, or more accurately, an amalgam of colors, including red.[4] Thus chance, noise, and decoration in all three panels become relative to a new context once a viewer begins to seek significance and discover sets of relationships, i.e., patterns, among lines and patches of paint.

A developing pattern of relations is much more than—and quite separate from—its discrete elements. A pattern is more than a linear relation among micro-constituents just as temperature, magnetism, and states of matter like solidity emerge from atomic particles or consciousness emerges from brain matter.[5] Additionally, the temperature of a gas emerges from a collection of atoms that do not themselves have a temperature. Most striking is the fact that time and three-dimensional space are not themselves fundamental, but are an illusory effect emerging out of something remarkably different on the order of quantum entanglement of bits of *information*.

A pattern requires new concepts and theories to explain its potential for interactions. It interacts in novel ways with other such patterns at a meta-level because it represents a specific selection among the properties of elements existing on a lower level. A subset of these lower-level elements is being collected in the pattern in ways that are not dependent on simple, linear adjacency. What makes for "emergence"—i.e., new and significant properties appearing at a higher level—results from a reconfiguration of the causal possibilities of constituent parts on a sub-level. That is, a temporal or spatial mosaic is created at a new, higher scale that allows one to perceive new sorts of relationships, combinations, and interactions.[6]

Is it possible that elements can exist entirely isolated and void of all relationships, defying all attempts at being (re-)framed and (re-)related? Chance may play a role in the making of an artwork, but once folded into object-hood for a beholder, randomness disappears into higher-level crisscrossing patterns, however inconsistent, obscure, or partly realized. Most importantly, patterns have a history in one's perception.

In this light, Lichtenstein's painting moves from incipient theory and possibility at left and middle to a concrete realization of a scene at right. An important theme addressed by Lichtenstein's diptych-as-triptych concerns the general conditions—both material and conceptual—that make for being, for the emergence through process of a figurative pattern. This is also a central issue for the present book whose focal point is the use of color in movements of narrative and figurative cinema. How should salient patterns be constructed and analyzed when looking? How will patterned color *necessarily* emerge out of a background seemingly composed of chance elements, noise, and decoration? Which backgrounds are relevant in producing specific patterns?

To have taken shape, a color patch must have a shape, must be delineated by a border separating it from whatever is adjacent and different. That is, a line will appear, no matter how blurry, implicit, or arbitrary, to frame the presence of color on a surface. This power of line is highlighted in the middle portion of Lichtenstein's painting. For example, one can trace the mutation of diagonal lines from Dagwood's clothes, hair, and background into the middle section and then into the curved white lines appearing within paint splotches at left, as if these curves were attempting to straighten themselves into diagonals. Compare also the three white curves within a black splotch (left panel middle) to Dagwood's curved body in white shirt and black pants. The diptych thus becomes a triptych by representing in the middle section the disciplinary regime of *line* in the transition from color patches and the lines within brush strokes (at left) to figuration (right). Or, equally, it might be seen as the reverse trajectory: the existence of areas of color has created a strong sense of line (at middle) as one moves from predominant figuration (right) to seemingly disordered color patches (left).

A line defines and confines a spot of color. But a line must have a color to be seen and a spot of color has a mass that defines the line that borders. We will encounter theoretical accounts in this book that debate the priority to be accorded to either color or line with attendant consequences for aesthetic appreciation. Notice that the power of figurative *line* extends to the representation of invisible realities since the lines around Dagwood's shoulder and head depict his feelings of startled alarm. I will argue that color, too, may make visible the invisible, e.g., a feeling or tone. Moreover, we will encounter theoretical and practical situations in which color itself vanishes from the screen into a spectator's mental image of color that is the trace of thought (Chapters 5 and 6). This is analogous to the way that a *theory* appears to disappear into—gives way to—a person's experience of an object when experience acquires concrete *significance*.

Like color and line on canvas, philosophy (theory) works to elucidate aspects of our world that are visible and invisible. Philosophy investigates the conditions under which something comes to exist, the possibilities of its forms, and how one marks varieties when form transforms into new form.[7] To accomplish its goals, philosophy must draw lines to mark out and name various areas of phenomena by employing "concepts."[8] Like color and line, one may debate which comes first in thinking philosophically: the presence of a distinction—a line, a felt difference—or a general concept about finding and locating areas of such differences. Distinctions and concepts work to give shape to philosophical claims while networks of concepts provide philosophy with large-scale structures and patterns for its "picture" of the world. For the most part, I see no need to decide which comes first—distinctions or concepts, lines or colors—and will instead concentrate on outlining the myriad phenomena of cinema's color as a diptych: a philosophy of color together with the experience of color as an aesthetic.

TRIPTYCH: COLOR, CINEMA, REMEMBERED LANGUAGES

This book is a diptych where theory and practice, principles of color design and figurative cinema, absorb each other. Is there a hidden part in the book analogous to Lichtenstein's disciplinary "line" (in the painting's middle section) that acts as a catalyst between theory and practice, color and cinema? The mediating process that, in effect, makes this book a triptych will be the activity of "language," especially the *remembered* languages and insistent mental images a perceiver utilizes to draw lines and frame his or her viewing of screened color. The book will treat both philosophy and aesthetics as rooted in the many different ways we *choose to talk about*—and recount stories about—the features of a world that pique our interest and remain in mind. Such talk is not random but organized into procedures that are used to map and navigate a world by bringing forth relevant patterns. Narrative discourse is one of these fundamental procedures we use to create patterns in order to understand a world (as noted by Sartre in the epigraph). I will argue that color sensations do not emerge from a vacuum to become sensuous spots, but come already imbued with diverse sets of expectations, which, in turn, are tied to what might be called language-games and perception-games. In cinema, and in life as well, these games and routines intermingle with the stories we wish to recount about the actions we wish to pursue, the patterns we desire to find.

Thus I will claim that not all instances of color are solely sensation(al) or tied to emotions. Color may function cognitively or, if you wish, provoke the feelings of ideas that are tied to habitual ways of talking about being in a world. I believe that one can hear in color the languages used to express how we are cognitively situated in an environment. In this way color may become a medium for thought. I will argue that to picture color as merely sensation, spectacle, or symbol is reductive; doing so ignores important cognitive processes that act to mediate and bring forth patterns (as in Lichtenstein's middle section). I believe there are numerous ways for color to acquire the mobility of consciousness and adopt the fluidity of cinema. Color, mind, and cinema may be synchronized to become an integrated stream of acoustic, diagrammatic patterns that are intermittently moving and interrupted, permeated with decision and thought.

I will propose a theory of color that relies on a flow or "tracking" of a perceiver's working memory as opposed to a semiotic or "standing" theory of color based on symbols/signs retrieved from long-term memory. I will emphasize the *event* of color, not merely the fact of patches of screened color. This theory will be anchored in research from cognitive science and philosophy of language. The theory will benefit from the work of Ludwig Wittgenstein, who near the end of his life began to chart some of the language-games that give color movement.

Just as Lichtenstein's painting is divided into an abstract or theoretical part (on the left) coupled with a figurative part portraying a character's action within colored

space and within a pop cultural discourse (on the right), so this book includes a theoretical examination of the nature of color coupled with a proposed set of terms for analyzing colored patterns that emerge as concrete figuration. As indicated above, there is a third part lying within both philosophies of film color and vocabularies for analyzing specific color patterns that acts to meld philosophy and aesthetics, film theory and cinema's color. The third part brings to the fore the background languages of mind—our remembered languages—that constitute the expressive potential realized in a community's sense of both theory and color aesthetics.

GREY GRAY

Color, colour, colur, colore, cooler, culler are some of the names. A person is certain, though, that he or she knows what color is despite the spelling. Or, is there doubt? Is it only a matter of properly distinguishing, say, a more silvery shade of grey from a darker gray?[9] When attending to patches, do we simply acquire a saturated feeling about the degree of gray or greenish stuff or redness, and that's it, all of it? My point of departure in this book will instead be that our sense of colour is dependent on general patterns and schemes of colurs interacting with other kinds of configurations in film, such as the rise and fall of narrative actions, intermittent tensions among characters, dialogue, music, camera movements, rhythm, and changing backgrounds due to character movement. Colore is both sensitive to, and impinges upon, contexts. A viewer's consciousness of cooler springs not just from a patch on the screen, but from networks of developing culler patterns superimposed upon other aesthetic patterns.

As we shall see, color designs may operate to guide a spectator toward themes, emotions, beliefs, and insights not directly visible in film just as many things are not visible inside and outside of Lichtenstein's Dagwood painting. Aesthetic disputes, however, have arisen over how color works as well as the appropriate manner in which it ought to work. In weighing these arguments, we will need to examine a third part within theory and practice, namely, the concealed languages, rhetoric, and assumptions from which theory and aesthetic claims emerge. Claims about color arise from diverse and competing stories earnestly being told by critics. Extending Sartre's assertion in the epigraph, one might say that color comes alive when its story is being told. Accordingly, our task will be to analyze some of these stories.

NOTES

1. Jean-Paul Sartre, *Nausea*, trans. by Lloyd Alexander (New York: New Directions Books, 2007), p. 39.
2. For something along the lines of *Two Paintings: Dagwood* (1983), but more complex, see Lichtenstein's painting, *Reflections on "Interior with Girl Drawing"* (1990).

Concepts about pattern, relationship, and emergence are central to the arguments of the present book. An essential essay on the notion of "pattern" and its connection to various interpretive stances is Daniel C. Dennett's "Real Patterns," *Journal of Philosophy* 88, 1 (January 1991), pp. 27–51. Dennett sketches five theories of realism and advocates an intermediate approach called "mild realism." I use Dennett's ideas together with three other middle-level approaches to film theory in *Projecting a Camera: Language-Games in Film Theory* (New York: Routledge, 2006). For more on the problems of patterning, see Chapter 8, section, "Reidentification, Radial Association, and Derrida." For an introduction to similar ideas in "relational realism," see the special issue on "New Realism: What's Really Real?," *Philosophy Now* 113 (April/May 2016). For my comment on the relationship of New Realism to language, see "Letters," *Philosophy Now* 114 (June/July 2016), p. 36.

3. For an excellent survey of some of the historical and theoretical issues of film color, see Eirik Frisvold Hanssen's 1999 essay, "The Functions of Colour," available through open access at <www.academia.edu/292978/The_Functions_of_Colour> (accessed Feb. 19, 2015).
4. On the curious properties of brown, see Chapter 5 below, subsection "Non-spectral hues."
5. David Papineau proposes that certain mental images may function as labels or "phenomenal concepts"; for example, to function as a mental template for a notion of redness that has been constructed from seeing red things. A phenomenal concept names a pattern, in this case a prototype or model for kinds of redness in things. Note that this approach amounts to a semantic, but not metaphysical, dualism between mind and material. The approach may have been inspired by Wittgenstein, who argued that two different language-games—different sorts of assembled concepts or different language-fields—are applicable when talking about mental/phenomenal as opposed to physical/observational events. Papineau's phenomenal concepts would seem to arise in perceptual and episodic (first-person) long-term memory as opposed to a fundamentally different type of long-term memory known as semantic or generic (third-person) memory. These differences in types of language and memory regions, however, do not sanction a metaphysical distinction between mind and matter. In effect, Papineau applies the concept of "concept" to aspects of phenomenal consciousness (i.e., when arguing for the notion of "phenomenal" concepts) in order to argue that consciousness is no more and no less physical than material things, like the brain, described by other sorts of concepts. Consciousness emerges from brain with new physical properties. David Papineau, *Thinking About Consciousness* (Oxford: Oxford University Press, 2002). For additional details on thinking about a thing (e.g., film) within different frames and employing different vocabularies—i.e., different regions of language or language-games—to describe kinds of mental versus physical causation, see Branigan, "'Motion and Movement' and 'Sustaining and Other Causes'" in *Projecting a Camera*, sections pp. 152–160, 178–191.
6. If there is a mathematical analogy to the notion of emergence, it may lie with nonlinear equations. This suggests the possibility that a series of elements in an artwork might form an emergent pattern that *subtracts* from the whole of the artwork toward a sturdy minimalism, rather than the more usual case whereby an emergent whole is greater than the sum of its parts.

   An emergent whole has acquired a significant new *use* or *function* from that of its individual elements. Simple examples of emergence and levels at work are the "levels of narration" in a fictional text and the relations between a son and his father. Other examples: a list of lists of things to do and repeated identical embedding, e.g., a fractal. Consider also the five levels posited by Josiah Ober in "Explaining Hellas' Wealth" in *The Rise and Fall of Classical Greece* (Princeton, NJ: Princeton University Press, 2015), chap. 9, pp. 101–122. See also Chapter 7 below, notes 3 and 10.

A provocative analogy is to consider so-called deep neural computer networks (also known as parallel distributed processing) in certain artificial intelligence programs containing multiple levels/layers that search for patterns of patterns of patterns. . . . Such programs are designed to mimic human learning and have demonstrated astonishing results. Another way of describing the phenomenon of emergence is to define higher "levels" that "supervene" on lower ones. For an introduction to these issues, see George Musser, "Is the Cosmos Random?," *Scientific American* 313, 3 (September 2015), section "Do Your Level Best," pp. 91–93. For a heady philosophical introduction, see Verity Harte, *Plato on Parts and Wholes: The Metaphysics of Structure* (Oxford: Oxford University Press, 2002).

7. The importance of *theory* as distinct from a practitioner's "true belief or judgment" (expertise, know-how, *practice*) has been recognized at least since Plato's *Theaetetus*, where Socrates draws a powerful distinction between a "rational account" and "true belief" (187a–210d). What, then, is a "rational account" or theory? Though far from settled today, one might begin by saying that a philosophical theory is a network of assumptions, inferential procedures, and justified beliefs that address basic issues of ontology, epistemology (mind, language), ethics, and aesthetics. Theory provides an explanatory, often causal, context for practice. For various approaches to film theory, including my own, see Edward Branigan, "Teaching Film Theory" in *Teaching Film*, ed. by Lucy Fischer and Patrice Petro (New York: Modern Language Association of America, 2012), pp. 26–39.
8. Is there a relationship between concept formation and "pattern"? Are concepts formed by exposure to a series of examples that begin to coalesce into an implicit pattern based on felt resemblances and carefully chosen relationships? If so, a *complex* concept based on disjunctive, defining attributes will name not one, but a set—a pattern—of related patterns, as perhaps with the concepts "good" and "art." See the notion of "radial concept" in *Projecting a Camera*. See also Edward Branigan, "Introduction (II): Concept and Theory" in *The Routledge Encyclopedia of Film Theory*, ed. by Edward Branigan and Warren Buckland (London: Routledge, 2014), pp. xxi–xl. The notion of "radial association" is further discussed in Chapter 8 below.
9. On silvery grey as opposed to dark gray, see <www.greyorgray.com> (accessed Aug. 17, 2015). For more on gray, see Chapter 2, note 38.


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Chapter 2 **Living with Chromophilia**

ABSTRACT

What are the components of a robust chromophilia—a love of color—or its opposite, chromophobia? What brings forth approbation or disapproval in the form of a "glorious" or "gaudy" judgment about a color pattern? Such evaluative conclusions, which I call the "g-words" (there are eleven of each) lead toward an analysis of what lies behind various favored or disfavored aesthetic strategies. That is, one needs to determine which sort of story is being told about an aesthetic strategy, such as, color restraint and the pursuit of "delicacy." The coloring of the human face provides a test case, even when the face is simply described using words, since words create images and images words. The interaction of words and images is one source of the power of color to organize literary and media texts using the top-down resources of a perceiver's working memory.

The fluidity of color also leads to a rethinking of many theoretical concepts, such as, fiction, figuration, illusion, spectacle, impossible hues (e.g., bluish yellow), and the endless debate about color versus line, specifically, the *colore* versus *disegno* controversy in Renaissance Italy, which was a debate between Venetian and Florentine painting techniques. These concepts, including the notion of fluidity itself (flow, movement), must be removed from the metaphorical and made literal. Indeed the metaphorical must be made concrete. Consider: do fictions and metaphors, whether true or false, have no connection to our concrete world? The present chapter begins to open these issues.

> If perhaps they [painters] manage to think up something so new that nothing remotely similar has ever been seen before—something which is therefore completely fictitious and unreal—at least the colours used in the composition must be real. By similar reasoning, although these general kinds of things—eyes, head, hands and so on—could be imaginary, it must at least be admitted that certain other even simpler and more universal things are real. These are as it were the real colours from which we form all the images of things, whether true or false, that occur in our thought.[1]
>
> —René Descartes

COLOR IN MIND

Color is highly addictive, or at least habit-forming, when noticed and savored. Unfortunately, most films do not nourish a graceful chromophilia. Though colored, they fail to be in color. An object is merely color*ed* and lifeless when hues seem to have been pasted onto it, whereas being truly *in* color hints at a vibrant authenticity informing—*form*ulating—experience. Sergei Eisenstein explains the distinction:

> I advisedly say *in colour* and not *coloured*, to preclude any association with something coloured, painted.
>
> We must not allow the plastic austerity of the screen to turn all of a sudden into a piece of gay printed calico or gaudily painted postcard. We do not want to see such postcards on the screen. We want this new screen to show us colours in organic unity with the image and the theme, the content and the drama, the action and the music. Together with these, colour will be a new potent means of film impressiveness and film idiom.[2]

Notice that Eisenstein associates being "coloured" with stasis in film (clothing and "postcards") as opposed to "drama, the action and the music." Adrian Cornwell-Clyne seems to say much the same thing:

> Because, unless the coloration actually contributes some further value to the power of the impression made upon the mind by the film as a whole, its presence is unnecessary, and even disturbing and undesirable.[3]

But it may not be quite the same thing despite Eisenstein's mandate that color achieve "organic unity" with all the elements of film and Cornwell-Clyne's insistence that color contribute to the film "as a whole." Cornwell-Clyne is demanding that color contribute something "further" to the "impression" upon the mind made by a film as a whole; otherwise coloration is "disturbing and undesirable." It may be that Cornwell-Clyne has slightly extended Eisenstein's aesthetic directive and is thinking of film as *essentially* a black-and-white medium unless color can be specifically justified. This assertion raises philosophical issues about the degree to which black and white themselves may be colors as well as about the nature of realism in film. A discussion of these issues will be deferred to later chapters. Instead, we must first examine what it means for a film or other artwork to be *in* color (Eisenstein) by asking what lies *inside* color (Cornwell-Clyne) that is capable of functioning with other aesthetic elements to make a meaningful contribution to the artwork "as a whole."

The present question may be rephrased: what general considerations surround any kind of talk about color? How have writers sought to anchor their beliefs about

the world within vocabularies that include a role for color? What can be achieved with color?

For the most part, film scholars have not been helpful in delineating what Eisenstein calls a color "idiom"—that is, specifying a working method or language for color. Nor has there been much signaling of color's potential for creating a striking "impression" (Eisenstein, Cornwell-Clyne) on a spectator. The majority of film criticism and theory until recently has proceeded as if all films were made in black and white. Nonetheless, there do exist a diversity of possible color principles, oftentimes conflicting and contradictory. Such an irregular mosaic may be a fortunate situation for filmmakers since it leads to patterns of enormous variety. For a chromophiliac, experimentation is an exciting prospect.

This book will tackle the intricacies of color principles along four main dimensions:

1. *Conventions* for representing color (techniques, technologies);
2. *Languages* by which to express a felt consciousness of color;
3. *Memories* of color in culture by individuals; and,
4. *Sensations* of color.

I will briefly sketch each of these areas.

To fully understand the *conventions* surrounding filmic color, a series of links must be traced among cultural practices, commercial/business imperatives, available technologies, and aesthetic norms. Conventions result from an evolving matrix that can best be appreciated by sampling points of convergence or "knots" where forces crisscross, reinforce, or diminish one another, producing an event of color. The focal points chosen in this book will represent detailed theoretical and historical cases meant to illustrate the processes at work rather than chart an exhaustive history.

Here is an example of forces converging, where Eisenstein's aesthetic view in 1940, mentioned earlier, meets the view of a business leader seeking profit. In his autobiography, Herbert Kalmus, one of the co-founders of the Technicolor Motion Picture Corporation, reflects on the period 1927–1928, and has this to say about having color in mind:

> Producers still did not realize that a color film is not just a black-and-white film shot in color. If a script has been conceived, planned and written for black-and-white, it should not be done in color at all. The story should be chosen and the scenario written with color in mind from the start, so that by its use specific effects are obtained, moods are created, beauty and personalities are emphasized, and the drama

> is enhanced. Color should flow from sequence to sequence, supporting and giving impulse to the drama, becoming an integral part of it and not something super-added, a novelty. The use of color demands an aesthetic, just as any other form of creative endeavor. Producers, anxious to hurry their products to market, were not worried about quality, but we at Technicolor were hurt by criticisms on color.[4]

This does not mean that Eisenstein shared a set of aesthetic precepts with Herbert Kalmus and his wife, Natalie (who earned hundreds of film credits for color design from 1928–1950),[5] nor were their views historically congruent. It is meant only to suggest that these sorts of ideas from filmmakers and entrepreneurs did circulate and at times reinforced one another producing recognizable effects in films. There was a firm conviction among a few persons that principled color designs could be artistically and financially successful and that a film should therefore not appear as a series of postcards of haphazard color.

The second topic in this book investigates how color emerges from an environment *into consciousness*. Here we will need to sample ideas about the functioning of human physiology, psychology, and emotion, and how various writers have imagined links among these facets of mind. It will be especially important to look closely at the functioning of ordinary language. For John Lucy, our language use is the key to understanding the functioning of color:

> To conclude, just as we now recognize that color is not "out there" in the light but in our perceptual interpretation of light, it is time to recognize that the communicatively relevant encodings of visual experience do not lie "in there" in the biology but out in *socially anchored linguistic systems*. It is time we. . . started describing these linguistic systems in their full complexity. Then, and only then, can we actually address the issue of what regularities there are across languages and how they relate to thought.[6]

Language is the defining trait of humans, and not, for example, the ability of sight, which is shared with many other living creatures. P.M.S. Hacker argues that language embodies our true nature as human beings:

> There is nothing trivial about language and its uses, or about the confusions into which we are led through our failure to have an overview of a domain of grammar [a way of talking] that causes trouble. We are the kinds of creatures we are because we possess a language. Our distinctive capacities, e.g. our rationality, our knowledge of good and evil, and our possession of a conscience, our self-consciousness, our capacity for apprehension of necessary truths, are all functions of the fact that we are language-using creatures.[7]

A key assumption of the present book is that the significance of color for humans derives from "socially anchored linguistic systems" that embody—for good or ill—our species' "distinctive capacities" as defined within "encodings of visual experience" from specific environments. How we sense, select, point to, sample, touch, and manipulate properties of objects is tied tightly to our talk, to our daily grammars (Hacker), i.e., enmeshed within our present vocabularies for negotiating cherished plans and projects, for imagining what to do next, for making decisions. Importantly, projects are tied to our values. Thus examining how styles create artful objects with accompanying states of mind will reveal an array of sources within our cultural and political sensibilities. As Jeff Scheible remarks:

> When we. . . begin to ask more complicated questions about style [in artworks], we can begin to learn more about culture—because the ways in which we decide to fashion things tells us about our sensibilities, not only aesthetic, but also political.[8]

A third theme of the book revolves around the question of what makes color *memorious*. Are color spots alone responsible for what becomes memorable? One approach to this question derives from George Lakoff and Mark Johnson's classic treatise, *Metaphors We Live By*, which has had a profound impact on the fields of linguistics, cognitive science, and philosophy of mind.[9] Since the 1980s a series of books has explored how mental schemata located in various memory systems make a world for us by providing doses of ready-content to live by. Chapter 3 will propose some types of schemata that generate *metaphors to see color by*. Chapters to follow will probe the ways in which color perception has been modeled on a global scale by an assortment of literal and figurative geometries (e.g., color circles; hot versus cold colors) as well as analogized to painting, music, gardening, architecture, and even a film's soundtrack. Analogies, of course, are created by abstracting certain properties from two things, then selectively comparing them. Specifically, I will argue that film color is micro-managed by a spectator moment by moment through metaphors, stylistic prototypes, narrative, and interest. One does not look at film in a vacuum, but through a series of memorial layers of mind.

Color is also being modeled by the presence of history; that is, a spectator is subject to multiple conventions and traditions that have been institutionalized for rendering color. This is historical memory and its influence. For example, the particular look imparted to an image by the rare Technicolor triad, gold-green-violet, has a history against the backdrop of other standardized Technicolor triads and dyads. Depending on how much one wishes to know, there are levels upon levels of history with complexities radiating in all directions.

Color flashed upon a screen energizes both our perceptual abilities and memory/knowledge resources. (Memory and knowledge are virtually the same as Plato

saw clearly in *Theaetetus*.) What is crucial to a color's significance is how initial appearances *will have been memorialized* once the film ends—once all the local sensations and events of the film have been narrativized, compressed, revised, resituated, and made available for reminiscence and retellings through the normalized forms that make up speech, indeed oblige us to speak to ourselves and then others. From an initial myriad of sensations, personal significance will linger. This book will therefore delve into the nature of those processes that make possible a spectator's talk about the colors that he or she remembers having seen (con)figured. This will amount to a new way of theorizing the connection between language and film.

It may also happen that filmic color becomes *constitutive* of a vital object, event, or idea while being itself "absorbed" into the entity and hence becoming lost to immediate recall. Remember, for example, how the memory of someone dearly loved may soak into places, objects, travels, and events, and thus often be brought to mind; or perhaps a thought comes to mind for no apparent reason, as in Freud's notion of the uncanny. This "melting" and disappearance of color *into* something else—to return later—is a special case where a hue acquires significant associative value. This situation should not be confused with the exercise of coloring a thing with forgettable, superficial hues, such as when one speaks of "coloring the world" or "coloring the truth." The important point is to appreciate that there may be a disparity between an initial presentation of a physical color appearing on the screen and, as noted by Eisenstein and Cornwell-Clyne, its final "impressiveness"—including a final cognitive and emotional effect that may be hidden within something else, such as the film "as a whole."

Memory makes color as movable as many other aesthetic features of an artwork. Frank Sibley argues that there are important differences between the physical and formal features of an artwork that lead to differences in how an artwork is understood.[10] If the formal features (relations, patterns) of an artwork are *not* indissolubly attached to its physical features, i.e., they are movable, then for Sibley at least some aesthetic values of an artwork are also *transferable*. This touches on a deep philosophical problem, which is explored in the context of a film's soundtrack by James Lastra, who divides sound theories into those asserting that a perfect identity can never exist between an original sound and its reproduction—its copy—as opposed to theories asserting that what is important is that a particular reproduced filmic sound must be *believed* by an auditor to be *typical* of such sounds and sufficiently *familiar* so as to elicit an intended meaning or response.[11] If the latter state is achieved, then the original, physical acoustic situation of the recording has been shifted (moved!) into a new physical context and the imperfections of copying are irrelevant. Similar considerations apply when a copied sound is again copied. The approaches of both Lastra and Sibley are congruent with the notion of fully movable aesthetic features, like color.

Are meanings and responses, then, to sound, composition, color, etc., merely illusory, intangible, unproven, fleeting, and *not* really real, despite our feelings of intimacy, possession, and mental reliance upon them?[12] This question is meant to cloud, and raise doubts about, our strict use of the words "illusory" and "real," immaterial and physical, and thus to inquire what is meant by the word "identity," or rather, to determine what use can be made of the word "identity." (This topic will be further discussed in Chapters 7 and 8.) Perhaps what is important is to map a series of *connections* and relations between, say, moving color and meaning. It may be that at least some set of connections may be detached from a physical site and shifted to a different physical site.

The type of theory that stresses immobility is premised on ontology and strict identity. Sibley and Lastra's type of theory that allows mobility is premised on epistemology and meaning. There are obvious parallels here to color, namely, to the question of how much aesthetic importance should be ascribed to the degree to which color has been faithfully reproduced, given that filmic color is being copied from the world and also will never be copied exactly to a particular film and to copies of the film, not to mention the depredations wrought by the passage of time on the specific viewing contexts that influenced an audience's original involvement with, and expectations about, color. For an ontological theory, aesthetic values are irretrievably and continually being lost. For an epistemological theory, however, the question of loss remains dependent on the manner in which sensory qualities are being shifted as well as dependent on the architecture of a perceiver's memory and responses.

My approach in this book embraces an epistemological theory. Hence, I will need to examine the cognition of color, broadly conceived to include both its visible and invisible forms as it moves through a perceiver's memory and knowledge.

Finally, a fourth theme will involve color in relation to all the other impressive *sensations* of film. There are many potential sensory thrills, starting with:

> shape, size, mass, texture, edge, line, angle, orientation, illumination, geometry, projection, contraction, loudness, pitch, rhythm, attack, noise, balance, tension, harmony, decoration, overlap, transparency, implicitness, depth, perspective, motion, speed, acceleration, direction, scale, progression, and transformation.

What is the place of sensory qualities and how are they placed in our discourse about film? I believe there is something to be learned about the power of film spectacle while studying the example of color.

We might begin an inquiry into a spectator's sensoria by gathering common accusatory adjectives that are inevitably called forth when color appears on the wrong side of a critic's aesthetic judgment, i.e., when a critic senses that color has too much of spectacle within itself, causing his or her critical distaste to mount like

a fountain. Spectacle is said to encourage gratuitous ornamentation, insincerity, and confused emotion at the expense of reason, restraint, refinement, clarity, good sense, and thus proper good taste. Recall that two of the following words, gaudy and gay, appear as negative characterizations in the Eisenstein passage quoted earlier.

gaudy ~ tawdry, showy
garish ~ livid beyond vivid; lurid, florid, flowery
glaring ~ harsh, blatant
glossy ~ alluring on the surface; false, shallow
glittering ~ glinting, showy, deceptive guise
gay ~ frivolous
gilded ~ gilded lily, flowery
gilt ~ glitzy
gauche ~ louche
graceless ~ ungainly, gawky
ghostly ~ ghastly, gloomy, grayish, pallid, pasty, ashen, weak, dead

The mistrust and fear of overwrought emotion and spectacle is deeply rooted in Western ethics—as with the Hellenistic and Roman Stoics—as well as aesthetics. For Aristotle:

> The Spectacle has, indeed, an emotional attraction of its own [in itself], but, of all the parts [of Tragedy: plot, character, thought, diction, song, and spectacle], it is the least artistic, and connected least with the art of poetry. For the power of Tragedy, we may be sure, is felt even apart from representation [public performance] and actors. Besides, the production of spectacular effects depends more on the art of the stage machinist than on that of the poet.[13]

Rosalind Galt adds that an "empty spectacle" is denigrated as the merely "pretty."

> The rhetoric of film theory has insistently denigrated surface decoration, finding the attractive skin of the screen to be false, shallow, feminine, or apolitical. . . . This impetus is naturally not universal. . . but the suspicion of "prettiness" nonetheless remains strangely resilient. We might think, for example, of the commonplace of "empty spectacle" as a figure of critique in film writing from journalism to theory. I would suggest that this critique itself must be interrogated.[14]

Faint praise for the pretty is to brand color as "charming."

It is true that certain bodily and experiential sensations may linger in memory to one's detriment. Rudolf Arnheim:

> After I had seen my first color film and left the cinema, I had a terrible experience—I saw the world as a color film. The Alban hills stood, a common soft lilac color behind the chain of dark green pines, topped by an emerald green sky—everything was blatant in its poisonous color, and presented a chaotic, fiendish, discordant picture. This lasted a few minutes, and then, without the light having really changed, my usual evening landscape was there again in all its placidity: the far-off mountains, the solemn trees, the Roman sky in the last light of day.[15]

Arnheim does not say whether the color effects that he experienced might have been due to the uses made of color by the particular narrative of the film he watched, i.e., what lingered in his perception may have been colors-as-narrativized-badly. Perhaps in other circumstances, cinema's color would alter perception in more beneficial ways.

On the positive side of critical language and pleasant memory are these g-words:

glorious ~ fulsome
gorgeous ~ beauteous
glamorous ~ enchanting
glowing ~ gleaming, glimmering
glassy ~ gossamer
grand ~ elegant
graceful ~ fluid, lithe, gracious
genteel ~ gentle, delicate, subtle
gilded ~ golden, glazed
gilt ~ gilt-edged
ghostly ~ ephemeral, delicate, spiritual

Typically, g-words are offered to delineate what is naughty and nice about color. When used in an evaluative context, they operate as *conclusions*. This means that the fine grain of a critic's argument will need to be carefully investigated in order to determine precisely which quality of a hue under which aesthetic norm, and which interaction between a hue and some other sensuous or narrative feature, has caused offense or delight. Moreover, some g-words may be applied to both positive and negative attributes (as "gilded," "gilt," and "ghostly") or used ironically.

Thus to properly evaluate evaluative statements, one will need a broad understanding of the functioning of color conventions within states of mind and across community languages, which, in turn, requires an appreciation of the four dimensions of cinema's color, namely, conventions and practices, expressive languages, memories, and the status of sensations. Acquiring this expertise will allow one to

make informed judgments about the presuppositions that drive a critic's claims and thereby may open the way to revising a critic's judgment.

OF DELICACY AND NOT (CORNWELL-CLYNE)

Let us tackle the details of a specific aesthetic claim to illustrate how a critic constructs an argument. Adrian Cornwell-Clyne complains not only about garish color in films but also about a superabundance of glorious color. He urges instead a restrained, subtle look that promotes "delicacy."

> "Gorgeously glorious" colour cannot equal the intense pleasure the average individual discovers in a sober, cool and mirror-like palette. We ought to study Vermeer and Velasquez rather than the Baroque Rubens and Titian. It is easier to compose colour massed like the fruit on a market stall [or the fruit on Carmen Miranda's headdress?] than to match the delicacy of a Girtin watercolour. It is grossly untrue to dismiss subtle colouring as wasted upon an apathetic public. Beautiful things have a singularly final way of winning all hearts and all minds, whether they want to be won or no. Let your tones be difficult to describe and difficult to remember.[16]

I wish to focus on Cornwell-Clyne's advice to cinematographers about designing a palette that spectators will find "difficult to describe and difficult to remember." A norm of delicacy makes color enter the mind by almost vanishing into a "cool and mirror-like palette" that seems to reflect a quiet world of color surrounding a beholder. For Cornwell-Clyne the tones are to be difficult, though not forgotten. Too often in judging a color palette, he says, one overlooks its special memorial character, which is the "singularly final way" of "beautiful things." A major theme of this book will involve the varied and diverse impact upon—the shifting balance between—bottom-up compositions and top-down predispositions as a perceiver wrestles with color designs that are meant to be remembered, whether quickly or with difficulty, but that do not fit easily into schemata or stereotypical patterns.

Should a filmmaker wish to create a delicate, though not brazen, tonal structure that lingers—one that is perhaps judged to be subtle, subdued, intricate, precise, mysterious, strange, or novel—a filmmaker must know how an audience understands, and bears in mind, displays of color. What patterns can be seen? How will an audience react? Obviously there are many variables, not least the color conventions of a community that provide a backdrop for audience expectation. A filmmaker must be keenly sensitive to cultural practices in order to construct the requisite feeling. Importantly, the norm of delicacy depends on exploring the resolving power of sensory and cognitive systems, particularly the *threshold* between two hues or between two degrees of lightness.[17] The emphasis will fall upon noticing

small, subtle differences where a given difference may be detected as often as it remains undetected. That is, these kinds of aesthetic differences are not certain to be recognized and hence present demanding tasks for spectators.

An elaboration of Cornwell-Clyne's norm of delicacy would find that perceptual thresholds have a dual nature: at a basic level, a person may detect the sheer presence of something while a more complete identification of the thing, if it occurs, will happen at a higher threshold. Artists, of course, can work with both of these kinds of thresholds. Often subtle discrimination is required to sense relevant *differences* among two or more closely related percepts or ideas, themes, feelings. . . . Judgments about the grading of stimuli along a spectrum will be affected by context, expectation, and nearby stimuli. This is the idea of "theme and variation," which itself has many variations,[18] including two nearly identical colors or two successive similar patterns of color. Also, the difference between two percepts may occur inside a single color, i.e., there may be very slight, almost imperceptible, variations in hue, lightness, texture, or saturation within an ostensibly single swatch of color or monochromatic design.

To make matters more complex, a hue may slide into and temporarily hide within another hue! This disguised or implicit delicacy is difficult to envision since it requires one to hold onto the memory of a hue and to await its (possible) return after another transformation. This situation will be explored more fully in Chapters 3 and 8 in the work of Sergei Eisenstein and Paul Coates. For now, a passage from Christopher Knight offers a preview of implicit delicacy. Knight observes a series of subtle nuances within "translucent veils of vaporous color" in the paintings of Richard Diebenkorn's "The Ocean Park Series" (1967–1985):

> The narrative in these paintings is a story of their making—of a vertical blue line that gains a violet-rose shadow as it tracks [like a tracking camera?!] down the edge of a canvas, turning off at an angle like a refracted ray of light and then sliding beneath a wash of luminous gray, only to emerge at the other side as a little wedge of canary yellow. Or, of a field of brushy green that, the longer you look, slowly gives up layers of underpainting in hues you can't quite put your finger on. A shape here echoes one over there, a line pulls your eye across to a surprise waiting on the other side, a mark of color that seems to hang in midair vivifies complementary colors in your peripheral vision.[19]

To return to explicit forms of delicacy, two components may be operative: first, a testing of one's perceptual acuity for the dual thresholds of a thing (presence and identification); and, second, discerning the difference among several percepts arising from separate areas or else from within a single area, such as an area of "white on white" (as in the 1967 song by Procol Harum, "A Whiter Shade of Pale,"

or in Seijun Suzuki's 1966 film, *Tokyo Drifter*). Indeed, a viewer may be required to decide whether there are distinct areas that form a pattern or only a single area responsible for nearly identical percepts; gray on gray, red on red. . . . An aesthetics of delicacy is designed to increase the density and number of possible responses of a viewer within a given perceptual range. The viewer is being prodded to make finer and finer discriminations and to create possible *new networks*. Moreover, lying within the interstices between two perceptions may be vague intuitions and partially formed uncertainties—the space of the true and "nearly true"—that may be explored and shaped by an artist.[20]

Expanding Cornwell-Clyne's approach to delicacy, one finds that the vague intuitions and faint memories that may arise from testing thresholds of color, plot, theme, emotion, and point of view are suggestive of why subtle color may be described as "muted" using a sonic metaphor. Color that is muted is quiet or silent as opposed to "loud" and "gaudy" as well as—by applying two closely related, venerable aesthetic principles—nonintrusive and unobtrusive.[21] Goethe, in discussing "Pathological Colours," offers a highly dubious sociological reason for muted color:

> It is also worthy of remark, that savage nations, uneducated people, and children have a great predilection for vivid colours; that animals are excited to rage by certain colours; that people of refinement avoid vivid colours in their dress and the objects that are about them, and seem inclined to banish them altogether from their presence.[22]

Again, garish colors are disfavored, but for spurious reasons. More soberly, one might say that what is at stake in an aesthetics of delicacy is an attempt to encourage a viewer to attend ever closer to an artwork in order to foster a sense of liminal or near-liminal states, whether of percepts or concepts, concrete stimuli or narrative and structural abstracta. There is a belief that encouraging spectators to engage with sensory thresholds and slight differences will lead to experiences of boundaries of new types and to the conditions that define boundaries. In this way, a spectator may become aware of what it means generally *to frame* something (within boundaries), to bring a thing into prominence, to announce its existence. Clearly, a color's delicacy and "lightness" of "touch" leads toward many basic psychological, aesthetic, and cross-modal issues.

But can't we say that similar issues will arise with other types of color displays less delicate? One hopes a close examination of any film's matrix of color will provide insights about its functioning and about the interpretive schemes we choose to deploy when explaining meanings we claim to have found. The following color configuration, postulated as a "cycle" and arrayed in tabular form by Cornwell-Clyne, begins to suggest a range of inquiry.

Primitive: Monochromatic. Low saturation. Sober but strong.
Early: Simple contrasted hue intervals of medium saturation. Effects of increasing vitality.
Middle: Three or more hues (chords) of high saturation, vigorous and powerful.
Mature: Compound groups of many hues and covering a wide range of saturation but combined with utmost subtlety. Learned and distinguished.
Late: Extremely elaborate combinations exhibiting utmost richness. The period of magnificence.
Decline: Simpler but over-emphasized and vulgar in effect. High saturation but hue contrasts discordant and "blaring."

Cornwell-Clyne summarizes the cycle in the following way: "simple oppositions of marked contrast at the primitive stage—subtle and multiple combinations at the height of power; at the end over-elaboration, over-emphasis, over-richness, vulgarity, ostentation and universal extravagance."[23]

All four themes of this book have appeared in our brief dissection of Cornwell-Clyne's aesthetic claims. He winds together descriptions of a supposed *convention* or cultural norm for color combinations (e.g., creating a monochrome with low saturation) using a psychological *language* or rhetoric (e.g., a certain monochrome producing feelings of soberness and strength) based upon the *memory* of a human life cycle projected onto the present (i.e., various phases of primitive stages—birth—through decline and death). Metaphors and analogies with body and mind abound, especially when seeking a place for *sensation*, which Cornwell-Clyne worries may easily expand into a spectacle that will usurp mind, replacing rationality with "blaring vulgarity."

In effect, Cornwell-Clyne selects those qualities of color he deems most important and then organizes them into a pattern that develops through time by mapping qualities onto the stages of life—where maturity is aesthetic delicacy—and onto qualities of being—sober, strong, increasing vitality, vigorous, powerful, learned, distinguished, and vulgar. Surely, however, other qualities and mappings are possible. Cornwell-Clyne's approach illustrates quite well that explanatory schemes can be created to chart distinctive features, affinities, tendencies, and large-scale patterns of color organization, including the possibility of matching aspects of these schemes to, for example, diverse bodily and mental states, narrative structures, themes, and sonic properties ("chords," "blaring").

The present book will offer an approach that differs from that of Cornwell-Clyne in charting the effects of color on the mind. There will be less rigidity in determining the *intrinsic* effects of color patterns. The reason is that the effectiveness of an

explanatory account must involve the actual employment and dynamic unfolding of color in particular situations, i.e., how color functions among all the systems of a text and how it impacts spectators moment by moment whether delicate or vivid in design. Thus in appropriate circumstances even the "vulgar" and "discordant" color that marks Cornwell-Clyne's phase of "decline" may have important uses and not simply indicate a premature deterioration of one's sensibility. Nor is aesthetic delicacy necessarily a return to the state imagined by one writer as "a rageless, fleshless, colourless whiteness."[24] Instead, descriptions of color design must be fit to larger contexts and themes, and to specific problems that the analyst wishes to solve. Color must be minded in a variety of manifestations within specific frameworks rather than colored by prescription.

As Cornwell-Clyne indicates, color engenders emotions and may thus be said to itself possess a "body." Color acquires a body when one focuses on its relationship to a spectator's sensibility. For Jacqueline Lichtenstein, sensibility links directly to feeling and flesh:

> The pleasure of color is, certainly, a pleasure of the eye; it is so to the supreme degree. It emerges from the spectacle of flesh as a desire for touch. . . .
>
> Understandably, nothing is more difficult to paint than flesh. The artist must not only capture the fleeting tones, master the delicate glazes, evoke the subtle shades, catch the imperceptible transparencies but also provoke an almost phantasmal perception in the viewer. He must give the viewer the illusion of feeling this matter's depth, of touching this supple flesh and seeing it quiver, of seizing all the impalpable movements of this body, as if painting were accessible to all the senses.[25]

Let us next examine what conventional representations of flesh can teach us about the body and allure of color.

FLESH-COLORED: OF TEARS AND DIAMONDS (KALMUS, UPDIKE)

Being simply "glorious," while remaining free of Cornwell-Clyne's intensified and negative "gorgeously glorious," was a frequent description applied to the special look of Technicolor.[26] There are many complexities that must be recounted in a story about how a series of cunning management decisions enabled the Technicolor Motion Picture Corporation to create a Hollywood product that was able to earn such universal praise from the 1930s into the 1950s. One of the vital parts of the foundation for the

commercial and aesthetic dominance of Technicolor was the primal *fact of flesh* and its distinctive coloring. For Herbert Kalmus the face was an article of faith:

> I argued that the purpose of most motion pictures is to tell a story and that people are the means of telling that story and that faces and expression are of greatest importance in the telling. I felt that it was precisely there, in the beauty and expression of human faces, that Technicolor did its most useful work.[27]

The "Technicolor tan" was how critics referred to a unique coloring of faces achieved through the specific reproductive capabilities of the Technicolor process (see Figures 6.4, 6.5, and 6.8). Technicolor's pre-production research and imbibition printing[28] were meticulously calculated to bring out a hardy, suntanned tonality to the face—a warm amber tone sometimes verging on burnished leather. Imbibition (dye-transfer) printing—which produced greater color saturation and softer, waxier tones for faces due to the lateral dispersion of dyes—along with the associated style of a "Technicolor tan" for faces were introduced in 1927 in a two-color process known as Technicolor Process Number Three. Softer tones also resulted from contact printing, rather than optical printing. Competing two-color systems from other manufacturers used red and green or red and blue, while Technicolor used red-orange and blue-green (cyan) to achieve its facial coloring. Beginning in 1932, in addition to imbibition printing and warm/cool shot compositions, the three-color, three-strip process of Technicolor Process Number Four painted actor's faces by employing special films, filters, optics, and prisms along with precise lighting, exposure, costumes, sets, wigs, hair colorings, and specially designed Max Factor pancake cosmetics.

Technicolor had available five standardized shades of gray that could be used strategically to keep hues and flesh tones pure, since gray absorbs or minimizes ambient reflections from nearby hues. For example, a white shirt without a light gray component will reflect the color of the sky onto a character's face. Reflections may also be cast from an actor's costume. Other light colors and neutrals could be employed to reduce reflections. A second important use for Technicolor gray was to enhance the appearance of delicate surface texture in whites. (Recall the earlier discussion of the aesthetics of delicacy.) In addition, only Technicolor's imbibition printing allowed in-shot adjustments to color timing. Eastman Kodak's integral tri-pack print stocks, which were used by Technicolor's competitors starting in the 1950s, permitted corrections only between shots (prior to the use of digital intermediate in 1998).

There were exceptions to the "Technicolor tan," notably for children and heroines, whose faces were typically a hardy and deep but creamy pink as if they were feeling permanently "in the pink." Maureen O'Hara's natural coloring of red hair, fair skin, and blue eyes was perfect for Technicolor. These renderings of flesh should be contrasted with the actual color of Caucasian faces, which generally are light

grayish-pink as well as contrasted with the facial colors achieved by Technicolor's competitors, which ranged from chalky white—as if a character's demeanor had been sapped by perpetual fear—to a cold, receding bluish-gray to a murky brown.

Most importantly, the warm advancing tones of Technicolor's faces were perfectly matched to Hollywood's concentration on character-driven stories, where facial expressions—more so than any other expressive part of the body—were used to telegraph emotions and make motives instantly salient. Even a lack of expressed emotion was attached to a character's state of mind; for example, to signal concealment of a secret or a treacherous failure of compassion. In short, Hollywood demanded that a lively visage become a manse with windows onto the soul. Technicolor well understood that there were hefty profits to be reaped by making actors' faces attractive and legible.

Focusing a viewer's attention on communicative faces already suggests that color can serve in more ways than merely describing the visible gleam or dimness of a surface. Color may be drawn resolutely into an expressive flow to become a catalyst for many sorts of dramatic actions and relationships. Let us explore the unfolding of such a process—step by step—by considering a few examples of how the invocation of color can foster complex psychological effects even where the effects are being generated through verbal methods that bring imagery to mind rather than through visible patches on a screen or painting.

To this end, let us consider the following five and a half lines from a John Updike unrhymed sonnet, "Perfection Wasted":

> . . . . those loved ones nearest
>
> the lip of the stage, their soft faces blanched
>
> in the footlight glow, their laughter close to tears,
>
> their tears confused with their diamond earrings,
>
> their warm pooled breath in and out with your heartbeat,
>
> their response and your performance twinned.[29]

The poem focuses on body parts as the measure for its meaning, explicitly and even implicitly (foot/light, ear/rings, heart/beat). The poem uses the face ("soft faces blanched") as a hub for the emotions that will be modulated and extended as the sonnet progresses. The line, "their tears confused with their diamond earrings," introduces a key emotional complication to the depicted event. What is it that motivates the elliptical two-way metaphor between "tears" and "diamonds" that become fused, con/fused? We make this connection because of an image summoned by our visual imagination. In reality tears and diamonds aren't at all alike: one is

liquid and tied to the body, the other solid and inorganic—they have many dissimilar properties—but our memory sees an idealization that brings them together, i.e., we imagine the same shape and size (as in teardrop diamonds) as well as sharing properties of color, such as apparent hue, lightness, clarity, shine, and translucence. Tears and diamonds are also related to one another by their proximity to significant sensation and emotion, i.e., to the eyes and ears (ear/rings) of a face. Moreover, the "loved ones" of the performer press up against the "lip" of the stage as if to (re)join with the performer's own lips, face, and body: their "breath in and out" synchronized to the performer's "heartbeat" until the moment when "laughter" and "tears," "response" and "performance," are "twinned" through identical movement (conjoined, "warm pooled") together with a mutual response/release.

We continue to wonder about the tears, since tears of laughter or orgasm ("in and out with your heartbeat") may also be bracketed by sadness (*post-coitum triste omni est*). We see the faces of the loved ones as pallid and "blanched"—made "soft" in a low (lower?) "glow"—and we realize that very near the words "confused" and "lip" (provoked by the words "nearest" and "close") are other words not directly in the poem that would bring forth an image of trembling lips in a "blanched" face that would spill into tears of grief: *la petite mort* projected onto a future, literal death, the death of the poet. This would transform the scene from portraying several kinds of pleasures of the flesh (flushing) into a new occasion: that of the poet's funeral where the "stage" is a casket and the poet is giving a final, miraculous performance filled with both joy and sorrow for entertainer and "loved ones" alike. The temporal ambiguity of the word "loved" marks the duality of "tears" and "diamonds," since the word may refer to ones presently loved and present for the poet's "performance" or to ones from the past who were once loved and present only within the poet's memory at death. Similarly, the ambiguity of "close to" in the phrase "laughter close to tears" may mean either laughter producing tears in reaction to the poet's performance or laughter followed by tears at his absence. Various possibilities rise and fall with fluctuations in the poet's memory.

The end of the sonnet asks who could perform again this special act. The conclusion: "That's it: no one; / imitators and descendants aren't the same." The import of the final line of Updike's sonnet, titled "Perfection Wasted," is that a perfect performance of those unique skills that have been hard-won by a person during life—specifically, this poet's aesthetic prowess and power—is to be wasted in death. One of Updike's recurring themes as a writer is the idea that an individual is fundamentally solitary but restless, and that he or she alone must encounter (i.e., face. . .) transience, transition, and death. Again, the focal point for eliciting these ideas and feelings is the face, both radiant in its silvery color (tears, diamonds) and simultaneously "blanched."

Before reaching the conclusion to the sonnet, however, a reader must see and appreciate the poet's link between "tears" and "diamonds," and that link is forged to an

extent by our imposition of the shared qualities of color between tears and diamonds (the two con/fused) that, while false to nature, is true to culture and true as well to a convention of seeing; that is, we *remember* how tears and diamonds are *said* to display certain qualities of color when they "glisten" and "sparkle" and what it means—what it feels like—to cry at, and to (un)dress for, a special occasion. We understand, too, that it is usually women who wear diamond earrings and who are thus being singled out as "those loved ones" who are "nearest the lip of the stage" and whose inviting lips are within reach of the (male) poet or else conjured by the poet's memory as having been reached and touched in a past time. The diamonds serve as the fantasy of making the ephemeral everlasting; the tears serve as loss, the color as catalyst.

The words of the sonnet push us to see a common color binding diamonds and tears—a sparkling white or luminescence—that, in turn, allows the exchange of other properties between diamonds and tears, including the exchange of similarities between *occasions* (glamorous women, sex, memory, death) in which diamonds, tears, or both are present, literally or metaphorically, thus expanding the sense of the sonnet by expanding implicit, related scenarios. Furthermore, if a thing or relationship is said to be like, or to rest upon, shared color, then to a degree color becomes *like* that thing or relation, and part of a thought process. Color condenses scenarios, or equivalently, radiates outward to be absorbed into scenarios.

To put it another way: a casual appearance of color can be made to work causally and in so doing can participate in a complex set of ideas that are not immediately visible in words or on a screen. One can, however, imagine the depicted event(s) of the sonnet being transformed into film with whatever detail and rhythm necessary to capture any given interpretation of the sonnet. That is, a film might be poetically reverse-engineered from the sonnet's mental images, which is exactly the point of thinking about color flashing among word, screen, and mental image. The paradox is that in the case of film viewing, where colors are quite literal, i.e., not created by words, color is especially apt to be overlooked because the picture in our mind of viewing a film is so strongly associated with a productive *surface*, namely, the screen. We simply tend to look through the screen toward a scene of action and objects, leaving behind colors unseen. But even when this is true, colors will continue to exert an influence on our interpretive activities as we attempt to frame and reframe within inner languages—and, above all, with explicit retellings to ourselves and others—the significance of what is being remembered as having been seen through the screen. Color is not forgotten in the stream of thought because it is not seen or is overwhelmed by fiction.

It may be that color continues to work nonconsciously when unseen. For Plato, the being of color is closely connected to feeling. He believed color to be the result of a titanic struggle between two streams of fire—one emanating from an object and the other from the eyes. For Plato, specific hues emerge from four basic ones—white,

black, red, and "bright"—as the dual firestorm is variously extinguished in the eyes by tears.[30] Eyes, face, feeling, and color link closely.

FICTIONS PAINTED IN LANGUAGE (UPDIKE REDUX)

> Everything in fiction is a tool. Pain. Betrayal. Even Death. These are. . . you know. . . these are like different colors on a painter's palette. You need to use them.[31]
>
> —Ted Cole

The fictional Ted Cole is correct that everything in fiction comes from a familiar tool box—locales, actions, obstacles, coincidences, friendships, dialogue, character traits, glances, gestures, entrances, emotions, enigmas, endings—everything imaginable, and not least, Ted Cole himself along with an immense range of stylistic and formal happenings. These are the many instruments of narrative, "like different colors on a painter's palette," as Ted Cole, or someone or something speaking through him, says. However, as we've just seen with John Updike's sonnet, this dictum may equally be *reversed* so that the flow is from different colors *to* the instruments of narrative; that is, color itself may function as a tool to shape and evoke ideas, emotions, desires, qualms, doubts, doubts about doubts, and much else, perhaps all else in fiction.

Color theorists have long been deeply enamored with the emotional implications of color—how a specific color sensation may be psychologized, engender a mood, or even be possessed innately by a specific mood. And filmmakers have created designs on this basis. Does such a design prove that the mood was all the time *inside* the color? Is it not possible that a fiction created with one idea in mind may be taken to mean something else, since a fiction may be fit to many [quite real] contexts? For now, I wish simply to explore how color may enter into language by being transformed into an invention of language. In this manner, color simply becomes an agreed-upon fiction or device capable of operating at quite abstract levels of discourse.

Let us begin with the idea that critics have largely misunderstood the nature of "fiction," thinking of it as a made-up fantasy or outright lie apart from real life. This has led to further devaluations of color, where gaudy and garish colors are said to be appropriate only for musicals, nonsense comedies, swashbucklers, historical romances, mythological settings, exotica, and fantasy sequences in films. The belief is that while insistent or excessive color is intensely real on the screen, it is only seemingly so because of its appearance within an invented scenario and so is best suited to circumstances where realism is not required. The hesitation over the nature of color—indisputably real, yet illusory and suspicious in a fiction—makes

of it something not exactly belonging to this world, neither of the flesh (for the screen is inanimate) nor ascending to pure thought and spirit because indisputably sensuous. Instead, color seems to be a thing fabricated for the screen, corrupted and corrupting, both too real and not real at all.

The problem with such beliefs is that the notion of fiction has been wholly misconceived. Fiction should be thought of as merely a device (like a counterfactual supposition) for talking about life and attitude by posing generalities and weighing contingencies. When color is taken up by language and fictionalized, it should be seen to be no different than any other device used to fashion a view of life, not feign a view. A fictional discourse is neither deceit nor simulation; it is an elaborate sequence of hypotheticals taken from (and when true, returning to) life and experience. Indeed, one might argue that representation in general is premised on an implicit counterfactual: is there a tree that stands behind the word, "tree," or the photograph? The ultimate truth, falsity, relevance, or usefulness of a given representation or fiction, of course, is a separate matter since a hypothesis may turn out to be inadequate to the issue at hand.

Moreover, fiction should not be confused—as it often is—with narrative structure and narration, which are concerned with particular sorts of *organizations* or configurations of information, e.g., creating characters and linking scenes of action, defining motifs, measuring temporal progressions, and dramatizing or concealing an observer of events. Fiction and narrative are not the same and so they can neither be compared nor contrasted. Nor is nonfiction any closer to narrative even though the two may continue to appear together, as in historical narrative, biography, and the stories we tell about our daily lives.

A proper conception of fiction leaves the nature of color untouched as a discursive tool. It matters not at all that something interpreted fictionally complicates a spectator's mental state. As described by Christian Metz:

> Here is the spectator, oscillating between belief and disbelief, between awareness and oblivion in front of the spectacle, between complete alertness and a state of reverie, a very particular reverie, piloted by real perceptions.[32]

Although a spectator's mental 'oscillations' when interpreting fictionally may not be conscious, Metz's characterization points toward the psychological struggles that underlie acts of construing fiction.[33] These kinds of competing mental states need to be taken seriously when analyzing how color impresses. "Reverie," for example, suggests daydreams, fancies, and fears that are linked to expectations and memories. A reverie may well be guided ("piloted") by the practices of everyday color perception and by conventions for displaying physical color on screen. In this

manner, color may become a tool for thinking whose value depends on the vocabulary available for talking about present and anticipated experiences.

Along with a spectator's 'oscillations,' Metz mentions film theorizing, which would include conjectures about color. He says that theorizing is an enterprise invested in weighing hypotheses while searching for a meta-vocabulary to yield accounts of still greater generality. The power of film theory, he implies, "comes from such poetico-didactic fictions [as does also, he says, the notion of an 'enunciatee' in filmic enunciation], from these *manners of speech* (as when we speak of table manners)." Metz adds that when well understood, these poetico-didactic fictions "give theory's walls a fresh coat of paint."[34]

But, one may ask, is "paint" the right metaphor for color in expressing the connection between concept and folk practice, between theory and talk of what we value? (cf. Figure 3.1). Though somewhat enigmatic, Metz's comment, I believe, points to the shaping power of the conventions and tools of everyday language to figure (configure, confine, define) states and objects; in this case, to theorize the state of film viewing and film as an object—perhaps to contain film within a set of rooms ("painted walls") in one or another sort of building. Here, a given film theory is like a building or a set of rooms in a building. In this approach, theorizing depends upon selecting among routines (routes for thought, "manners of speech") in order to interact in a mannerly and suitable fashion with particular films as themselves objects (tables, rugs, lamps) in painted rooms. There are numerous ways to recruit paint and color, tints and tones,[35] into an image of theoretical discourse that aims to construct (hypothesize, frame) the nature of film: not by applying the discourse of scientific objectivity, but by adopting certain manners of speech appropriate to human interaction with those objects we choose to surround ourselves with, as at a dinner table!

Whether occurring in a film or novel, or in a critical or theoretical essay, color comes to be known within (the specific walls of) a common rhetoric of life experiences derived from a folk theory of color, that is, from the way in which a community commonly understands its understanding of colorful appearances. Color is converted through a rhetoric into an instrument, a convenient fiction or hypothesis able to speak to a community. One of the tasks of this and the next chapter is to set forth some of these folk theories of color—the accepted manners of speech—that undergird the familiarities we make for ourselves in fictional narratives. Folk theories of color bear on all four themes of this book: conventions, languages, memories, and sensations.

As an example, consider how we are being asked to see color through the rhetoric of James Wood, who is reviewing the career of John Updike.

> In Updike's defense it is often maintained that these are the thoughts of his characters, not necessarily of their creator. But obsessions of this kind have recurred and overlapped thickly enough in his work to constitute, now, the equivalent of an artist's palette: this is how Updike chooses to paint the world. . . . In [Updike's short story] "The Women Who Got Away," for instance, the narrator tells us of his old lover, and how "her voice and its quick inspirations of caustic perception painted the world, which seemed to me rimmed with a vague terror, in bright fearless colours." But is this perfect sentence, with its delicate deferral so characteristic of Updike ("painted the world. . . which seemed. . . in bright fearless colours"), the expression of a man who really felt the world to be rimmed with a vague terror? Or does the terror not seem a little too vague, as if the narrator [like Updike himself on many occasions] were paraphrasing a novel for a *New Yorker* review?[36]

Wood states that Updike's "obsessions" are "the equivalent [!] of an artist's palette." Wood has selected a passage from one of Updike's short stories in order to paint for the reader "how Updike chooses to paint the world" through the words of a narrator of the story who is recounting how a character's *voice* "painted the world" while she herself remained unaware that her "bright fearless colours" actually revealed "a vague terror." This suggests that Updike's narrator is able to see that the woman's "fearless" colours are concealing—presumably only partially covering—their opposite internal cowardly colours. But what is truly being painted by an author of the narrator's words, asserts Wood, is a terror that is itself a touch too vague, being delicately deferred—too subtle even for Cornwell-Clyne's color delicacies—in a sentence too perfect, and hence not a terror at all. For Wood, it is as if Updike (as the author or as an author of the author) had mistakenly lapsed into summarizing the rhetoric of some novelist in the course of writing a review for *The New Yorker* magazine in the guise of a narrator's speech in a fiction—a review perhaps not unlike reviews actually written by Updike about novelists. Perhaps Wood's observation about Updike's disguised prose applies to the review that the novelist Wood is writing about the novelist Updike, for Wood is also using precious language to color his rhetoric.

Color in Wood's passage about the color in Updike's passage appears and circulates, then disappears, like a pattern slowly developing in a tapestry, only to suddenly reappear in several new senses, none literal. The lesson here is that it is a mistake to believe that our thoughts about color are confined to the hues of physical surfaces, rather than already generalized into rhetorical schemes that fatefully steer our thought this and that way in accordance with half-hidden designs in a tapestry.

Wood concludes his review by remarking that even in one of the best of Updike's short stories "often the prose relaxes into a kind of grey chunter." This final word,

"chunter," which means "to mutter in a low inarticulate way," brings color decisively into the sonic realm of a novel's verbal narration and character dialogue[37] as well as suggesting that grey is a color that has relaxed, let go, slackened, and sadly shunned its responsibilities to be colorful. In Wood's rhetoric, black and white are not fundamental (as some theorists believe), but rather faulty and incomplete colors. We are meant to *see* grey in a certain way, as we see a "grey day" and through that to see Updike's prose. Moreover, Wood's accusation that Updike's prose engages in a "delicate deferral" is a reminder that an aesthetic of delicacy (discussed earlier in connection with Cornwell-Clyne) may reduce to a disingenuous evasion rather than to an ephemeral sublimity.

But circumstance is everything. In the genre of "mumblecore" film, for instance, chunter in the form of muted dialogue has great value, and in a fine black-and-white nitrate print with low contrast where gray has many shades, the shades may be said to achieve a delicately sublime, even transcendent, quality.[38] As Cornwell-Clyne argues, "muted" color, including gray, may connect favorably to an aesthetic of delicacy.[39]

WHAT IT'S LIKE TO SEE FILM COLOR (DELEUZE)

Fictions of color painted in literary criticism may also appear in the language of film criticism. For example, Gilles Deleuze opposes a merely "coloured image," like Eisenstein (above), and instead extols the "colour-image." Deleuze praises Jean-Luc Godard's film work:

> The fact is that, in Godard, sounds and colours are attitudes of the body, that is, categories: they thus find their thread in the aesthetic composition which passes through them, no less than in the social and political organization which underpins them.[40]

We have seen with the Technicolor tan how faces may be specially lit and colored to highlight emotion. According to Deleuze, it would seem that Godard extends these possibilities into an expression of the entire body, its "attitude," and into a corporeal situation, including the traversal of social and political contexts. Godard accentuates a viewer's top-down perception, which is a mental process utilizing task-oriented cognition and long-term memory, such that color becomes capable of exemplifying *categories* of thought and especially categories of analysis. Deleuze provides an illustration:

> Categories can be words, things, acts, people. *Les carabiniers* [Godard, 1963] is not another film *about* war, to glorify or attack it. It films the categories of war, which is something quite different. Now, as Godard says, these can be specific things, armies

> of sea, earth and air, or "specific ideas," occupation, countryside, resistance, or "specific feelings," violence, rout, absence of passion, derision, disorder, surprise, void, or "specific phenomena," noise, silence. It will be noted that colours themselves can fulfil the function of categories. Not only do they affect things and people, and even written words; but they form categories in themselves: red is one in *Weekend* [Godard, 1967]. If Godard is a great colourist, it is because he uses colours as great, individuated genres in which the image is reflected.[41]

According to Deleuze, the fact of war is understood in many of Godard's films through the language and categories it has spawned. Godard is not representing war in the first instance as a drama (of horror, sacrifice), but rather as a cluster of micro-genres, stereotypes, formulas, and ready-made visions at one remove from drama. Godard articulates the effects of war that may appear in individual behavior (e.g., crying, cowering, defiance) or are inarticulate in a person (e.g., a scream or blank face) by showing these effects *as transposed* into the suffering of a collective body through an array of community concepts characterizing that collective body. For Deleuze, color may become one of these categories capable of performing an analysis of war and other matters. Color is functioning cognitively for Godard, not realistically, because war is not being represented realistically. Conventions of realism are ignored in favor of filming colors that embody the psychological categories we live by, or in the case of war, die by.

Filmmakers who arbitrarily confine color to a physical patch that is meant only to be perceived bottom-up on the screen and then discarded are arbitrarily cutting off the potential to depict color as a holistic and structural part of fictions and life experiences from the top down. Bottom-up perceptual processes are automatic, mandatory, and stimulus-driven. In film, such perception is tied to the irreversible movement of shot-lengths through a projector, which acts much like a clock. (It is a clock that possesses a bit of slippage since the mind feels the "present" to be about two and a half seconds, which is the span of unconscious attention.) But is this projector-time the only sort of time that can be told through narrative film? I believe that other sorts of mental time schemes act from the top down like a blueprint or *pattern*. These "spatial" (conceptual, strategic), goal-driven processes, in turn, create new types of temporal and causal relations for a spectator—beyond, say, "push-pull," clock-like causation[42]—and hence are able to create striking new roles for color.

Godard demonstrates that color may function top-down as a category tightly enmeshed in an intellectual and critical cinema.[43] As we shall see, other filmmakers, too, have found uses for color beyond insistent sensation. What, then, does color look like when seen top-down? I believe that to ask *what it is like* to actually experience experiencing a particular color—what it is like to see—is to ask what a color

is made *to be like* in a situation of viewing, i.e., how memory and mind are making of color sensation a metaphor from the top down—a link—to be fit to, onto, into, and for an occasion. To ask what a color is like qua color, however, is to reduce, say, red to a distressing and uninformative tautology where it is what it is: red is red is red. The impression of red is merely an impression—in effect, to have no real question in mind—as opposed to investigating what the presence of red is *like*: how red may lead toward, and merge with an idea or category. . . red functioning as a part of one's body and thought, and hence not apart, but a part of situational awareness.

I am here seeking to revise and recast a classic debate about "what it is like" to be aware, to have a sensation, to feel a percept, to experience a phenomenal state, like happiness or redness. This still lively debate was touched off by Thomas Nagel's essay, "What Is It Like to Be a Bat?"[44] To repeat, rather than asking *what it is like* to actually experience experiencing a particular hue, I believe it would be more productive to ask what a hue is being made *to be like* in a particular situation of viewing, i.e., to *literalize* and *conceptualize* the metaphor, to make seeing be fully literal for the eyes within an alert and active mental state that interrelates with one's knowledge and experience. If that were the case, would there be anything left to discover about what color innately secretly is like?

In the next chapter, I will re-formulate "what it is like" to see color by introducing the notion of mental "tracking." Subsequent chapters will further examine what color looks like when seen top-down through working memory.

NOTES

1. I will quote the sentences leading up to the passage in the epigraph because I believe that Descartes's argument should be interpreted more expansively about what can be *real and existent* in a fiction than what is offered by his conclusion. Moreover, perhaps what the conclusion (quoted in the epigraph) refers to as "other even simpler and more universal things [that] are real" need not themselves be visual.

> Suppose then that I am dreaming, and that these particulars—that my eyes are open, that I am moving my head and stretching out my hands—are not true. Perhaps, indeed, I do not even have such hands or such a body at all. Nonetheless, it must surely be admitted that the visions which come in sleep are like paintings, which must have been fashioned in the likeness of things that are real, and hence that at least these general kinds of things—eyes, head, hands and the body as a whole—are things which are not imaginary but are real and exist. For even when painters try to create sirens and satyrs with the most extraordinary bodies, they cannot give them natures which are new in all respects; they simply jumble up the limbs of different animals.

Scholars in the humanities should take seriously the implications of this argument, including the impact of a theory of "embodied cognition" on the claims of the first two sentences above.

The remaining sentences in this passage would seem to owe a debt to Epicurus and Lucretius. "First Meditation" in *Meditations on First Philosophy (with Selections from the Objections and Replies)*, ed. and trans. by John Cottingham (Cambridge: Cambridge University Press, rev. ed. 1996), pp. 13–14.

Undoubtedly, Descartes draws upon Aristotle's strong link between perception/sensation, which can be misleading but never false, and phantasia, which is much broader than its usual translation as "imagination" and is essential to all thought; *On the Soul*, 427a17–429a9. On the nature of "fiction," see section below, "Fictions Painted in Language (Updike Redux)." On embodied cognition, see note 9 below and Chapter 7 below, note 24.

2. Sergei Eisenstein, "Not Coloured, but in Colour" in *Notes of a Film Director*, ed. by R. Yurenev, trans. by X. Danko (New York: Dover, 1970; orig. 1940), pp. 118–119 (Eisenstein's emphases).
3. Adrian Cornwell-Clyne, "Preface to the First Edition" in *Colour Cinematography* (London: Chapman & Hall, 3rd ed. 1951), p. ix (1936).
4. Herbert T. Kalmus with Eleanore King Kalmus, *Mr. Technicolor* (Absecon, NJ: MagicImage Filmbooks, 1993), p. 65. Herbert Kalmus apparently wrote his sections of the book in 1960–1963; he died in 1963 with the manuscript possibly unfinished.
5. On Natalie Kalmus, see Chapter 6 below, subsection "*Two or Three Things I Know about Her*; Natalie Kalmus and Technicolor Style."
6. John A. Lucy, "The Linguistics of 'Color'" in *Color Categories in Thought and Language*, ed. by C.L. Hardin and Luisa Maffi (Cambridge: Cambridge University Press, 1997), p. 341 (emphasis added). See also C.L. Hardin, "Color Subjectivism" in *Readings in Philosophy and Cognitive Science*, ed. by Alvin I. Goldman (Cambridge, MA: MIT Press, 1993), pp. 493–507.
7. G.P. Baker and P.M.S. Hacker, *Wittgenstein: Understanding and Meaning, Vol. 1 of An Analytical Commentary on the Philosophical Investigations, Part I: Essays* (Oxford: Blackwell, 2nd rev. ed. by P.M.S. Hacker, 2005), p. 301.
8. Jeff Scheible, *Theories of Media Textualities: On Prepositions and Media* (Santa Barbara: University of California, unpublished 2008), pp. 11–12.
9. George Lakoff and Mark Johnson, *Metaphors We Live by* (Chicago: University of Chicago Press, updated 2003). On core cultural metaphors, see, e.g., Lakoff and Johnson, *Philosophy in the Flesh: The Embodied Mind and Its Challenge to Western Thought* (New York: Basic Books, 1999); Gilles Fauconnier and Mark Turner, *The Way We Think: Conceptual Blending and the Mind's Hidden Complexities* (New York: Basic Books, 2002); Tony Veale and Mark T. Keane, "Conceptual Scaffolding: A Spatially-Founded Meaning Representation for Metaphor Comprehension," *Computational Intelligence* 8, 3 (August 1992), pp. 494–519.
10. Frank Sibley, "Why the 'Mona Lisa' May Not Be a Painting" in *Approach to Aesthetics: Collected Papers on Philosophical Aesthetics*, ed. by John Benson, Betty Redfern, and Jeremy Roxbee Cox (Oxford: Clarendon Press, 2001), pp. 256–272.
11. On James Lastra's argument, see Chapter 3 below, note 19.
12. A related question concerns whether perceptual illusions are illusory or whether they are perfectly real for a perceiver, since one can't not see them. Are artworks deceptions or are they statements arising from, and impinging on, facts? See further discussion in Chapter 7, section, "The Reality of Illusions and the Illusions of Reality."

13. Aristotle, *Poetics VI*, trans. by S.H. Butcher; see also XIV. Aristotle says that "spectacle" is exemplified by dramatic scenes laid in Hades (XVIII).

    For Aristotle, mixtures of black (earth) and white (fire) produce all the hues, with minor contributions from air and water (involving lightness). Color is thus a *substance* for Aristotle, not an evanescent quality, not a subjective secondary quality, nor some literal or figurative surface "paint." *Sense and Sensibilia*, 439a6–440b25.

    The following single sentence from *Poetics VI*, trans. by Ingram Bywater, has been interpreted by writers on color in several ways: "the most beautiful colours [in painting] laid on without order will not give one the same pleasure as a simple black-and-white sketch of a portrait." For Aristotle, this is strictly an analogy. He is *not* discussing the aesthetics of color nor saying that a black-and-white design is superior to a color design. Aristotle is merely recruiting color to an argument meant to show the superiority of plot over character. For me, the key question in interpreting this sentence would be to investigate what counts for Aristotle as a proper "order" for color that would make it analogous to the tight ordering principles of a plot, thus promoting "beautiful colours" to a place of esteem. As the sentence stands, "beautiful colours. . . without order" (random colors?) are inferior to a black-and-white character portrait.

14. Rosalind Galt, "Pretty: Film Theory, Aesthetics, and the History of the Troublesome Image," *Camera Obscura* 71 (2009), p. 2 (footnote omitted). Cf. James Lastra, "Film and the Wagnerian Aspiration: Thoughts on Sound Design and the History of the Senses" in *Lowering the Boom: Critical Studies in Film Sound*, ed. by Jay Beck and Tony Grajeda (Chicago: University of Illinois Press, 2008), pp. 123–138.

15. Rudolf Arnheim, "Remarks on Color Film" in *Film Essays and Criticism*, trans. by Brenda Benthien (Madison: University of Wisconsin Press, 1997), p. 21 (orig. 1935); reprinted in *Color, the Film Reader*, ed. by Angela Dalle Vacche and Brian Price (New York: Routledge, 2006), p. 55.

16. Cornwell-Clyne, *Colour Cinematography*, p. 654. For an example of what Cornwell-Clyne calls "the delicacy of a [Thomas] Girtin watercolour," consider *Bamburgh Castle, Northumberland* (1797–1799) at <https://en.wikipedia.org/wiki/Thomas_Girtin#/media/File:Thomas_Girtin_005.jpg> (accessed June 26, 2016).

    For Cornwell-Clyne, "The film show is a commodity, like ice cream, prepared to titillate the jaded anatomies of pert shop-girls." He says that his book will instead be concerned with those films that merit discussion as "art and science." Then, remarkably, he questions his own distinction between entertainment and art (p. vi; cf. p. viii).

17. In the text I am generalizing signal-detection theory, also called sensory decision theory, which is a mathematical model of the psychology of detecting physical stimuli based on statistical measurements. See C.L. Hardin, *Color for Philosophers: Unweaving the Rainbow* (Indianapolis: Hackett, expanded ed. 1993), pp. 170–182.

18. See, e.g., Nelson Goodman, "Variations on Variation—or Picasso Back to Bach" in *Reconceptions in Philosophy and Other Arts and Sciences*, ed. by Nelson Goodman and Catherine Z. Elgin (Indianapolis: Hackett, 1988), pp. 66–82. A film requiring exceptional powers of discrimination is James Benning's thirty-minute *Two Cabins* (2011), which is composed of two fifteen-minute static shots containing two variations on nearly identical color and flickering, barely changing light,

accompanied by a truly intricate whispering of a soundtrack. See Julie Ault, ed., *(FC) Two Cabins by JB* (New York: Art Resources Transfer Press, 2011).

19. Christopher Knight, "Diebenkorn Crystallizes an Era," *Los Angeles Times* (March 3, 2012), section D, pp. 1, 7.
20. See also discussion of the "nearly true" in Chapter 1 above, section "Process into Pattern," and Chapter 7 below, subsection "The Otherwise of Texts."
21. For iterations of the two aesthetic principles, the nonintrusive (vs. unwelcome/uninvited) and the unobtrusive (vs. disagreeably or unduly prominent), see Edward Branigan, *Projecting a Camera: Language-Games in Film Theory* (New York: Routledge, 2006), index entry, "unobtrusiveness principle."
22. Johann Wolfgang von Goethe, *Theory of Colours*, trans. by Charles Lock Eastlake (Cambridge, MA: MIT Press, 1970), para. 135, p. 55. David Batchelor concludes his book on a forceful tone by quoting Goethe's claim reproduced in the text. Batchelor follows Goethe's statement with his own claim that color is still subject to the prejudice of "the cold light of refinement" that leads to "a world banished of colour," a world only of whitescapes; *Chromophobia* (London: Reaktion Books, 2000), p. 112.
23. Cornwell-Clyne, *Colour Cinematography*, p. 650.
24. Batchelor, *Chromophobia*, p. 112. An aesthetics of delicacy and subtlety in color comes in many forms. A stunning, classic example is the *Isenheim Altarpiece* (Niclaus of Haguenau and Matthias Grünewald; 1512–1516). See especially details of the painting at <www.wga.hu/frames-e.html?/html/g/grunewal/2isenhei/index.html> (accessed Nov. 12, 2015). Consider some of the late works of Picasso where graded colors are at war with shape and line, e.g., "Buste" (1970), "Couple" (1970), and "Personnage" (1971). A range of possibilities is explored by Jasper Johns in *Numbers*, ed. by Roberta Bernstein and Carter E. Foster (Cleveland, OH: Cleveland Museum of Art, 2003). Cindy Sherman's delicate gradations of hues are shattered and scattered by disturbing subject matter, as in *Untitled* #153, #156, #175, #188, and #190. Instructive examples may also be found in the paintings of Anne Slaughter.
25. Jacqueline Lichtenstein, *The Eloquence of Color: Rhetoric and Painting in the French Classical Age*, trans. by Emily McVarish (Berkeley and Los Angeles: University of California Press, 1993), pp. 167, 168.
26. On the glory of Technicolor, see, e.g., Fred E. Basten, *Glorious Technicolor: The Movies' Magic Rainbow* (Camarillo, CA: Technicolor, 3rd ed. 2005). Previous editions of this book are also worthwhile (1980, 1994), as well as the beauteous book by James Layton and David Pierce, *The Dawn of Technicolor, 1915–1935*, ed. by Paolo Cherchi Usai and Catherine A. Surowiec (Rochester, NY: George Eastman House, 2015).
27. Kalmus, *Mr. Technicolor*, p. 53.
28. The "Technicolor tan" can be seen in fine form in *The Adventures of Robin Hood* (Michael Curtiz and William Keighley, 1938) and as recycled in *Sense and Sensibility* (Ang Lee, 1995). On adding a component to increase the warmth of a hue generally, see, e.g., Edith Anderson Feisner and Ron Reed, "The Dimension of Temperature" in *Color Studies* (New York: Fairchild Books, 3rd ed. 2014), chap. 9, pp. 98–108. A subsection on Natalie Kalmus in Chapter 6 details an elaborate set of aesthetic principles developed by Technicolor to guide color designs in foregrounding character and character action.

29. John Updike, *Collected Poems, 1953–1993* (New York: Alfred A. Knopf, 1993), p. 231.
30. On Plato's theory of color and tears, see *Timaeus*, 67e–68a. For a detailed analysis, see Katerina Ierodiakonou, "Plato's Theory of Colours in the 'Timaeus'," *Rhizai: A Journal for Ancient Philosophy and Science* II, 2 (January 1, 2005), pp. 219–233.
31. The words are spoken by the fictional writer and painter Ted Cole (Jeff Bridges) in the film by Tod Williams, *The Door in the Floor* (2004).
32. Christian Metz, "'Crossing over the Alps and the Pyrenees. . .,' an Introduction to Francesco Casetti" in *Inside the Gaze: The Fiction Film and Its Spectator*, trans. by Nell Andrew with Charles O'Brien (Bloomington: Indiana University Press, 1998), p. xi. For more on fiction, see note 1 above and Chapter 4, note 11.
33. See, e.g., Murray Smith, "Film Spectatorship and the Institution of Fiction," *The Journal of Aesthetics and Art Criticism* 53, 2 (Spring 1995), pp. 113–127.
34. Metz, "Crossing over the Alps," p. xv (Metz's emphases).
35. I elaborate on the type of film theory described in the text that focuses on Metz's notion of "manners of speech" in "Teaching Film Theory" in *Teaching Film*, ed. by Lucy Fischer and Patrice Petro (New York: Modern Language Association, Options for Teaching series, 2012), section "Language, Tropes, Topoi," pp. 30–32. On tints and tones as metaphors for film theory, see Branigan, "How Frame Lines (and Film Theory) Figure," section "Color as Container" in *Projecting a Camera*, chap. 4, pp. 127–132. Moreover, color will appear differently within each of the fifteen frames described in chapter 4 of *Projecting a Camera*. That is, my claim is that color will be perceived not just as identical to itself—whatever philosophers might mean by such a dubious notion—but relative to particular frameworks for looking at things.
36. James Wood, "Gossip in Gilt," *London Review of Books* 23, 8 (April 19, 2001), pp. 30–31.
37. I believe it is always fruitful to examine the aesthetic connections between the sensory pair, color and sound, for their cognitive relations. See Appendix below.
38. In general, both the concept gray and an aesthetic of delicacy have many uses, none intrinsic. Aris Fioretos has written a quasi-treatise and meandering literary memoire about grayness, *The Gray Book* (Stanford, CA: Stanford University Press, 1999). On valuable forms of grayish inarticulateness, see David Denby, "Youthquake: Mumblecore Movies," *New Yorker* 85, 5 (March 16, 2009), pp. 114–115. Nuri Bilge Ceylan shows what can be done and undone when sharp green, desaturation, and gray become a key pattern in the neo-noir of guilt and resilience, *Three Monkeys* (2008). In Lars von Trier's *The Element of Crime* (1984), the imagery and voice-over remain murky in several senses along with the plot, which follows the effort to find Harry Grey, a serial killer of young girls. Consider also the film characters David Gray (Carl-Theodor Dreyer, *Vampyr*, 1932), Vincent Grey (M. Night Shyamalan, *The Sixth Sense*, 1999), E. Edward Grey (Steven Shainberg, *Secretary*, 2002), and Christian Grey. The latter film is based on E.L. James's novel *Fifty Shades of Grey*.

    See also David Batchelor, *The Luminous and the Grey* (London: Reaktion Books, 2014), esp. pp. 90–96. Consider that an overcast, gray day actually promotes sparkling color because clouds filter out the infrared wavelengths and diffuse the sunlight that washes out color. On the impossibility, or perhaps possibility of a luminous gray, see Chapter 5 below, note 27.
39. On the notion of an aesthetic of delicacy, see section 2 above and see Chapter 5 below, note 27 where a muted gray is said to become "luminous" in an aesthetic of delicacy.

40. Gilles Deleuze, "Cinema, Body and Brain, Thought" in *Cinema 2: The Time-Image*, trans. by Hugh Tomlinson and Robert Galeta (Minneapolis: University of Minnesota Press, 1989), chap. 8, p. 195. On the "colour-image" or "colorism" as opposed to the merely "coloured image," see Gilles Deleuze, *Cinema 1: The Movement-Image*, trans. by Hugh Tomlinson and Barbara Habberjam (Minneapolis: University of Minnesota Press, 1986), pp. 117–120. For more on Deleuze's notion of color, see Chapter 4 below, section 4.

    Deleuze defines "movement-colour" as being unique to cinema and also finds an "absorbent form of colour" in cinema (p. 118). It is unclear from his examples how these two color states operate. Perhaps absorbent-colour may be recast as a colour that assumes "any-hue-whatever," for which see Chapter 4 below, note 17. Absorbent-colour may be an expanded version of the standard *being-colored-by* metaphor for perceiving color, i.e., diverse things being nominally colored have paradoxically been themselves absorbed—taken up/in/on—by one or more visible hues. Absorbent-colour is thus released from a hue's singular symbolic role in space and plot. Absorbent-colour may be a target that is capable of absorbing many other things or hues to reach a purified state of affect. On *being-colored-by*, see Figure 3.1.

41. Deleuze, "Thought and Cinema" in *Cinema 2*, chap. 7, pp. 186–187 (Deleuze's emphasis; footnote omitted). *Le Petit Soldat* (1963) is another noteworthy Godard war film.

42. See, e.g., Lakoff and Johnson, "Events and Causes" in *Philosophy in the Flesh*, chap. 11, pp. 170–234; Lakoff and Johnson, "Causation: Partly Emergent and Partly Metaphorical" in *Metaphors We Live By*, chap. 14, pp. 69–76. I discuss standard and non-standard forms of causation in narrative film in *Projecting a Camera*, pp. 178–191, and in *Narrative Comprehension and Film* (New York: Routledge, 1992), pp. 26–32. The next chapter will investigate methodological issues that flow from bottom-up versus top-down processes.

43. See my analysis of a Godard film, "The Articulation of Color in a Filmic System: 'Deux ou trois choses que je sais d'elle' " (expanded version) in *Color, The Film Reader*, pp. 170–182; and see Chapter 6 below, subsection on Godard's *Two or Three Things I Know about Her*.

44. Thomas Nagel, "What Is It Like to Be a Bat?," *Philosophical Review* 83, 4 (October 1974), pp. 435–450.

Chapter 3 **To Stand in Place or to Track?**

ABSTRACT

Two top-down approaches to color experience may be contrasted, though there are ways in which they are complementary. A semiotic framework is based on the idea that an element, A, present to our senses in a text, is meant to stand for, or to stand in for or stand up for, another element, B, that is absent. A particular standing theory relies on learned codes that are resident in long-term memory. By contrast, a tracking approach to color derives from a perceiver's judgments about ongoing mutual changes and emerging patterns that appear through tracking a flow of textual events in present time. A tracking approach relies on the reidentification of elements and evolving patterns in working memory. (Reidentification will be examined more fully in chapters 7 and 8.)

The experience of tracking may be analogized to three major effects produced by a moving camera that lead to four modes of color tracking. Tracking may also be analogized to the movements of irregular wavy lines in relation to each other. A discussion of a passage from Susan Blackmore illustrates the movement of consciousness within working memory. A consideration of passages from Adrian Cornwell-Clyne and Sergei Eisenstein illustrates some aesthetic consequences of color movements within a spectator's consciousness.

When color is properly seen as *relational* and moving, the problem of its archival restoration is put into a new context since the issue no longer reduces to ontological considerations of seeking to restore an Original. (For example, restoring what an original audience saw *in* the colors is a very different matter than restoring the colors.) Andy Warhol's series of "Do It Yourself" paintings leads to an investigation of ten fundamental schemata that I believe underlie our talk and thinking about the nature of color. The key question is epistemological, not ontological. Therefore, the analyst should ask which schema has been chosen on an occasion (present or past) in order to see, feel, and describe a hue as such-and-such within one or another *pattern*, rather than to find which precise name should be given to a colored spot in order to determine whether the spot is perfect and pure and will fit to a chart of names and samples. It is not finally about correct "naming"—if a name even comes to mind—or correct

"restoring," but about methods of conscious and unconscious *description*, and *their* restoration, i.e., how we will talk about temporal progressions we believe we have seen through using approximations and guesses to convey some idea. What is "in mind" is already ahead of a named identification.

> The activity of sense-perception in general is analogous, not to the process of acquiring knowledge, but to that of exercising knowledge already acquired.[1]
>
> —Aristotle

TOP-DOWN PERCEPTION: STAND IN PLACE OR TRACK?

The argument in the preceding chapter moved from a consideration of gaudy or glorious colors in mind toward delicate and subtle, sensuous colors (Cornwell-Clyne) that anchor an attractive coloring of flesh and body, especially of an individual's face (namely, the "Technicolor tan"), and, finally, to colors being painted through manners of speech (Metz) and interpretive activities, including the device of fiction where a perceiver may oscillate between alertness and reverie (Metz). A state of reverie includes entertaining precious hypotheticals as in Updike's sonnet and Wood's critical response in which subtle color is disparaged. In the case of film, Godard's coloring of concepts is said to em*body* a person's experience of life—not a single person's experience (symbolized through his or her face), but rather numerous persons as they live through generic situations of a specific community; for example, as a person lives through commonly held "categories of war" or else as he or she experiences familiar concepts and stereotypes that are, as Deleuze says, being translated through images and sounds into personalized mini-genres of a community's collection of often colliding norms and values. In all cases, the inert question about what it "should be like" to experience experiencing color is replaced by what color is made by mind to be like.

Aristotle insists that sensing and sensation are not primary, but informed by prior knowledge (see epigraph). The question then arises, which sorts of acquired knowledge are important on which occasions? How do we make use of knowledge when looking? Specifically, which top-down model will provide an appropriate framework for analyzing the cognitive aspects of color?

One possible framework is semiotics. Here the idea is that an element, A, that is *present* to our senses in a text (e.g., a word on a page or a color patch on a screen) is meant to *stand for*, or to stand in for or stand up for, another element, B, that is *absent*: A → B.[2] The element, B, will be absent in a variety of ways. For example, what is being referred to and absent may be an emotion, an intangible object, an

abstraction, a relationship, or be unconscious, ideal, fictional, or simply located elsewhere in the world in space or time. In addition, parts of an object B summoned to mind by element A may remain permanently absent and unknown. For example, though an actor is present, his or her height as shown on screen may be different than the actor's true height. In a semiotic approach, what may be missing about an object is missing because what is present on the screen (A) is only a sign or representation, not the actual thing, and therefore the sign (A) may misrepresent or lie about the thing being represented (B). Thus a film actor may be seen to be taller or shorter than he or she would be, if not for, say, camera positions that make all upper bodies and faces the same size in medium close-ups or an angle that hides the fact that the actor is standing on top of a box or in a special trench to make him or her appear equal in height to a co-star. (Alan Ladd is made to look taller in George Stevens's *Shane*, 1953.) In general, for a semiotic approach there are always qualities associated with absent element B that are permanently lost in representation A.

Color furnishes another example: some aspects of original colors may be permanently absent because their exact qualities have been distorted through successive stages of technological reproduction as well as degraded through successive projections and the aging of a film's celluloid. Nonetheless, some kinds of distortions are highly desirable and may be exploited for profit; for example, plywood was painted orange and photographed in black and white to make it appear as mahogany for a desk in the U.S. Senate in Frank Capra's *Mr. Smith Goes to Washington* (1939). In general, every color technology (notably, Technicolor) possesses unique characteristics that modify the qualities of hues in distinctive ways.

Making meaning in a semiotic framework takes place when something is experienced as a present signifier (A) that, in turn, brings something *else*—an absent signified (B)—to mind. A sign is a signifier joined to a signified by a code, e.g., through a rule, convention, cause, or resemblance. A sign is a mental entity, though there is a material component to the signifier (e.g., a word that appears in ink) and a material component to the signified (i.e., something in or *about* a real world is being referred to). There are many semiotic theories and many remarkable nuances. For example, the "arrow" may be reversed between presence and absence: A ← B. In this situation, a mental "category" or interpretive presupposition acts to bring into being something new, or to highlight the recognition of something present in a text. Nevertheless, I will cautiously propose that the focus of semiotic approaches is generally on "symbols standing for things" or "symbols standing on top of other symbols standing for things," as in structures of connotation and metalanguage. More simply, I will refer to semiotic theories as *standing* or *stand in place* approaches to the making of meaning.

I wish to offer an alternative to semiotics that might better bring to light the dynamism of top-down color. Instead of *stand*, I propose *track* or *tracking* as a

framework for color. The idea is to incorporate movement or flow into an account of color cognition. Attention switches from the search for a presumed link from A to B, or from B to A, A → ← B, and from an inventory of the types of connections that reside in long-term memory (involving codes that have been learned), toward the delineation of ongoing mutual changes and emerging patterns that appear through tracking a flow of textual events in present time: A^—^B^—^C. . .

One of the fundamental differences between the two approaches, I believe, lies in the area of cognitive processing. Each approach invokes a different balance among the kinds of memory employed during processing. A semiotic approach relies on a priori connections and links, i.e., signs (icons, indexes, and symbols) that are learned and reside in long-term semantic memory. By contrast, a tracking approach relies on singling-out, identifying, following, and then reidentifying a presented quality, object, or pattern against a chaotic background of other ongoing changes. Tracking imposes a form of constancy (or near similarity or simple variation) upon a discovered pattern despite changes in individual qualia due to sometimes wide differences in the conditions of observation and context, much like, I believe, the well-known constancies of shape, size, velocity, melody, spoken word, brightness, and color, which is known as chromatic adaptation. Tracking therefore depends crucially on the components of working memory.[3] These components are located in distinct areas of the brain and function quite differently than long-term episodic and semantic memory stores. A standing approach relies directly on applying already existing knowledge from long-term memory, while tracking involves continuous mental operations of working memory under the pressure of immediate past and present online experiences guided by a spectator's evolving expectations. Long-term memory and knowledge does play a role in working memory, though less directly than in a semiotic theory.

How might one visualize the operation of a tracking model? Imagine for a moment a tracking shot in film as a camera moves to the left while objects are seen to move on screen to the right at speeds dependent on their distance from the camera. Then imagine that the camera moves forward and objects enlarge at differential rates dependent on their distance. Next, replace the moving camera in your imagined film with a moving color—i.e., a color is now tracking among objects and interacting with them in a locale. There are at least three ways to think about what is happening in such a situation.

1. The color-as-camera reacts to, inflects, changes, and combines in various ways with the individual colors of objects in a locale *just prior* to the scene being registered on the film's emulsion. Note that the color-as-camera is not itself directly seen, but must be inferred and felt through its *effects* on the various hues of objects, i.e., through color shifts among the objects. For example, in looking at

color mixtures (assuming which primaries and color circles?!) and hybrid hues (as opposed to pure hues) that are the result of a prior interaction with the color-as-camera, a spectator may be encouraged to separate an object's visible hue into component elements, creating new patterns within the scene. A spectator may also be encouraged to mentally carry a concealed sub-element forward to form a new pattern with hues in succeeding images.[4] Furthermore, there may be rare situations in which a color's absence due to a movement of the color-as-camera—an absence that is assumed, implicit, suppressed, or repressed—becomes important as a latent presence in comprehending a manifest color design.

2. The color-as-camera through its movements is able to high*light* a manifest color design by actions that select, juxtapose, emphasize, arrange, and consolidate objects and hues within a locale to create a sequence of compositions and framings. These compositions include perceptual effects arising from the manipulation of color contexts giving rise to, for example, simultaneous and successive contrasts, afterimages, spreading effects, phantom colors, and other color illusions and constancies.[5]
3. The color-as-camera through its movements is able to continuously *alter* its relationship to the objects and colors in a locale by moving closer or further away, affecting, for example, the amount of a colored area that appears on screen, which, in turn, will produce effects on the other colors. The color-as-camera may stop to set up at various distances allowing for temporary mis/matching among colors and objects, including repositioning a hue out of sight, i.e., offscreen, in order to later bring it back.

These are the kinds of effects and alternatives that a spectator may sense within a color design when *tracking* patterns on the screen and between shots. Again, the idea is that colors are conceived as being in movement, creating dynamic patterns, parallels, and contrasts.[6] In tracking, color is not attached to a rigid semiotic rule, fixed by an immovable sign or symbol, or placed where it can only *stand* in stillness in long-term memory. Think of the color-as-camera as a new type of language-game: what can be said about color using its grammar?

Here's a second way to picture color tracking.[7] Imagine two irregular wavy lines. Sometimes the lines are close together and roughly approximate each other's shape; at other times they are further apart and the shapes are only vaguely similar or else diverge in an assortment of ways. The relationship between the lines is constantly changing; they may even become opposite to one another, as a warm hue opposes a cool one, or one line may cross over and move above the other line for a time. Suppose now that only one of these lines is taken to represent a set of colors as they may be dynamically changing in a series of compositions through several scenes in a film. The other line, then, can be taken to represent some variable within

either the film's narrative or narration, or even within a spectator's changing mood or beliefs. We can then say that the line representing the color design *tracks* at various times and to various degrees some narrative aspect or cognitive state.

Notice especially that a tracking model does not commit an analyst to hypothesizing, or searching for, a specific *causal* connection between any two points on the curves. By contrast, a semiotic model attempts to match up every pair of points from one curve to the other, such that one point will *stand* for, or lead to, the other, say, by a rule or convention. A tracking model encourages us to think of a text not as a mass of discrete particles that must be correlated point by point (in terms of "signs" or "cues") to achieve meaning, but rather to think of a text (expanding the two irregular lines into areas) as a set of layers, manifolds, or fields marked by internal disturbances and cross-connections. These fields will be moving into and out of temporary and diverse equilibria with one another, thereby disclosing collections of relations, i.e., patterns that are interacting, evolving, and in transition.

TRACKING THROUGH WORKING MEMORY (BLACKMORE)

The preceding two pictures of a tracking framework—involving an imagined color-as-camera and a pair of irregular lines or fields—operate to draw the traits of color nearer to actual movements of the camera, characters, and plot trajectories, and especially nearer to the temporal relations that are put into play through a soundtrack; for example, through noise, music, ambient background, beat, accent, voice, tempo, tone, loud tone, pure tone, intonation, asynchrony, harmony. . . . With character behavior, dialogue, and causally linked events in a film, one has the makings of a range of abstractions that are available to form *mobile relationships* with color. Later chapters will examine what can be learned about the flow of color by exploiting these and other analogies to features of film. For now, let us be a bit more literal and explicit about how one may conceive color tracking.

There are at least four modes of color tracking, and all must be weighed when analyzing color:

1. The interrelations among colors in a shot. Broadly, relations among colors may be continuous or discontinuous. An example of continuity is a stretch of "analogous" hues that produce a "passage" of continuous gradation where there is blending and flow along discrete compositional lines linking the hues (see Figure 6.4). Examples of discontinuity are a pair of complementary hues as well as strong contrasts (see the red-green opposition of Figure 6.2 and the contrasts in Figure 6.8). More extreme are forms of disharmony, such as an irregular scattering of many hues in an image (Figures 6.9 and 8.1).

Discontinuous color may function like sonic disharmony, sonic noise (pink noise and white noise), and forms of asynchrony between sound and image. Particularly evident in this mode of tracking are effects of what psychologists term "weak synesthesia," which is a type of cross-modal interaction (lateral processing) that can be experienced by most people; for example, an exchange between a vivid (loud) color and a loud sound.[8] Color and sound share several descriptive terms, such as tone and harmony. Thus some analogies between the roles of color and sound would seem to have a firm psychological basis.

2. The interrelations among sets of colors in successive shots or as revealed during movements by a camera or character (Figures 8.2–8.6).
3. The interrelations among a set of colors in *non*-adjacent shots—a heterarchy produced by hyperdiegetic narration.[9] In this mode of color tracking, a filmic temporal articulation has intervened (as with 2 above) to break up or reanimate a color complex. Less explicit and less visible are the temporal interventions with 4 below.
4. Most critical are the interrelationships put into place among a set of colors (appearing in 1, 2, or 3) and other narrative or cognitive elements being tracked, as suggested by the metaphor in the preceding text concerning two moving irregular lines or else two moving textual layers, manifolds, or fields.

Clearly, these modes of tracking will need additional specificity. Chapter 5 will outline some of the basic color relationships that are available to measure evolving patterns within particular grammars of color. Chapters 6, 7, and 8 will illustrate the four modes and consider the philosophical bases for tracking.

For now, I would like to offer a loose analogy in order to attempt to convey a feeling for what is involved in thinking about tracking as a special form of interplay between a film and a spectator's working memory. This analogy will tilt toward a subjective description of tracking as opposed to the two analogies offered above that stressed objective descriptions. Susan Blackmore justifies a subjective description in the following way:

> Colours are the quintessential philosopher's qualia; those supposedly basic, private, indescribable, raw feels that make up all our experiences; the "what it's like" of subjective experience; the awfulness of pain or the redness of red. This is what a science of consciousness is supposed to explain; how can the objective workings of a brain give rise to these qualia?[10]

Color perception is one of the entry points to a more complete examination of the status of sensation in our thinking and viewing. What is it like to be me, to feel

being me? That is, to be sensing, to have a view, to be in a conscious moment of inhabiting a unique body while situated in place, in this *very* place? We might call the following description of the movements of consciousness an example of inner tracking. With its tireless unfolding, tracking has become film-like. I have taken the following passage by Blackmore out of context:

> Am I conscious now? Yes.
>
> Ah, here's a new question: Can I stay this way?
>
> A funny thing happens, again and again. I ask the question. I answer yes. I am fully conscious now, I have woken up to this present moment. Right. This is easy. Here I am. But before I know it I am far away in distraction, thinking about something else, being angry with someone, being miles away in the past or the future or something completely invented and troubling and annoying.
>
> The question appears again (from where?). I sigh. Lost again. Yes, I am conscious now, but where was I? Forget that for a moment. Steady. Ask the question.
>
> Am I conscious now? Yes.
>
> It troubles me that I seem so often to be unconscious. I wonder what this unconsciousness is. I cannot believe I spend most of my life in a kind of darkness. Surely that cannot be so. Yet every time I ask this question it feels as though I am waking up, or that a light is switching on. All the more troubling is that this light is so rare. . . . Was this why I so often felt that nothing was real; that nothing was clear, as though something I couldn't place obscured the view and made my head swim?[11]

In reading these six short paragraphs just now, a moment ago, you might have wondered what accounts for the paragraphing, if not the sequential ordering! Is this an analogue to what Blackmore refers to as the unconscious, a "kind of darkness" that structures and guides thought while the view seems "obscured" and suspect? In this analogy, your unconscious has been partially displaced onto "Blackmore" (and "Branigan").

While reading, you were doubly tracking: the time of the sentences, the accumulating sense *on* a page, along with your own sense of consciousness *inside* as meanings were translating into mentalese, though perhaps you were slightly tired or momentarily distracted or needing to re-read the. . . what was her name? have you read anything by her before? or what was it like to read this, think *that* about her words? was something from elsewhere breaking your thought? think back, was that after or before? before what? where were you then? Maybe there was no intrusion that you can remember, only the suspicion of intrusion, or at least there need not

have been an interruption, since thoughts—necessarily being merely possible and unthought, and potential before being not—still always come. . . when needed. . . .

You do not ask where all these questions have come *from*—you focus on answers or hedges, and re-focus, backtracking to. . . to re-read the Susan Blackmore quotes in order to take *from* your consciousness an answering idea, a tight pattern, an evolving pattern as the paragraphs finish, the feeling of an idea coming into focus, something that was already there as a good, poor, indifferent, or partial answer in memory. . . before slipping, appearing, moving onward to the next, despite a flash thought about a nonconscious *there* where one is presently not or no longer.

One of the reasons we track ceaselessly is to try to prepare for what will be next, to project possibilities on the basis of what has been and what is desired—to ensure that what there *is*, we will be *conscious of*, in control of, and ready. Though the personal moment Blackmore depicts—is depicting?—is past, even past on the present page, we make it present by projecting the presence of a self who projects our *own* self sensing in order to follow, to watch unfold, to hold in mind, alert and aware, the next. For Blackmore, the answer to the question, "Am I conscious now?" lies in the fact of asking: "Asking means waiting to see."[12] And, to see, I believe, means not to be stopped short by an inner sight, but to feel the spread of connections as memory works. To see is to be expectant, to track. To see is to experience what something is made *to be like* for an occasion.

COLOR SPLITTING AND FLOW

I have been arguing that analyzing color in terms of tracking avoids being limited to an explanation of color in terms of expressive symbols, or what I will call "color-as-symbol." This is one of ten root metaphors about color. Cultural metaphors set in place descriptions of what color is and how it may be perceived (see Figure 3.1). The semiotic, standing approach to *color-as-symbol* might conclude that a patch of red in an appropriate place evinces anger or passion; clothes that are black indicate a bad or commanding character type; plot events are signaled, "*If It's Purple, Someone's Gonna Die*";[13] themes are telegraphed as in Martin Scorsese's film title, *The Color of Money*; or an atmosphere is implied as in Michelangelo Antonioni's *The Red Desert*. A given spot of color, however, does not always exist in a one-to-one relationship with an "x-factor," nor do several spots necessarily aggregate to make a larger "X." As argued earlier, a hue may be present to our senses on the screen and simultaneously be a component within a top-down fiction imposed upon the screen by a spectator who is sensitive to the flowing of color in relation to other flows in film.

Many film analyses so far have suffered from an inability to formulate certain dynamic qualities of color; for example, dynamic temporalities created through

color patterns across successive shots or through effects of sudden juxtapositions of character and camera movement or as the result of intermittent emerging arcs of narrative. A number of writers, however, well understood this problem, including Herbert Kalmus and Natalie Kalmus at Technicolor.[14]

According to Adrian Cornwell-Clyne:

> Whatever law might be deduced from a scientific analysis of the immense history of applied colour will relate to static patterns, for colour relationships have been conceived by the artist in the past without reference to any extension in time. The motion picture in colour represents the first opportunity for the creation of colour compositions in which the combinations are subject to continuous change, the emotional significance being built up in sequential form, so exploiting the basic principles of music and the drama.

He expands on the idea of color "music":

> What may be good static composition at one moment may be bad a moment later, so that the motion picture is not to be conceived as a successful series of *pictures* but rather as an unfoldment of an idea as expressed in terms of visual motion.

Cornwell-Clyne recommends that filmmakers create elaborate storyboards:

> The artist should prepare a series of compositions in colour, illustrating vital moments of the action, and between these can be inserted further outline drawings indicating important movements of colour masses.[15]

Sergei Eisenstein, who proposed a theory of "dialectical" color, offers a suggestive example of color flow from shot to shot:

> What must be "separated" in the present instance are the colouring of an object and its "colour sound," which form an inseparable whole in our notion of colour.
>
> Just as the creaking of a boot had to be separated from the boot before it became an element of expressiveness, so must the notion of "orange colour" be separated from the colouring of an orange, before colour becomes part of a system of consciously controlled means of expression and impression. Before we can learn to distinguish three oranges on a patch of lawn both as three objects in the grass *and* as three orange patches against a green background, we dare not think of colour composition.
>
> Because, unless we develop that ability, we cannot establish the colour-compositional connection between these oranges and two orange-coloured buoys floating on the surface of limpid greenish-blue water.

> We cannot follow the *crescendo* presented by the movement, from piece to piece, from pure orange to reddish orange, and the greenness of the grass, through the bluish green of the water, to the orange-red patches of the buoys glowing like red poppies against the sky preserving a tint of the greenness but recently seen as the dominant note in the waves of a bay, to which we had been led by the scarcely-perceptible blue hues in the lush green grass.
>
> For the orange does not become a poppy by going through a buoy.
>
> And grass does not become sky by going through water.
>
> But the orange colour, going through reddish orange, finds its consummation in red, and azure is born from bluish green engendered by pure green with a spark of blue in it.
>
> For some reason or other we feel the need of a series of objects: the three oranges, the two buoys and the poppies are blended by one common movement of colour supported by the tints of the background.[16]

Eisenstein stresses that thinking in color requires the *separation* of a "colour sound" from a colored object, followed by its sonic-like travel through a film in concert with other separated colors, like a melodic line of advances, returns, tensions, attractions, dominants, overtones, crescendos, and releases. Eisenstein would also seem to endorse the kinds of values discussed in Chapter 2 associated with an aesthetic of delicacy and subtlety when he speaks positively about the experience of having been led through a series of images by a few "scarcely-perceptible" hues, e.g., the movement of orange "going through reddish orange." Indeed, when Eisenstein speaks of "buoys glowing like red poppies," the use of the word "like" could almost suggest that the red he mentions has come from his memory, from a red imagined on poppies perhaps moving with the wind in a green meadow as if bobbing on a "greenish-blue" sea, thus already separated from an object being photographed and within a stream of connected mental images that resound on a higher level, no less true for being metaphoric.

As we shall see in Chapter 5, color has a number of qualities besides the three main ones of hue, lightness, and saturation; qualities like surface shine, highlights, translucence, reflections of a hue onto another, and kinds of mixtures, e.g., Eisenstein's detection of a tinge of blue within a cyan. (Perhaps you have noticed the blue within the green, actually a mosaic cyan, of the standard traffic light that signals "go"? If you have failed to see the blue, is it because the light is always referred to as simply "green"?!) In addition, a color may have countless relationships with nearby and distant objects, whether the objects are material or immaterial. Thus there are

a great many ways to "separate"—or split off—an aspect of color from its colored object for purposes of movement along unfolding lines through a film where it may become temporarily attached, even multiply attached, to new places and relations. As Eisenstein suggests, learning how to split up color while splitting apart other objects is a precondition for relocating the cognitive significance of color.

In an important study, Paul Coates elevates Eisenstein's notion of separating color from its object into a major principle of aesthetic theory and the basis for a history of filmic color. Coates develops an intricate system of allied concepts: separation, splitting, doubling, mirroring, oscillation, dialectic, alienation, abjection, ambivalence, ambiguity, absence, abstraction, indeterminacy, polyvalence, fantasy, fluidity, and suture. Evidently, the drive to separate color from its object has many implications and creates many sorts of parts that may engender "ambiguity," which for Coates is the principal goal of aesthetic texts.[17]

If a given color has many possible *parts*, then which part is relevant in a given context? Tracking theories reveal a color bonding with various invisibilia, such as narrative elements, thus increasing the number of parts in a provisional whole. It is vital to note that a narrative part—e.g., the narrative phase "in medias res"—is not at all visible on the screen in the same manner in which color is visible. Narrative is intangible and trans-media, an arrangement of relationships between objects and actions imposed by a mental schema. These factors concerning the relations of parts, patterns, and invisibilia can make interpretation a difficult art while making color ghostly.

THE ARCHIVE DILEMMA

There is, however, a bright side to the complexity of color. These same considerations of splitting and flow militate against the need for a precise reproduction of each individual color when viewing a film. Within reasonable limits, one does not need the presence of the exact *original* color that first appeared on celluloid during initial projection in a theater. There has long been much hand-wringing and despair in the literature on film color concerning the fact that we cannot see the *original look* of a given color, or in many cases even know how to recapture that look because it was due to a lost technology of reproduction. I will refer to this as the "archive dilemma."

I am certainly not arguing against archives and their invaluable work of preserving and restoring films. I am merely making an aesthetic point. Even semiotic theories do not always require the presence of the original color—any red within a passable range will be sufficient to signify or symbolize anger or passion. There is little point in studying film color while afflicted by a distressed nostalgia that can never be lifted because the exact color of a lost object can never be re-found. The Original with its special aura is irretrievably past.[18] Fortunately, tracking theories significantly reduce,

though do not entirely eliminate, the toxicity of the archive dilemma since their focus is on functioning patterns. In the case of tracking theories, a certain variation in color is often acceptable. For instance, any of several hues with a modest range of lightness and saturation values will serve to establish a warm versus cool pattern of oppositions; similarly, any values within a modest range will constitute a *relationship* among secondary hues. A tracking theory relies especially on relationships and patterns, rather than on the specificity of individual color.

One can glimpse in the archive dilemma the beginnings of a divide between two types of film theories: those that emphasize ontology—an inquiry into the essential nature of a thing, its determining features, its origin and uniqueness, its originality—versus those theories that emphasize epistemology and meaning, how we come to know, interpret, and appreciate a thing, where certain kinds of (sometimes rough) copies are sufficient to answer aesthetic kinds of questions in certain contexts.[19] The divide is between what we perceive or believe is constitutive of an object—of its essence, of whatever makes it the type of thing it *is* (ontology)—as opposed to what we perceive or believe an object to be *capable of*, how it functions to produce belief effects (epistemology). Theories in the former class attempt to draw up a definitive list of core, defining properties for an object so as to certify its authenticity, or lack thereof, while theories in the latter class emphasize causal and dispositional properties within situations as well as an object's ability to point to things, to discriminate, classify, and categorize, thus making the object's emergent properties and performance *variable* according to circumstances of viewing, rather than predetermined by essence. A perfectly restored film still says nothing about how it was seen by an original audience.

A similar divide informs how we think about the nature of color in the first place. Should we be measuring the continuous magnitude of various qualities or dimensions of a hue to judge its legitimacy, as in an ontological standing theory, or should we be explicating relational structures and topological spaces, as in an epistemological tracking theory? The former theories are substance-based, the latter function-based. In short, is color to be analyzed as being locked into a sensuous patch or spot of some type—perhaps now lost—or is it movable, relational, and interactive with its original or present surroundings, not just interacting with other colors, but functioning with other concreta and generica, especially with elements of narrative?

Once the relevant characteristics of colors have been described by a given standing or tracking theory, a method of analysis must be crafted in order to display these colors and to evaluate them as being appealing or in poor taste. In order to understand how analytical methods are fashioned from color theories, we must first investigate the typical ways we are led to think and speak about color. As Metz suggested in

Chapter 2, "manners of speech" provide well-worn paths for thinking, experiencing, and formulating theories. Let us now examine some of these accepted manners.

WHAT'S *IN* A COLOR? (WARHOL)

"What's *in* a color—*of* a color—that of necessity makes it unique, giving it the power to be attractive or unsightly?" is the wrong question to pursue.[20] A better question is "what can a color *do*?" or "in these circumstances, what is a color doing for a perceiver?" The first question, which searches for special internal qualities and their effects, has received a variety of compact, but reductive, answers. The second, which calls for measuring actions and circumstances, requires long, messy, and incomplete responses that have no definite center and exhibit no hierarchy, no predetermined predilections. I think the second question is to be preferred because it focuses upon our continuous and changing engagement with objects and with the commitments we daily renew to a familiar world. The second question leads to an inquiry about how we have invested our world *with familiarity*. What do we expect from ourselves when we expect something from a colored object? In answering the second set of questions, it will be helpful to probe why the first question, "what's in a color," seems so comfortable and tempting. This will require an examination of the ways we are inclined to talk about the (purported) nature of color. Which mental images about color are available? Which mental images will best serve a tracking theory?

In the present and previous chapters, the words "paint," "painted," "painting," and "painter" appear nearly one hundred times. This is no accident. The everyday way in which we think of color is *as paint*. It is not that we are visualizing something that is in a paint container or on a palette. Our image of paint is instead of how we might use, manipulate, and control color—how we could make something *possess* a color—how we, in fact, imagine objects to have literally or figuratively *received* their color. In short, we tend to think of all color as having been *applied*. We visualize a thin film appearing on a surface and hence we think of color as merely a surface phenomenon; in a word, *as-superficial*. The real essence of an object—its thingness—apparently resides below down deep. True substance lies out of sight—and it is solid, bare, and perfectly colorless.

A closely related metaphor to *color-as-paint* that captures the distinction between merely colored surfaces versus deeper truth is *color-as-cosmetic*. The "paint" in the metaphor has been transformed into a beauty aid and a disguise. Food coloring is a frequent cosmetic for what we eat and drink. Beginning with Plato, color has been theorized as a kind of cosmetic, indeed, as a form of sophistry. According to Jacqueline Lichtenstein:

> Flattery, cosmetics, artifice, appearance. . . all the terms of this metaphorical chain linking the critique of painting, of sophistry, and of rhetoric, also qualify the effects of color as effects of seduction; they are the effects of illusion and pleasure.

Further, the equation of color with cosmetics derives from cultural beliefs.

> An entire tradition, of which we are the heirs, takes cosmetics to mark an original defect, to veil an ugliness always sensed beneath the virtuosity of masks, to signal an imperfection that art seeks to dissimulate.

The result is that color becomes "feminine, indecent, unnameable, illicit."[21]

A more positive master metaphor is *color-as-jewel*. This rhetoric derives from the look of valuable jewels, gems, precious stones, and stained glass windows, which seem to radiate an interior light of particular intensity and purity; that is, one can see into the depths of the jewel and see inside that there is only color *simpliciter*. It is a step toward *color-as-spectacle*, discussed in Chapter 2 in conjunction with evaluative g-words. In this way of thinking, color seems to emerge directly out of precious jewel-like objects to be applied to things. The analogue for film would be the film projector or the screen out of whose depths emerge glowing images. While wrestling with the question of why so many theories that seek to explain color "get caught in the spell of gems and precious stones," David Batchelor remarks parenthetically: "but isn't the mesmerizing illuminated screen of the cinema something like a modern equivalent of the sparkling gems of older times?"[22]

But what does the screen screen? Not all writers adopt Batchelor's line of thought about magic gems and dazzling images. For Adrian Cornwell-Clyne, a different core metaphor obtains:

> We are here concerned with the art and science of casting moving coloured shadows upon a dead screen in such a manner as to deceive the eye by simulating reality. But perhaps these shadows have no more and no less substance than that reality they set forth to reproduce.[23]

There is more than an echo here from Plato's cave and theories of art based on imitation, simulation, and illusion.[24]

We have moved from *color-as-paint* and *color-as-cosmetic* to glowing gemstones and finally to an illuminated screen in front of a mesmerized spectator. Notice that the language of Batchelor and Cornwell-Clyne—through a chain of metaphors—is leading inexorably toward deeper theoretical commitments about art and toward a *mental picture* that determines how we will think about a thing that has color so as make a series of *inferences* about its effects and then *project* ideas onto new things (through further leaps of "seeing-as"). The final result of this chain of metaphors

suggested by Cornwell-Clyne is not far from an inactive, semiotic standing theory of film color: shadow, absent signified, deception, absence, and dead screen.

By contrast, if we replace the film projector or screen with the activity of a camera, and begin anew with a metaphor of *color-as-camera*—thus installing a tracking theory of color—we will discover while moving backwards along a chain of metaphors from cinema to color that we will need a replacement for *color-as-paint-or-jewel*. Shortly, we will find just such a replacement metaphor illustrated in the work of Andy Warhol. But first, let's examine a few more consequences of the paint-and-jewel rhetoric.

Allied with the *color-as-paint* and *color-as-jewel* metaphors is the omnipresent, and constantly mutating, passive rhetoric of *being-colored-by*, which takes a wide variety of intriguing forms. For example:

> "His words were colored by envy."
> "She is colorful. She has a colorful personality."
> "He has a colorful past."
> "The primacy effect describes how initial information colors the processing of all subsequent information."
> Thinking of the sky as being painted or colored blue may have led ancient philosophers to create a cosmology based upon an array of tangible cosmic spheres, which in some versions were devised to explain the movements of celestial bodies as a form of (colored) music.
> As pointed out by a well-known film theorist, our "special intuitive relationship with film-thinking. . . is also one that is coloured by knowledge and context."

The fact that color is being conceived as a trace element in the *being-colored-by* rhetoric allows anything at all to be subjected to coloring by anything else! Neither thing needs to be an actual color (e.g., "our thoughts color our perceptions"), though one or both may be colors. *Being-colored-by* is a larger class than first appears. When we say that a person is "green" with envy or a new recruit is "green"—perhaps a "greenhorn"—or someone looks a bit "greenish" (i.e., feels nauseous) or is "in the green" (i.e., has money), we are using the color green as a one-place predicate, i.e., as an inherent property, and saying, in effect, that a person's physical or mental state *is-colored-by* green.[25] Also, characterizing an object as a "green object" may be a twice-true metaphor: green-colored as well as beneficial to the environment, i.e., possessing the growth/life-affirming properties attributed to springtime—a time of vitality, renewal, and rebirth; and so also grass may be literally and figuratively greener on the other side of a fence.[26]

The image of *color-as-paint* that allows even an abstract object to *become colored-by* would seem to fit well with a semiotic standing model, especially since color can be taken to be what is present, a sensuous signifier, while the true, core

substance of a thing remains unseen and absent, a signified and referent, a secret to be uncovered. Color that stands out, like the redness of measles, is seen to be a symptom of—to be standing in for—something else. This handy rhetoric, however, over-generalizes and obscures other possible causes and uses for color sensations. Furthermore, it installs (*in*stills?) a closely related picture in our minds that has a deleterious effect on our thinking about the character and analysis of artworks. This related mental picture, is based on an artwork's fabrication, its history of production, on the multiple ways in which an art object has come to be "painted" in the widest sense (to appear as an author's measles). We might call it *colored-as-intended*. This sort of picture can make the first cause of an artwork seem to be the most important (e.g., its scripting, time of production, technical features, and an author's explanation), and perhaps the artwork's only real cause. The rest—an artwork's circulation in a community, its interaction with contexts, and its production of individual responses, indeed its future—is merely downstream flotsam. If a *colored-as-intended* artwork is a tool for understanding, it is so only in the hands of its creator, the master maker who applied the literal or figurative paint. Thereafter the art remains inert—showing or saying the same thing over and over, fit only for exegesis.

The world, however, is filled with movements, unpredictable by degrees, and is not neat. It seems to me that art should be considered a part of this same world—in a word, "public." Art is about ways of seeing, attitudes, and arguments directed at community life—older traditions are not "replaced" as they are in science where new experiments and theories supersede older ones. A tracking theory provides a way to model the working of color that is not limited to searching for a first and single cause.

I believe that a helpful corrective to the semiotic *color-as-paint* image may be found in Andy Warhol's series of five "Do It Yourself" paintings in 1962—composed of acrylic paint, pencil, wax crayon, and Letraset printing—which are based on a "painting-by-numbers" kit sold to amateurs and children.[27] Each of Warhol's paintings depicts a generic scene, e.g., of a landscape, flowers, seascape, or sailboats, covered by a map of detailed contours, each contour being marked by a number corresponding to a color that should be filled in to make it a proper painting. Leonardo da Vinci is known to have directed his assistants to fill in numbered sections of his paintings, and compression software, which is ubiquitous in contemporary life, works the same way.

Warhol's versions of "Do It Yourself" paintings, however, contain surprises. Some of the numbers are already filled in with colors, though the hues are not always consistent with the numbers. Furthermore, the numbers sometimes continue to show through the paint even though the chosen hue is supposed to be opaque. Large sections of a scene remain unpainted and incomplete, filled out only by numbering and outlined shapes. Warhol's scenes are barren of human figures. Although

a large sailboat is moving in the foreground of one painting, no one is in it. Objects in these paintings appear luminously alive, colored like Life Savers and Popsicles mostly in vivid, stark, and pure hues that pop.

Warhol's paintings suggest that the objects of a scene are basically empty until they are filled up by giving them color; a world under construction like our own. In effect, the surface/depth paradigm of standing theories is dissolved. There is no longer a secret or absent core or first cause that is covered by layers of distorting translucence and opaque veils. Warhol's colors themselves appear to be mobile quantities such that different colors might fill an object at different times despite the numbering. The pervasive numbering across the entire canvas and the division of things into many parts—including fragmented water and sky—is a constant reminder of a certain manufactured, cultural relativity beyond the manufacture of artworks. Objects stand not alone, but acquire, and then perhaps change, their significance. This is not a classic naturalism in which painted chiaroscuro lighting motivates a drama of subtle color variation. This is painting that emerges from culture and adapts to changing circumstances. Its utility exists under uncertainty, always provisional.

Further, there is a sense in Warhol's partially unfinished renderings within the paintings that chance and noise have played a role, and that much of life is being lived through fairly simple stereotypes fit for quick decision-making, like the generic, easy-to-remember shapes of an unpainted paint-by-numbers (which, according to Deleuze, is Godard's point, too; see Chapter 2). Shapes emerge from spaces that seem otherwise blank, where viewing has been lost to daydreams and *temps mort*. Warhol's "Do It Yourself" paintings—based on pre-sketched numbered drawings—work to dispel the mystique of an "original" versus a "copy" in favor of a sense of contingency and circulation.[28] (Recall the earlier discussion of the problems of originality and authenticity in relation to ontological theories and the "archive dilemma.") These paintings are consistent with a tracking theory, which, I believe, replaces the standard and static, semiotic metaphor of *color-as-paint* with a richer, more dynamic image for the possible workings of color, including meanings to be added by future viewers. This richer image might be called *color-as-context-sensitive* or *color-as-causative*.

Warhol is not pursuing what Adrian Cornwell-Clyne would term an aesthetics of delicacy and subtlety. At least not obviously! That is, one cannot find delicacy in Warhol's painting by looking in some of the usual places: color, shape, space, and subject matter. Delicacy, however, may be found in less obvious places in the "Do It Yourself" paintings where Warhol provokes oblique reflections on culture and the status of art. With respect to the narrow issue of Warhol's bold and basic hues, one can find in modern painting and films aesthetic uses for a range of simplistic, neon, lurid, shocking, discordant, and even chaotic color designs, since confusion and multiple contingencies may have their uses. This should not be a surprise, since art often negotiates tensions that aim

toward resolution or a resolution of measured irresolution. In the best art, color does not stand by itself, but tracks and acts to visualize other features and provide links to forms of culture.

The conclusion is that it is not finally what's *in* a color that matters. Rather it is which mental picture of color is being summoned as an explanatory framework for the inferences that will be made about what is being seen.[29] Figure 3.1 summarizes ten core metaphors for thinking about color. Each metaphorical frame may appear in a verbal or visual format and be blended with either words or images to generate a person's expectations about color and provide a ground for inferences and projections. These mental pictures may appear to be static, but like gravitational potential energy, they may be rapidly converted into kinetic energy and a stream of meanings designed to embrace what is being seen. In this sense, all ten of the schemata make color move.

My assumption is that color is capable of becoming an active force in moving a spectator toward new ideas and feelings. Color is capable of dynamically organizing both top-down and bottom-up perception. It need not be reduced to a record of reflected ambient light from a spot, but, like art itself, it may become artful. Color may track a world of ideas and events starting from splendid sunsets to grays, spanning the extravagances and minimalities of human culture.

SUMMARY: THE STORY SO FAR

I would like to summarize the arguments of the first three chapters and the assumptions they put in place for what is to follow. I have argued that the condition of chromophilia may enhance one's life when warranted by a proper design. Many writers have drawn a sharp distinction like that of Sergei Eisenstein between two sorts of color designs: a film or painting that appears merely color*ed*, gaily or dimly, as opposed to an artwork that is designed and structured to be *in* color and capable of true "impressiveness." (Note that the first distinction places our sense of color in the past tense while the second makes color present.) Unfortunately, until the recent decade most film theorists and critics acted as if all films were in black-and-white. Worse, it was often forgotten that black and white are themselves colors. Still further behind are novelists, and surprisingly even poets, whose words typically paint with the most insipid and rudimentary of colors.[30]

The present book aims to introduce some of the concepts and tools through which one may begin to see more of color. Four major themes will be addressed: (1) the conventions and norms governing color, including cultural, commercial, institutional, and aesthetic practices; (2) the distinct languages—the socially anchored linguistic systems—presently available for expressing our consciousness of color, including its expression through physiology, psychology, and emotion; (3)

the processes by which colors may be memorialized—that is, established in memory and made emblematic through core cultural metaphors and mental schemata, especially in those situations where color melts into important concreta or abstracta; and (4) the problem of *color-as-spectacle* and the general nature of sensation in film.

Critics have been easily moved by film's sensoria to offer up laudatory or accusatory judgments about the appearance of a color design by employing their favorite g-words. I count twenty-two such words from "glorious" to "gaudy." Which presuppositions about the nature of film and color are available to set the conditions for applying these sorts of statements and epithets? One value that many critics appreciate is delicacy and subtlety. Underlying this value, however, are facts concerning the mind that are applicable to all aesthetic judgments. In measuring the effects of color on a spectator, one must understand how memory as a whole—not just the eye—deals with, and transforms or conforms color sensation into memorial traces through the resources of language. A given color may be completely forgotten after having been seen; or, it may continue to exist in some altered state apart from its source as a quale capable of recall; or, in special cases, it may infuse a concrete object or abstraction precipitating a memorial act. Whether a tonal structure is meant to be slight, subtle, or "difficult," it will play upon and within both the thresholds and the variations among the sensory and cognitive systems that create our sensitivity to an environment. Thus it is necessary to inquire how color patterns are fit to patterns of mind.

There will be significant disparities among individual responses to thresholds and variations, but not an infinite number, and not every response will be deemed relevant to characterizing the totality of an artwork. As a public object, an artwork is designed to elicit something approximating a statistical distribution of responses falling along—one may imagine—some sort of bell-shaped curve. (Although science and art may be used to analyze each other, neither reduces to the other.) Any individual response is only more or less probable. A theory of color, therefore, should aim to register psychological and aesthetic *tendencies* across a spectrum of possibilities. As Adrian Cornwell-Clyne's six-part "cycle" of color patterns suggests, many schemes and effects are possible that incorporate an assortment of contributions from norms of culture, commerce, narrative, mind, and language.

One conspicuous norm is the color of flesh. Technicolor understood that commercial success depended on orienting their technology toward reproducing a particular tonality of flesh—the "Technicolor tan"—not as flesh actually appears in reality, but as an audience may *desire* it to look, according to a cultural *ideal*. Furthermore, glorious-looking faces expressing deep emotions perfectly fit the character-driven stories that Hollywood wished to market.

But color need not be limited to the surface of a body. A John Updike sonnet demonstrates how our visual imagination for color and its qualities, e.g., glisten and

sparkle, is able to supply, for example, a link between "tears" and "diamonds," thus generating, as it were, a poetic metaphor of memory mapped onto color to hold *events* of the body by. This demonstrates that in the right circumstances color can function as a causal agent and need not be imagined merely as a lifeless coating upon an object—where an object's real nature is believed to lie somewhere out of sight beneath a thin and filmy, painted façade. Imagine color sinking into memory. For the same reason color should not be confined to the surface of a film screen where spectators peer through it to a story world as if it were merely tinted glass. Instead, color is fully capable of assuming a dynamic role in a spectator's network of thought.

In fact, every element in a fiction is capable of becoming a tool for thought. Theorists of color, however, have been too easily charmed by a hue's emotional resonances—which some claim are semi-intrinsic—to the exclusion of a hue's possible cognitive tonalities. Many theorists fail to recognize that color may be invented anew within a given text in order to participate within language-games that convey dispositions and values. When speaking—displaying our comprehension—we must first make sense to ourselves, and we do this by selecting a community language. Broadly conceived, a language-game or language *field* is a distinctive system of signs or vehicles together with rules, conventions, linkages, and associations of meaning that are adjusted to particular human goals and problems. A language-game may appear in the form of strings of words, images, plot actions, motivated behaviors, gestures, stylistics, and in other ways.

Examples from John Updike and from the literary critic James Wood show that the cognitive potentials of color are undiminished whether appearing in fictional or nonfictional languages. The reason is that the differences between fiction and nonfiction do not lie with a special poetic rhetoric or in narrative structure. A fiction is neither inherently false nor deceitful nor an imitation/simulation/pretense, but is merely a discourse that is put to a special *use* by a spectator, just as counterfactual conditionals are employed to state scientific laws. Color retains the power to inform abstractions of all sorts, whether fictional or nonfictional, narrative or nonnarrative, verbal or pictorial. Indeed, as Christian Metz implies, the concepts of film theory themselves are indebted to "poetico-didactic fictions," "manners of speech," and metaphors of color.

Once color is seen as a tool with many uses, it is no surprise that a filmmaker like Jean-Luc Godard can—as described by Gilles Deleuze[31]—exploit colors as attitudes of the body, using hues to form categories and conduct analyses of social situations. Thus color moves from being a physical patch that is typically perceived bottom-up as a gleam painted onto an object toward the possibility of its functioning top-down to imbue an object with attitude.

How is it possible to display color become argument? Can top-down, cognitive potentials of color be drawn out and analyzed? One possibility is to employ a semiotic framework of signifier and signified that is premised on a dialectic of presence

and absence whereby an element, A, present in a text, *stands in* for something else, B, that is absent from the text in accordance with a code or rule: A → B. Although there are many important nuances to this framework of *color-as-symbol*, I will refer to the class of semiotic theories as a *standing* or *stand in place* approach.

An alternative framework is a *tracking* approach to color. A tracking approach must go beyond expressive symbols and the hunt for one-to-one relationships between a color spot and an "x." Tracking aims to incorporate movement or flow into an account of how color fashions meaning moment upon moment. The emphasis is not on connecting A to B, but rather on measuring dynamic fields and mutual patterns: A^—^B. The approach may be visualized by imagining a tracking camera and then substituting a specific hue for the camera, resulting in three sorts of possible interactions between the moving *color-as-camera* and objects in a scene: (1) a pattern of color mixtures and shifts may be made evident among objects in a mise-en-scène that reflects a prior (unseen) movement; (2) various effects become manifest based on explicit movements whereby a sequence of colors is being selected and arranged within a shot or several shots; and (3) the continuous framing and reframing of colored objects at various distances make a specific hue's area exert an influence on nearby hues as well as enable color matches or mismatches employing off-screen space.

A second way to visualize color tracking is in terms of two irregular moving lines that assume various relations with one another at different times. One of the lines may be imagined to represent a field of colors while the other line represents some important concreta or abstracta in a given film. In this second visualization of color tracking (as in the first visualization of *color-as-camera*), the emphasis falls upon analyzing shifting patterns and parallels among textual layers and relational structures, together with their interactions within a spectator's *working memory*. The search for symbolic and psychological links among elements—a feature of *standing* approaches dependent on *long-term memory*—is less important than describing the changing shapes of color combinations in relation to changing shapes in an evolving narrative. A tracking approach brings to light fascinating analogies between the flow of color and temporalities on a soundtrack.

Four modes exist for tracking the flow among interrelated colors, depending on whether the colors appear in a single shot, in successive shots, in non-adjacent shots, or in conjunction with a relationship to some narrative or cognitive feature being tracked. The flow, as Susan Blackmore suggests, is simultaneously of external stimuli accumulating sense on a screen and of an inner, already film-like stream of consciousness in working memory that is compiling meanings and a sense of self for those meanings. If one must always be conscious *of* some thing, so a thing seen is always *for* a self.

A tracking approach attempts to map developing color patterns as described by, for example, Adrian Cornwell-Clyne and Sergei Eisenstein as well as Natalie Kalmus

in Chapter 6. Eisenstein separates the "sound" of a color from the "colouring" of an object. "Colour sound" is able to move much like the flow of music with particular sorts of advances, returns, tensions, attractions, dominants, melodies, overtones, crescendos, and releases. For Eisenstein, some of these color movements are "scarcely-perceptible," which would seem to summon—at least on some occasions—an aesthetic of *color-as-delicacy*, though not necessarily the delicacy advocated by Cornwell-Clyne.

Color that flows like music has many intrinsic parts as well as many extrinsic relations to its context, especially to the *in*visibilia of narrative. This complicates a spectator's interpretive activity, but, happily, somewhat reduces the "archive dilemma," which is a paralyzing fear caused by the fact that one can never completely restore on screen the color originally photographed, but only its pale shadow. The archive dilemma is made less severe by a tracking approach because of the strong focus on constructing patterns and relationships among groups of colors, rather than focusing on the quality of each individual color (and how it *stands*) apart from any context. Nonetheless, and importantly, the archive dilemma hints at a deeper issue that divides theories and methods along ontological and epistemological lines, as described for example by James Lastra in the context of sound theories.

"What's *in* a color?" is the wrong question for a theory or analytical method because it leads toward metaphysical speculations about supposed hidden internal qualities—the true *standing* of color—and away from tracking movements, patterns, and spectator responses in the context of epistemology, i.e., responses born of perception, cognition, emotion, and the influence of contexts, inside and outside of film. The sensuous properties of color are easy to fasten upon and thus perhaps mislead us into thinking that color is simply like *paint* on a palette—that is, color may seem to be an entity casually applied to a surface, covering over the real substance of an object beneath like a *cosmetic*. This common image for color shows it to be readily accessible but superficial; hence the everyday rhetoric in which some abstraction is said to *be-colored-by* something else. The semiotic *color-as-paint-or-jewel* image over-generalizes and obscures other possible causes and uses for color. It also encourages one to overvalue details of a film's production and history of production—its first cause—leading to a rhetoric of *colored-as-intended*. A better image to have in mind, consistent with a tracking approach, might be called *color-as-causative*. Ten master metaphors of color are summarized in Figure 3.1.

An illustration of a tracking approach is furnished by Andy Warhol's "Do It Yourself" paintings, which portray color as mobile and in the process of forming objects by providing them color and outline as well as exemplifying an attitude toward the kinds of social conditions out of which these paintings have materialized. In Warhol's works the dilemma of an "original" versus a "copy"—authenticity versus situated use—becomes less urgent. Warhol employs a condensed folk idiom—generic types of

objects and blank areas, rather than symbols, tokens, or striking objects—to literalize and satirize the notion of *being-colored-by*. The effect is to flaunt the power of color to conceive.

In the best art, cinema's color does not stand alone, but tracks variously with other features to explore tensions and contrasts, to conceive ideas and drive feelings. Overall, the present book promotes an image of color—whether delicate, simplistic, bold, irreal, discordant, or chaotic—that shows color dispersing like an invisible luminiferous aether, moving like waves from place to place, soaking into objects, condensing out of objects, triggering thought and emotion, and, finally, giving to abstracta a sensuous embodied quality engendering in the viewer a state of chromophoria. **FIGURE 3.1** Color-As: Ten Root Metaphors for Perceiving Color

FIGURE 3.1 Color-As: Ten Root Metaphors for Perceiving Color

In each of the following ten cases an objective feature of color has been selected and highlighted through comparison to, and blending with, other things or phenomena. Color is made to be pragmatic in the form of a creative metaphor, i.e., it is being metaphorized. Once an objective feature of color and its associated metaphor have been chosen, then successive inferences, elaborations, and projections by perceivers will produce a lattice-like folk theory congruent with the master metaphor that specifies the nature, perception, cognition, aesthetics, realism, analysis, criticism, and cultural politics of color and specific hues. Through a process of back projection of a given explanatory metaphor, various histories and folk histories of color practices may be constructed. The root metaphors of color may also be formally analyzed as "blends."[1]

ESSENCE

1. COLOR-AS-PAINT

In the metaphor, *color-as-paint*, all color is identified with a typical way in which we furnish it to objects, make them have it. Color is being conceived from the bottom up as a special liquid that spreads out and hardens in place. It is visualized as a final, thin surface feature—something external that has been added to or coated onto an object and remains static. Thus the color of paint, dye, or ink appears as superficial to what is real, namely, the solidity, shape, and three-dimensionality of an object. Hues can be altered without changing an object, thus making any hue accidental because we believe that shape, material, and mobility are what truly define objects. Color is often seen as accidental, not necessary, to the true substance of a thing: a rose by any other color

would smell as sweet. (There may, however, be exceptions for color that is believed to be thoroughly imbued with "jewel-ness" or "author-ness"; see below numbers 3 and 7.) Even when an object possesses a natural color, e.g., wood, we tend to think of its hue as the result of a sort of applied varnish or else the result of a paint being mixed with an object's "preexisting paint." It is layers of paint all the way down to a real object.

We think of color in terms of paint even when it is perceived without the usual grounding of a solid object; for example, with sky, smoke, clouds, rainbows, fog bows (white rainbows), liquids, light sources, shadows, colored shadows, reflections, mirrors, video monitors, afterimages, glories,[2] auroras, coronas, ice blinks, screened images, and those colors seen while remembering and dreaming as well as those colors appearing on token materials, such as the canvas of a painting. It is not relevant that we actually know the true origin of a non-surface color, such as smoke, because we continue to imagine and project color as essentially something painted, applied, static, and arbitrary. Color is something like a sock that covers the real thing that moves.

Color-as-paint supports the complicated and much-debated philosophical distinction between primary and secondary qualities, where solidity, shape, extension, mass, relative motion, and number are primary and color, sound, touch, taste, and smell are secondary. The *color-as-paint* schema is as forceful and compelling as many of the fictions and images we daily choose to live by. It is the default option through which objects are seen to have color.

2. COLOR-AS-COSMETIC

In *color-as-cosmetic*, color is a painted disguise, artifice, and mere cosmetic appearance, even if the appearance is important, e.g., a label, flag, or clothing. Like sophistry, color is taken to be seductive and illusory—a form of dazzling embroidery, though deceptive and distracting. There are many similar uses in jurisprudence, e.g., "under color of law," "under color of title."[3] The phrase "showing its true colors" suggests that a cosmetic or paint has been removed to disclose a thing's "real" nature in the form of a (painted) undercoating or a new, metaphorized sighting of color. That is, one may continue to ask about the status of any newly found "real" color underneath the old and how it was produced. Each new color will be seen as simply another layer of cosmetic and paint that covers up the real object, which may ultimately be immaterial! See, e.g., the final paragraph of "The Whiteness of the Whale" in chapter 42 of Herman Melville's *Moby-Dick*.

3. COLOR-AS-JEWEL AND COLOR-AS-RAINBOW

In the closely related metaphors *color-as-jewel* and *color-as-rainbow*, color is imagined—imaged—as strong, deep, and pure, radiating from the innermost points

(Continued)

FIGURE 3.1 *(Continued)*

of precious gems or stained glass windows. One's perception stops with the hue and gives no thought to the light source behind the object that has created the colors as *luminous*, rather than as appearing flat and reflected from an opaque surface. In fact, luminous color gives the impression of not needing a light source: it simply radiates itself, appearing as a mysterious emanation from the soul of things. A perfect illustration of this way of thinking is Stan Brakhage's film, *Black Ice* (1994)—the title of which poses color as an alluring mystery, as simultaneously black and (icy) white, an impossible dark luminosity or opaque transparency. Nor are color theorists immune from such rhapsodic experiences. For example, theorists who privilege dream states as a model for film may have in mind *color-as-jewel* since colored objects in dream are self-luminous—there being, of course, no light in the brain by which to see, making dream states an occasion for extra-ordinary in-sight.

If a rainbow is appropriated as the dominant image, it is being seen, in effect, as a necklace of jewels lit from behind by the sun.[4] The resultant glowing ensemble of hues obeys the ancient aesthetic maxim, "unity in diversity," whereby unity is carefully created "without uniformity" and diversity "without fragmentation."[5] Rainbow hues derive their beauty from the revelation of a source in nature's life-giving sunlight. Gemstones, too, are of nature, which seems to distinguish these sorts of hues from paints and cosmetics.

There may, however, be a darker side to the rainbow. For example, Plato in *Republic* describes democracy as a form of government that is but one step from tyranny and despotism, which is the fifth and worst sort of political arrangement. Socrates describes democracy as follows:

> Possibly, said I, this is the most beautiful of polities; as a garment of many colors, embroidered with all kinds of hues, so this [democracy], decked and diversified with every type of character [character type or person], would appear the most beautiful. And perhaps many would judge it to be the most beautiful, like boys and women when they see bright-colored things.
>
> (557c; tr. Paul Shorey)

"Like boys and women" adds a wily, backhanded touch to the deceptive rainbow-like beauty of democracy.

Provisional Conclusion

What is color, then, when viewed through these three ontological metaphors? Not really itself. Instead, color exists to reveal, however deceptively or displaced, the existence of a radiant mass or substance or force or spirit lying behind, below, beyond, or with—an alien thingness. Color merely indicates a presence that exists elsewhere, a mysterious extent infused by light.

AESTHETICS: TWO PROMINENT METAPHORS[6]

4. COLOR-AS-SPECTACLE

Color-as-spectacle is seen as glorious and dazzling or else garish and gaudy, but always intense. So intense and plentiful that the sensation may easily become excessive and exhausting, driving cautious reflection from mind, leaving only unpredictable—unpremeditated—emotion to rule thought. The assumption here is that color is inextricably and directly linked to emotion. Though emotions must not be absent in viewing color, the claim is that they must be regulated and controlled. Like a rainbow, a human being should exhibit a balance between emotion and reason, a "unity in diversity."

As an illustration of the tension within *color-as-spectacle*—glorious but not garish—consider the twice-true metaphor, "colorless voice," which refers to sound that is both literally and figuratively without color and so rather lifeless. The positive obverse of such a vapid colorless-ness asserts the intrinsic mating of color with living emotion. The idea is that color is only ever a *feeling* of some secret sort that opens onto an intuition beyond the merely cognitive, albeit that such deep feelings, like color, must still be carefully controlled and restrained so as to be glorious, vibrant, and valuable (resonant) without becoming garish, unruly, or dull.

5. COLOR-AS-DELICACY

When *color-as-delicacy* becomes the norm, color is seen to solicit a fascination with liminal states by demanding finer and finer discriminations among sensations. Patterns emerge through subtle and slight variations. Borders may become indeterminate, color cloudy. Harmonies unfold slowly, often creating the sense of a faint, holistic reality or provoking an unusual state of mind. The world seems to be a place of delicate, piquant secrets requiring all our powers of acuity, like a connoisseur of fine cuisine. Color is marked by deliberate and profound good taste.

The reverse of delicacy is *severity*. Here lies stark contrast, denial of transition, pop-out hues and borders, even discordant juxtapositions. Discrimination is erased in favor of intensity and stark simplicity. See, e.g., the palette of André Cadere in Chapter 5, section 7, subsection 4.

PRACTICE

6. IS-COLORED-BY OR HAS-BEEN-COLORED-BY

How does color perform when conceived as an object of use that *acts upon*, or along with, other objects; that is, when color is being predicated of a thing or things?

(Continued)

FIGURE 3.1 *(Continued)*

Is-colored-by and *has-been-colored-by* are exceedingly powerful schemata extracted from our perceptual experiences of movement and action. Anything at all, concrete or abstract or even a hue may be said to be *colored-by* a color or, indeed, to be colored by anything else not a color, i.e., a thing already invisibly colorized may act as a color to color something else! In these schemata, a targeted thing is conceived to be absorbent – i.e., a permeable substance able to take on, soak up, or incorporate another. (A related process involves an *ad*sorbent thing that is able to take on another thing, which is *ad*sorbable, by adding it as a thin surface film.) A target thing is absorbent for something else absorbable – i.e., a substance, a colored thing, able to soak into another, capable of itself being swallowed up. The absorbent entity receiving absorbable "color" is only slightly altered – tinged, tinted, tilted, tainted, or toned. In this manner, a schema for understanding color is able to travel – is imported – to/into another (targeted) realm to organize or newly characterize it. Note the past tense of this metaphoric process of being "color*ed*-by": something prior to color is the first and true object or target. The absorbent-thing is privileged over the absorbable-color-thing. Color is overshadowed by the absorbent thing.

An example: "His *manner* colored my *reaction* to the ideas." In this formulation, a person's manner (which is absorbable) alters my reaction (which is absorbent), which then affects the ideas. The prior ideas—which are ultimately being colored by my reaction—may be his own ideas, the ideas of a person he was quoting, my own ideas, the ideas of a person I remember, and so forth. . . . The ideas themselves may be said to be directly colored as in Rita Felski's "the idea of [literary] critique contains varying hues and shades of meaning," and in Noam Chomsky's famous sentence, "Colorless green ideas sleep furiously."[8] False ideas may also be colored as in Claude Chabrol's *Au coeur du mensonge* (1999), released in the U.S. as *The Color of Lies*. In this case, what doesn't exist may still have a color.

Further instructive examples:

(1) "An abrupt thought colored my perception of the color." "The same factors [involving ideas of narrative closure] seemed to colour my perception of the rhythms and patterns of history."[9] Note that these sorts of formulations and the ones in the previous paragraph have the merit of suggesting that thoughts and ideas may work directly upon perception, rhythm, pattern, and color, as if an invisible beam has emerged from the mind through the eyes to light upon an object giving it a distinctive hue.

(2) Consider blue. A thought or thing that has appeared suddenly may take on a hue "out of the blue" or vanish "into the blue." Something not "true blue" may be seen as wrongly colored and become "off-color," even if literally blue, as in

a mismatched hue in a design. Feeling unwell is "feeling blue." Music about this state covers many situations in life by employing numerous tones of "the blues." An improper remark that is considered indecent may be referred to as "blue," thus coloring a non-colored thing an off-color color. If the words are especially steamy or ornate, the blue darkens toward "purple" prose.

(3) The *is-colored-by* metaphor may be reduced to the adjectival: "rainbow democracy" and "black ice" (examples from above); an award-winning California high school mock trial team is the Oak Park Black Team; and, "luminous joy," "luminous idea," and—why not?—"luminous gray," which makes an impossible color combination provisionally possible in mind.[10] The chameleon-like adjective "colorful" consigns specific hues and implications to each hearer's imagination while sticking easily to many things: "colorful talk," "colorful ideas," "colorful personality," "colorful trip," "colorful life."

(4) An important example: René Descartes has asserted that certain universal real things "are as it were the real colours from which we form all the images of things, whether true or false, that occur in our thought."[11] Note that in Descartes's argument, *both* the nature of certain invisible universals (which colour all thought and imagery) and the nature of colours in nature have been modified, even entirely transmuted, by their conjunction and then mapped into the real.

David Batchelor's book on color, *The Luminous and the Grey*, indulges the *is-colored-by* metaphor when he asserts the following: that there are "several shades of each" type of a certain set of philosophical positions; that a certain remark is "in the shadow of" the 1960s counterculture; and that a given grey color actually has "many shades of grey," including "that which it is not." The latter statement would seem to assert that grey is complex, subtle, ambiguous, and may be quite otherwise than grey, including being invisible or nothing, such that both literally and figuratively, it is true as he says, "in grey there are grey areas."[12]

It may be that the powerful *being-colored-by* metaphor derives from the phenomenon of perceiving color *transparency* where certain hues, but not all,[13] are shown to be permeable by, in effect, thinning themselves out to partially take on a new hue appearance coming from a hue on the other side or else to provide another hue on the other side with a new and changed apparent hue. Color is being colorized before our eyes and a hue is being shown to have a new hue. (These colored appearances themselves are non-surface colors.) In this manner, transparency acts as an image for a verbal proto-metaphor, i.e., transparency acts as a *two-way filter* that characterizes the operation of metaphor in general.

As an illustration, consider the following two-way dissection of a verbal metaphor. If "man is a wolf," then so also, to a lesser degree, is a wolf a man. There is always the possibility in a given metaphor of a slight "backscatter" effect or of "transparency"

(Continued)

FIGURE 3.1 *(Continued)*

whereby the original target domain (man/absorbent) is transformed to a degree into a source domain while converting the original source domain (wolf/absorbable) into a target with the result that a wolf becomes to an extent humanized while the man is animalized. The rhetoric of anthropomorphism is a strengthened version of the flip side of this metaphor of "man is a wolf."

In general, the *being-colored-by* rhetoric demonstrates how a person's sensation of various color qualities may become a source for understanding a more abstract, target domain. This special type of metaphor is known as a "conceptual metaphor" or "cognitive metaphor" and is an example of embodied cognition at work.[14]

ANALYSIS: HOW WE COME TO KNOW

7. COLORED-AS-INTENDED

The idea that a thing has been *colored-as-intended* directs one toward a search for the shallow or deep purpose-in-mind of a Creator and toward a need to know the full production history of a colored entity. How has color arrived to be? What thought has been brought to light by light? The assumption here is that color has been made human according to the humanity of its maker and making. Note the past tense of color*ed* and of intend*ed*: something or someone prior to color is the true subject. In effect, the colors of an artwork are *being-colored-by* an absent Author. A variant of *colored-as-intended* works to anthropomorphize hues, giving them certain "feelings," which have been transfer*red* or communicat*ed* by an author-originator to a beholder later in time, who is beholden to the Artist. Another variant is *being-colored-in* whereby the perceiver becomes his or her own Author viewing an immediate colored presence on a private, mental screen.

8. COLOR-AS-SYMBOL

When thought seizes on *color-as-symbol*, a hue is being analyzed as a thing that *stands* (in) for, identifies, or points out (up), something else that is absent, e.g., an idea, ideal, expression, emotion, intention, personality trait, atmosphere, or symptom. An object is simply *colored-by*, becoming noun-like, by something else that is often abstract. (The *is-colored-by* metaphor discussed above also informs the next two verb-like metaphors, *colored-by-a-camera* and *colored-by-a-context*.) A prominent type of such a *standing* theory is a semiotics of signs that relies on learning a system of arbitrary codes drawing upon long-term semantic memory.

Less formally, a symbolic condition is manifest when a critic encourages spectators to look for covert or ironic "references" or "quotations" in a text or promotes a

text's "self-reference." The game is to "remember when." Thus color's presence as a signifier or symbol is only preliminary to that which is real, referred to, absent, and located elsewhere or located somehow nowhere, as with fiction. In some semiotic systems, the entity that is "elsewhere" is an impossible otherness, a thing real but never reached, as in Lacanian psychoanalysis.

9. COLOR-AS-CAMERA

I have named this metaphor, *color-as-camera*, in order to highlight the familiar idea of colors being able to become fluid and plastic in space and time. Colors may affect one another, combine, juxtapose, shift their qualities, and rearrange adjacency from moment to moment. Sergei Eisenstein argued that colors could detach themselves from objects to become mobile (see Chapter 3, section 3). The question arises, leaving behind what in order to become what?

In the *color-as-camera* metaphor, color is analogized to a tracking camera movement. The goal is to describe how a spectator finds and follows developing patterns against a chaotic background of ongoing changes and noise. This approach to seeing colors aims to measure how a flow of stimuli that incorporates scattered colors comes to assume one or more definite shapes in consciousness through the top-down processes of working memory. The spectator is attending to ripples and surges of microsequences in real time. In such an environment, a perceiver seeks to assess how a text is modeling—narrating piece by piece—some aspect of a world. This approach to color perception seeks to *track* the online integration of color with other mental functions and states, such as inferences, hypotheses, expectations, feelings, desires, beliefs, and values. (See Chapter 3, section 1.)

In general, the *color-as-camera* metaphor serves to remind us about arrangements, mixtures, shifts, and framing effects involving color patterns that appear in a single shot, in successive shots, in non-adjacent shots, or in conjunction with some narrative or cognitive feature being tracked (see Chapters 7 and 8). In tracking, color becomes part of a holistic, embodied theory of mind where a person's experiences with color create immediate templates for far-reaching thoughts about an interactive environment.

10. COLOR-AS-CONTEXT-SENSITIVE AND COLOR-AS-CAUSATIVE

The metaphors, *color-as-context-sensitive* and *color-as-causative*, emphasize that color is taken to be mobile and relative to thought, culture, and use. It may be efficacious in promoting design at all levels of a narrative text. A causative, tracking theory focuses on a perceiver's construction of important relationships and patterns, including possible patterns and the "nearly true." This approach maps sequences of *reidentifications*, radial associations, and transformations, rather than drawing upon a priori knowledge

(Continued)

FIGURE 3.1 *(Continued)*

through static semiosis (Chapters 7 and 8; Chapter 3, section 1). Despite appearing in a fictional setting, color is humanized according to how it reasserts, re-makes, or imagines the potentials of objects in relation to actual language-games being lived by a perceiver.

11. COLOR-AS-PIXEL?

Does digital media require a new conception or image of color imagery? If so, which theory of new media will do to map the metaphor *color-as-pixel*?

[1] On the theory of "conceptual integration" or "blending," see, e.g., John A. Bateman, "Intermediality in Film: A Blending-Based Perspective" in *Film Text Analysis: New Perspectives on the Analysis of Filmic Meaning*, ed. by Janina Wildfeuer and John A. Bateman (New York: Routledge, 2017), pp. 141–168; Christian Quendler, "Blending and Film Theory" in *The Routledge Encyclopedia of Film Theory*, ed. by Edward Branigan and Warren Buckland (London: Routledge, 2014), pp. 56–62.

[2] Here is a description of a glory seen in 1738:

> A cloud that covered us dissolved itself and let through the rays of the rising sun. . . . Then each of us saw his shadow projected upon the cloud. . . . What seemed most remarkable to us was the appearance of a halo or glory around the head, consisting of three or four small concentric circles, very brightly colored. . . . The most surprising thing was that, of the six or seven people who were present, each of them saw the phenomenon only around the shadow of his own head, and saw nothing around other people's heads.

H. Moysés Nussenzveig, "The Science of the Glory," *Scientific American* 306, 1 (January 2012), p. 70.

[3] See entries under "color" in *Black's Law Dictionary*, ed. by Bryan A. Garner (Eagan, MN: Thomson West, 10th ed. 2014).

[4] On hues envisioned as a rainbow, see Chapter 5, section 2, and also the subsection, "Spectral Hues." Figure 8.1 illustrates what I call a "bad rainbow." For more on the ideology-infused rainbow that paradoxically represents both circularity and linearity, see Paul Coates, *Cinema and Colour: The Saturated Image* (New York: Palgrave Macmillan, 2010), chap. 7, "Colour and/as Fantasy: The Rainbow and the Jewel," pp. 139–152.

[5] The description of unity in diversity comes from Lalonde Roxanne. See "Unity in Diversity" <http://en.wikipedia.org/wiki/Unity_in_diversity#CITEREFLalonde1994> (accessed Sept. 23, 2014).

[6] I mention only two prominent metaphors for perceiving color-as-aesthetic. Barthes suggests an interesting alternative whereby color appears through its discoloration; see Chapter 4, section 3, but compare Chapter 8, subsection on *Little Dutch Mill*, where colors clean and bright represent the original and proper state of the world and its hues. Does the image on the cover of the present book represent (partially) a discoloration within hues?

[7] On the schemata *is-colored-by* and *has-been-colored-by*, see also Chapter 3, section 5, and Chapter 8, section 2. This schema partially overlaps the critical technique of employing a "stylistic metaphor." As an example of the overlap, consider this claim about a character in Eric Rohmer's *The Green Ray* (1986):

> The scene and room are awash with shades of red: of Delphine's jacket, bedspread, on a nightlight, in sheets of wallpaper. The deep concentration of one colour emphasises the singular fixity of Delphine's mind, whilst the redness aggravates—she must flee now.

Here a literal red as a stylistic element has colored her mind and actions. Steven Peacock, *Colour* (New York: Manchester University Press, 2010), pp. 56–57.

Here is another example from Peacock of style creating the metaphor *being-colored-by* to fortify a claim about Yasujiro Ozu's *Equinox Flower* (1958): "The permeation of red hints at how Yukiko's youthful vitality is beginning to colour Hirayama's more usually black-and-white mindset" (p. 40). Thus one character's colorized (red) state colors another character's already colorized (black-and-white) state. For more on stylistic metaphors, see Chapter 6, note 20.

Melinda Szaloky (personal correspondence) notices that sometimes one might interpret *being-colored-by* as itself based on a double metaphor: first, the color that is being applied with the metaphor of *being-colored-by* is taken to evoke a new, apparent surface hue of flesh (e.g., red), and then the flesh is applied to an object or abstraction to enliven it, to give it a new sort of flesh coloring and hence a "life" in the spectator's mind because vividly "fleshed out."

Gottlob Frege asserts that "poetic eloquence" involves a poet who is "coloring and shading" a perceiver's mental conceptions and private experiences. Thus, in this situation, *being-colored-by* will occur in language and image with *no* actual color having to be mentioned or be visible. Moreover, in the case of colored images, it would follow that the colors being seen need not be the ones working to invisibly color a perceiver's subjective impressions. The limitless power of color! Or, perhaps, it is Frege's limitless power of imagination to conceive that the power of sensing things may become the material for reconceiving things. The fictional Ted Cole and the writer James Wood have the same idea as Frege about a meta-function for color; see Chapter 2, section 4. Cf. the ascendant power of the Artist in *colored-as-intended* discussed in the text below. Frege, "Sense and Reference," *Philosophical Review* 57, 3 (1948), pp. 213–214.

[8] Ideas that are both colored and colored colorless green appear in Rita Felski, *The Limits of Critique* (Chicago: University of Chicago Press, 2015), p. 2, and in Noam Chomsky, *Syntactic Structures* (The Hague: Mouton, 1957), p. 15.

[9] Laura Mulvey, *Visual and Other Pleasures* (New York: Palgrave Macmillan, 2nd ed., 2009), chap. 14, "Changes: Thoughts on Myth, Narrative and Historical Experience," p. 165.

[10] On the impossibility or perhaps possibility of a luminous gray, see Chapter 5, section 8, and Chapter 5, subsection, "Non-spectral Hues."

[11] Descartes's quote is the epigraph of Chapter 2, and see further note 1.

[12] David Batchelor employs the *being-colored-by* rhetoric in *The Luminous and the Grey* (London: Reaktion Books, 2014), pp. 53, 71, 74; see also pp. 59, 76, 94. For more examples, see Chapter 4 below, and also Chapter 6 on *Vertigo*. On visually coloring one hue by another, see Chapter 8, sections 2 and 3. On the quality of "luminous," see Chapter 5, subsection, "Non-spectral Hues."

[13] Not all hue combinations may lead to color transparency in nature even though artists can create false transparencies that cannot occur naturally. What is the status of such falsity and illusion, since it can be clearly seen? The fact of a visible, false transparency in an artwork borders on a range of philosophical issues involving impossible hue combinations (e.g., bluish yellow) and the nature of fictiveness. On illogical transparency effects, see Paul Zelanski and Mary Pat Fisher, *Color* (Upper Saddle River, NJ: Prentice Hall, 6th ed. 2010), pp. 136–137; Edith Anderson Feisner and Ron Reed, *Color Studies* (New York: Fairchild Books, 3rd ed. 2014), subsections, "The Illusion of Transparency," "Translucency," "Volume Color," and "Film Color," pp. 133–138. On related matters in the present book, see Chapter 2, section 4; Chapter 4, note 11; Chapter 5, section 8; Chapter 6, note 4, and accompanying text; and Chapter 8, subsection on *Little Dutch Mill*.

[14] On cognitive metaphor, see George Lakoff and Mark Johnson, *Metaphors We Live By* (Chicago: University of Chicago Press, updated ed. 2003). On embodied cognition, see Chapter 6 below, note 12.

NOTES

1. Aristotle, "Sense and Sensibilia" ("On Sense and the Sensible" or "De sensu et sensato"), *The Revised Oxford Translation*, 441b22.
2. I have elaborated on the semiotic formula, A → B in *Projecting a Camera: Language-Games in Film Theory* (New York: Routledge, 2006), as follows: We see A as-is and present, and also A as-if B (or "about" B—where B is blank and not present, i.e., B remains to be filled-in and filled-out by a beholder who is imagining possibilities for A's quality of "about-ness"), when circumstances are C under description D_1, which intersects or jumps to radial descriptions D_2, D_3, D_4 . . . (pp. 223–224; see also chap. 3). For more on radial descriptions, see below Chapter 8.
3. Working memory is a middle-level memory system with a cycle of about 150 seconds. It has a number of components: a specialized language processor, phonological rehearsal loop (inner speech), visuo-spatial sketchpad (inner scribe), phonological and visual stores, episodic buffer (which binds the phonological loop and sketchpad to episodic and semantic long-term memory systems), sensory buffers, general workspace (consciousness and a short-term memory store) along with executive functions (i.e., the self-control, attention, saliency, and planning functions necessary to track and use time efficiently). There may also be a hedonic detector and cache that rapidly evaluates positive and negative associations of an object, representation, or array held within the episodic buffer. See Alan Baddeley, *Working Memory, Thought, and Action* (New York: Oxford University Press, 2007).

   Short-term static memory has a capacity of seven bits of information, plus or minus two, and a half-life of ten to fifteen seconds when interrupted or interfered with. However, a person's dynamic

short-term memory span for material presented in a continuous stream (e.g., while watching a film) is more severely limited: three items of information, plus or minus one! The limitations and processes of working memory have enormous implications for filmmaking practices, and the perception and interpretation of films by spectators.

Working memory can expand its workspace by outsourcing its computations and memory stores to a film being watched or to the environment. Also, there is strong evidence that the imagery of silent movies excites the phonological loop in a spectator. If true, this provides a novel way of thinking about the "language" of film as well as making silent films sound in mind. See generally Yadin Dudai, "Enslaving Central Executives: Toward a Brain Theory of Cinema," *Projections* 2, 2 (Winter 2008), pp. 21–42.

A serious study of the types of human memory along with their unique operating methods has far-reaching consequences for the construction of film theories and for attempts to delineate the shifting boundaries of media. See Edward Branigan, "If-Then-Else: Memory and the Path Not Taken" in *Transmedia Frictions: The Digital, the Arts, and the Humanities*, ed. by Marsha Kinder and Tara McPherson (Berkeley: University of California Press, 2014), pp. 45–79.

The concept of "suture" is designed to address, or perhaps conflate, the interplay of attention and awareness (consciousness), memory, narration, and ideological belief. Paul Coates proposes an analogy between the afterimage of a hue and the "reverse shot" called for by suture theory and presumably also an analogy between a word like "low" being summoned by its opposite, "high." *Cinema and Colour: The Saturated Image* (New York: Palgrave Macmillan, 2010), pp. 9–10, cf. pp. 8, 84. See generally "Suture" by Sean Cubitt and "Interface" by Warren Buckland in *The Routledge Encyclopedia of Film Theory*, ed. by Edward Branigan and Warren Buckland (London: Routledge, 2014), pp. 268–272, 453–457. See also George Butte, "Suture and the Narration of Subjectivity in Film," *Poetics Today* 29, 2 (Summer 2008), pp. 277–308. For more on suture, see Chapters 5 and 8 below, notes 55 and 29.

4. For an example of this form of *color-as-camera* tracking, which moves among and inflects other hues in a field, see Chapter 2 above, text accompanying note 19.
5. To elaborate on the color effects mentioned in the text: In simultaneous and successive contrast, two adjacent hues or two hues seen consecutively act to accentuate their differences. An afterimage is a faint, complementary hue produced in the eye through the steady viewing of a color that will appear to be overlaid on the next hue to be seen. Spreading effects influence adjacent colors to appear/become weaker. Finally, there are a great many types of perceptual illusions and constancies involving color.
6. An example of color tracking may be found in Christopher Knight's description of paintings by Richard Diebenkorn in Chapter 2 above, text accompanying note 19.
7. Eisenstein was the inspiration for me to imagine color tracking as the movements of two irregular wavy lines. See, e.g., Sergei M. Eisenstein, "'Synchronization of Senses' and 'Form and Content: Practice'" in *The Film Sense*, ed. and trans. by Jay Leyda (New York: Harcourt, Brace & World, 1942), pp. 67–109, 155–216. A more recent, definitive translation of these two essays is "Vertical Montage" in *S.M. Eisenstein: Selected Works*, *Vol. II, Towards a Theory of Montage*, ed. by Michael Glenny and Richard Taylor, trans. by Michael Glenny (London: BFI, 1991), pp. 327–399.
8. See, e.g., Carl Plantinga, "Affective Trajectories and Synesthesia" in *Moving Viewers: American Film and the Spectator's Experience* (Berkeley and Los Angeles: University of California Press,

2009), chap. 5, pp. 140–168; Clark Farmer, "'Every Beautiful Sound Also Creates an Equally Beautiful Picture': Color Music and Walt Disney's 'Fantasia'" in *Lowering the Boom*, pp. 183–197; C.L. Hardin, *Color for Philosophers: Unweaving the Rainbow* (Indianapolis: Hackett, expanded ed. 1993), pp. 132–134.

9. On editing schemes that produce a heterarchy (parataxis) through hyperdiegetic narration, see Branigan, *Projecting a Camera*, pp. 122, 287–288 n. 63, and index entries "editing" and "heterarchy." Indeed editing itself cannot be defined apart from far-reaching conceptual and rhetorical decisions made within a given film theory; for discussion, see *Projecting a Camera*, pp. 19, 61, 122, 238–240 n. 32, 278 n. 38, 297 n. 109. See also Chapter 8 below, note 42.
10. Susan Blackmore, *Ten Zen Questions* (Oxford: OneWorld, 2009), p. 73; on color, pp. 69–76. On the so-called hard problem of consciousness involving "what it's like," mentioned by Blackmore, see Chapter 2 above, section 5.
11. Blackmore, *Ten Zen Questions*, pp. 43–44.
12. Blackmore, *Ten Zen Questions*, p. 45.
13. Despite the title, Patti Bellantoni's book is quite useful, *If It's Purple, Someone's Gonna Die: The Power of Color in Visual Storytelling* (New York: Focal Press, 2005).
14. Herbert Kalmus urged that "color should flow from sequence to sequence." See the quotation in Chapter 2 above at note 4. Natalie Kalmus stressed that color should move in her aesthetic scheme. See Chapter 6 below, subsection "*Two or Three Things I Know about Her*; Natalie Kalmus and Technicolor Style."
15. Adrian Cornwell-Clyne, *Colour Cinematography* (London: Chapman & Hall, 3rd ed. 1951), pp. 651, 653 (Cornwell-Clyne's emphasis); see "Colour Scores," pp. 659–660.
16. Sergei Eisenstein, "Colour Film" in *Notes of a Film Director*, ed. by R. Yurenev, trans. by X. Danko (New York: Dover, 1970; orig. 1940), p. 127 (orig. 1948; Eisenstein's emphases).
17. Coates, *Cinema and Colour*, esp. chaps. 1 and 2, pp. 1–42; "the essence of colour is its ambiguity of meaning, and interesting colour use acknowledges that ambiguity" (p. 141).
18. Marshall Deutelbaum's remarks concerning the attempt to re-find an original [Original] sound track apply equally to color. "We must recognize how illusory any hope is of restoring a [physical] sound track to what it was like as it was experienced by an audience at the time of the film's original release. And yet, sound tracks must be restored and preserved." One might add that a significant part of what is "experienced" by an audience will be relative to the historical situation and original world of that experiencing, now lost. Marshall Deutelbaum, "Essays in Cinema Sound: Rick Altman's 'Silent Film Sound' and Nasta and Huvelle's 'New Perspectives in Sound Studies'," *New Review of Film and Television Studies* 4, 1 (April 2006), p. 69.

    David Batchelor in *The Luminous and the Grey* (London: Reaktion Books, 2014), argues that Andrei Tarkovsky in a scene from *Andrei Rublev* (1966) searches merely for an *equivalent* to a lost original experience (p. 95):

> It is as if Tarkovsky is not simply trying to describe to the viewer what the experience of seeing such a [religious] icon may have been like in the fifteenth century, but to instill an equivalent of that experience in the viewer's own present. It is as if Tarkovsky is asking the viewer to imagine not just what it might have been like back then to have such an encounter, but what it might be like now.

19. In terms of the soundtrack, James Lastra has expertly delineated the opposing ontological and epistemological approaches, which he calls theories of nonidentity (where a copy of a sound is always fatally deficient) and identity (where the key question concerns what a sound is used to identify, how it is used to point to something, i.e., not what a sound is, but what it is about). "Sound Theory," *Sound Technology and the American Cinema: Perception, Representation, Modernity* (New York: Columbia University Press, 2000), chap. 4, pp. 123–153. See Edward Branigan, "Soundtrack in Mind," *Projections* 4, 1 (Summer 2010), pp. 41–67. In the text, I am extending this idea about sound theories to two types of color theories. See also Chapter 2 above, text accompanying note 10.
20. Part of what's wrong with the question, "what's in a color?" is the word "in." See Edward Branigan, "Of Prepositions: Lost and Found," The Velvet Light Trap 64 (Fall 2009), pp. 95–98. See note 29 below and text accompanying note 44, Chapter 2 above.
21. Jacqueline Lichtenstein, *The Eloquence of Color: Rhetoric and Painting in the French Classical Age*, trans. by Emily McVarish (Berkeley and Los Angeles: University of California Press, 1993), pp. 53, 42, 195 (Lichtenstein's ellipsis); see chap. 1, "On Platonic Cosmetics," pp. 37–54.
22. David Batchelor, *Chromophobia* (London: Reaktion Books, 2000), pp. 73, 77; see also p. 79. The sparkling vigor of gems has provided an occasion for the veneration of the special powers of natural light. See, e.g., "Light and Color in the Middle Ages" in *History of Beauty*, ed. by Umberto Eco, trans. by Alastair McEwen (New York: Rizzoli, 2004), chap. 4, pp. 98–129.
23. Cornwell-Clyne, *Colour Cinematography*, p. vi.
24. Plato's cave parable is in *Republic VII*, pp. 514–518; see also *Republic X*, pp. 595–608b and *Sophist*. On the many valuable lessons Plato teaches and the few that are taken up by film theory, see Edward Branigan, "Apparatus Theory (Plato)" in *The Routledge Encyclopedia of Film Theory*, pp. 21–33.
25. Equivalent to the rhetoric of *being-colored-by* is the hoary oratory derived from motion sensation where a person's words, say, are given a new "spin" or "slant."
26. A green-colored environment may also be absorbed into a too-conservative form of life; see Nagisa Oshima, "Banishing Green" in *Color, The Film Reader*, ed. by Angela Dalle Vacche and Brian Price (New York: Routledge, 2006), pp. 118–120.
27. Warhol's "Do It Yourself" paintings offer insight into many of his other works in which a preliminary image is first screened onto a canvas to serve as an under-drawing followed by Warhol's selective (re)coloring of various sections—doing it himself from, in effect, his personal "paint-by-numbers" kit; for example, the *Marilyn* silkscreens based on a publicity photograph for Henry Hathaway's 1953 film *Niagara*. The paint-by-numbers scheme closely resembles the operation of computer graphic compression algorithms. See Daniel C. Dennett, "Real Patterns," *Journal of Philosophy* 88, 1 (January 1991), p. 33 n. 11.
28. One might say that Warhol partially reverses the value placed on "authenticity" by intending his original painting to look like a fake or copy, something executed by an amateur, who is merely "painting by the numbers." In "Do It Yourself (Narcissus)" and "Do It Yourself (Flowers)," graphite and crayon have been used to suggest the haste and sloppiness of an amateur by leaving very thin white streaks of unfilled color within a colored area and by sometimes coloring an area in patches where each patch is colored-in with lines oriented in a different direction. The latter tactic makes line (*disegno*) subservient to color (*colore*). Chapter 5 will discuss the assertion by many art historians and theorists that *disegno* is more important than *colore*. Further, the white streaking

may be seen as a *texturing* of a most delicate and subtle kind, not of a flower's surface, but of the graphite or crayon substance that has materialized the hue.

"Narcissus" and other "Do It Yourself" works may be found in Heiner Bastian, *Andy Warhol: Retrospective* (Los Angeles: Museum of Contemporary Art, Tate Publishing, rev. ed. 2002), pp. 116–119. The "Flowers" version done on paper may be found in Batchelor, *Chromophobia*, p. 50, and discussed on pp. 61–62. Additional "Do It Yourself" works may be found in Ann Temkin, *Color Chart: Reinventing Color, 1950 to Today* (New York: Museum of Modern Art, 2008), pp. 70–71.

29. The next chapter will examine the perils of wondering what's *in* a color.
30. A survey of seventeen best sellers showed that, of a total of 4,416 color terms, 4,081 were occurrences of just twelve basic terms, and half of the total number were of "white," "black," and "gray." Hardin, *Color for Philosophers*, p. 183.
31. On Deleuze on Godard, see Chapter 2, section 5. For more on Godard, see Chapter 6, section 5.

Chapter 4 **What's *in* White?**

ABSTRACT

This chapter considers the philosophical implications of "whiteness," which posits an essence to that particular hue, versus the notion of "white," which treats that hue as a nominal presence in language use and so need not be of that particular color when named, nor even be visible. White may be an elaborate figurative condensation. Film theorists have certainly seen the light, i.e., have been busy seeing the whitish light on screen in many different ways. I argue that ordinary language as well as theoretical language—and the community life they spring from and support—are forms of *manufacture for exchange and use* like other technologies in society. It should not be surprising, therefore, that language colors white, colors the other hues, and generally colors color. The essence of a thing is not a timeless and unchanging thing-ness, but lies in our uses of language. Color-being-seen moves through life as do we.

The present book treats the notion of "essence" as merely a shorthand way of talking about a thing of interest when one wishes to shine a spotlight on a part of some process, cutting off before and after, i.e., making the thing *stand* alone out of time, out of its natural or aberrant thread of meaning in the midst of its transformation. Not just time may be cut off, but also what lies above, below, and alongside, i.e., proximate contexts for the thing involving juxtaposition and association. Since the issue of essence involves the significance of physical or quasi-physical matter, a better description of the process than "transformation" is perhaps "trans-*figuration*." Things are made to be—to be seen for what they are, i.e., appear to be in the moment—within or under present *figures* of language.

> While the perception that there is white before us cannot be false, the perception that what is white is this or that may be false.[1]
>
> —Aristotle

OF WHITENESS; OR, ABOUT WHITE

Are physical materials, technical processes, primary colors, and photoreceptor cells in the eye enough to give color to our experiences? As Wittgenstein notes in the

epigraph of Chapter 5, there is more to having experiences than physics. We move within a world connecting experiences to experiences.[2] From these links emerge new experiences. We are deeply embedded in a world when searching for how to live as we interact with others to advance our plans. Most particularly, we are immersed in a *language* that speaks to our involvements and commitments. Language allows us to experience more than a single point at a time, to see from point to point to accumulate points, making emergent patterns. How does embodiment in culture drive an ability to speak about the colors of things we think we have seen—the colors we have seen while thinking about things? Let us consider the color "white" as an example.

But first, what is this ability we have with words? Look around the room you are in—or just *imagine* looking around the room. You are aware of your body and who you are, your unique history, what these things are that you see, why you are there, who else you know, what your plans might be tomorrow and next week. You don't need to speak appropriate words to yourself, but you know you *could offer* verbal explanations if need be, muttering to yourself or giving an account to others. With language, it is as if you were skating across a frozen pond, knowing that the icy language was there below—and networked in memory—providing the requisite mobility for thought. Cracks in the ice are only temporary, for the weather is very cold. . . the ice normally impenetrable.

Incidentally, can ice be separated from its color? Is it white? Naturally white? Conventionally, we think of it as a blanket of white, or less comforting, a white blank. If there were a specific reason to look more closely, however, further descriptions and further hues might easily be found. A range of possible responses to ice points toward the general issue of how a person's use of language interacts with, and summarizes, a particular context or problem that shapes his or her behavior toward color. Perhaps language use and purpose has as much to do with discriminating varieties of whiteness as does sunlight.

We confidently speak of the "whiteness" of sunlight, or, when feeling sentimental, envision it as a golden glow of yellow rays. But there are different sorts of natural light depending upon the time of day and weather, not to mention differences between fluorescent, incandescent, halogen, and LED artificial light. A light source may even be said to be "black," as with a lamp that radiates invisible ultraviolet energy. We clearly see the hues that are strangely lit by it, though we do not see ultraviolet directly, unlike many birds and insects. These differences in the quality of whiteness are important when employing the materials of an art medium, especially in photography. For example, in the case of photographing human flesh, great care is required when converting ambient light into a series of reflections off skin while managing the skin's subsurface scattering in order to exactly fit the resultant light

to the specific sensitivities of photo-sensitive materials as well as to the sensitivities of what a perceiver expects of an ideal, memory-familiar flesh tone.

What hue is the sun itself? Is it really light yellow? And what color is its light? White? But why do we say that light is white? Does light in a bright room create a white fog? When light from a window appears in the form of a highlight spotted on an object, it is *transparent* while a white hue is—and must be—*opaque*, as with porcelain, snow, a glass of milk, or whitewater rapids in a river. Painters may use a highlight to represent shine and texture on an object by using opaque white paint, relying on a viewer to imagine the requisite degree of transparency.[3] Even so, in our imagination, the "light at the end of the tunnel" is always opaque as is—in another sense of "opaque"—both the cover up of misdeeds in a "whitewash" and its harmless opposite, a "white lie." Although the water in a glass is transparent like an actual highlight, we do not choose to call this water or the glass "white." Yet again, wine in a glass is called "white" that is actually clear and yellowish. This, however, does accord with Aristotle's view of yellow—the lightest of all hues—as simply being a *form* of white.[4]

Perhaps whiteness is more certain in a concrete object. White chess pieces, however, need not, in fact, be white at all, and a "white knight" need be neither a chess piece nor a knight. The "light of my life" is neither a light nor a color. Air in daylight, though transparent, is not said to be white, unless suffused by opaque fog or snow, as in a "whiteout." Is static noise on a television set snowy white? In the first thirty years of color television an "acceptable white" had a distinctly yellow tinge; now it is more bluish.[5] Notably, "white skin," which is actually grayish-pink, may be tied up with invidious racial classifications and other deeds tending toward black.[6]

Let me tackle the status of white from another direction. Can one decide by simply looking whether white is meant to be seen as containing all the other hues of the spectrum or whether it is to be seen as having been drained of all color and therefore "wasted" and "empty"? Consider as an analogy a room in a building. What criteria would determine whether the room is full or empty? Is it relevant whether there is furniture in the room? enough furniture? of the right kind? Is it relevant whether there are people in the room? whether the lights are on or off? If the room were only bare walls, would that necessarily make it empty? Wouldn't a bare room still be a distinct *place*. . . ? When do we say that a location is empty and when not?[7] The point I'm trying to make is that "white" and the "room" can be seen to be many things depending on the context for the viewing. In this sense, both white and the room are *less visible* to us than might at first be imagined.[8] In an important way, the status of the color and the room exists in the language and actions through which we conceptualize, occupy, and use them at a given time.

I would like to stress that I am not asserting that color lies only in language and description. Language interacts with physical and non-physical objects in a world. One must determine *which* language game is being played—which field of language rules is in play—on a given occasion in order to properly match a word or description or photograph to our behavior with objects that are said to have color in some sense or senses. Language connects to objects because we have many different uses for objects that require us to talk about them in a variety of ways and to organize appropriate behavior.

Clearly, "white" has a great many consistent and inconsistent appearances that are mediated and assimilated through our ways of talking and acting.[9] It is as clear as day(light) that different realities, abstractions, and hues are being held together by using the single word "white" scattered among a variety of grammars fit to different situations and settings. Does white have a final core definition in "whiteness" or must one instead inquire about the *need* to describe a thing using the word "white"? In Aristotle's terms (see epigraph), it may be that on occasion our need for an object influences how we call the thing and see it.

LIGHT, WHITE, AND FILM THEORY

The association of whiteness with the action of light or lighting creates the expectation that with white a person should be *able to see things clearly* as if turning on a lamp. This need to see a thing in cognitive terms, not just experience its sensation, is the source of a pervasive metaphor in culture that equates the act of understanding with the presence of light being cast on it or illuminating it. A sudden realization of a thing's reality amounts to "seeing the light." No doubt no actual light is present to be seen in such understandings, even in imagination. Nor is light meant to be seen, since one is supposedly seeing a new thing through, by, or with the light. Paradoxically, as light itself becomes less visible and less noticeable, more mental in tone and less white in hue, our view becomes clearer—whereas as a literal hue, like all the others, it is as opaque as a glowing white lampshade and requires a series of distinct edge contrasts—relationships to adjacent colors—in order to highlight a scene containing shapes and movements. White lace curtains, for example, only seem to be a "transparent white" as a person constructs distant shapes of objects out of many tiny gaps in the fabric that act somewhat like pixels. Moreover, gazing through lace curtains is not unlike looking at a mirror's surface and seeing lighted shapes like spotted highlights on/in the mirror's illusory depth—a false depth that forms a distant space through a myriad of tiny reflected edges and differences. Color, no less than other percepts, has no firm and certain foundation other than its presumed

and usual relationships, thus giving it the same reality *and mobility* in our thoughts as other percepts.

Curtains, veils, mirrors, and reflections are fascinating objects for artists because these objects seem to catch and form an image in transit—to model light and sight itself—to represent a sight that is being seen together with the act of seeing (understanding) it. It is as if an aperture has suddenly opened inside space to afford a singular view. The slight effort to see through curtains or into mirrors has a self-validating allure: the sight and the seer are represented together, including the possibility of the seer doubled—seeing itself reflected back as both subject and object of view. Various psychic states of character and spectator may thus be served, e.g., narcissism, doubling, the uncanny, projection, dissociation, alienation, and more. Once an image has thus been seemingly split off or split up, it may be metaphorically shifted onto new terrain.

Closely related to curtains and mirrors are screens. And where artists have gone, film theorists soon follow with carefully calculated metaphors about movie screens lit up with white light, which is actually colorless, or else with metaphors featuring white's binary twin, black (as in "life is not black-and-white" as a retort to those who believe it is), followed dutifully by such metonymic extensions of a light-and-dark screen (where "screen" becomes a synecdoche for "cinema") as "shadow," "moving shadows," "ghostliness," "irreality," "absence," and "death."[10] As with the notion of whiteness—and the metaphoric extensions of its negative, blackness, or its gradations in gray—there is a fruitful ambiguity built into the reductive notion of "screen" as a symbol for cinema that serves to embrace a duality of transparency and opaqueness. Being both transparent and opaque allows a theorist great freedom in argumentation. As a verb, "screen" may refer to showing forth or projecting something into view, such as to screen a film; or, it may refer to the exact opposite, to hiding or concealing something from view, such as to screen out (screen off) the sight of a thing by placing it behind something else. Similarly, "blackness" may signal either absence, a profound nothingness, or else concealment by screening out something on the screen as if by a black curtain or optical fade.

It is also possible to do both at once, like the blanket hanging between the beds of the characters portrayed by Claudette Colbert and Clark Gable in *It Happened One Night* (Frank Capra, 1934) that displays their bodies in shadow, both screening and screening out. A mirror functions in a similar way. It brings things into view at the same time that it conceals by placing them in a false space, seemingly behind the mirror's surface (from where they do not actually reflect light) while reversing left into right. Notably, the viewer is relocated to this semi-unreal space. A film theorist can exploit the partial unreality of a screen or mirror by freely selecting among degrees of ambiguity

(transparency/opaqueness) and illusion whichever assertion best fits his or her rhetoric when constructing a theory. For example, it might be claimed that cinema has the power to engage us by showing us something: what was once in front of a camera is now re-presented in light. Alternatively, it might be claimed that what cinema presents is only what was once arranged by a filmmaker using the materials of the medium while the original event, if it existed, is entirely concealed or remade. This allows a theorist to freely use the metaphor of film as a screen or a mirror that reflects the world while dismissing what is unacceptable as being caused by minor distortions or omissions.

More powerfully, a film theory may define ways in which revealing and concealing can work together to reinforce each other. For example, film may be compared to a "mirror" that reflects the *mind* that faces a world—either the mind of an author or of a spectator. Adopting a psychoanalytic framework, it can then be said that a repressed desire or dread—that which cannot be, and dare not be, shown—is, after all, indirectly shown to be present because of its leaking through, lying behind (below, beyond), or distorting what is literally being seen. This entails a two-part reality: (1) a surface with (2) a partly concealed depth, as if one is looking into water but can see only part way down or looking into the semi-unreality of a mirror. Such a disguised or censored presence appearing on a surface forms a temporary hiding place for semiotic symbols or a "language" of the unconscious. In this approach, transparency and opaqueness are being mapped onto a dialectical play of presence and absence, whiteness and blackness, where whiteness itself is ambiguously both transparent and opaque.

Another example of an interactive surface/depth may be found in the insightful work of Stephen Heath who, employing a psychoanalytic method, asserts that "Like fetishism, narrative film is the structure of a *memory-spectacle*, the perpetual story of a 'one time,' a discovery [of sexual difference] perpetually remade with safe fictions."[11] Here the white film screen acts to display the film and alternately act as an opaque white curtain or semi-transparent veil of defense, i.e., act as the disavowal and displacement associated with fetishism that will facilitate a sequence of safe fictions through which a spectator may partially glimpse a deeper unconscious memory as if in faded shades of gray. One can also imagine that an image on screen has come from a distant point and has partly squeezed through the white lace curtain of the screen. In both cases, whiteness stands in a double relation to the spectator. The screened image stands for a denial of insight—a protective blankness, veil, or white slate exhibiting a nominal image that has become a safe location for most-any-fiction-whatever—at the same instant that the image becomes the illuminated white ground of in-sight—bringing the real into sight by casting light onto, and seeing into, some deeper and darker traumatic with-in lying below conscious awareness—pointing to a specific-few-fictions—and requiring as a condition of its visibility that visible fictions be made nominally "safe."

HOLLIS FRAMPTON'S WHITENESS (WINCKELMANN, COATES, CUBITT)

Hollis Frampton finds a new way to recruit the fluidity of our language about white into the language of a theory about our experiences watching film. He explicates the fundamental nature of film and whiteness through a series of brief comments during a lecture-demonstration, starting with the illuminated light thrown onto a screen by a projector:

> Please turn out the lights. . . .
>
> [*The projector is turned on.*] . . .
>
> Our rectangle of white light is eternal. Only *we* come and go; we say: This is where I came in. The rectangle was here before we came, and it will be here after we have gone. . . .
>
> It is only a rectangle of white light. But it is all films. We can never see *more* within our rectangle, only *less*.
>
> [*A red filter is placed before the* {projector} *lens at the word "red."*]
>
> If we were seeing a film that is *red*, if it were only a film of the color red, would we not be seeing more?
>
> No.
>
> A red film would *subtract* green and blue from the white light of our rectangle.
>
> So if we do not like this particular film, we should not say: There is not enough here, I want to see more. We should say: There is too much here, I want to see less.
>
> [*The red filter is withdrawn.*]
>
> Our white rectangle is not "nothing at all." In fact it is, in the end, all we have. That is one of the limits of the art of film.

As soon as light from a projector has replaced the room lights, Frampton announces that "Our rectangle of white light is eternal." Shouldn't he have simply said, "Let there be light," thus making clear the religious overtones? For Frampton, the eternal source of all film is whiteness with never an end in black.

Frampton in his lecture goes on to explain that all films, first of all, are really *about the physical ribbon of celluloid* moving through the projector as each photographic frame acts as a filter to subtract from white light. Frampton states that

film is neither about the "self-expression" of a filmmaker nor about "the world at large." The true filmmaker "is more interested in recovering the fundamental conditions and limits of his art." The aesthetic value of reflexivity or self-(critical-)consciousness sought by some critics, along with whatever is meant by the redundant (self-redundant?) term 'self-reflexivity,' lies not far off, as if a spectator might forget to remember that he or she is watching a film of light and dark shapes.

Frampton is enamored with the sheer presence of white light on the screen: "Perhaps its sheer presence has as much to tell us as any particular thing we might find inside it."[12] Not only does a hue (e.g., the "red film" he mentions above) *subtract* from the purity of white, but so does the appearance of a line (*disegno*), shape, or figure. Frampton is attempting to reduce white to its most literal and fundamental, cut off from all else. Actually, he wishes to go further: to escape *language* entirely. White will no longer link or compare, be the subject of, stand as a metaphor for, or become a safe fiction. White will simply be left to be. Without language, thought, or intent.[13] It will be reduced to a single point that simultaneously starts and ends. As he declares, any particular object will be found only "inside" white. Symptomatic of Frampton's longing to escape language is the peculiar use here of the word "inside": 'inside white' is what? Another 'deeper' white? A truer whiteness? And what would this more perfect 'white' look like? Or, does the essence of white have itself no color, no light?

One may, as it were, accept half of Frampton's argument. Whiteness may be deemed superior to all the other hues, but still remain subject to the powers of line. That is, the *colore/disegno* debate (see Chapter 5, section 1) may entirely reframe the nature of white. As Matthew Gurewitsch notes:

> In the 18th century, the pioneering archaeologist and art historian Johann Joachim Winckelmann chose to view the bare stone figures [of ancient Greek and Roman statues] as pure—if you will, Platonic—forms, all the loftier for their austerity. "The whiter the body is, the more beautiful it is as well," he wrote. "Color contributes to beauty, but it is not beauty. Color should have a minor part in the consideration of beauty, because it is not [color] but structure that constitutes its essence." Against growing evidence to the contrary, Winckelmann's view prevailed. For centuries to come, antiquarians who envisioned the statues in color were dismissed as eccentrics, and such challenges as they mounted went ignored.[14]

Frampton's approach can be adapted to the art of painting and employed to avoid the *colore/disegno* debate by celebrating works that juxtapose simple blocks and stripes of color in abstract patterns. The result is perfect and primal color-ness apart from line. Presumably, these paintings (pace Winckelmann) can be said to explore a fundamental reality prior to the divisions, subtractions, encrustations, and restraints imposed by the appearance of shapely lines (*disegni*) that will only

be further damaged by the emergence out of those lines of outlines, figures, objects, and depths. Nevertheless, Frampton does not take whiteness to its very limit, to the abyss and void, which would be to speak of white as a color drained of all color, blinded by the light, or else to be pallid, ashen, depleted, or wasted, as when someone is said to be sickly pale, white with fear, white as a sheet, or near dead (his or her last light).

Many film theorists have focused on a tangible presence on the screen—white or black, or some other sensation, like motion or sound (or two-dimensionality, shape, shadow, frame/framing, style, cues, signs, and more. . .), that will bring us firmly (back) to our senses by reasserting brute sensation. One sort or another of physical material has been a starting point for a large number of phenomenological, formalist, and semiotic approaches to film. From such a starting point, all else (meaning, feeling. . .) is seen as *absent*, as a *loss* against which one must struggle to reclaim a presence not unlike Frampton's "sheer presence." But must there be a *starting point* for film? Even some realist approaches have imagined a starting point by conceiving of the screen as a mirror or window, possibly tinted or curtained, enabled by photography; or else, have conceived of the screen—against all appearances—as being *black* so as to provide an opacity that captures and brings light. When metaphors derived from properties of light and glass or from optical instruments—a microscope, telescope, kaleidoscope, or camera obscura—are chosen to model film, the whiteness of the screen is no longer opaque like milk but becomes strangely semi-transparent like cloudy water in a glass. The white screen acts as a barrier (like black) to be penetrated in favor of an elsewhere that may or may not finally appear. Frampton, however, does not venture in these directions, but instead exploits the screened whiteness for its supposed ontological implications, i.e., for what the light sheds on, he says, "the fundamental conditions and limits" of all film art. The white rectangle, he insists, is "eternal" while we are transient. Nonetheless, the implication is that a black rectangle by contrast would be a sign of loss or death.

The whiteness of the screen may be writ directly onto death. Paul Coates mounts an extraordinary argument that black-and-white, light-and-dark, is the primordial mover of all color. He asserts that "black-and-white is the basis of film narrativity as such. Colour replaces dialectic with [mere] difference." For Coates, black-and-white is driven by the deep realities of life while, for example, "colour denies the founding fact of the universe of terror, which is death."[15] This argument draws upon how we are choosing to see our immediate world, specifically, drawing upon the connotations of daylight color vision with retinal cones—which would seem to represent the mere (?) pursuits of the everyday—as opposed to nighttime black-and-white vision with retinal rods—which conventionally ties to shadows, dreams, phantoms, and to hidden forces, the acutely undefined, and the unknown hereafter. Day is

illusion, night is real. Logically, the argument might work better with an analogy based upon the gray tones of saturation, which are present in traces in all color (as death is present in life), than with the solid blacks of night. Saturation, however, is too subtle a perceptual quality when one seeks an argument of stark contrast between illusion and reality, day and death.

The film theorist Sean Cubitt treats black in the way Frampton and Coates treat white. Cubitt announces on the first page of his massive and rich treatise that blackness is the primal/primary color of all media: "Absolute black is no color: it is an absence that nonetheless weighs like a presence. . . . Black has the specific quality of being only ever virtual."[16]

At first, it would seem that Cubitt's "absolute black"—absent but still primodial and weighty with its "specific quality of being only ever virtual"—is the obverse of Frampton's Platonic "white." (Is Cubitt's "virtual" the *idea* of black?) Nonetheless, all three theorists fit within a general approach to philosophizing about color that Charles Riley calls the "rhapsodic tradition." This tradition includes Nietzsche, Spengler, Adorno, Derrida, and Barthes, who locates color within *jouissance*, which is a kind of (rhapsodic) bliss-transgression, where the fact of color is best seen in its discoloration. Barthes would seem to be rejecting aesthetic theories of color based on spectacle, dull unobtrusiveness, transcendental delicacy, and rational severity in favor of depleted hues that show how their objects have been used—how labor, exertion, and time have left an indelible mark. Cubitt's "absolute black" is more extreme, since it would seem to be the end point of an 'absolute discoloration' of white.

According to Riley, an opposing tradition, the "linearists," includes Kant, Goethe, Hegel, Wittgenstein, Westphal, Hacker, Hardin, and Rorty. When an ontology for color is sought, especially among the rhapsodics, the focus variously falls upon a single primal hue: black (Adorno, Cubitt; cf. the paintings of Odilon Redon), shadow (Goethe), gray (Hegel), white (Derrida, Frampton), black-and-white (Coates), brown (Spengler), red (Kierkegaard), gold (Derrida), and perhaps Deleuze's 'any-hue-whatever.'[17] The weighty presence of the ontological absence of black, however, is not quite the whole story for Cubitt. Matters of what he calls "mattering," as opposed to mere physical matter, make it all much more complex. In somewhat parallel fashion, the theme of the present book is that in matters of mattering, language is what matters.

GILLES DELEUZE'S WHITENESS/BLACKNESS

Gilles Deleuze manages to encompass both whiteness and blackness within a new, fundamental entity that accounts for the potency of screened light. The governing principle would seem to be one of high compression to achieve unity in diversity.

In a very demanding passage—which renders my interpretation merely tentative—Deleuze seems to find a series of unusual cross-connections among two kinds of close-ups, two kinds of style, and two kinds of white in films by Pabst, Eisenstein, Griffith, Lang, Murnau, and Sternberg.[18] These three basic pairings represent extreme poles that a film may slide between, conceivably in the same shot. In order to understand the place of whiteness/blackness in this theory, I will attempt to delineate the connections among the three pairings.

With respect to a close-up shot, one extreme is represented by the "reflexive" face of a character, i.e., *reflective* in a double sense relating to character introspection as well as the reflection of a reflected unity by/of white light from the character's face. The reflexive face depicts a character explicitly being subjectivized by narrative. The other extreme of a close-up is an objective, "intensive" face, which initiates a narrativized *sequence* of a character's changing emotional states (states of the face) that, in turn, emphasize and clarify an interval of developing action in a film.

The second duality embraces two kinds of film style: the first is Expressionism, in which "the face participates in the non-organic life of things," and the second is "lyrical abstraction," which amounts to "the other aspect of [Goethe's] theory of colours: light no longer has to do with darkness, but with the transparent, the translucent or the white."[19]

Finally, whiteness itself has two kinds of associated white. First, there is the Expressionistic fade into darkness that creates a sense of light through its gradual absence by means of shade, shading, and shadow. Thus, for Deleuze, this first kind of white becomes evident through light's progressive play with opaqueness, loss, and negation. The result is that "all the degrees of shadow. . . have value as colours."[20] This is consistent with one of the dominant, though unacknowledged, metaphors that structures Deleuze's two-volume theoretical study: film as moving shadows. He finds two master modes of cinema: movement and immobility. The latter is a special sort of shadow stillness in which movement has been perfectly distilled to become an ideal—the essence of a *disposition* to move and change, time frozen or crystallized.

The second kind of white for Deleuze is a *luminous cloudiness* displaying degrees of translucency. Here white is made manifest through light's difficulties rather than through gradations of shading and absence (as in Expressionism). Such "cloudiness" also includes certain "blurred" shots by soft focus.[21] Perhaps one could think of Deleuze's two kinds of white on the analogy of different kinds of primary colors. For example, the subtractive primaries of paint mix through a series of states toward black opacity (phases of Expressionism?) while the additive primaries of natural light mix through a series of states toward white opacity (phases of "lyrical abstraction"?). That is, a shadow's darkness either descends further into blackness (Expressionism?) or else elevates to reveal the original light source (whiteness) that has created the shadow.

For Deleuze, the three basic dualities interface with three additional dualities: the "reflection" of light from an object and the "refraction" of light, which he says is close to a "form of Impressionism"; the colors red and blue; and open and closed diegetic worlds.[22] Interface may be the appropriate concept, since Deleuze ends a five-page argument about the six dualities of whiteness/blackness with the statement that "the same effects" may be produced "by opposite principles."[23] The elements of the six dualities as well as the dualities themselves would seem to exist in a state of dialectical tension, if not continual transformation. This suggests that Deleuze's initial set of distinctions have bent back around to form an enclosed unity. The power of Deleuze's approach is that he avoids a formulaic compartmentalization of whiteness by instead allowing white to migrate through a film to join initially disparate elements—such as a narrative technique (the close-up), a film style, and color—in order to generate overall a dynamic, unified textuality.

Let us proceed one step further with Deleuze on whiteness/blackness. He remarks that strictly speaking "there is no close-up *of* the face, the face is in itself close-up, the close-up is by itself face." He says that non-human objects, too, may have a face and look back, or stare, at us.[24] Deleuze is thinking of imagery from the standpoint of a spectator's reception and affect, i.e., when a spectator is face-to-face with an image of a human or non-human face. How might this idea apply to the color *of* an object? For Deleuze, distinctions between subject and object, quality and object become permeable. He states that a "pure quality"—which implies the existence of *im*pure versions as measured against a definition of purity—can be expressed in film. His example:

> The reflecting faces of young women in Griffith[']s films] can express white, but it is also the white of a snowflake caught on an eyelash, the spiritual white of an internal innocence, the dissolved white of a moral degradation, the hostile and searing white of the iceberg where the heroine will wander ([D.W. Griffith's 1921] *Orphans of the Storm*).[25]

My question: where have these various forms of white formed? Do they come from an image or feeling, or instead from a way of talking in a community about a way of seeing and living occurrences in a narrativized white? In the case of Griffith, film becomes an explicit medium for narrativizing our talk of perception and social value. This means that our language lies not in an image, but within the many facets of a particular social visualizing of the image.

To conclude, the focus of theories proposed by Frampton, Winckelmann, Coates, Cubitt, and arguably Deleuze[26] remains steadily on the *is* of art, the fundamental being of art and color, rather than on their about-ness or their status as an instrument for knowledge. Deleuze's ontology of film and its connection to Frampton's spirituality of whiteness may be glimpsed in the context of the theory

of the reflective close-up as Deleuze extols the presence of "a divine operation when mind is reflected in itself, in the form of a. . . photogramme which would eternally consume itself in reaching the luminous internal life."[27]

For Coates, Cubitt, and Deleuze, ontological whiteness is leavened somewhat by certain epistemological elements. For Frampton, however, whiteness is simply a substance that answers to the siren call of simplicity and innocence. Eventually whiteness elicits a kind of mental surrender to a feeling as exemplified by the feelings associated with a white flag, white wedding dress, white rose, and white lie. Whiteness for Frampton becomes the irreducible building block of cinema while being itself built by drawing a conceptual thread through notions of simplicity, sincerity, purity, and uncomplicated virtue ("lily white"), as symbolized by, for example, Snow White. Importantly, whiteness points to cleanliness, to the absence of darkish dirt.[28] I believe the whiteness of white for Frampton corresponds not to the color of snow or ice, to a frigid blankness and sterility, but merely to something more inviting, like a white bed sheet, white sheet of paper, or a porcelain surface that is cool, clean, hygienic, perfect, inhuman, immortal, and ready to be of service.

THEORETICAL EXCURSUS (I): WHITE AS LANGUAGE

Previous chapters have detailed searches similar to the one of Frampton for purity and for an indivisible, unique monad of cinema. One such discussion of purity appeared in the context of what I termed the "archive dilemma" concerning the notion of an original—the lost or regained presence of an Original. Recall also how the metaphor of *color-as-jewel* fits with the idea of a film projector's light as it disperses into glittering, attractively colored and moving shadows upon a screen. In these cases, there arises in a viewer either anxiety or elation over a lost or found original object—an object, in the present instance, envisioned by Frampton as the pure dazzling whiteness of the fundamental ground for film imagery—much like the staging of John Updike's nostalgic glitter of diamonds meant as an embodiment of the troubling allure of women he has known together with those original women who were lost, "who got away."[29]

What, then, will count as truly "being white"? It would seem that no single quality makes for white on all occasions. Instead, different criteria for whiteness are applied depending on the circumstances and needs of a given description or theoretical argument. These criteria result in a series of radial meanings for the word "white."[30] There is nothing natural or inevitable about stringing together ideas about white, other than the practical situations that arise in our lives that demand expression and action. Thus I believe that white does not possess whiteness. White has no immutable quality, no certain essence, but is rather a summation of our

present uses for the word. There is no stable core for being white—there is only a series of different occasions, loosely connected, for its appearance in descriptions.

Film theories do not spring from physics or science. A film theorist selects and arranges descriptions of how things can be called using descriptions drawn from life experiences, including experiences from films. The proof of a film theory lies neither in the design of its argument nor in its invocation of physical facts about materials, say, of silver halides in a photographic emulsion or of colored patches on a screen, but rather in what the argument *can do*, i.e., in what an argument adorns and on the cultural situations it addresses. A film theory remodels ordinary language about an ordinary object (film) in order to point toward understandings, values, and new uses. We perhaps should simply declare that ordinary language—and the community life it supports and springs from—is a form of *manufacture for exchange and use* like other technologies in society, and that the language of a particular film theory, accordingly, is itself an object manufactured for exchange and use.[31]

It is time to say that language colors color. Language also colors the arguments of film theory. A word gives us something to see this way and that. Thus it makes no sense to select a color, such as white, or to select a group of colors, such as a rainbow, and assert that it or they are somehow *basic* or that we must start with *sensation* in order to build ideas upon a firm platform. There is no privileged starting point. Color acquires its effects when we act upon it by speaking and thinking. Even saying that a color sensation has *no meaning* in itself—that it starts with a sober demeanor exclusive of meaning—is entirely vacuous and white (as it were) and is already to assume a starting point and *to use language* in trying to think about color outside of language.

Let's simply ask this question: is there really a difference between pointing to—pointing out—the color white as opposed to handing someone a paint sample? Or saying the word "white" to someone as opposed to acting appropriately in response to a situation involving white?[32] Once language and, for that matter, color are seen to arise from *actions* in response to circumstances and plans, there is no need to divide language-actions from, say, behavior, gesture, and visible color. Since language depends on consensus and behavioral situations, it is as valid to employ words bundled into a description as it is to handle a paint sample or gesture in a direction. The notion of a bare essence or perfect blankness that can escape language so as to exist in a ready state of pure experience is merely a tenuous tenet in a theoretical doctrine. A doctrine merely represents a point of view that seeks to foster a view while obscuring the fact that there has already been a selective pointing to, and framing of, the view itself. The fact remains that the "pointing to" or framing of a thing in a theoretical doctrine is equally an artifact of language-*action* (witness Frampton's lecture-demonstration) and arises from exactly the same sources. A theory implicitly announces about its author, "I've

thought it through and reached a conclusion." *Through* what? Which framework? Not surely from within a vacuum. . . ex nihilo.

Hence we must carefully mind which languages, metaphors, and vocabularies have been chosen by a theorist when thinking about color and claiming about an object that its sensation alone somehow exceeds language when it is said to be bare and basic. As Aristotle notes in the epigraph, though an object be indubitably white, the nature of the object—its whiteness—remains still to be evaluated. To understand whiteness, one must inquire about the set of current applications of the word "white" to objects. That is, how white is being consciously or nonconsciously framed by and for an act. The meanings engendered by words, language, and theory are not a sort of second sight built upon a raw sensation; they are the sight. The priority of the present gives weight to the meanings we think we see in whiteness when white appears. The meanings, however, reside not *in* whiteness, but come *from* a context said to be white for some reason, no matter the hue.[33] The pertinent question should be: which sort of white from which language will characterize an object productively? More generally: what stories might white, or another hue, be made to tell?

NOTES

1. Aristotle, "On the Soul" in *The Revised Oxford Translation*, trans. J.A. Smith, 428b21.
2. Though an experience may supervene on the physics of matter and brain, an interrelation of experiences in mind is another matter. As Wittgenstein suggests in the epigraph to Chapter 5, an emergent quality may arise from experiences connected to experiences.
3. In a virtual environment created by, for example, Autodesk Maya software, a user is able to select the degree of transparency of highlights as well as manipulate such variables as reflected, ambient, and specular color, incandescence, translucence, irradiance, and many other features. The paint company, Benjamin Moore, lists 157 varieties of white: <www.benjaminmoore.com/en-us/color-overview/find-your-color/color-families?gclid=CjwKEAiAyanCBRDkiO6M_rDroH0SJAAfZ4KL5Z9YSwaGWxew_SmyEDxgOBtfdo9mF2PhQFZb2B5lOBoCvGfw_wcB> (accessed Dec. 9, 2016).
4. Aristotle, "Sense and Sensibilia" ("On Sense and the Sensible" or "De sensu et sensato"), *The Revised Oxford Translation*, trans. J.I. Beare, 442a22.
5. Jonathan Westphal, *Colour: Some Philosophical Problems from Wittgenstein* (Oxford: Basil Blackwell, 1987), p. 37.
6. The racial aspects of "whiteness" are evident in the Goethe quotation in Chapter 2, section 2, and in Blanc's hierarchical "scale of being" quotation in Chapter 5, section 1. See also David Batchelor, *Chromophobia* (London: Reaktion Books, 2000); Richard Dyer, *White* (New York: Routledge, 1997); and Tyler Stallings, *Whiteness: A Wayward Construction* (Laguna Beach, CA: Laguna Art Museum, 2003). On the languages of whiteness in film, see Paul Coates, *Cinema and Colour: The Saturated Image* (New York: Palgrave Macmillan, 2010), pp. 11, 18–19; and Richard Misek, "The

Invisible Ideology of White Light," *New Review of Film and Television Studies* 8, 2 (June 2010), pp. 125–143.

7. Can a place be simultaneously full and empty? or neither? Consider, for example, the photography of Eugène Atget and also Roy Lichtenstein's painting, "Artist's Studio No. 1 (Look Mickey) [I've Hooked A **Big** One!!]" (1973) in which the artist of the busy studio and the artist of the painting of the studio are both "absent," though in different degrees. In addition, note that Lichtenstein has rendered reflections and highlights on the painting of a mirror within the painting of the artist's studio as black geometric shapes and graded black dots, i.e., here "black" doubly *stands for* white on Lichtenstein's painting and on the painting within the painting. Thus black is being represented as no longer strictly "empty" or "absent," but has taken on a specific role as something non-black and "reflective." Also, one may recognize the mirror painting by knowing that Lichtenstein has executed other paintings of mirrors in this style. The partial re-painting of the painting of "Look Mickey" on the studio wall is only one of several references to original (and absent) works by Lichtenstein and others.
8. On the need, or not, for a color's visibility, see Chapter 5, section 8.
9. The mysterious metaphysics of white have encouraged many speculations: Is it a color? the sum of all colors? the sum of the colors of natural light? the sum of two complements (if so, which two)? the sum of three additive primaries? *any* combination of wavelengths that equally stimulate the three types of cones in the eye (where the number of such combinations is vast)? the total absence of color (anti-color)? completely neutral (ante-color)? pure illumination? a blending element to augment/lighten other colors? a thing that destroys—bleaches out—other hues? a thing *opposed* to a hue by being opaque and defiantly non-transparent? and more. . . . Matters become still more vexed when one considers white within the trio, white-gray-black. If gray (with a greater or lesser trace of white) is a hue, it is also an ingredient to describing both lightness and saturation, thus merging three distinct language games.

   One of the problems here is that in searching for an essence, one is trying to find the parts, particularities, and unique properties of a thing. What is *inside* white defining it as such a thing and making it distinct from other colors? What makes it up and makes it be? Are we not led toward the idea that the essence of white is simply anything that something said to be white *can do* for us or that we can (reasonably) imagine for it? For a range of speculations about white—keeping in mind the epistemological emphasis of the present book—along with some contextual values and associations for white, see Batchelor, "Whitescapes" in *Chromophobia*, chap. 1, pp. 8–19, and *Chromophobia*, chap. 2, pp. 20–49; and Steven Peacock, "Three Colours: White" in *Colour* (New York: Manchester University Press, 2010), chap. 1, pp. 7–28 (on the 1994 Kieślowski film). See also note 7 above and accompanying text, and note 15 below.
10. On the association of death and film in the rhetoric of film theorists, see Edward Branigan, "Epilogue: Death in (and of?) Theory" in *The Routledge Encyclopedia of Film Theory*, ed. by Edward Branigan and Warren Buckland (London: Routledge, 2014), pp. 494–504.
11. Stephen Heath, "The Question Oshima" in *Questions of Cinema* (New York: Macmillan Press, 1981), p. 154 (Heath's emphasis). Metz describes the psychoanalytic interplay between fiction and disclosure as follows:

    > I shall say that behind any fiction there is a second fiction: the diegetic events are fictional, that is the first; but everyone pretends to believe that they are true, and that is the second;

> there is even a third: the general refusal to admit that somewhere in oneself one believes they are genuinely true.

Perhaps there is even a fourth fiction: the conviction that a story (or dream) is not already made of parts from the world and so may be genuinely true emotionally, and true for contexts that would elicit such emotions and not merely an entertainment. Christian Metz, "The Imaginary Signifier" in *The Imaginary Signifier: Psychoanalysis and the Cinema*, trans. by Celia Britton, Annwyl Williams, Ben Brewster, and Alfred Guzzetti (Bloomington: Indiana University Press, 1982), p. 72. See also Murray Smith, "Double Trouble: On Film, Fiction, and Narrative," *Storyworlds* 1 (2009), pp. 1–23. For more on fiction, see Chapter 2, section 4; Chapter 2, Descartes epigraph and note 1; and Chapter 7, note 12.

12. Hollis Frampton, "A Lecture," in *On the Camera Arts and Consecutive Matters: The Writings of Hollis Frampton*, ed. by Bruce Jenkins (Cambridge, MA: MIT Press, 2009), pp. 125–126, 129–130 (Frampton's emphases; the square brackets are his, the braces are mine). Frampton's "A Lecture" appears as a 1968 performance piece on the Criterion Blu-ray, "A Hollis Frampton Odyssey."

13. On living within, or else attempting to evade, ordinary and theoretical languages, see Edward Branigan, *Projecting a Camera: Language-Games in Film Theory* (New York: Routledge, 2006), pp. 98–102, 113–115, 145–148, 221–224.

14. Matthew Gurewitsch, "True Colors," *Smithsonian Magazine* (July 2008) at <www.smithsonianmag.com/arts-culture/true-colors-17888/> (accessed Sept. 16, 2016).

15. Paul Coates, "'Fragments of a Theory of Cinema,' subsection 'Notes on Colour'" in *The Story of the Lost Reflection: The Alienation of the Image in Western and Polish Cinema* (London: Verso, 1985), chap. 2, p. 49.

    Coates may take inspiration from a two-sentence paragraph in chapter 42, "The Whiteness of the Whale," in Herman Melville's *Moby-Dick*:

    > Though neither [Ishmael and a young colt] knows where lie the nameless things of which the mystic sign [of whiteness] gives forth such hints; yet with me, as with the colt, somewhere those things must exist. Though in many of its aspects this visible world seems formed in love, the invisible spheres were formed in fright.

    In the next paragraph, Melville asks why whiteness "should be as it is, the intensifying agent in things the most appalling to mankind." Whiteness is thus indissolubly, and surprisingly, linked to conventional blackness—at least in this sort of language-field. The light of day hides horror.

    Coates has perhaps modified or changed his view on monochrome versus color in his recent book. See "Colour and/as Monochrome" *Cinema and Colour*, chap. 3, pp. 45–68; see also, pp. 16–18, 22–27, 69–70, 84, 89, 169–171.

16. Sean Cubitt, *The Practice of Light: A Genealogy of Visual Technologies from Prints to Pixels* (Cambridge, MA: MIT Press, 2014), chap. 1, "[black]," pp. 21–44; but see pp. 42–44, 114–115, 142–144, 149–151.

17. Charles A. Riley II, "Color in Philosophy" in *Color Codes: Modern Theories of Color in Philosophy, Painting and Architecture, Literature, Music, and Psychology* (Hanover, NH: University Press of New England, 1995), chap. 2, pp. 20–69, cf. p. 4; on Goethe, p. 22; on Barthes, pp. 59–60. I have imagined "any-hue-whatever" as a projection from Deleuze's "any-space-whatever," for which see

note 24 below. "Any-hue-whatever" may be related to Deleuze's notion of "absorbent-colour," for which see Chapter 2, note 40.

18. Gilles Deleuze, "The Affection-image Face and Close-up" in *Cinema 1: The Movement-Image*, trans. by Hugh Tomlinson and Barbara Habberjam (Minneapolis: University of Minnesota Press, 1986), chap. 6, pp. 91–95. The close-up shot has been a favorite target of numerous film theories. See generally Melinda Szaloky, "Close-Up" in *The Routledge Encyclopedia of Film Theory*, pp. 92–97.
19. Deleuze, *Cinema 1*, pp. 92, 93; see also pp. 111–117.
20. Deleuze, *Cinema 1*, p. 92.
21. Deleuze, *Cinema 1*, p. 93.
22. Deleuze, *Cinema 1*, pp. 94–95. Eight more dualities are listed on pp. 90–91.
23. Deleuze, *Cinema 1*, p. 95.
24. Deleuze, *Cinema 1*, p. 88 (Deleuze's emphasis); see also p. 97. Deleuze argues that a close-up is neither an *enlargement* of an object nor something *partial* that has been torn away from an object; pp. 95–96. He deploys his famous concept of "any-space-whatever" (*un espace quelconque*) to describe how a close-up of a face may abstract *away* from its particular spatial and action context to achieve "pure affect" independent of space and cause; hence, in this situation, space can be any space; pp. 96–97, 109, 111, 120–122. A character and the spectator will suddenly experience an inward turn of intense feeling, and become lost in space, lost in thought, uncertain how to act. See also Deleuze, *Cinema 2: The Time-Image*, trans. by Hugh Tomlinson and Robert Galeta (Minneapolis: University of Minnesota Press, 1989), pp. xi, 1–3.
25. Deleuze, *Cinema 1*, p. 90; cf. p. 94. This passage should be compared with Deleuze's remarks on colors as "attitudes of the body, that is, categories" in the work of Godard; see Chapter 2 above, section 5.
26. I discuss Deleuze's reliance on ontology in "Of Prepositions: Lost and Found," *The Velvet Light Trap* 64 (2009), pp. 95–98.
27. Deleuze, *Cinema 1*, p. 93.
28. A notion similar to Frampton's cleanly whiteness is deployed in the film of Chapter 8, *Little Dutch Mill*.
29. On Updike's nostalgic glitter of diamonds and James Wood's commentary, see Chapter 2, sections 3 and 4. On Frampton's distinctive nostalgia, see his 1971 film, *(nostalgia)*.
30. On radial meanings and their associated frames, see Branigan, "How Frame Lines (and Film Theory) Figure" in *Projecting a Camera*, chap. 4, pp. 97–149; and Chapter 8 below, section 2.
31. On the manufacture of film theories for exchange and use, see Branigan, "Introduction (II): Concept and Theory" in *The Routledge Encyclopedia of Film Theory*, pp. xxi–xl.
32. See, e.g., G.P. Baker and P.M.S. Hacker, *Wittgenstein: Understanding and Meaning, Vol. 1 of An Analytical Commentary on the Philosophical Investigations, Part I: Essays* (Oxford: Blackwell, 2nd rev. ed. by P.M.S. Hacker, 2005), chaps. 1, 5, and 6; and Part II: Exegesis §§ 1–184, chaps. 1 and 2.
33. On the theoretical implications of the preposition "in," see also Chapter 3, section 5.

Chapter 5 **Making It Color-Full—Relations and Practices**

ABSTRACT

Is there a necessary starting point to think about the nature of color? One might believe that a starting point casts a negative light on the possibilities for color, or a positive light, or that imagining a starting point for color is a mistake. Theorists have taken all three approaches. Wittgenstein pursues the latter approach, which rejects an absolute grounding for color. He argues that simply looking at colors does not reveal the conceptual structures and processes that have been activated in seeing them on an occasion and in finding relevant patterns and relationships. Instead, he insists that our talk about color is often a vague blend of *grammatical norms* about primaries, contexts, and systems, coupled with particular experiences and expectations. As Wittgenstein observes, "Someone who speaks of the character of a colour is always thinking of just *one* particular way it is used."

This chapter surveys many of the relationships among hues that are available as material to be formed into particular grammatical and aesthetic norms that govern patterning. Examples of hue relationships include: physical relationships (e.g., Newton's optics, rainbows, various color circles, and other geometries), non spectral hues, warm versus cool hues, color versus line, scales of luminance values, scales of focal colors, key colors, dedicated aesthetic systems, and, perhaps surprisingly, mentally vibrant impossible hues (e.g., bluish yellow).

> It would. . . be wrong to say, "Just look at the colours in nature and you will see that it is so." For looking does not teach us anything about the concepts of colours. . . . The colour of a blood-shot eye might have a splendid effect as the colour of a wall-hanging. Someone who speaks of the character of a colour is always thinking of just *one* particular way it is used.

* * *

> I think that it is worthless and of no use whatsoever for the understanding of painting to speak of the characteristics of the individual colours. When we do it, we are really only thinking of special [particular] uses. That green as the colour of a tablecloth has this, red that effect, does not allow us to draw any conclusions as to their effect in a picture.
>
> * * *
>
> And don't I have to admit that sentences [about colour] are often used on the borderline between logic and the empirical [causal], so that their meaning shifts back and forth and they are now expressions of [grammatical] norms, now treated as expressions of experience?
>
> For it is not the "thought" (an accompanying mental phenomenon) but its use (something that surrounds it [i.e., the "thought" is surrounded by a grammar employed to situate or affect the perception of a thing]), that distinguishes the logical proposition from the empirical one.
>
> * * *
>
> Psychology connects what is experienced with something physical [empirical, causal], but we connect what is experienced with what is experienced [in a grammar, a norm].[1]
>
> —Ludwig Wittgenstein

COLOR DEGREE ZERO

Wittgenstein in the epigraph argues that our sense of color depends on the norms by which we arrange and make connections among our experiences, linking experiences into a network of experiences. The "network" amounts to a grammar or norm. How do we talk about color and act? For Wittgenstein our consciousness of color does not depend upon physics or physiology, or upon sensations of single hues. Rather, our sense of color emerges from customary ways of thinking about color when we are engaged with *patterns* in determinate contexts. The effect of color in a picture is not due to single colors, but to patterns; and the effect of a pattern, in turn, is due to its place within a chosen grammar of use, a way of seeing.[2] Simply "looking," as Wittgenstein says, is not enough.

Rudolf Arnheim also focuses on patterns and experiences. He notices that as the number of possible hues in film increases, the number of possible combinations

of hues likewise increases leading to greater expressive powers as well as to greater dangers of confusion.

> As the possibility that two objects have the same tone in color film is narrowed down [i.e., as the number of distinct hues for objects grows], color causes a considerable enlargement of the grammar, or optical range—from which it follows, on the one hand, that much more complicated relations among several objects can be expressed in the image. On the other hand, however, the composition of the image becomes enormously more difficult. It enters into the field of polyphony—a comparison only approximately correct. The possibility of discord arises.[3]

If Arnheim is correct that color in film should be seen as a "polyphony," not as a series of individual sensations, then filmmakers must learn to orchestrate the tones and spectators must learn to see the relationships within norms and grammars. Our perception is anchored firmly to contexts. The present chapter and the next are designed to survey some of the important conventions and concepts that govern color relationships. The aim is to create a resource to be consulted when searching for and examining specific patterns—those we expect to see and those that become visible.

The basic problem in film is to discover which color combinations are functioning rhythmically across time and space with other narrative matters. That is, what "complicated relations among several objects" are being brought into prominence for a spectator? Arnheim's mention of these "complicated relations" amounts to an expansion of his initial reference to the "tone" of an individual color. The reason is that a network of tones may contribute to movements in a film's plot that give 'voice' to a filmmaker's vision and 'speak' to a spectator's emerging mood and attitude. An assemblage of tones may thus play a role in generating a film's changing 'point of view.' To put it differently: a pattern of colored objects—a framework of tones—may work to establish the 'tone' of a narrative. As Arnheim warns, however, the result may be dissonance and disruption instead of harmony and clarity. Even so, as we shall see, filmmakers sometimes wish to create friction and disharmony among objects in a story.

Color relationships among persons, objects, actions, and themes in a film can be constructed only through a disciplined knowledge of the workings of types of color patterns combined with a filmmaker's strategic selection among the possibilities. Color need not function simply as a point-sensation—as a pleasant or unpleasant gloss to narrative. Instead, it may place narrative elements into complex motion along lines of development. To appreciate this new sort of motion in film, one must become sensitive to a wide variety of color relations and practices.

Let's begin by asking a general question about the nature of color: is there a necessary ground or grounding for its existence? Is there some intrinsic quality that locks color in place and limits its movement? Opinions differ.

1. NEGATIVE

For Roland Barthes, color is merely perceptual, superficial, and sometimes anomalous and odd. Barthes seems to believe that the solidity of an object is the pivotal factor in its existence and that solidity can be marked perfectly well in grayscale, as if one were to live exclusively with night vision. The belief is that by systematically stripping away color and certain other properties, one eventually will be left with the essence of what it is to be that very 'object.'[4] Thus for Barthes the real nature of an object has nothing to do with color. Some sources for this view were mentioned in Chapters 2 and 3 along with the folk beliefs responsible for the tetrad of metaphors: *color-as-paint, color-as-cosmetic, is-colored-by*, and *colored-as-intended* (see Figure 3.1).

Barthes confesses:

> I am not very fond of Color. . . . I always feel. . . that. . . color is a coating applied *later on* to the original truth of the black-and-white photograph. For me, color is an artifice, a cosmetic (like the kind used to paint corpses).[5]

One wonders what will count as "original truth." What does Barthes believe a photograph is being asked *to do*? Is there a relevant context for an object seen pictured in a photograph? Perhaps for Barthes there is no context at all, only the apparent presence of a depicted object being naked in black-and-white *there then*.

Earlier we saw that one danger of color being in mind was that it would become a spectacle or fantasy pushing aside rational thought. At the other extreme, might color be *too real*? In 1926 Gerard Buckle argued:

> The art of insinuation in drawing or painting is to suggest; the insinuation is immediately lost if the suggestion is made tangible. Tinting is the most perfect form of insinuated colour in photography. It is because the *raison d'être* of the film is illusion that reality in its fullest sense must be guarded against, and perfect coloured photography is not an illusion of colour. By its very perfection it will throw up into strong relief one of the limitations of the screen and, by so doing, will introduce eye strain.[6]

If color is not associated with the dead, spectacle, or too real realism, then in this negative view, it resides with the lower forms of nature. Charles Blanc in his 1876 book asserts:

> Intelligent beings have a language represented by articulate sounds; organized beings, like animals and vegetables, express themselves by cries or forms, contour, carriage. Inorganic nature has only the language of color. It is by color alone that a certain stone tells us it is a sapphire or an emerald. . . . Color, then, is the peculiar characteristic of the lower forms of nature, while the drawing becomes the medium of expression, more and more dominant, the higher we rise in the scale of being.[7]

Blanc points to the dangers of color:

> But the taste for color, when it predominates absolutely, costs many sacrifices; often it turns the mind from its course, changes the sentiment, swallows up the thought. The impassioned colorist invents his form for his color, everything is subordinated to the brilliancy of his tints. Not only the drawing bends to it, but the composition is dominated, restrained, forced by the color. To introduce a tint that shall heighten another, a perhaps useless accessory is introduced. In the "Massacre of Scio," a sabre-tache has been put in the corner solely because in that place the painter needed a mass of orange. To reconcile contraries after having heightened them, to bring together similars after having lowered or broken them, he indulges in all sorts of license, seeks pretexts for color, introduces brilliant objects; furniture, bits of stuff, fragments of mosaic, arms, carpets, vases, flights of steps, walls, animals with rich furs, birds of gaudy plumage; thus, little by little, the lower strata of nature take the first place instead of human beings which alone ought to occupy the pinnacle of art, because they alone represent the loftiest expression of life, which is thought.[8]

Blanc's lengthy listing of examples of inert objects and lower strata of life ("gaudy plumage" and other items like so much fluff-stuff) itself speaks to his fear that a human being can easily be overcome by a plethora of things that don't think, thus becoming himself or herself unthinking and dead. It is as if "a mass of orange" color indeed "swallows up the thought." Worse, an uncontrolled appetite for color impels the artist to create a plethora of useless objects, "bits of stuff," as vehicles for "the brilliancy of his tints." One has to wonder whether Blanc believes that persons, too, may be overtaken in their daily lives by a surfeit of things made attractive by color.

In his next paragraph Blanc worries about the danger of "spectacle" to overwhelm human presence, action, and heroic deeds, and he returns with new lists of threats, including "cushions, slippers, narghilehs, turbans, burnous, caftans, mats, [and] parasols" that would seem to point toward a veiled fear of (France?) being inundated by a foreign culture (North African? Middle Eastern?) in the same way that a "composition is dominated. . . forced by the color."

Blanc's two references in the preceding quotations to "drawing" (the Italian *disegno*) draw on a line of reasoning in art history about line where it is said that color (*colore*) must be subordinated to the *rationality* of the "line" or "outline."[9] This ideological slant is perfectly illustrated by the painting, *Disegno e* [*and*] *Colore* (ca. 1640) by Giovanni Francesco Barbieri, who is known as Guercino or Il Guercino. Melinda Szaloky notes that in the painting, *Disegno* is personified as a wise old master, who is male, dressed in drab clothes, and sitting at the left instructing his pupil, *Colore*, a woman clothed in bright red, blue, and white, who is painting an angel.[10] The woman is depicted as an ornate and opulent figure, but only a copyist and embellisher. A drawing, the design for the painting of the angel, is the brainchild of the male artist, who is holding out a paper with a sketch of the angel for the woman to imitate. Moreover, the dull colors of blue sky and clouds visible out of a window behind the two figures is shown to be entirely inadequate as a match for the artificial and contrived brightness of the woman's blue and white clothes.

Three senses of the noun *disegno* have been subtly connected in this kind of argument so as to culminate in the achievement of "rationality": (1) an act of drawing upon a surface that results in (2) a drawing, design, pattern, composition, or (con)figuration through which (3) an intention, plan, design, or thought is made manifest by drawing forth an artist's idea. The compression of these three strands of meaning in the word "drawing" is fortunate for the argument and perhaps reveals the form of rationality that pits *disegno* against *colore*.

The strong bias favoring line and drawing over color may derive from the notion that a line may coil into a recognizable "shape" or "figure," which, like represented motion and depth, suggests three-dimensionality and the realism of (an object's) volume and mass (object-ivity?). *Colore*, by contrast, seems alone in being merely pasted onto a place to glisten, being weightless in stasis, evincing at best the allure and flatness of sophistry. (Notice how I have recruited embodied, sensory metaphors in the previous sentence to carry an argument.) In this way of thinking, perceptible features (line, shape, color—*disegno*, *colore*) are being projected onto intelligence and rational expression (lines of thought, weighty ideas), which, in turn, are being interpreted in relation to a biological theory of life forms seen as arrayed on an ascending scale from primitive (rocks, plants) to advanced, from unthinking to thinking. For Blanc, color is mapped to the bottom, becoming at best a "sapphire or an emerald," attractive, but empty and static (see Figure 3.1, *color-as-jewel*).

Is the alleged superiority of "line" an old-fashioned view or is it available to be recycled for use? Consider Gilles Deleuze's comment:

> It will be noted that [Alain] Robbe-Grillet. . . was even harsher: he renounced not merely the tactile, but even sounds and colours as inept for the report, too tied

> to emotions and reactions, and he kept only visual descriptions which operated through lines, surfaces and sizes.[11]

2. POSITIVE

David Batchelor, fighting mad, states the case for color being fully cognitive and valuable:

> The notion that colour is bound up with the fate of Western culture sounds odd, and not very likely. But this is what I want to argue: that colour has been the object of extreme prejudice in Western culture. For the most part, this prejudice has remained unchecked and passed unnoticed. And yet it is a prejudice that is so all-embracing and generalized that, at one time or another, it has enrolled just about every other prejudice in its service. . . . It is, I believe, no exaggeration to say that, in the West, since Antiquity, colour has been systematically marginalized, reviled, diminished and degraded. Generations of philosophers, artists, art historians and cultural theorists of one stripe or another have kept this prejudice alive. . . . As with all prejudices, its manifest form, its loathing, masks a fear: a fear of contamination and corruption by something that is unknown or appears unknowable. This loathing of colour, this fear of corruption through colour, needs a name: chromophobia.
>
> Chromophobia manifests itself in the many and varied attempts to purge colour from culture, to devalue colour, to diminish its significance, to deny its complexity. More specifically: this purging of colour is usually accomplished in one of two ways. In the first, colour is made out to be the property of some "foreign" body—usually the feminine, the oriental, the primitive, the infantile, the vulgar, the queer or the pathological. In the second, colour is relegated to the realm of the superficial, the supplementary, the inessential or the cosmetic. In one, colour is regarded as alien and therefore dangerous; in the other, it is perceived merely as a secondary quality of experience, and thus unworthy of serious consideration. Colour is dangerous, or it is trivial, or it is both. . . . Either way, colour is routinely excluded from the higher concerns of the Mind.[12]

Recall that I argued in Chapter 3 that an ontology based on the metaphors *color-as-paint* and *color-as-cosmetic*—in which color is seen to exist on a surface as superficial—tended to evoke aesthetic and psychological effects of either *color-as-spectacle* (gaudy, garish) or else *color-as-delicacy* (glorious, liminal).[13] By contrast, if one begins not with ontology, but with an epistemology of *color-as-causative*, then aesthetics will remain much more fluid in its judgments, and color will instead be evaluated within a set of contexts and measured according to its

actions and deliberations, rather than against immutable values (positive or negative) tied to a supposed essence: alien or superficial or both. When color is viewed as causal, it acquires a use within a setting—it is mobile, emergent, and fully cognitive, as illustrated by, for instance, Andy Warhol's "Do It Yourself" paintings (Chapter 3) or the intersection among Action Painting, Color Field Painting, and Pop Art, as in Roy Lichtenstein's *Two Paintings: Dagwood* (Chapter 1).

3. INTERREGNUM

> When we're asked "What do the words 'red,' 'blue,' 'black,' 'white' mean?" we can, of course, immediately point to things which have these colours,—but our ability to explain the meanings of these words goes no further! For the rest, we have either no idea at all of their use, or a very rough and to some extent false one.[14]
>
> —Ludwig Wittgenstein

Wittgenstein notices that if one tries to *define* a color word in terms of what we experience when confronted with, say, redness, one ends at a loss for words. The problem is that we are trying to define a given hue by capturing some private mental state or sensation—a phenomenal patch glimmering somewhere in the darkness of the brain. This is hopeless. A color word must be talked about and employed as part of a larger way of describing the use of many reds and all colors. To have meaning, a color word must be placed within a vocabulary and a situation. Whatever "red" may secretly be, or not, can be known only when the concept or thought of "red" appears within a chosen language, i.e., within a way of looking at, and acting upon, a world.

Charles Riley states exactly what is at issue for Wittgenstein with color:

> In Wittgenstein's work color questions yield to more general observations on how we make sense of anything, and these departures from the line of color research and reasoning are typical of the philosophical approach to the problem because it has been, since Aristotle, a gathering place for the whole tangled skein of relationships among the mind, body, and language.

Riley concludes that for Wittgenstein's philosophy:

> The result is a color world. . . that excludes no conceivable effect—physical, intellectual, accidental, or artistic—supported by an enmeshed root system of causes and concepts interconnected through language.[15]

Wittgenstein was writing the treatise on color at his death. It exists only in partial form in Part I while Parts II and III are earlier drafts. An important aspect of his work

following *Philosophical Investigations* was directed at the interface between sensations and language(-games/-grammars). How far "down" in cognition does language go? How far into perceptual computation, into what we think we see and come to believe we have seen? (Compare: "what colors and relationships do you *see*?" with "do you *see* what they mean?") To put it differently, the issue concerns the ways that ideas and values extend into sensuous aesthetic perception. Wittgenstein famously stated that "aesthetics and ethics are one." He argued that aesthetic appreciation involves drawing attention to a series of "descriptions" that seem "right." This leads to questions about where right aesthetic descriptions have come from and what ideas about rightness might justify them. For Wittgenstein, talking—i.e., describing, elaborating, and "knowing how to go on"—about a work of art as a "right" arrangement of sensuous materials involves a process whereby *lived ideals* are fit to an object, rather than a process aimed at discovering the causes of a person's mental state of aesthetic pleasure.[16]

It is clear that Wittgenstein in his later work meant to replace the idea of a world defined by logical simples. In *Remarks on Colour*, Wittgenstein questions whether hue, lightness, and some other color qualities, along with certain fixed relationships in, say, a color circle, can be fashioned into a complete and determinative logic based on simple units that can explain all the ways we talk about, describe, and judge color sensations. His earlier view in *Tractatus Logico-Philosophicus* would have led toward a single logic for color, as if color reasoning were arrayed in a firm and regular, crystalline structure. Wittgenstein's new approach is that there exist distinct, smaller-scale uses (games, grammars, fields, practices) of language, each with its own set of procedures designed to achieve particular goals.[17] The existence of many language-games does not mean that all of them are equally good: games are fit to goals. The sum of these many language-game interactions with the world produces collections of judgments, including sets of color judgments that would be rather less Copernican and more Ptolemaic in structure; that is, color judgments would possess assorted complicated epicycles and adjustments adapted to specific practices and tasks in life. There would be no need for precision or for consistency among the various language-games.[18] The complicated Ptolemaic whole would reveal a "form of life" in a particular world-situation or stance comprising many related sorts of 'rightness.'

There is, of course, a physics of color. The physical facts are facts under a particular form of scientific description, yet they do not determine the use we make of them, i.e., which relations we see and choose to utilize in ordinary life and art. There is also a psychology of color. For example, two different light hues are more alike than if one were light, one dark. A person can see "inside" an orange hue both red and yellow, but not purple. Mixing blue and yellow paint produces green, though neither can be seen "inside" green. And so forth. What remains undetermined are the innumerable combinations and possible norms (algorithms, harmonies, disharmonies) that may be

created out of such perceptual facts. We select, arrange, and imagine colors in various combinations according to operative language-games and goals. These languages are templates available to construct, project, and discover patterns in what we have chosen to see and according to how we live. We make new experiential things out of the physical parts and facts.[19] We also make fictions out of the physical parts and facts. Can fictions ever entirely escape physical parts and facts to live a life apart from life?[20]

Thus, to conclude, in thinking about color theories today, perhaps one should pause between positive and negative assessments in order to simply untangle *which contextual elements* are being selected to *blend with color* in order to interpret its general nature and function. How is color being colored? Elements that may be blended with color to give it importance and worth may be chosen from science, biology, the study of mind, politics, culture, aesthetics, commerce, technology, and more. Whatever is *presently* important in our lives—and what we believe to be true and worthwhile—may become a background for understanding and valuing color appearances.

The approach taken by this book involves identifying the various components that are woven into a particular color theory, not from external sources—for what would its counterpart, an "internal" source, be?—but from selected, descriptive frameworks devised by humans and found to be useful in pursuing some endeavor. The aim is not to somehow find a first or final "cause" for color, or to "purify" or "objectify" color into a feeling, or quantify a luminous spot, but rather to clarify our rhetorical commitments when we speak about, and *act toward*, color experiences in one way or another. There is no privileged starting point for color. There are only projects and behavior.

Moreover, it is false to say that color is only color—somehow standing apart from mind and our concerns—and false to believe that to truly know color one must experience it in a vacuum when it can somehow be only itself lighting up a secret place in the mind. Instead, reflection on the nature of color—specifically, on the nature of a particular theory of color—is a matter of weighing the interactive contexts within which one has chosen to experience and explain it. To understand color, one must analyze how it enters our talk about our tasks. Color, like language, is therefore neither fundamentally negative, positive, *nor* neutral. It is simply a process, like change.

A FEW COLORS FROM NATURAL LIGHT

> The theory of colours, in particular, has suffered much, and its progress has been incalculably retarded by having been mixed up with optics generally, a science which cannot dispense with mathematics; whereas the theory of colours, in strictness, may be investigated quite independently of optics.[21]
>
> —Goethe

Let's start by placing color within the context of physics and optics. Which physical aspects should be enlisted to describe color? The so-called *hue* of a color refers to a perceptual category, e.g., the hues yellow, green, and blue. The hues of natural light from longest (least energetic) to shortest wavelength may be seen spread out rainbow-like by passing light through a prism, yielding:

Red
Orange
Yellow
Green
Blue
Indigo
Violet

Taking the first letter of each of these hues will spell out the name, "Roy G. Biv." Or, another way to keep these hues in mind is through the mnemonic: "Richard Of York Gave Battle In Vain."

The often reproduced picture of white light being transformed through a prism into a rainbow has unfortunately become entrenched for many persons as a mental image of the basic logic of color (see Figure 3.1). It is a mode of logic, however, that applies only to very limited situations; for example, in scientific applications. One must be careful that a rainbow image does not dominate one's thinking or become exclusive of other ways of conceiving a logic of color. The prism spectrum makes colors appear to be continuous and complete, with each hue bleeding imperceptibly into the next with no discernible gaps. Nevertheless, there are several major colors missing from this array (see below on "non-spectral hues") and important qualities (to be discussed) remain unaccounted for.

There is no reason to imagine that hues must seek the regularity, intensity, and specific ordering of the natural light spectrum rather than another sort of continuity or discontinuity, especially in aesthetic contexts. For example, natural light need not become the prototype for thinking about "whiteness." Visualizing white in this manner would make it the sum of all hues and both a majestic first cause and essential final aspiration—a grand symbol of perfect unity. The rainbow image assigns the color white to natural light prior to entering a prism. But light is actually colorless. White has a variety of attributes depending on the situation, including being the sign of a profound emptiness or nothingness (see Chapter 4). The same may be said for the other hues: there is no definitive number of hue categories or types (seven by tradition), no privileged combination of hues, and no essential unity; there is also no color hierarchy, no linearity, no intrinsic identity, and no ideal state that a hue must obey in order to be.

1. PRIMARIES AND MATERIALS

In some physical situations three of the hues of natural light become special: red, green, and blue. They are termed "primaries" because various mixtures of them will generate (almost) all the other hues while the primaries themselves cannot be created through mixture. What counts as a primary, however, depends on the operative color system. The natural light primaries match the sensitivities of the three kinds of daylight photoreceptor cells in the retina, i.e., the three kinds of "cones" in the eye that respond to long-, middle-, and short-wavelengths of a certain range of electromagnetic radiation. In actuality, the maximal responses of the cones are not to red, green, and blue, but to greenish-yellow (long), green (middle), and violet (short). Cones do not carry direct information about the color of light. Instead they respond to a mixture of wavelength *and* intensity, producing an ambiguous response. The brain computes the *ratio* of the responses among the three kinds of photoreceptor cells in order to produce a sensation that will be experienced in consciousness as a particular hue. This is known as a reflectance theory of color perception, which depends on the relative mixtures of wavelengths and intensity in the reflection, transmission, and emission of light.

It is important to remember that the selection of a set of primaries is strictly relative to materials and purpose. Colors are commonly fashioned through the chemical properties of various paints, inks, and light-sensitive silver halides within the respective media of painting, printing, and photography. These three types of materials have primaries that include red (bluish-red/magenta), yellow, and blue (cyan), though these hues are *not* the same from medium to medium. For example, the "red" primaries of light, paint, and ink are *different* shades of red. In addition, there are four so-called psychological primaries. Some commercial paints have four primaries. Moreover, the chemical properties of paints differ depending on whether a given paint is based in oil, enamel, acrylic, water, aluminum, copper, charcoal, graphite, or a fluorescent material. Other material, perceptual, cognitive, logical, and aesthetic systems use 5, 6, 7, 11, or 12 primaries. The latter two represent universal focal colors (see next section).

Furthermore, there are two different processes for "mixing" the primaries. With natural light, one mixes "reflectances," which add up toward "white light." That is, when all the wavelengths of natural light are reflected equally, no one wavelength will stand out and the result will be experienced as "white." The situation is different with paint and ink; here, one is mixing the absorbances of molecules, which subtract toward "black." That is, when all the wavelengths have been absorbed by the chemical material, no wavelengths are left to be reflected and the result will be experienced as the absence of light or "black." Mixing hues is sometimes counterintuitive. Mixing red and green light produces yellow, which is a lighter hue than

either red or green. Mixing blue and yellow paint produces green, which cannot be perceived as a constituent in either blue or yellow. This demonstrates that the "logic" of color depends crucially upon specific materials and processes, and is not simply based on what one notices or imagines about individual hues. There are, thus, a number of prominent color logics.

The exact qualities and range of a given hue experienced by a person are dependent on the properties of the materials and the nature of the processes used in creating the material. For this reason, when thinking about color in aesthetic contexts, it is vital to understand the technology driving a given medium. The human eye can discern about two hundred colors in the natural light spectrum, i.e., within the rainbow of colors generated through the medium of a prism. Variations of these colors, principally through gradations in lightness and saturation, produce a total of about ten million tints, shades, and tones that may be discriminated through (the medium of) the neural circuitry of humans. Commercial paints can produce about 500,000 colors.

Too often we forget that while seeing color is natural—other animals can see more or fewer hues—many of the colors we do see are *manufactured and distributed* and thus have a commercial basis. The production of such visible color will possess a very particular range of tints, shades, and tones, i.e., a range of white, black, and gray (saturation), as well as a set of limitations that often are determined by choices made in designing the relevant technology and by profit considerations. (Case in point: in spite of many advantages, the use of organic dyes in Eastmancolor in 1952 led to disastrous fading to which the inorganic dyes of Technicolor were resistant.) It is important to think of color not as defined by its object or resident merely in mind, but as a manufactured material entity, a commodity that has a place and circulates in culture.[22] A complete description of an aesthetic use of color must include an examination of how the manufactured colors have come into existence and why their makers have succeeded or failed. The reverse is also true: manufactured colors are designed to be meaningful and being meaningful derives from considerations of rhetoric, discourse, communication, and expression.

CIRCLES OF COLOR: CIRCUITOUS THOUGH (SOMEWHAT) CONVENIENT

> Colours can appear in places in the visual field without any physical change in that place, but only a relational one to the illumination. The colour is induced into the place, but in some undefined sense it isn't really there. But then what is? The appearance of *all* colours depends on the relation of the eye to the light and to darkening or spectral absorption, to contrast, edges and adaptation.[23]
>
> —Jonathan Westphal

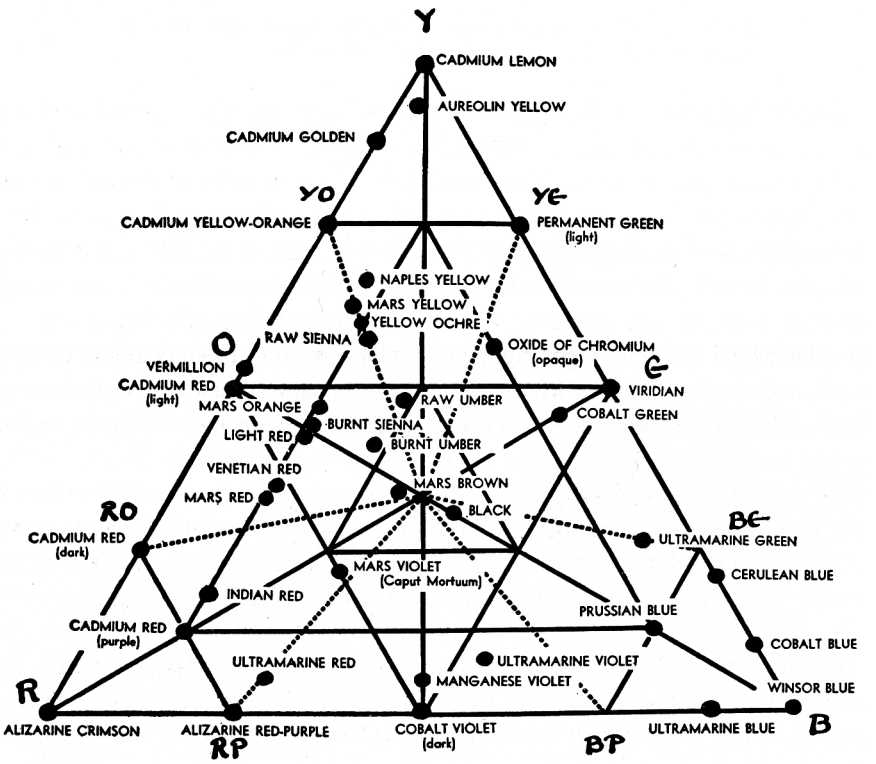


FIGURE 5.1 The Dudeen Color Triangle introduced by Charles A. Winter, taken from the diagram, "Pigment Colors," in John Sloan, *Gist of Art: Principles and Practise Expounded in the Classroom and Studio* (New York: American Artists Group, 2nd ed. 1944), p. 121. The vertices of the large equilateral triangle are the painter's primaries (red, yellow, blue); midpoints are the secondaries resulting from subtractive mixtures between the primaries while midpoints between the secondaries and the vertices are the tertiaries. The larger upright inner triangle contains semi-neutral hues; the smaller triangle, very neutral hues. Complements are opposite one another. Note that the complement of red-purple is between yellow-green and green and is not yellow-green as inaccurately shown on color circles. Rotating a 3-4-5 triangle inside the large triangle produces the so-called *inherent harmony* of six major and six minor triad chords. This rotation generates one of the analogies between the twelve basic hues of the outer triangle and the twelve notes of a musical scale.

1. SPECTRAL HUES

Circles or wheels provide one method for creating a color language with sufficient power to narrate a world. Circles may be arranged in various ways to single out relationships among major variables. They are usually created in order to display a few hues of the light spectrum along with some potential combinations.

A color circle, however, is an abstraction. It is a description that has come into view from a particular standpoint, from a theory or set of theories. A circle should only be thought of as a summary of certain properties in relation to a particular task, i.e., as a depiction of special relationships deemed important. For example, different numbers of colors may be chosen (72 for Chevreul's circle in 1864), and colors typically will have different qualities from circle to circle depending on which materials and interactions are being modeled. Mixing natural light will require a color circle different than when mixing colored paints or inks; still other hues will be selected when representing dyes and chemical emulsions used in printing and photography. Ogden Rood's circle and Albert Munsell's tree were developed on the basis of *afterimages* created in the eye. Also, circles typically divide the hues into equal areas whereas reds and yellows constitute about 40% of the light spectrum. Brown, black, gray, and white are among the hues usually excluded from a color circle.

Although a circle is the most common shape chosen to display color relations, a great many other shapes have been devised with important advantages based on triangles, squares, pentagons, chessboards, charts, stars with varying numbers of points, cubes, spheres, distorted spheres, cylinders, double cones, double pyramids, tetrahedrons, octahedrons, irregular trees, lozenges, toy tops, and more. (For one sort of triangle containing many other triangles, see Figure 5.1.) The circle shape, of course, has the advantage of simplicity and smoothness, where the hues seem to blend seamlessly with no beginning or end and with no angles or points where change is abrupt. This simplicity, of course, is useful only when highlighting particular kinds of relationships.

Geometrical shapes impart a quasi-mathematical rigor to those qualities selected as relevant. Circles, for example, are usually designed so that when any two hues are mixed that are directly opposite one another, the result will be achromatic. Another approach places unitary or pure hues on opposite sides. The type of mixing, and hence the exact color relations, will depend on which material is being modeled by the circle, e.g., ink, paint, light, photographic dye, or even interior, "psychological sensations." The hues opposite one another are called complements and are joined by a straight line through the center of the circle. Straight lines give relationships the appearance of being direct and basic. Primaries, which can be mixed in various amounts to generate all the other hues in a particular system, are typically located at the points of an equilateral triangle inside the circle.

Hues that are adjacent to one another on a circle are said to be *analogous*, or continuous, because each contains something of the other, e.g., yellow, yellow-orange, orange, orange-red, red. . . . A *dyad* or *triad* is any pair or grouping of three hues, sometimes employed to refer to harmonious or discordant combinations. The

terms dyad and triad derive from the analysis of music intervals and chords. In fact, Isaac Newton's decision to divide the natural light spectrum into seven basic parts (see section 2 above), rather than nine or eleven (his first choice) or twenty or the two hundred that can be readily seen, was influenced by the fact that a *musical scale* has seven notes. He may also have been influenced by Aristotle, who believed that all hues were mixtures of a basic seven—black, white, yellow, crimson, leek-green, deep blue, and violet. Hues producing the most pleasure were due to mixtures containing simple, linear proportions. Aristotle connected the seven basic hues to musical harmonies and, further, to seven basic food tastes.[24]

Hermann Hesse, with thoughts of a rainbow and a musical scale firmly in mind, penned a poem asserting not only a connection between color and music, but also between "all the colors of the rainbow" and "all knowledge" (see epigraph of Chapter 6). Still, Newton decided that two of his seven basic rainbow hues—orange and indigo—were somewhat less than basic, calling them "semitones." (Does this make orange and indigo *half* non-spectral?) Furthermore, most people now refer to Newton's indigo as "blue" and Newton's "blue" as cyan. If Newton had settled on nine divisions for the natural light spectrum instead of seven, the next two hues, which are easily seen in the spectrum, might have been a darkish red at the far end of the spectrum and an aqua between blue and green, amounting to a light blue with a greenish tinge. These possibilities hint at an issue to be explored later in the chapter involving a degree of arbitrariness about color that a perceiver seeks to overcome through (evolving) language practices and schemata that manipulate a wide range of cultural connotations.

A color circle may be filled in from its circumference to its center by a series of incremental steps that are meant to depict each hue changing according to the amount of white ("tint") or black ("shade") contained in the hue, which is termed *lightness*, luminosity, or value—there are, however, slight differences among these three terms relating to their measurement. A circle may also be constructed to represent variations within each hue according to the amount of gray ("tone") contained in the hue, which is termed *saturation*, purity, chroma, or intensity. The so-called brightness of a color is due approximately to its degree of lightness and/or saturation.

When not paying close attention to our surroundings or to a film, we tend to think of color mostly in terms of hue alone. But consider:

> We probably think of colour mainly as hue, but this may not be true of all societies. It has often been noted that Homer referred to the sea as "wine-coloured," which is very odd if we think of its hue, but completely understandable if we think of its luminosity and saturation, which are very similar to those of a deep red wine.[25]

Today it seems that we generally do not take special notice of hues unless they are dissolving in twilight or need to be mentally recovered from a black-and-white film. Does one normally remember the exact hues of objects encountered daily without making a special glance?

2. NON-SPECTRAL HUES

There exist non-existent colors. For example, these three: red-non-red, red-green, and yellowish blue, including variants of the latter two, such as bluish orange, lilac green, and yellowish violet. Light and its absence can also be problematic: there is no luminous gray and no blackish yellow. Glaring and dazzling blacks do not exist. Although there are glaring and dazzling whites, there are no white eyeglasses that tint dark objects whitish in the way that rose-colored glasses, and glasses of a few other hues (not all), present a changed view.

What logic or rationale organizes these issues and describes the regularities of color? Perhaps one should instead ask what is meant by a *logic*. Consider the following. There is no reddish green hue and no object that is red all over and green all over at the same time. There is, however, a bluish green hue and there are bluish green objects, but still no object that in the previous sense is blue (all over) and green (all over) at the same time. Or, at least, the blue would not be the same in the two situations. The reason is that a thing that is two hues simultaneously (or is red-non-red) is impossible logically. Nevertheless, a new sort of logic will be needed to describe the existence of bluish green in order to be compatible with the first logic. This second type of logic will attempt to specify the possible and impossible internal relations among all the colors within some abstract hue space: a bluish green combination, for example, will exist, but not a bluish orange, reddish green, and yellowish violet, which are impossible hues. There is even a third sort of logic that must be considered when one talks about things that can possess color under various lighting conditions—things tangible and intangible that are being colored or given color, such as hues reflected onto a hue, colored lights, afterimages, phosphenes, semi-transparent luminous objects, and colored volumes (as in a fog, the depth of a liquid, or a dark room). It is not just the surfaces of solid objects that display color.[26]

A useful way to think about color generally is as follows. Visible color depends on wavelengths of light in the range of 390–700 nanometers. Thus one must first consider the quality and, importantly, the position of the source of light. Next, one must consider the object that acts as a medium for the light and that comes to possess a color due to the action of light. This object may be solid, liquid, or gas; opaque, semi-transparent, or transparent. The physical properties of the medium determine how the light is broken up in reaching the eyes. When the source of light is positioned so as to shine through the medium from behind, the resultant hue of

the object seems to sparkle with light for a perceiver and be "luminous." One must also account for interactions among light rays reaching the eyes from nearby colored objects (e.g., effects of simultaneous and successive contrast). Finally, and crucially, one must analyze the many perceptual, cognitive, and schematic operations that transform light stimuli into a meaningful colored image in consciousness. Are such mental images exclusively private, or do public perceptions exercise a hidden influence?

In dealing with filmic images, one is mostly concerned with the last two kinds of color logic, that is, internal relations among colors and entities appearing to have color under certain conditions. The present book will usually focus on an expanded version of the logic of internal color relations including how cultural practices and interpretive routines bear on how color is being seen. For example, although luminous gray does not exist, could we *imagine* it: an evolving, grayish *morning* light? a bright gray *day*? an expansive gray *sky*?[27] What is a person trying to describe about his or her day by combining the color gray with a sense of impending (luminous) change? This sort of talk based on a folk psychology of color becomes relevant when we experience film as more than a surface of moving color patches on a screen and instead a fictional space of objects possessing spectral and non-spectral hues emerging from lives and places we can envision. At the conclusion of the chapter, we will consider whether color must be visible in order to be seen.

A closer examination of non-spectral hues may suggest the importance of thinking more broadly about our responses to color. Pink, though a basic focal color (see next section), is non-spectral; it is a desaturated, light red with a small blue component. Non-spectral hues also include one form of white that is only "mixed" in the eye when all the spectral hues appear together as well as magenta—a deep purplish red—that is mixed in the eye out of the two "ends" of the natural light spectrum, red and blue. These sorts of mental facts encourage us to imagine colors as existing in an array more complex than a circular relationship and less continuous, for example, than a step-by-step increase in frequency, one cycle at a time, of electromagnetic radiation stretching above and below the visible spectrum even though a linear, step-by-step code also governs the firing of neurons in the visual system.

Then there is brown, which is a special and quite unusual color that is created by surrounding orange light by something (anything!) lighter. Brown is produced through a *contrast effect* with a surrounding, brighter color. If one begins with, say, yellow—which is the lightest of the hues—and surrounds it with something brilliant, the yellow will be darkened by comparison and become brown. This brownish color should not be confused either with a dark yellow or with a dim yellow, i.e., a yellow that is simply dark for a yellow or else a yellow that is seen in low luminance. Brown is a unique color produced by the ratio of—the contrast between—the reflectance of the yellow

and the reflectance of a surrounding field. Thus there is no distinct physical stimulus for brown. That is, there is no brown sort of reflected light, no light that shines brown, and no special electromagnetic radiation coming from a spot that defines brown.

Brown is a very darkish and muddy, even blackish, de-saturated orange-yellow. Different brown tonalities emerge depending on what sort of light is surrounded by something brighter. Red produces a reddish brown or rich brown; yellow, a chestnut brown or light brown; green, a murky olive. In general, there seems to be a sort of heaviness or inertia to the varieties of brown. Because of its indefinite qualities and sources, brown is considered to be a "neutral" hue in contrast to the spectral hues. There is no clear or transparent brown and no saturated or desaturated brown, which also distinguishes it from most hues. Significantly, there is no bluish brown or brownish blue. This may be due to the fact that yellow, and related hues, are relatively light and hence may be darkened by a bright surround, while blue is already fairly dark leaving insufficient range to darken it further in order to make a brown or bluish brown before simply creating a blackish blue. Since blue-browns do not exist, blue and brown hues appearing adjacent to one another in a display *do not mix*. They must stand apart and be distinct in our perception, a fact carefully utilized by painters and filmmakers (see Figure 6.1).[28]

Finally, there are a number of unique colors produced in ways other than the simple reflection of natural light or its refraction through a prism. For example, a double reflection from two reflecting surfaces, one behind the other (interference) as well as reflection from a finely textured surface (diffraction). Such "structural colors" are due to the spatial arrangement of atoms and molecules. Examples include butterfly wings, beetle wings, pearls, opals, hummingbird and peacock feathers, soap bubbles, oil slicks, oyster shells, and the surface colors of ordinary compact disks. In recent decades, more colors for cars are being designed not as "flat," but as "travel," i.e., where a hue looks very bright from one angle and dark from another.

In addition, some colors have special visual qualities that cannot be described solely by hue, lightness, and saturation; for example, transparency effects (one color seen through another, which changes the look of both) and certain metallic and fluorescent appearances. The luster, gloss, or polished shine of a metallic appearance—as opposed to a matte finish—derives from the fact that the reflected light comes almost entirely from the surface of the object and possesses a hue that is not the same as the white light of a highlight.

It is evident that the material production and visible manifestation of color may be much more than would at first appear. Moreover, there are many sorts of relationships among colors, only some of which will be salient when represented within a geometric form. One should especially be a wary user of color circles, and one should be careful to evaluate claims that are advanced about any so-called

natural ordering of hues or a natural harmony among certain hues. Though colors themselves are already visible, readily visible in fact, one should keep this form of visibility distinct from another sort of visibility that is created when colors are projected onto a geometric shape to display their "logic." The first visibility is apparently context-free (most-any-context?) while the second visibility is dependent upon a rigid set of rules and assumptions whereby each hue is implicitly being measured by degrees of (some sort of) quality against others. The second sort of visibility of a hue—as seen or imagined, say, within a color circle—emerges from the invisibility of a logic or language or folk theory that strictly selects and arranges hues and determines the sizes of samples. Nonetheless: could a hue, or two adjacent hues interacting, be sensed outside of all contexts as in the first form of visibility? That is, could there exist just a hue with no connections or assumptions? Does the mind simply become blank, emptied out, and open to a stinging sensation of hue-ness? A blank mind would be colorless. Such a peculiar state would return us to the claims of Hollis Frampton (Chapter 4).

The conclusion would seem to be that we must choose among an array of languages, color arrangements, and norms to find a way of talking appropriate to a task. Let us continue to inventory possible arrangements.

TACTILE HUES: WARM AND COOL

A language of color is fashioned by selecting and arranging positions and differences among a set of colors within an overall architecture followed by rules, algorithms, or guidelines for their combinations and uses. The previous section considered several geometric models as basic structures for a color language. In this section I will elaborate a tactile model that starts by setting up a primary division between warm and cool hues and then projects that binary onto a large number of cultural distinctions. Equally, one might begin with cultural distinctions and map them onto two sets of hues. However, one is soon led to consider deeper issues: the nature of relationships, critical analysis, the expression of thought, and the "language" of cinema.

1. A BALMY BINARY: INVISIBILITY AND TEXTUAL ANALYSIS

Hues have been divided into two groups based on their perceived "color temperature."

Warm Hues	*Cool Hues*
yellow	yellow-green
yellow-orange	some greens

orange	green-blue [*coolest*]
orange-red [*warmest*]	blue
red	blue-purple
red-purple	purple
white, light gray	black, dark gray

Neutral
some greens
magenta (deep purplish red)

Ambiguous/Chameleon-like
brown

Depending on the context, brown may be considered to be warm, cool, both, neither, or neutral. Green and magenta are usually neutral. The contrast between warm and cool hues is frequently stated as a cultural contrast between binaries, depending on the context:

Warm	*Cool*
red	blue
dawn/sunset	clear blue sky
warm	cool
hot	cold
fire	ice
dry	wet
blood	water[29]
advancing	receding
near	far
active	passive
striving	apathetic
stimulant	sedative
excitement	inactivity
warning	confidence
alarm	optimism
caution	openness
analytic	creative
danger	adventure
violent	tranquil
earthy	airy

material	ethereal
dense	rare
thick	thin
heavy	light
opaque	transparent
sun	shadow
day	night
light	dark
shadow	light
up	down
rising	falling
light-hearted	heavy-hearted
positive	negative
happy	sad ("feeling blue")

Notice that there exists some fluidity within the dichotomy depending on the specific contrast: "light" is warm when compared to "dark," but cool when compared to "shadow," and "shadow" is warm when compared to "light," but cool when compared to "sun." It may perhaps be surprising to learn that the warm-cool contrast has been shown to be a cross-cultural phenomenon, a universal of sorts.[30] Warm-cool is an extremely important relationship and appears as a structuring element in many color designs in film and other media. (See Figures 6.8, 8.3, and 8.6.)

Note especially that although red and blue hues are visible, the *relationship* between them, like any relationship defined by a color circle, is *invisible*. According to philosophers, relationships—though not at all exotic (in fact, they are profuse and teeming: to the left of, in love with, before, caused by, often caused by, after, intend. . .)—are, in general, strange beasts neither quite physical nor mental. For example, whatever lies *between* red and blue—what makes for their meaning against one another, where the relation warm-cool is only one such possibility—is intangible, abstract, invisible, untouchable. One cannot *see* the fact, necessity or probability of a relationship. Yet a person's mere thought about the relationship between red and blue does not alter their appearance. Relationships—of any sort—are not directly on the screen, but in a way of looking, inferring, remarking, and acting, upon an event. There are many possible relationships between red and blue (more on this shortly), which is not to say that all are equally possible. Interpretation and critical thinking are indispensable in assessing the presence and meaning of a relationship or a nexus of relationships in an image or text.

If a spectator's developing sense of aesthetic form is tracked through distinguishing a particular sequence of (likely, unlikely) relationships, it will be seen that

neither so-called internal nor external significant relationships are displayed on the screen. This fact about the nature of form has been too little noticed by some film theorists and narratologists who imagine that internal form is before the eyes, complete, objective, and the most important fact about the working of art. Furthermore, there may be unusual situations where a relationship exists that brings to mind hues that otherwise are not visible in the world (see final section in this chapter).

The long list of cultural associations of the warm-cool binary recalls E.H. Gombrich's use of a "ping-pong" heuristic in exploring a cross-cultural consensus embracing a vast range of binaries in semantic space.[31] By imagining how quite diverse things can be reliably divided into "ping" or "pong," Gombrich is led toward a major idea that I am exploring in this book. He contends that "the very metaphors of our language that we use in describing this picture [Lorenzo Lotto's *Allegory*] still preserve the basic *relationships* on which its symbolism is grounded." In this direction lies a new way of conceiving the "language" of cinema in terms of the collective ordinary language responses of spectators living amid cultural discourses.[32] In particular, I believe that the metaphors and rhetoric employed to describe an image may be intimately connected to the grammar of color that has been selected to map a structure of color experienced in the image (or only heard in dialogue).

How has the entrenched status of the warm-cool relationship been explained? Here are six common theories.

1. Long-wavelength light (red) focuses behind the retina in the same way as light that comes from a distant object (as in far-sightedness), while short-wavelength light (violet) focuses in front of the retina in the same way as light that comes from a close object (as in near-sightedness). Red seems to come closer or advance toward the viewer because the lens of the eye tries to bring the red light "nearer" in order to focus it directly on the retina, while violet seems to recede from the viewer because the lens tries to shift the violet light "farther away" in order to focus it directly on the retina. Red and violet represent the two ends of the natural light spectrum after white light has been passed through a prism. Thus the tension engendered by red and violet appearing together in a scene begins as an optical stress in the musculature of the eye as the lens alternately tries to focus first one, then the other hue. For this same reason, the spatial cue of "aerial perspective" (e.g., the blueness of distant mountains) acts to independently produce a heightened sense of distance. In contrast to red and violet, green and magenta focus directly on the retina and are usually felt to be neither warm nor cool, neither advancing nor receding, neither near (caution, attention) nor far (tranquil, airy), but "neutral."

2. The human eye is more sensitive to—is more active and responds more strongly to—the reflective energy of yellow (warm) than of blue (cool).
3. Infants, but not older children, have a distinct preference for red. Persons viewing warm hues show elevated vital signs while persons viewing cool hues show lowered vital signs, e.g., blood pressure, pulse rate, body temperature, and respiration rate.[33]
4. Recent scientific studies have found that persons working against red backgrounds performed better on tests of recall and attention to detail, like remembering words or proofreading. Blue backgrounds promoted imagination on tests that required persons to invent creative uses for a brick or to create toys from shapes.[34]
5. Red and yellow are associated in human experience with fire and violent movement (so red appears "fast"), while blue and green are associated with water and tranquility ("slow"). In this scheme, black may appear as an extreme form of passivity, as if it were a mysterious, limp sort of curtain acting to keep a person calm and "in the dark." This sort of blackness may be envisioned as hiding something (through its opaqueness) or simply as the sign of absence, or not even a *sign*, simply nothing or nothing yet.[35] In none of these cases of blackness are we confronted with something that requires an immediate reaction. By contrast, a "red herring" in a plot is perhaps called "red" because it solicits urgent attentiveness, even if only to finally distract or mislead.
6. The first four chromatic hues to appear in a language are the basic primaries red, yellow, green, and blue. Red is always first, blue always last, as if they were opposed (see section "Universal ['Landmark'] Focal Colors").

Notice that *cultural* beliefs about the warm-cool opposition—red is hot and energetic while violet is cold and inert—are the exact reverse of the physical facts. In terms of the physics of light, red possesses much less energy than violet because the wavelength of red is longer (i.e., fewer peaks and troughs per second) than violet. Thus the warm-cool dichotomy depicts not physics, but *commonly felt* polarities.

In Vincent van Gogh's last letter from the asylum in which he was confined, he writes to Émile Bernard about color combinations and their effects. He comments on the issue of visible red in reality as opposed to the reality of feeling redness.

> At present am working in the olive trees, seeking different effects of a gray sky against yellow earth, with dark green note of the foliage; another time the earth and foliage [is] all purplish against yellow sky, then red ochre earth and pink and green sky. See, that interests me more than the so-called abstractions. . . . My ambition is truly limited to a few clods of earth, some sprouting wheat. An olive grove. A cypress. . . . You'll understand that this combination of red ochre, of green

> saddened with gray, of black lines that define the outlines, this gives rise a little to the feeling of anxiety from which some of my companions in misfortune [in the asylum] often suffer, and which is called "seeing red."

2. "THE MASQUE OF THE RED DEATH"

Edgar Allan Poe's short story, "The Masque of the Red Death," features seven rooms, each decorated in a particular color with a stained glass window of the same color, except for the seventh room.[36] The colors from the first to seventh rooms are:

1. Vivid blue
2. Purple
3. Green
4. Orange
5. White
6. Violet
7. Scarlet and black: stained glass windows of a deep blood scarlet with black walls, black velvet tapestries, black velvet carpet, and a large black clock.

Notice that in terms of the warm/cool contrast, the progression of the hues is:

1. Cool
2. Cool
3. Neutral
4. Warm
5. Warm
6. Warm
7. Hot (scarlet) and cold (black).

In terms of hues, the first and seventh rooms recall the painter's primaries (blue; scarlet) while the secondaries (purple; green; orange) appear together in the second, third, and fourth rooms without transitions. The fifth and sixth rooms also represent strong contrasts, namely, white versus violet, where white combines and cancels all the hues versus violet as a combination of the two extremes of the light spectrum, red and blue. Thus the hues of the seven rooms, in general, are arranged to maximize contrast. In addition, the seventh room contains an internal contrast: scarlet (hot) versus black (cold).

The title of the story suggests what interpretation should be ascribed to red, namely, death, and how the last room becomes the end, even though a person might try to hide during his or her lifetime behind the "mask" of another color in another room. The idea of redness as one's "lifeblood" is reversed by Poe to become the color of one's

living—slowly toward, room by room—death. Poe's story is set at a masquerade ball with the revelers being described as dressed to resemble "dreams" and "fantasms." Death is figured as an infection that comes from within one's body, within one's own blood, even as the revelers believe they are safely shut up in a fabulous castle and safe from a devastating plague ravaging the kingdom known as the Red Death. Even artful dream seems part of death's plan.

3. A BINARY UNDONE BUT ENDLESSLY REMADE: THE FLUID POWERS OF CULTURAL RESEMBLANCE

The warm-cool binary arises from circumstances being lived and felt. Our sense of touch (warm, cool) is applied to color perception as a way of thinking tactically about tactility. Color is related to other facets of life through the use of metaphors in elaborating responses to situations. One should not think that experiences and metaphorical ascriptions to experiences are separable from circumstances and memory. When reacting aesthetically, we feel a larger presence than the physics of sensation—larger than discovering relations among sensations and more profound than seizing a phenomenological sheet of sensations in conscious awareness.

The twin sciences of embodied and situated cognition explore how meanings are generated out of lived contexts.[37] A fundamental fact is that our bodies are present always in environments and especially so when watching a fiction. Where else could we be in a fiction? out of body, out of mind? Though the warm-cool binary is cross-cultural, its application depends on being situated.

Thus it is not feasible to limit color combinations to the laws of physical interactions among materials, or to limit the logic of color to reflections off surfaces or to the mental counterpart of reflections that paradoxically require a steady concentration of mind to bring the phenomenal spots into view. Shouldn't color combinations reflect the way we perceive, think, and feel at specific times about the world as evidenced in the earlier two-column list of cultural qualities associated with warm and cool hues?

If there is a cultural component to perception that is situated, then what should color itself resemble? Should it be compared to materials like paint, cosmetics, or precious stones? Or, should color resemble a causative force responsive to concerns and feelings about life in our world? If the latter, color would be seen to acquire the power to forge links with how we live and talk.

The warm-cool binary can be employed to multiply link items, not just in one-to-one relationships, as in red is fire, blue is ice. Perhaps our default image for red and blue is as polar opposites, for example, placed on either end of a line. The default image relies on a simple spatial relation, say, left and right. As mentioned above, the mental image may be visible, but not the relationship. There may, in fact,

be a great many possible relationships between red and blue. First of all, *any*thing in the universe resembles anything *else* to some degree or in some respect (red and blue share the letter "e"; I think I remember seeing red and blue together in a quilt in a store window). One merely needs to find the way—the ways!—and acquire the requisite credulity under persuasive rhetoric.

Thus if polar opposites, e.g., red and blue—the two ends of the natural light spectrum or, in another framework, the extremes of hot and cold—are variously to. . .

> attract each other (drawing close but remaining distinct, perhaps bordering one another) or else to mix (becoming magenta) or simply to neutralize (each moving halfway to the midpoint of the spectrum and becoming green) or perhaps to dissolve (becoming white, whether opaque or blank or clear) or merely to exchange appearances (each becoming the other) or to cancel (becoming achromatic, each retaining their characteristic lightness or darkness) or to interact and transform according to some other set of rules

. . . then a method of *looking* and speaking must be found that fits with a scheme of representing the requisite feeling and idea. That is, one must deploy a persuasive rhetoric—whether expressed in words, pictures, or actions—to institute a rule for how red and blue are meant to interconnect to achieve a new state. When this is achieved, we will see red and blue apart or at a common border, or as magenta, green, white, blank, clear, light, dark, or as each other, or in another way under a specified scheme, when we look.

For example, consider the powers of resemblance. Michel Foucault has studied four distinct forms of resemblance that controlled the nature of knowledge in Western culture up to the end of the sixteenth century. Using a form of resemblance that he calls "the play of *sympathies*," he demonstrates how the idea of red was transformed into blue within a larger cosmology:

> Fire, because it is warm and light, rises up into the air, towards which its flames untiringly strive; but in doing so it loses its dryness (which made it akin to the earth) and so acquires humidity (which links it to water and air [associated with blueness]); it disappears therefore into light vapour, into blue smoke, into clouds: it has become air.[38]

In this sixteenth-century way of thinking and seeing color, the world continues as always—changing, to be sure, but essentially untouched, always re-finding a harmonious equilibrium in which everything has a permanent place beneath superficial change, where even the apparent opposites red and blue become deeply resemblant (appearing "in the same light"). Warm and cool act to balance one another, exchanging and shifting places, thereby restoring a balance in which a re-found symmetry becomes the mark of stability.

One might wonder what can be said about color linkages in the cosmologies of today. What new ways of thinking can be located in the movements of color that are responsive to contemporary realities? When and how do we make fresh resemblances by figuring and re-figuring colors? The answer lies in examining contemporary practices by closely tracking preferred color relationships.

THE NATURAL SCALE OF LUMINANCE VALUES

Interacting with a world provides us with expectations, color included. These expectations are woven into a given language-game or language-field and are fruitful, at least until they conflict with another language we like to use. This section describes a natural pattern in the *perceived lightness* of hues and how it enters thinking.

The luminance value of a hue at full saturation depends on which hue has been chosen; that is, different hues reach maximum saturation at different degrees of lightness. For example, even though the amount of green and blue light may be the same in a given situation of daylight illumination, an observer will see the green as twenty times brighter than the blue. One consequence is that a naturally darker color, say, blue rather than yellow, has less latitude for using gray to create a range of saturations. Hence, there are fewer degrees of saturation for naturally darker hues of low value than lighter hues of high value. In effect, one can see more tones of gray (saturation) in a lighter hue.

In addition, due to the specific neural response curve of the eye, one can discriminate many more degrees of lightness in a hue than darkness, i.e., there are more light yellows than dark yellows. There are also more degrees of yellow (which is naturally lighter than blue at full saturation) than blue at high levels of illumination, but more degrees of blue than yellow at low levels of illumination. Thus the possibilities for an aesthetic of "delicacy" are not spread equally among the hues.

Yellow is the lightest hue at full saturation (actually a very slightly greenish-yellow is the lightest at 5% gray[39]) while purple (violet) is the darkest, and between them—arranged in descending semicircular gradations on a color circle—are the other hues, as follows:

Lightest

white

yellow

yellow-green yellow-orange

green orange

green-blue orange-red
blue red
blue-purple red-purple

purple (violet)

brown

black

Darkest

The values of natural sunlight as well as the coloration of plants and animals follow this order of luminance values. Consequently, there is a powerful expectation in a spectator that the light values of a composition will follow the order in which they appear in nature. Any deviation will be noticed. For example, an olive brown (darkened yellow) paired with a light, baby blue will seem "unnatural"; similarly, an orange paired with a pink, or a dark green paired with a pink.[40]

One of the first tasks in analyzing a combination of colors is to determine which relations among the hues, luminosities, and saturations are normative or familiar and hence come laden with expectations. These presuppositions are not necessarily a hindrance to be resisted or overcome since efficient perception is designed to be rapid, forward-looking, and directed by hypotheses. In fact, only a few of our myriad expectations moment by moment while tracking a film's plot are denied for expressive reasons.[41] Filmmakers build on expectations, default assumptions, and background knowledge as well as on the typical ways in which inferences are drawn from events (e.g., by relying on judgment heuristics). This allows the psychology of the everyday to be employed *against* spectators when it is important that something not be seen or fully understood during the telling of a story; for example, when creating effects of mystery and surprise.[42] Expectations, combined with imagined alternatives, direct what a spectator looks to find.

UNIVERSAL ("LANDMARK") FOCAL COLORS

How do colors figure in our talk when we explain what we've seen? What colors do we expect to see? Brent Berlin and Paul Kay in their classic study, *Basic Color Terms* (1969), discovered the following remarkable set of rules based on a study of 110 languages:

1. All languages contain terms for black and white. "Black" encompasses black, green, blue, and other "cool" colors. "White" encompasses white, red, yellow, and other "warm" colors.

2. If a language contains three terms, then it contains a term for red.[43]
3. If a language contains four terms, then it contains a term for either green or yellow, but not both.
4. If a language contains five terms, then it contains terms for both green and yellow. "Green" includes green and blue, and is called "grue" by Berlin and Kay.
5. If a language contains six terms, then it contains a term for blue.
6. If a language contains seven terms, then it contains a term for brown.
7. If a language contains eight or more terms, then it contains a term for purple, orange, gray, pink (a pale, whitish red, sometimes with a slightly bluish cast), or some combination of these. English uses eleven terms. Russian and Italian add a twelfth term—a special "light blue" parallel to the distinction in English between "red" and "pink." Hungarian also has twelve terms with two types of "red," including a special "dark red." Some women through genetic mutation have a fourth cone type in their eyes that generates additional discriminations among reds, oranges, and yellows.

Black and white are each precisely focused on single prototypes; for the other hues, there are slight variations from language to language in what will count as the best example of each hue. Focal hues correspond to those color frequencies that produce a maximal neural response *within the selective vocabulary* of a given language. However, the boundaries of hue categories remain highly variable even within a given language and even with repeated tests with the same person.[44] All the hues, of course, are being "seen," but ones not yet named are grouped into existing categories. Perhaps this is a general principle of cognition and demonstrates the power of language to create schemata and *maps*.

For what it may be worth, I have assembled a somewhat larger list of color terms. The following set contains twenty basic color terms divided into seven sets based on the Berlin-Kay study and on Level 1 of the *Universal Color Language* of the Bureau of Standards and the Inter-Society Color Council (which lists 7,500 color names on Level 3 in 267 equivalence classes), together with the final set, and the colors "tan," "beige," and "taupe." Some additional variations are included below.

white	red	brown	purple	gray	violet	turquoise
black	green	tan	orange		olive	chartreuse
	yellow	beige	ice pink			lime
		blue	taupe			mauve

Here are specifications for these colors:

White, black = the colors used to lighten or darken the hues.

Red, green, yellow, blue = in various combinations and in various tints and shades these colors are "primary" in color circles as well as being "primary" in explanations of the physiology and psychology of color vision; a dark red is *maroon* or *puce*; a moderate red is *cerise*; a moderate to strong red is a *madder*; a bright red is *scarlet*; a special pink scarlet is *kurenai*, for which see, e.g., the final few scenes of Yasujiro Ozu's *Equinox Flower* (1958); a deep red is *carmine*, also called *crimson lake*, often derived from crushed insects and found in many foods and drinks as coloring; a deep purplish red is *crimson*; a vivid reddish orange is a *vermilion*; an *ocher/ochre* is an earthy red or yellow, often from impure iron ore; a bright green or greenish-blue is *verdigris*; an intense metallic yellow is *gold*; a grayish-greenish yellow is *sallow*; a vivid blue is *ultramarine*; a pale, transparent blue to light greenish blue is *aquamarine*; a sky-blue is *cerulean*; a strong greenish blue is *cobalt blue*; a dark greenish blue is *teal*; a deep reddish blue is *indigo*.

Brown = a blackened orange or blackened yellow or blackish-reddish yellow (although it is not known for certain what "brown" really is); *raw sienna* is brownish yellow; *terra-cotta* is brownish orange (clay-colored); *burnt sienna* is orange-red or reddish brown and perhaps the earliest hue since employed by Paleolithic Man.

Tan = a light yellowish brown; a very light and pale tan or light gray is *pumice*; a light and pale gold brown is *blond*; a dark yellowish orange or light brownish yellow is *amber*.

Beige = a light and pale grayish-yellowish brown; a light, pale grayish yellow is *ecru*.

Taupe = a brownish gray.

Purple = any of a group of colors that lie about midway between being red and being blue, perhaps darker and more bluish than reddish.

Orange = any of a group of colors that lie about midway between red and yellow; *saffron* is a moderate orange to orange-yellow.

Ice pink = a light, shiny de-saturated red that is very slightly bluish.

Gray = the color used to de-saturate the hues.

Violet = a spectral color that is a dark reddish blue.

Olive = a blackened greenish yellow.

Turquoise = a pale, light greenish blue; less pale and a bit more blue is an *aqua*.

Chartreuse = a brilliant yellow-green.

Lime = a yellowish green.

Mauve = a moderate purple, violet, or *lilac* (which is a light pinkish purple); a pale mauve is a *lavender*; a vivid reddish purple is a *fuchsia*.

The natural focal colors vary in number from two to twelve across languages. Evidently there is a series of causal links among biology, phenomenal appearance,

language, environment, and culture that accounts for the remarkable stability of these focal colors in spite of the differences among societies. Various reductive accounts may be offered of this universal fact when the question being asked is itself limited; otherwise, one must respect complexity.

I believe that the idea of focal colors should be extended beyond the simple identification of a hue. I would like to suggest that there exists a series of related foci within larger semantic fields working to orient the significance we attach to color, i.e., what we purport to find with/in color. These semantic foci address when a color may come to our notice, the gist of it, its connections within a field, and what we will make of it. Figure 3.1, for example, lists a number of basic metaphors and mental pictures (foci) that anchor our thought about the nature of color, its aesthetics and analysis. Mental pictures and their projections provide landmarks that support a broader rhetoric.

PROMINENCE: HOW A COLOR BECOMES A "KEY COLOR"

How might the presence of color act to shift and develop our expectations within a semantic field and thereby come to notice? Color may organize our thinking about any aspect of a film—about character motive, theme, action, narration, narrative structure, or important objects. Such a mobile color is said to be a "key color." Adrian Cornwell-Clyne insists that a "colour director" establish one or more key colors in the effort "towards achieving unity" in a film. "The feeling must be conveyed to the audience that a certain colour is the pivot, or axis, towards which all other tones converge."[45]

Most often when a composition is described, only the key colors are mentioned. For example, if it is said that a shot is triadic and based on the painter's primaries (red, yellow, blue), what is being asserted is that a pattern created by these three colors is the most *prominent*, which does not mean that the three take up the most *area* in the image since there may be many additional hues and shades in the shot. What follows are some of the techniques that a filmmaker may use to push a color into prominence. Once prominent, a color may initiate a cascade of links to other matters.

1. AREA

A color that takes up a significant amount of area within a frame may become a key color. The relation of a hue to the area it occupies is sometimes called the "weight" or "force" of the color. Theories abound on the problems of assembling areas so as to be harmonious and useful in developing connections to aspects of narrative.

A major issue for an artist involves managing the perceptual effects of areas that are nearby in space (simultaneous contrast) or nearby in time (successive contrast), or, indeed, combinations of both proximate space and time. The reason is that the human

visual system perceives a hue as being tinged with the complementary hue of its neighbor. The effect is most intense when the nearby hues are themselves complementary. For instance, when the complements red and green are adjacent, the red hue tints the green with green (the complement of red) and the green tints the red with red. The result heightens the red-green contrast, driving the two colors further apart. The same effect holds for shadows of a hue: the shadow of a red area will not simply be black, but black tinged with green, which will add green to the hue over which the shadow falls. Hence in designing a grid-work of colored areas, a filmmaker must evaluate the perceptual effects of interactions among the areas as well as the interaction when one color replaces another in time, e.g., through editing or camera movement.

As a sample of such theorizing, here are Cornwell-Clyne's three rules for colored areas in film along with his commentary. I will illustrate the rules in the next chapter in a discussion of *Gentlemen prefer Blondes* (Figure 6.3), and examine the nature of such "laws" when thinking about mise-en-scène.

Cornwell-Clyne attempts to balance a number of factors.

> 1. The area of a given colour should be inversely proportional to the saturation of the colour, so that the greater the degree of saturation the smaller the area.
> 2. The area of a given colour with relation to its hue should be such that the relative area will be greater [and] broadly proportional to the diminution of the wavelength. (Less for the long wavelength end of the spectrum [red] and more for the short wave end [violet].)
> 3. The area of a colour should be inversely proportional to its lightness, so that the greater the lightness the smaller should be the area.
>
> Since all three variables [saturation, hue, and lightness] are often present in combinations of any degree of complexity, it may be said that the first law is the most important one to conform to, and that it provides the governing factor. It follows that if saturated colours are used as a means of creating centres of interest upon which the visual attention is focused, backgrounds should be relatively desaturated, such that the percentage of the dilution with white [i.e., gray, not white?] should be approximately proportional to the area of the background as compared with the contrasting centre of interest.[46]

The fact that Cornwell-Clyne's first "law" relates to saturation is, paradoxically, symptomatic of a fear of color.[47] It is a plea for restraint and classical decorum endorsed by many writers together with a warning to avoid the deleterious effects of becoming a garish "spectacle." His three laws represent a preference for quiet order and balance to allay worries about excess and chaos. Nevertheless, his approach

represents only one type of "unity" in color design. As examples in the present book will demonstrate, prominent color patterns may work to unify and make coherent various aspects of narrative structure in various ways without thereby acceding to a classical aesthetics of restraint and balance.

Cornwell-Clyne's three rules can be partly summarized by saying that in order to be equally impressive, a bright, vivid red, or other warm advancing hue, should be confined to a smaller area than a dim, pale blue, or other cool receding hue.

Cornwell-Clyne adds:

> For let there be no mistake, almost any conceivable combination of colours may be made pleasant, or duly expressive, providing the areas allotted to the respective tones be rightly adjusted.
>
> The commonest fallacy is to discuss the harmony of colours as if they were entities always destined to occupy equal areas, as in a pattern of strips equal in width.
>
> Another fallacy easily exposed is implicit in the theory that when unequal areas are allotted to a collection of various colours, providing the integrated light of these. . . is a neutral grey, the condition of harmony will be met. It would not be difficult to show that few, if any, of the acknowledged masterpieces of decorative art would meet this test.[48]

For Cornwell-Clyne, then, a central problem of color composition is to adjust the qualities of individual hues to their areas in relation to other colored areas. The key to this standard of unity lies not with individual qualities, e.g., saturation, hue, lightness, or uniformity of area, but with adjustments across a series of relationships.

One may consider in more detail the effects of area in relation to the *lightness* of hues, which is known as the problem of hue "proportion." Based on Goethe's influential 1810 book, *Theory of Colours*, Edith Anderson Feisner and Ron Reed assign weights to hues: "white-10, yellow-9, orange-8, yellow-green-7, green-6, red-6, magenta-6, blue-green-5, cyan-4, blue-4, blue-violet-3, and black-0."[49] So, for example, it takes three times as much area of blue-violet-3 to achieve the intensity of yellow-9; such a combination is said to achieve "harmony." Equal areas of identically weighted hues, e.g., red-6 and green-6 are equally strong, but if a small amount of red would appear with a large amount of green, the red would become "vivid" and "very active," working to "attract and draw the viewer into the composition" (see next subsection). Two hues appearing in their natural proportion, e.g., blue-violet-3 and yellow-9, results in "quiet, static effects." Are these "quiet" effects due to the fact that the hue weights remain consistent with the natural scale of luminance values? When the natural proportions are inverted, e.g., nine parts blue-violet to three parts yellow, the hues achieve a "perfect balance." Moreover, as high-intensity areas

are moved further apart in a composition, their intensity weakens, but "a greater sense of balance is created."

Clearly, several factors—lightness, complementarity, the warm-cool dichotomy, and composition—seem to be at work in assessing the effects of various combinations of hue weights in attaining one of four basic states: harmony, vividness, quietness, or perfect balance. Arguably, three of these states are types of *balance*, an important mental schemata. Hues that appear in a type of balance exemplify certain sensory forces in equilibrium much like (derived from?) our proprioceptive sense when standing upright or our sense of a succession of bodily equilibria when walking across a room or our evaluation of abstract forces as expressed in the strophe and antistrophe of the ancient Greek chorus.

The aesthetic rationale for attempting to adjust the lightness proportions among hues offered by Feisner and Reed is that "we must strive to have color and imagery seen simultaneously in a composition. In most visual situations, color is seen before imagery, so the artist must work to counteract this effect."[50]

The use of the word "simultaneously" in this aesthetic prescription suggests the norm of balance. The word "imagery" may suggest the perception of lines and outlines that mark out areas. If so, then Feisner and Reed's argument would be that since color (*colore*) naturally outweighs line (*disegno*), its intensity must be muted in order to properly balance color with line. They also argue that warm dark hues belong on the lower portions of a composition while larger areas of cool light hues belong in the upper portions.[51] This aesthetic prescription suggests the additional norm of verticality (lower/upper), e.g., ground and sky. Balance and verticality are two of a number of bodily and mental schemata that may be applied to generate a (familiar) aesthetics of color.[52]

2. MAGNETS, PINPRICKS, BLOTS, STAINS, AND ACCENTS

A color may become prominent if it stands out in relation to other colors in the frame; for example, if it is emphasized through the use of a high contrast harmony or through a dissonant color design (Chapter 6). Mary Beth Haralovich finds that small areas of red act as "visual magnets" in Douglas Sirk's *All That Heaven Allows* (1954),[53] and Russell Merritt locates a variety of films employing primary and secondary "pinpricks" of color.[54] This may be what Cornwell-Clyne means by his strictures on "saturated colours" when they "are used as a means of creating centres of interest upon which the visual attention is focused."

A center of interest, of course, need not coincide with a center of *narrative* interest. Magnets and pinpricks have many uses, including white sparkling pinpricks that testify to the presence of a light source as well as unmotivated dots of hues creating a measured amount of disunity in a composition.[55] Disunity, in turn, may be used

to create counterpoint, tension, irresolution, conflict, friction, distraction, dialectic, digression, or competing points of view. Disunity will be further augmented if "unconscious" processes are taken as sources for a magnet or pinprick. The relevant unconscious state may be that of a specific character or more broadly that of an ideological contradiction in society. For example, combining Lacanian theory as interpreted by Slavoj Žižek with the notion of a magnet or pinprick will yield a color "blot" or "stain,"[56] analogous to a Freudian slip of the tongue or disguised dreamwork.

If a color does not become a key color through area, strong visual contrast (magnet, pinprick), or mysterious appearance (blot, stain, puncture), then it may figure as an "accent," that is, as a small colored detail that draws attention in some way, not to itself, but to an associated key color, e.g., through complementing or supplementing the key color in some way. Accents may also be a trace of a past pattern or forecast the future (Chapter 6).

Some colors may be designed to be even less noticeable than an accent and may serve some general function, e.g., to identify a locale or character. These may be called "background" colors whether they appear in the background of a shot or not.

3. OTHER IMPORTANT FACTORS

Some additional factors that contribute to establishing a color as a key color, especially the first three, are as follows:

1. The color is located in an important place within a frame, e.g., in the center of the frame or in the foreground.
2. The color marks an important physical object in a narrative that a character may look at or handle; or an important part of a character, e.g., a heroine's lipstick.
3. The color is repeated within the frame or in successive shots. For example, "with El Greco, the same yellow lightens the edge of an angel's wing, the knee of a figure, the lines of a face, and the convexities of a cloud [i.e., of the heavens. . .]; the same madder red colors clothing, ground, or buildings."[57]
4. The color marks an object that is moving.
5. The color marks an object that has an unusual or unexpected shape.
6. The color marks an important spatial plane and/or develops volume (see next subsection).
7. The color is not appropriate. For example, in a Western, the hero's gun belt is pink, his gun a metallic lavender. Also, there are red-hot and white-hot flames, but, absent computer-generated imagery, there are no lower-degree, brown-hot flames or flickering gray ones.

4. MAKING PATTERNS PROMINENT IN AESTHETIC SYSTEMS

Not just an individual color, but a pattern of colors may become prominent and serve as a "key," pointing out important features of a narrative progression. One may blend the basic dimensions of color (hue, lightness, saturation) with the basic harmonies in specific ways, and then divide the result into *new* sorts of opposing color categories, complexes, or relationships in the composition of a shot or across a series of shots. In this way filmmakers may develop fresh sorts of binary oppositions and sets of color divisions. As discussed earlier, warm versus cool hues are an example of a binary scheme with ready-made cultural connotations.

Le Corbusier and Amadée Ozenfant propose three families of hues for artists who wish to represent volume in an architectural aesthetic.

> One can by hierarchy determine the *major scale*, formed of ochre yellows, reds, earths, white, black, ultramarine blue, and, of course, certain of their derivatives; this scale is a strong, stable scale giving unity and holding the plane of the picture since these colors keep one another in balance. They are thus essentially constructive colors; it is these that all the great periods employed; it is these that whoever wishes to paint in volume should use.
>
> Second scale.—The *dynamic scale*, including citron yellow, the oranges (chrome and cadmium), vermilions, Veronese green, light cobalt blues. An essentially animated, agitated scale, giving the sensation of a perpetual change of plane; these colors do not keep to one plane; sometimes they seem in front of the surface plane, sometimes behind. They are the disturbing elements.
>
> Finally there is the *transitional scale*, the madders, emerald green, all the lakes which have properties of tinting, not of construction.[58]

One might almost sense in the words used to describe these three scales a basic form of Aristotelian narrative being performed through the movement of types of colors: a beginning "stable" state providing "unity" and "balance" (strong anchor; major scale) is disrupted through a series of linked changes involving measured conflicts and obstacles ("disturbing elements"; dynamic scale) only to return in the end to a new stable state that appears as a rhyming closure inflected by time (mere "tinting"; transitional scale). A venerable form of such a three-part narrative structure is the story of a quest: yearnings/departure from home—leading to a journey through trials of adventure/war/marriage—followed by outcomes/returning home where the protagonist has been slightly (through mere "tinting"), though significantly, altered.

Kress and Leeuwen define another set of potential color patterns capable of charting narrative movements.[59] The following is my summary of their approach.

1. Purity versus hybridity. For example, green versus cobalt blue, i.e., pure green versus the color mixture, green + blue; red versus orange, i.e., red versus red + yellow; red versus violet, i.e., red versus red + blue.

   Notice that there is an ambiguity in Kress and Leeuwen's purity/hybridity scheme concerning what counts as a mixture or blend of two colors. Green is both a pure and *unitary* color (as opposed to, say, bluish green) and yet green is also the result of mixing blue and yellow pigments, which subtract to produce green. Furthermore, making a bluish green less blue does not bring out something yellow in the green. These two notions of a "blend" are incompatible and perhaps reflect two different ways—two language games—for perceiving and talking about color. White is another notable color that is both pure and an additive blend of the spectral hues.

   The pure or unitary colors in Kress and Leeuwen's scheme are red, yellow, green, blue, white, and black. In our perception or mind they simply appear to stand alone without any trace of another (standard) hue. Should brown be classified pure as well? *Some* browns?! Brown, of course, represents a third type of mixture or blend: a darkening of yellow, orange, red, or green by a brilliant surround. A fourth type of mixing is simultaneous and successive contrast and the effect of a shadow cast on a hue. Should something be said about afterimages, too?

   Despite complications, purity versus hybridity does seem to capture an intuitive sense we have about colors. Six of Aristotle's seven basic hues are "pure" (violet is the exception; see subsection "Spectral Hues"). Aristotle offers the following rationale:

   > It is easier to perceive each object of sense when in its simple form than when an ingredient in a mixture; easier, for example, to perceive wine when neat than when blended, and so also honey, and a colour, or to discern a note by itself alone, than in a chord; the reason being that component elements tend to efface one another. Such is the effect of all ingredients of which, when compounded, some one thing is formed.[60]

2. Modulation versus flatness. For example, a rich texture of tints and shades versus comic strip color.
3. Differentiation versus tightly bound or unified. For example, a high contrast and varied palette versus either a low contrast, monochromatic color design or an analogous set of hues (see discussion in the next chapter).

4. Light versus heavy color:

Light	*Heavy*
achromatic colors	high saturation colors
light value colors	dark value colors
cool colors of light value or low saturation	warm colors (of dark value?) or high saturation
low saturation light colors	high saturation dark colors (e.g., soft pastels)
light, inherent value (natural luminance) hues	dark, inherent value (natural luminance) hues

André Cadere employed a more austere color palette than Kress and Leeuwen. In his early artworks, which were designed as criticism from within the movement of Conceptual and post-Minimal art, Cadere specified that his *barres de bois ronds* ("round bars of wood") be executed in combinations of only eight precise colors:

> There are a limited number of colors that can be used: only those which give the greatest contrast: black, white, and the six colors of the rainbow: yellow orange red purple blue and green. However, these eight colors only constitute a reality in words. If we ask for green paint at a hardware shop the assistant will show us ten tins all containing different shades of green. The shade chosen must, in the context in which it is used, be different or distinct from the other colors: In this case, the most *obvious* green, the most *obvious* red, etc. . . . Subtle nuances must be avoided because they automatically make a work harmonic, aesthetic and subjective and so allow for an avoidance of discussion.[61]

THEORETICAL EXCURSUS (II): IS COLOR ALWAYS VISUAL? (WITTGENSTEIN)

Writers and artists have devised various groupings of colors in order to create potent differences that will be seen to create "boundaries." The boundaries can then be employed to articulate meaning and integrate aspects of narrative. Some important groupings are described in this and the next chapter. No doubt additional combinations await discovery and implementation. What is important to recognize is that colors may be divided into categories and then the categories may be used to create prominent patterns. When recruited to narrative concerns, the patterns may work to articulate the significance of character, action, theme, and cultural overlay. How do these fluid colors work? Must they always be visual to be seen? Let us start with the last question.

One of the ways a color may be narrativized is to make it directly visible as a symbol of something. Consider the title of a curious but interesting book, *If It's Purple, Someone's Gonna Die*.[62] Symbols depend on prior acquaintance and long-term memory, e.g., within a semiotic, "standing theory." Surprisingly, in some cases there may be no requirement that color be *directly visualized* when it comes to possess significance.

Consider Wittgenstein's comment:

> If someone described the colour of a wall to me by saying: "It was a somewhat reddish yellow," I could understand him in such a way that I could choose approximately the right colour from among a number of samples. But if someone described the colour in *this* way: "It was a somewhat bluish yellow," I could not show him such a sample.—Here we usually say that in the one case we can imagine the colour, and in the other we can't—but this way of speaking is misleading, for there is no need whatsoever to think of an image that appears before the inner eye.[63]

Wittgenstein in this passage is not thinking about the case where blue and yellow are *mixed*. If these were pigments, then green would be the subtractive result. Unequal amounts of the pigments would result in light green (only a little blue) or dark green (abundant blue). If one were mixing blue and yellow lights additively, as in television, then white would be the result. But what about mixing *mental* colors? I believe Wittgenstein is discussing the implications of the nonexistence of certain colors of which there are a number (see subsection "Non-spectral Hues"). So, although there is a red-tinged yellow[64] and a yellow-tinged red, there is no blue-tinged yellow or yellow-tinged blue. Might "impossible" colors nevertheless be visualized? Is visualization in all cases dependent on recalling an objective phenomenon? Does it follow that if there is no bluish yellow in the world that such a color cannot make sense, cannot exist in thought? To state it differently, does visualization always depend on remembering things seen in the world? Perhaps there are multiple color logics generating different ways of *seeing colors*, even apparently "impossible" ones.

Nonetheless, Wittgenstein's comment is a bit ambiguous. He may mean that there is no "inner eye" or inner theater where an image of reddish yellow can be conjured in the mind. He may be arguing that there is no special psychological and imagistic process that must accompany a color sensation to point out its proper identity by mentally re-screening it. The idea here is that instead of private sorts of feelings or inner efforts to match a sensation to a mental album of images there may exist only relevant behavioral actions utilizing, say, paint samples, color circles, and learned routines.

However, there is another way to interpret Wittgenstein's words: he may be alluding to a stronger claim that goes beyond the situation of the impossible-to-visualize "bluish yellow," namely, that there is no need of a remembered "image" of color at all! This is true, I believe, at least in some cases of language games where color is being

manipulated in a top-down manner, i.e., being driven by goals, plans, schemata, metaphor, narrative, and imaginative memory. Possibility and impossibility may make less sense in this situation or at least shade gradually in and out of each other. The test here does not involve physical possibility, but rather measuring the degree of usefulness of color terms when fit to a specific problem; for example, measuring the usefulness of color when interpreting a fictional narrative. In this book, I am pursuing an expansive approach to color and its meaning rather than relying exclusively upon the objectivity and physical presence of color as an immutable standard.

Wittgenstein uses the word "somewhat" to describe the two colors—somewhat reddish yellow and somewhat bluish yellow. Should we ascribe the impossibility of bluish yellow to its quality of being only "somewhat" bluish yellow, i.e., an indefinite sort of bluish yellow, something not quite fixed or defined, something movable, perhaps something that might edge impossibility nearer to possibility? Wittgenstein's previous example concerned a reddish yellow that was only somewhat this hue, which he states a person could perfectly well understand. Even though reddish yellow and bluish yellow are of different orders with respect to behavior—i.e., being able in the reddish yellow case to select "approximately" the right colour from among a number of samples, though not be able to do so in the case of bluish yellow—the word "somewhat" that qualifies the two colors does seem to suggest an *affinity* between them. The task with color samples constitutes one sort of rule-governed behavior, but not necessarily the only possible criterion.

Wittgenstein is suggesting, I believe, that the line between what is visible in the world and what is visible in mind is not sharp in whichever direction it runs—world to mind, mind to world. His use of the word "somewhat," I believe, functions as a bridge between the two colors, the two situations, and points to the idea that mental imagining is more supple and less constrained by sensation (in both cases of reddish yellow and bluish yellow) than first appears, because imagining may involve more than behavioral logic or visibility; specifically, it may involve our ability to construct a new grammar under pressure of a will to explore and know more fully or differently a colored world. To put it another way, the grammar that Wittgenstein is suggesting by using the word "somewhat" for both cases is a grammar that is working to annul the seemingly strict division between what is possible to see on a painted wall and what is possible to see (approximately) through the idea of a bluish yellow wall.[65]

To better situate the arguments of the present book, I will further elaborate the preceding two interpretations of Wittgenstein's claim, namely that (1) there is no inner eye and (2) there is no need for a remembered "image" of color. To begin: if reddish yellow is "the colour of a wall," then how are we to think about the wall when its color is instead described as "a somewhat bluish yellow"? Is a wall with no color—an impossible color—still a wall? Don't we continue to know the nature

of a wall and the *action* it performs as a sort of 'obstacle' or as a structural part of a building? If so, then, what are we to think of the possibility of its bluish yellow color? In this anomalous version of the wall, its color is apparently more blank than a blank wall since a mental image of the wall based on having previously seen the reddish yellow hue fails to form in the case of bluish yellow, i.e., the bluish yellow hue appears emptier than blank for nothing comes to mind to form the surface of the wall, like the description of a 'round square' or like the indefinite, empty space, say, on a page that surrounds a color chart. Bluish yellow is not, of course, analogous to a blank square *in* a color chart, since the mere presence of such a square would give it a location and "hue." Blankness outside a color chart is different from blankness inside a color chart. Still, the two blanks do *look* the same. Perhaps we should say that nothing comes to mind at all—not even the solid wall or a dissolved wall or an idea of a wall or a generalized wall. When we choose to think of color in terms of a chart, circle, rainbow, or other selective logic of ordering, then that mental image becomes an inner template based on paint, dye, or other physical manifestations. It would seem that when thinking materially, i.e., in material terms, the impossibility of bluish yellow simply cancels everything; or else, looks *like* other mixed hues.

What Wittgenstein is saying, I believe, is that a color-chart type of logic is only one way of talking about, and making use of, color experience. Furthermore, it is not the only way to talk about paint or dye. When we see reddish yellow on a wall or hear a wall described as reddish yellow, must we summon to mind an extensive internal color chart of which the wall color is an externalization? Do we scan through the internal chart page by page (or perhaps nowadays mentally scroll down it) to find the proper reddish yellow to match our sense of the color of the wall or our sense of someone's description of the color of the wall? Does one initially need to bring to mind a large chart containing reddish yellow, but missing bluish yellow, and then move among the many mental samples to find "approximately the right colour" designated by the description? Wittgenstein answers in the negative for two reasons.

First, our image of the imaging of the mental color chart is nonsense if it requires that the chart be projected onto a "screen" from somewhere else in mind to/for the eye of an inner viewer who, in turn, sees a chart in *its* inner mind being projected from elsewhere in its mind to still another inner, inner viewer's eye, and so on without end, without an image ever appearing in final form.

Second, for Wittgenstein, a description of color is locked into a way—a specific grammar—of *interacting* with a world. Acting in a world is what underlies the "number of [paint] samples" mentioned by Wittgenstein that have been created for someone to understand the words "reddish yellow." Here, a form of language—collecting

and creating paint samples to use in a game of matching—has a tight connection to a way of seeing and doing things. Nevertheless, there are many such forms and ways of acting because stimuli from the world (including color stimuli) do not lead irresistibly to a single response or to a single mental image or necessarily to any mental image at all, even in the case of something visible like a color; or, perhaps the words "reddish yellow" and "bluish yellow" lead to multiple, yet distinct, appropriate images. What the world is is not a priori certain; we must be content with a constructed and connected world within which to (usefully) speak and act. There is no inner eye and need be no inner color chart. The framework—i.e., *the descriptions under which* reddish yellow and bluish yellow are named and achieve mental 'visibility'—are thus not limited to the perceptually visible, but depend on accepted ways of acting that have found favor and are justified with descriptions emerging from stereotypes, folk theories, artworks, projections, projects, desires, manipulations, and values. The mind is a greater network than we often imagine as it struggles with a world of possibilities that exceeds our imagination.

Let us think about bluish yellow from the other end. Is reddish yellow always certain and distinct? Do we always see 'red' when it is mentioned or the 'yellow' in reddish yellow? Or do we simply know it is there to be summoned if necessary and put into a proper context, for example, if asked to pick out a paint sample for a room? Might we sometimes simply 'know it' without seeing it, even when looking at it or imagining it? Is 'redness' restricted to a sensation of it? Do we not know it or its significance without sensation? That is, if we have come to know its significance, do we need always to see it in front of our eyes or inner eye or inner chart? If we do not always need to see it to know it, then what of other related hues, even impossible ones?

Supposedly, a beholder who is thinking bottom-up senses the stimulus qualities of a hue first, and then may be directed to a symbolic meaning or connotation as if the hue were pushing upward through layers of possible meanings until landing on a meaning that seems relevant, makes sense—makes something *of* sense, completes the sensation. Seeing purple means someone's likely "gonna die," for example. Seeing red means anger or passionate love. Black clothing will signify a commanding presence or bad person or someone in mourning, and so forth. But can all color effects in a film be explained in such a bottom-up manner where behavior is an environmental reflex? Does a spot of color always exist in a one-to-one relationship with a symbolic "x," while several spots become merely a sum, an "X"? Aren't there other possibilities from the top down when thinking of how color makes meaning?

To pose the question another way: must a hue on the screen be mentally copied into cognitive space and then carried forward as a glowing spot into meaningfulness or else be mediated by, or become, a definite Symbol? Might it be possible for

a hue and its context to activate a mini-language network, as if bouncing somewhat randomly through a pinball machine of meanings? An example might be the activation of radial associations in working memory, e.g., within a "tracking theory." In this incarnation, a hue would become dynamic and mobile and not reduced to a pinpoint and not tied to moment-by-moment, bottom-up sensation. Associations spreading out—links becoming active—might themselves drive thinking in ways quite distant from both literal and symbolic color. One imagines that "nuances" and "adjustments" to a stereotypical color might come from subtle top-down evaluations and interpretations of *context*—from a person's intuitions and mood, for example—that need not themselves be "visual" or be reduced to the "visual," yet are about a particular color.

For instance, what of "seeing red" in the sense described earlier by Vincent van Gogh? This expression refers to a diffuse state of psychic disturbance and vexation. Is a color needed at all in understanding the expression? Do we need to visualize a specific "red" or any red, or some of the usual associations of red, when encountering the expression? Do we need to visualize some sort of red *as part of* the meaning of the expression? Would another color serve? Perhaps not, since a cultural nexus that is relatively entrenched surrounds the expression. Other examples would be "white lie" and "feeling blue." What needs to be in mind to use and comprehend these phrases? Why couldn't a text create a nexus of meanings and expectations for "bluish yellow"?

More generally, one might ask which color or colors, if any, come to mind in representing the word "color" itself. Any hue at all? All hues? A special set of hues? When thinking about colors, *relationships* among them come to mind as well; even if the colors are only imagined in a list or chart, they will appear in some order. Though relationships are crucial in defining the existence of patterns, they are *conceptual* and not themselves visible: "I see a red spot, now a blue spot" is not the same as seeing an abstract relationship between red and blue. Examples of such relations are a red spot *to the left of* a blue or seeing red appear *after* blue or seeing red *replace* and apparently destroy the blue. Moreover, relationships are thoroughly mobile. The same warm-cool significance of red-blue may obtain equally between other pairs, e.g., orange and purple. The warm-cool relationship is fluid and invisible, tied to a certain way of categorizing colors, and not tied to only red and blue spots. What relationships might obtain—and become significant for a perceiver—between blue and yellow in the impossible "bluish yellow"?

The point is that thinking top-down about relationships and patterns may not require that a standard hue combination be visualized in every case; conversely, a context may activate one of the many possible relationships that obtain between a pair of hues, producing a sense of the pairing even if the two cannot appear

together in the real world. If this is what Wittgenstein has in mind, his examples are well-chosen: reddish yellow is a possible relationship between two hues while bluish yellow is an impossible one, yet we should attend to the possible ways in which the two pairs of hues may be *similar* when thinking about them; namely, how is reddish yellow sometimes not visualized (and so impossible like bluish yellow) and bluish yellow sometimes "visualized" (along with, say, the impossible luminous gray mentioned in the subsection "Non-spectral Hues") and so possible like reddish yellow? Thinking in a top-down manner may allow us to see color where there is none; for example, seeing a boy as blond in a black-and-white photograph.[66] The question arises, can some types of thought manipulate previously experienced/learned hues and patterns, incorporate previously thought impossible combinations, and connect them to narrative elements, independently of consciously re-visualizing or failing to visualize the hues? In short, can one be too literal-minded when thinking about color and filmic images and miss a range of possibilities and calculations about relations?

Consider that if sound can be made visual,[67] then color may be made sonic within the speech-based phonological loop of short-term memory. Sonic color would entail that short-term memory be theorized not as a simple gateway for color to squeeze through as in a "standing theory," but as a workspace where many elements mingle to produce thought as in a "tracking theory." In some cases, the thought of "red" might exceed the sum of its components. My argument has been that patterns of colors and relationships among colors appear differently—and are differently visualized or not visualized—under a range of vocabularies and ways of talking. This does not mean that the materiality of color or the materiality of a world is ignored or superseded, only that patterns and levels emerge in a world with a measure of autonomy.

Taking these questions more broadly, one may ask how much of a film is actually *visible* on the screen when considering the larger context of top-down meanings, implications, projections, values, abstractions, possibilities, fictions, memories, desires, points of view, and emotions that are brought to bear on our experiences with film. Indeed, what is our experience of film? Is it an irreversible flow and limited to the sensuous? Is the "real film" only those visible and formal stimulations and simulations that appear on the screen while other experiences are merely add-ons or stray opinions? Does the contingency of thought make thinking and remembering less essential to the fact of film? Films may act in many ways to direct, guide, prompt, remind, or merely inspire a viewer to consider the shifting lines among possibility and impossibility, visibility and visuality. One should consider our experience with film not as a steady contraction onto sensation, but as opening outward and inward.

NOTES

1. Ludwig Wittgenstein, *Remarks on Colour*, ed. by G.E.M. Anscombe, trans. by Linda L. McAlister and Margarete Schättle (Berkeley and Los Angeles: University of California Press, 1977), Part I, §§ 72–73 (Wittgenstein's emphasis) and Part III, §§ 213, 19, 234; cf. Part I, §§ 22, 32, 40, 51, 71, and Part III, §§ 4, 156, 180, 232.
2. On the importance of pattern, see Chapter 1.
3. Rudolf Arnheim, "Remarks on Color Film" in *Film Essays and Criticism*, trans. by Brenda Benthien (Madison: University of Wisconsin Press, 1997), p. 20 (orig. 1935); reprinted in *Color, The Film Reader*, ed. by Angela Dalle Vacche and Brian Price (New York: Routledge, 2006), p. 54.
4. The "essence" of a thing is tied to questions one asks about the thing. What is elided in a search for essence is often an implicit question concerning an object's interactions with its environment and possible contexts. Compare these questions: (1) what will *hit* me—the mass of a banana or its color? (2) is the banana *ripe*? The two questions lead to different salient qualities of a banana. Is there a banana for all times and purposes?
5. Roland Barthes, *Camera Lucida: Reflections on Photography*, trans. by Richard Howard (New York: Hill and Wang, 1981), p. 81 (Barthes' emphases). Paul Coates espouses a similar negative view of color though elsewhere Barthes has offered more positive and intriguing comments on colorism; see Chapter 4, section 3. See also Chapter 8, note 38.
6. Gerard Fort Buckle, *The Mind and the Film: A Treatise on the Psychological Factors in the Film* (London: George Routledge & Sons, 1926), p. 87 (Buckle's emphases). Rudolf Arnheim's early objections to color are along similar lines. See *Film*, trans. by L.M. Sieveking and Ian F.D. Morrow (London: Faber & Faber, 1933), pp. 22–24, 76–83. Arnheim retranslated and abridged this book in *Film as Art* (Berkeley: University of California Press, 1957), pp. 14–16, 65–73. For more on Arnheim's changing ideas about color, see *Arnheim for Film and Media Studies*, ed. by Scott Higgins (New York: Routledge, 2011), especially the essays by Higgins and Jinhee Choi.
7. Charles Blanc, *The Grammar of Painting and Engraving*, trans. by Kate Newell Doggett (Chicago: S.C. Griggs & Co., 3rd ed. 1891 [orig. 1867; 3rd ed. 1876]), pp. 4–5. Blanc's hierarchical "scale of being" derives from Aristotle; see Arthur O. Lovejoy, *The Great Chain of Being: A Study of the History of an Idea* (Cambridge, MA: Harvard University Press, 1936).
8. Blanc, *Grammar of Painting*, pp. 168–169.
9. On the clash between *colore* (color) and *disegno* (drawing, line), see Jacqueline Lichtenstein, "The Clash between Color and Drawing; or, the Tactile Destiny of the Idea" in *The Eloquence of Color: Rhetoric and Painting in the French Classical Age*, trans. by Emily McVarish (Berkeley and Los Angeles: University of California Press, 1993), chap. 6, pp. 138–168; and see p. 51. Lichtenstein's thesis is extended in several directions by Brian Price, "Color, the Formless, and Cinematic Eros" in *Color, The Film Reader*, pp. 76–87. David Batchelor discusses the tradition described by Lichtenstein in which color is viewed negatively and offers an argument by Baudelaire that neatly reverses the traditional view. Baudelaire finds positive value in the supposed "artificiality" and "adornment" of color because these aspects furnish clear evidence of culture, meaning, and the power of human will. David Batchelor, *Chromophobia* (London: Reaktion Books, 2000), pp. 53–55. One could also imagine arguments from a psychoanalytic perspective that would treat "artificiality" in general as a convenient fiction or masquerade symptomatic of—genuinely responsive to—an existential condition.

Mike Leigh's film, *Mr. Turner* (2014), offers depictions of how diffuse color, rather than line, may take the lead in defining shape and depth, and how such color may have been connected to the world of the painter J.M.W. Turner. On Warhol's possible view on *colore* and *disegno*, see Chapter 3, note 28.

10. Melinda Szaloky (personal correspondence).
11. Gilles Deleuze, *Cinema 2: The Time-Image*, trans. by Hugh Tomlinson and Robert Galeta (Minneapolis: University of Minnesota Press, 1989), p. 12 (footnote omitted).
12. Batchelor, *Chromophobia*, pp. 22–23.
13. In Chapter 3, I argued that theories tend to divide according to whether they stress ontology or else epistemology/psychology. This decision affects how the "archive dilemma" is resolved.
14. Wittgenstein, *Remarks on Colour*, Part I, § 68.
15. Charles A. Riley II, Color Codes: Modern Theories of Color in Philosophy, Painting and Architecture, Literature, Music, and Psychology (Hanover, NH: University Press of New England, 1995), pp. 22–23, 34; see also p. 32. On the color theories of Wittgenstein and his successors, see pp. 25–45.
16. See B.R. Tilghman, Wittgenstein, Ethics, and Aesthetics: The View from Eternity (Albany: SUNY Press, 1991); Westphal, Colour: Some Philosophical Problems from Wittgenstein. On right aesthetic descriptions, see G.E. Moore, "Wittgenstein's Lectures in 1930–33" in Ludwig Wittgenstein, Philosophical Occasions, 1912–1951, ed. by James C. Klagge and Alfred Nordmann (Indianapolis: Hackett, 1993), pp. 103–108. According to Moore, Wittgenstein stated that "all Aesthetics is of the nature of 'giving a good simile' " (p. 107). A thought question: Where does a simile come from—life? culture? one's body?

    On "knowing how to go on," see the index entry in Wittgenstein's *Philosophical Investigations*, trans. by G.E.M. Anscombe, P.M.S. Hacker, and Joachim Schulte (Oxford: Wiley-Blackwell, 4th ed. 2009). I discuss and apply Wittgenstein's notion of right aesthetic descriptions in *Projecting a Camera: Language-Games in Film Theory* (New York: Routledge, 2006), pp. 208–224; see esp. note 69, pp. 322–323. For more on "rightness" as a criterion, see Nelson Goodman and Catherine Z. Elgin, *Reconceptions in Philosophy and Other Arts and Sciences* (Indianapolis: Hackett, 1988).

    The reader might enjoy testing the preceding ideas by proposing descriptions for the narrativized colors, events, and themes (values, ideals) of the following (similar?!) films: *The Descent* (Marshall, 2005); *The Element of Crime* (von Trier, 1984); *Damnation* (Tarr, 1988); *Post Tenebras Lux* (Reygadas, 2012); and *A.I. Artificial Intelligence* (Spielberg, 2001).
17. Cf. Wittgenstein, *Remarks on Colour*, Part III, §§ 35, 158, 160, 241–255, 278, 309–318.
18. Cf. Wittgenstein, *Remarks on Colour*, Part III, § 33.
19. See, e.g., the Wittgenstein epigraphs to this chapter; and cf. Wittgenstein, *Remarks on Colour*, Part I, § 5 and Part III, § 160 with Part I, §§ 22, 40 and Part III, § 303. Is there a "physics"—an objectivity—of patterns? No. See Chapter 8 below, section 2.
20. On the reality of fictions, even false fictions, cf. Descartes's view in note 1 of Chapter 2 above.
21. Johann Wolfgang von Goethe, *Theory of Colours*, trans. by Charles Lock Eastlake (Cambridge, MA: MIT Press, 1970), para. 725, p. 287.
22. On color as a physical and cultural commodity, see, e.g., Michael Taussig, *What Color Is the Sacred?* (Chicago: University of Chicago Press, 2009); Victoria Finlay, *Color: A Natural History of the Palette* (New York: Random House, rev. 2004); Finlay, *The Brilliant History of Color in Art* (Los Angeles: J.

Paul Getty Museum, 2014); Finlay, "The Meaning Behind the Many Colors of India's Holi Festival," *Smithsonian Journeys Travel Quarterly* (February 26, 2014). For stunning representations of color's materiality, see, e.g., *Daughters of the Dust* (Dash, 1991) and *Girl with a Pearl Earring* (Webber, 2003); and see, Diana Pozo, "Water Color: Radical Color Aesthetics in Julie Dash's 'Daughters of the Dust'," *New Review of Film and Television Studies* 11, 4 (December 2013), pp. 424–437.

23. Westphal, *Colour*, pp. 49–50 (Westphal's emphasis).

24. Aristotle, "Sense and Sensibilia," 439b22–440a3, 442a. For a concise overview of the complexities of creating color geometries along with their widely different assumptions, see Edith Anderson Feisner and Ron Reed, "Color Theorists" in *Color Studies* (New York: Fairchild Books, 3rd ed. 2014), chap. 3, pp. 16–31.

25. F.R. Palmer, *Semantics: A New Outline* (London: Cambridge University Press, 1976), p. 75. See esp. Ananda Triulzi, "Ancient Greek Color Vision," plus comments <www.serendip.brynmawr.edu/exchange/node/61> (accessed Feb. 6, 2012).

26. See Westphal, section "Two Types of Colour Concept" in *Colour*, pp. 90–95. Is it true that something cannot be at once all-over red and all-over green? This is an example of the first type of logic mentioned in the text above. See Arthur C. Danto, "Foreword" in *Color for Philosophers: Unweaving the Rainbow*, ed. by C.L. Hardin (Indianapolis: Hackett, expanded ed. 1993), pp. x–xii; see also p. xxix. Danto finds important general lessons in this issue of impossible color for philosophers. More recently, however, the story has resumed in a startling way: Vincent A. Billock and Brian H. Tsou, "Seeing Forbidden Colors," *Scientific American* 302, 2 (February 2010), pp. 72–77; <https://en.wikipedia.org/wiki/Impossible_color> (accessed May 25, 2016). See also section 8 below on the impossibility of bluish yellow. On the shifting line between philosophy and science, see Daniel Dennett, "Philosophy as Naïve Anthropology" in *Neuroscience and Philosophy: Brain, Mind, and Language*, ed. by Maxwell Bennett, Daniel Dennett, Peter Hacker, and John Searle (New York: Columbia University Press, 2007), pp. 73–95.

    What logic governs the sight of fictional colors? Is there no logic, not even narrative logic? It would seem that a fictional hue at least would have some minimal connection to seeing a hue or to the thought of a hue. Novelists have conjured about fifty fictional hues that can't be seen directly by humans; see instead their beguiling *descriptions* at <https://en.wikipedia.org/wiki/List_of_fictional_colors> (accessed Oct. 28, 2015). How do these descriptions manage to make sense? On the need for visibility, see section 8 below.

27. Cf. Westphal, *Colour*, pp. 60, 62–64. Surely "luminous gray" may emerge in the mind's eye; it figures in Christopher Knight's description of Richard Diebenkorn's Ocean Park Series of paintings, quoted above in Chapter 2, section 2. A *luminous* hue is one that shines forth with light, typically from a light source passing through the hue from behind, as with a stained glass window. A luminous hue appears radiant and dazzling as a result of the embedded light, whereas light that is merely reflected to the eyes from an opaque surface renders the surface hue flat by comparison. Watercolor, which is usually formulated to be translucent against a white background, such as paper, creates a faux luminous effect. For wonderful examples of luminous hues, see *Chihuly: Form from Fire* (Daytona Beach: Museum of Arts and Sciences in association with University of Washington Press, 1993). For more on gray and whether it can be luminous, see note 38 in Chapter 2 above.

28. Blue-brown is an important dyad in several kinds of aesthetic systems. It is an interesting fact that as a natural language develops distinctions among perceivable colors, blue and brown appear as

adjacent contrasts. If a language distinguishes a sixth from a seventh color, the sixth will be termed blue and the seventh brown. See section "Universal ['Landmark'] Focal Colors."

29. The warm-cool hue dichotomy maps to the cultural dichotomy warm blood versus cool water in the same way that it is said that "blood is thicker than water," which is an assertion that (red) blood is both warmer than a typical water temperature and warmer/closer than "non-blood" relations/distant persons. The scientific version of "blood is thicker than water" is known as Hamilton's rule of kin selection, which is a modification to the law of natural selection.

30. On the warm-cool hue contrast as a universal of sorts, see the first point in the section "Universal ['Landmark'] Focal Colors." Josef Albers argues that there are three fundamental color dichotomies that divide all visible hues into three different sets of binaries separated by neutrals: warm-cool, wet-dry, and light-dark. *Interaction of Color, 50th Anniversary Edition* (New Haven: Yale University Press, 4th ed. 2013), pp. 59–60, and see p. 66.

    Rudolf Arnheim observes that in color mixtures the effect of warmth or coolness is determined not by the main color, but by the trace color toward which it deviates. Thus a reddish blue looks warm whereas a bluish red feels cool. *Art and Visual Perception: A Psychology of the Creative Eye—The New Version* (Berkeley and Los Angeles: University of California Press, 1974), p. 369. Shortly, I will explore one of the ways that a binary norm, e.g., red-blue, can be transgressed.

31. E.H. Gombrich, *Art and Illusion: A Study in the Psychology of Pictorial Representation* (Princeton, NJ: Princeton University Press, 2nd ed. 1961), pp. 370–373, 381, 437. Is red "ping" and blue "pong"? With respect to "ping-pong," see also my examination of "tick-tock" in "Sound, Epistemology, Film" (revised version) in *Film Theory and Philosophy*, ed. by Richard Allen and Murray Smith (Oxford: Oxford University Press, 1997), pp. 95–125, section "Two Perceptions of Time and Two Types of Perception," pp. 102–106.

32. Gombrich, *Art and Illusion*, p. 371 (my emphasis). The analysis of film, like Gombrich's prescription for painting, is about an analyst finding equivalent ways of expressing within ordinary language the *experience of relationships* uncovered in image and sound. Stephen Heath argues that film analysis is not the search for a single system of a text or its coherence, or even the search for several systems—the sum of different readings—but rather is an effort to map a heterogeneous "*movement of relations*" across many cinematic and non-cinematic discourses mobilized by a film. "Film and System: Terms of Analysis, Part II," *Screen* 16, 2 (Summer 1975), p. 113 (Heath's emphases).

33. On the physiological reactions to warm and cool hues, see C.L. Hardin, *Color for Philosophers*, pp. 128–131, 166–167.

34. See Pam Belluck, "Creative Lift? Find Blue Room: Better Accuracy? Go with Red," *New York Times* (February 6, 2009).

35. It seems more natural to say "black and white," rather than "white and black," i.e., to move from a beginning in darkness and nothing—when thinking in this manner—into light and presence, as in "let there [now] be light [out of darkness and nothingness]" (Genesis 1:3). As discussed in Chapter 4, forms of whiteness frequently become associated with light, leaving "black" as the opposite. But doesn't "black" have equally many senses? And don't we see it as a color when the lights are on?

36. The exact color progression of "The Masque of the Red Death" is used as an homage to Poe's story by Dan Simmons in his book, *The Terror: A Novel* (New York: Back Bay, 2007), pp. 311–316.

37. There is a vast literature on embodied and situated cognition. See <http://en.wikipedia.org/wiki/Embodied_cognition> and <http://en.wikipedia.org/wiki/Situated_cognition> (both accessed May 29, 2014). See also Chapter 7, note 24.
38. Michel Foucault, *The Order of Things: An Archaeology of the Human Sciences* (New York: Vintage Books, 1970), p. 23 (Foucault's emphasis); see p. 25. The four forms of resemblance are as follows: entangled adjacencies, distant reflective emulation, relationship similitude, and the movement of qualities through sympathy into an assimilation perfectly counterbalanced by antipathy. A modern theory that refashions the notion of "resemblance" and demonstrates its power is Gilles Fauconnier and Mark Turner's *The Way We Think: Conceptual Blending and the Mind's Hidden Complexities* (New York: Basic Books, 2002).
39. Perhaps this is an opportune moment to note that the "-ish" notation is quite common in describing composite hues. The order in which the hues are given is important. In the present instance, a "greenish yellow" is not the same as a "yellowish green." Other common modifiers are as follows: light, dark, pale, vivid, bright, intense, strong, deep, weak, dim, and very. Perhaps the addition of a hyphen, as in "yellowish-green," serves to emphasize a deeper melding of the two hues. For a cornucopia of color descriptions, see David Grambs, "Light and Colors" in *The Describer's Dictionary: A Treasury of Terms and Literary Quotations* (New York: W.W. Norton, 1993), pp. 144–163.
40. See generally, the indispensible book by Margaret Livingstone, *Vision and Art: The Biology of Seeing* (New York: Abrams, rev. exp. ed. 2014), pp. 32–37. Many fascinating effects are possible by manipulating luminance values. See, e.g., Feisner and Reed, "The Dimension of Value" in *Color Studies*, chap. 7, pp. 74–87.
41. As discussed in Chapter 3, working memory is the source of a spectator's tracking ability.
42. I discuss and illustrate eleven types of judgment heuristics normally employed during film viewing, but employed against a spectator in *The Sixth Sense* (Shyamalan, 1999). See "Butterfly Effects upon a Spectator" in *Hollywood Puzzle Films*, ed. by Warren Buckland (New York: Routledge, 2014), Appendix, pp. 256–264, and note 16, p. 253, on the color red.
43. A recent discovery of a 100,000-year-old paint container in the South African Blombos Cave—older by 40,000 years than previous finds—revealed a red ochre pigment. There was also some evidence of a yellow paint mixture.
44. On universal focal colors, see Hardin, *Color for Philosophers*, pp. 155–186. For a skeptical reaction to the universality of focal colors, see Josh Berson, "Color Primitive," *Cabinet* 52 (Winter 2013/2014), pp. 41–49.
45. Adrian Cornwell-Clyne, *Colour Cinematography* (London: Chapman & Hall, 3rd ed. 1951), chap. 15, "Colour Harmony," p. 655.
46. Cornwell-Clyne, *Colour Cinematography*, pp. 644–645. Cornwell-Clyne slightly modifies his first rule, which stipulates that as saturation increases, the area covered should decrease, by saying that "this rule applies more rigidly in the case of the warm hues, purple, red, orange, yellow, than to green, blue and violet, which can occupy large areas without giving rise to unbalance" (p. 654). Is this adjustment to the rule because blue is cool and receding, and green and "violet" are neutral?
47. The fear of color was discussed in the first section above.
48. Cornwell-Clyne, *Colour Cinematography*, p. 652.
49. Feisner and Reed, *Color Studies*, p. 95.

50. Feisner and Reed, *Color Studies*, p. 95.

51. Feisner and Reed, *Color Studies*, p. 104; on hue proportion, see pp. 19–20, 95, 98–99, 102–104, 116–119.

52. On the schemata of "balance" and "verticality," see Mark Johnson, "'The Corporeal Roots of Symbolic Meaning,' section 'Image Schemas and Cross-Modal Perception'" in *The Meaning of the Body: Aesthetics of Human Understanding* (Chicago: University of Chicago Press, 2007), chap. 7, pp. 136–145. It would seem that the harmonies of the next chapter can be described using schemata; for example, link and source-path/passage-goal schemata for low contrast harmonies; balance schema for high contrast harmonies; center-periphery for key color and accent color.

53. Mary Beth Haralovich, "'All That Heaven Allows': Color, Narrative Space, and Melodrama" in *Close Viewings: An Anthology of New Film Criticism*, ed. by Peter Lehman (Tallahassee: Florida State University Press, 1990), pp. 64, 67, 70. The essay is reprinted in an abridged form and without its half-page, spectacular color images in *Color, The Film Reader*, pp. 150, 151. The color term "visual magnet" comes from William R. Holm, et. al., *Elements of Color in Professional Motion Pictures* (New York: Society of Motion Picture and Television Engineers, 1957), p. 41.

54. Russell Merritt, "Crying in Color: How Hollywood Coped When Technicolor Died," *Journal of the National Film and Sound Archive of Australia* 3, 2/3 (2008), pp. 4–5. The term color "pinprick" comes from Ron Haver, *A Star Is Born* (New York: Alfred Knopf, 1988), p. 155. A multiplicity of disturbing pinprick colors embedded in subtle discordant color schemes may be found in Gus Van Sant's *Psycho* (1998); for example, a scheme of lightish blue (a primary hue), lightish green (a secondary), and off-orange (a tertiary). A fine example of a non-realistic and unsettling use of red pinpricks is the June 23, 2014, cover of the *New Yorker*, available to view at <www.newyorker.com/magazine/2014/06/23> (accessed Sept. 26, 2014). By contrast, on the cover of the *New Yorker* for March 20, 2017, pinprick reds become visual magnets. On color discord, see Chapter 6, subsection on disharmonies.

55. For example, one might be able to adapt Roland Barthes's two terms for analyzing a photograph to the task of analyzing tension in a color display. He describes the *studium* as the specific background knowledge, i.e., knowledge presumably from long-term semantic and episodic memories, which affords a cultural commitment to, or rouses interest in, the subject matter of a photograph. The *punctum* is a sensitive point in a photograph acting as a pinprick or wound that disturbs, or is poignant for, the viewer, i.e., presumably an effect of working memory. Thus a color studium would draw upon cultural and emotional values while a color punctum would direct one's attention to a compelling matter. *Camera Lucida: Reflections on Photography*, trans. by Richard Howard (New York: Hill and Wang, 1981), pp. 25–27.

    Joel Meyerowitz's astute comments on why he prefers six color photographs over six black-and-white photographs of the same subject matter perhaps illustrates Barthes's approach. See "A Question of Color—Answered," available through open access at <http://lens.blogs.nytimes.com/2012/11/12/a-question-of-color-%E2%80%94-answered/> (accessed July 3, 2016).

    One might go further by adapting a version of "suture theory" to explain how punctum wounds may be closed or hidden, if not healed; the wounds, though, would doubtless be marked by "scar" tissue sufficient for critics to find and interpret. See my section "Caesura and Suture," *Projecting a Camera*, pp. 133–145. For more on suture, see Chapter 3 above, note 3, and Chapter 8 below, note 29.

56. On Žižek's notion of a "blot" or "stain," see, e.g., Branigan, *Projecting a Camera*, pp. 42–52. On the colored version of a "blot" or "stain," see Richard Allen, "Hitchcock's Color Designs" in *Color, The Film Reader*, pp. 138–139, and in chapter 6, "Color Design" in Allen's book, *Hitchcock's Romantic Irony* (New York: Columbia University Press, 2007), pp. 140, 205–208, 237, 239, 246, 249, 255. The photography of Marie Cosindas often combines dark, low contrast harmonies with strong visual magnets of red and/or white that seem suddenly to lay bare and magnify a facet of the subject matter.
57. Le Corbusier and [Amadée] Ozenfant, "Purism" in *Modern Artists on Art: Ten Unabridged Essays*, ed. by Robert L. Herbert (Englewood Cliffs, NJ: Prentice-Hall, 1964), p. 71. Ideas about color organization appear throughout Herbert's book.
58. Le Corbusier and Ozenfant, "Purism," p. 70 (orig. emphases). The ochres mentioned by Le Corbusier and Ozenfant that constitute their "major scale" are created through unequal proportions of all the primary hues. The result is known technically as a "broken hue" and the ones found in nature are known as "earth colors" or ochres. For example, nine parts of yellow, five parts cyan, and one part magenta creates the broken hue, pea green. Such hues have a range of special attributes. See Feisner and Reed, *Color Studies*, pp. 69–70. The lake colors of Le Corbusier and Ozenfant's "transitional scale" are created by combining an organic dye with an inorganic fixative to extend a hue's range and brightness. By contrast, the "dynamic scale" utilizes inorganic pigments based on metals.
59. Gunther Kress and Theo Van Leeuwen, "Colour as a Semiotic Mode: Notes for a Grammar of Colour," *Visual Communication* 1, 3 (2002), pp. 343–368. For another semiotic approach, see Gianfranco Bettetini, "Colour" in *The Language and Technique of the Film* (The Hague: Mouton, 1973), pp. 120–125.
60. Aristotle, "Sense and Sensibilia," 447a17–22. Feisner and Reed define twenty-four pure hues; *Color Studies*, p. 67.
61. André Cadere quoted in Matt Jolly, "The Barred Colors of André Cadere," *October* 144 (Spring 2013), p. 125 (orig. emphases).
62. Patti Bellantoni, *If It's Purple, Someone's Gonna Die: The Power of Color in Visual Storytelling* (New York: Focal Press, 2005).
63. Wittgenstein, Remarks on Colour, Part III, § 27 (Wittgenstein's emphasis).
64. When you think, for example, of a red-tinged yellow, do you think first of the red, maybe of a strong red, then "add" bits of yellow to a mental image, or do you start with a sheet of yellow and then create a reddish yellow? Do you next combine the red and yellow to see something orangish, but then realize you've gone too far and so "back up"?
65. On the impossibility of reddish green, see Chapter 6 below, subsection on "disharmonies."
66. Wittgenstein imagines seeing a boy as blond in a black-and-white photograph in *Remarks on Colour*, Part III, §§ 269–276. There are similar instances when viewing a film.

    Although William Wyler's original film, *Jezebel* (1938), is entirely in black-and-white before being colorized in recent years, there is an important scene involving the color of a dress—which is said to be brilliant red. The spectator, I believe, sees the red in the black-and-white image, and, moreover, is shocked that the Bette Davis character, Jezebel, has decided to wear the dress. Should we call the red a ghostly or nonexistent color? Contrast this example with the girl in a red dress shown actually as red within a black-and-white sequence in *Schindler's List* (Steven

Spielberg, 1993). Similarly, a girl in a red cloak appears within a black-and-white sequence in *The Great Train Robbery* (Edwin S. Porter, 1903). An intermediate case is presented by *Time Stands Still* (Péter Gothár, 1982). Within its monochromatic color schemes, a hue will seem to flicker or oscillate as it emerges from the murk only to sink back in. Ghostly real color? Still visible in black and white?

Alain Resnais in his documentary about Nazi death camps, *Night and Fog* (1955), sets up an alternation of black-and-white with color that corresponds to the contrast between past and present. The result is, arguably, a kind of monochromatic sense of color: the black-and-white documentary sequences of the past seem to bleed the color from the present time sequences because, afterwards, the scenes that one remembers are the scenic horrors in black-and-white.

Consider also the following comment by David Alan Black:

> The publicity for Fassbinder's *Veronika Voss* [1982] included a poster, printed in subdued pastel colors, showing the title character standing next to a piano. The film was in black-and-white. However—thanks, I now believe, to the unconscious memory of this poster—I literally saw the scene corresponding to the poster in color. I remember musing, during this sequence, about how interesting it was that Fassbinder had chosen to film only one color scene. I mentioned this later to a friend who had also seen the film, and who set me straight: the entire film, excepting none of its scenes, was in black-and-white.
>
> Much of my taste for textual relativism dates to that experience. . . .

David Alan Black, "'Homo Confabulans': A Study in Film, Narrative, and Compensation," *Literature and Psychology* 47, 3 (2001), p. 37 n.15 (Black's ellipsis points).

67. On visualized sound, see Edward Branigan, "Soundtrack in Mind," *Projections: The Journal for Movies and Mind* 4, 1 (Summer 2010), pp. 41–67. Vladimir Nabokov in his autobiography explains that he is a synesthete who sees colors when hearing letters and phonemes as well as when seeing their shapes. He provides extensive examples. *Speak, Memory: An Autobiography Revisited* (New York: Vintage, rev. ed. 1989), pp. 34–35.


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Chapter 6 **Musical Hues: Color Harmonies**

ABSTRACT

The present chapter surveys ideas about the concept of harmony and outlines some common types of harmonies and disharmonies. These design strategies together with many of the norms discussed in the previous chapter are illustrated in two photographs and eight compositions drawn from films. What will be demonstrated is that color is colorized under the influence of narrativizations of time, space, and action as well as the effects on color of technologies of reproduction. Especially important are the discourses of color criticism (discussed in conjunction with *Vertigo*) in which one can detect grammatical norms at work that are selecting color properties and eliding others to justify evaluative judgments so as to make us see this or that color *as*.

I spied the back and edges of a folio

Aglow with all the colors of the rainbow,

Its hand-painted title stating a decree:

The interrelationships of hues and sound:

Proof that for every color may be found

In music a proper corresponding key.

Choirs of colors sparkled before my eyes

And now I was beginning to surmise:

Here was the library of Paradise.

To all the questions that had driven me

All answers now could be given me.

Here I could quench my thirst to understand,

For here all knowledge stood at my command.[1]

—Hermann Hesse

CHOIRS OF COLOR

Certain types of color relations have traditionally been thought to achieve distinction because they promote a natural visual sense of rhythmic harmony or disharmony through manipulation of the *intervals* among colors. The intervals are typically defined geometrically, as on a color circle. The resultant material patterns are said to create an inner resonance or dissonance with a viewer. Some of these types of "melodies" and "discord" will be described and illustrated along with several interpretations that seek to explain the general nature of harmony and disharmony. One need not assume that the sum of harmonies and disharmonies exhausts all types of color combinations. It is likely that there are additional types of relationships and patterns affecting a viewer.

1. LOW CONTRAST HARMONIES ("COORDINATION")

Low contrast harmonies are adjacent, analogous, or densely packed hues on a color circle that are *continuous* or coordinated in some way. A linked series of closely related hues across a composition is said to facilitate a "passage" or flow along a line connecting the hues, which individually may be said to bleed, blend, or bend into one another. Such a harmony supposedly provides a route of least resistance for the movement of a perceiver's eye. This is what is meant by saying, for example, that a color design in a dress shirt or tie "picks up" the color of a suit. A low contrast harmony creates fluid lines that act to texturize an object and suggest tactile qualities. In effect, color is creating "line" (*disegno*). The opposite of passage may be called "blockage."

Although both eyes, of course, actually move, critics routinely speak of a beholder's eye in the singular. One reason is that critics are interested not in the eye or eyes, but in an artwork's exploitation of the movements of *attention* and *awareness*. What is at stake is not the recognition of a color pattern by eyes moving in tandem, but in how a perceiver is drawn to explore a series of immediate color relationships that connect to one another and to linked abstractions. Critics thus tend to favor overt patterns in the foreground at the expense of implicit, dispersed, ambiguous, or subtle color designs that emerge piecemeal over time and through connectives that must be inferred (for such designs, see Chapters 7 and 8).

Six low contrast harmonies are as follows:

1. **Analogous** color schemes are based on hues *adjacent* to one another on a given color circle. For example, this series: blue—greenish blue—blue green—green blue—bluish green—green—green yellow—yellow green—yellow. Some gradations may be omitted and the scheme will still be seen as analogous and flowing.
2. **Slightly warm** designs tend to be preferred over slightly cool hues.

3. **Monochromatic** or near monochromatic color schemes are based on any single hue accompanied by measured and simple, gradations in its lightness and/or saturation as well as the possible appearance of *neutrals*—wispy white, ivory, pearl, vanilla, silver, cool matte or shiny black, charcoal, gray, beige, ecru, eggshell, cream, caramel, tan, blond, various hazels (browns with greens and grays), and multiple sorts of browns. Giorgio Morandi created extraordinary paintings working throughout his career mostly in neutrals. Monochrome may escape a similarity to monotone by exploring the subtle thresholds of aesthetic delicacy (Chapter 2).

   How should one use the term "neutral"? Here I refer to colors in a range of white, black, gray, and brown. However, sometimes the term refers to hues that are neither warm nor cool, namely, some greens and magenta. It also sometimes refers to matters of mixing hues (neutralizing). What is the opposite of neutral, or the set of opposites of neutral? Should one define a set of positive and negative colors to accompany neutrals? For example, are rainbow hues the opposite of neutrals? Or, perhaps an opposite lies with pure hues that exclude white, black, and all hybrid hues. There are many possibilities depending on the starting point for a given color grammar.

   Two simple ways of creating a monochromatic color scheme—prevalent in silent film aesthetics—are through the processes of *toning*, whereby the exposed (blackened) silver in a positive emulsion is dyed in a single hue (leaving the lighter areas unchanged), and *tinting*, whereby either a positive black-and-white print is dyed in a single hue (leaving the darker areas unchanged) or else the film's images are printed onto stock made from colored celluloid.
4. **Common tint** occurs where a series of different hues appear as if they were all being viewed through a slightly colored, transparent glass or filter, i.e., each of the hues remains distinct.
5. **Common saturation** makes hues appear as if they all contained the same degree of grayness.
6. **Common luminance** makes hues appear as if they were all of the same degree of lightness or darkness. Blue and brown are both relatively dark and so many blue-brown dyads will be harmonious, though the hues remain distinct and do not mix, i.e., the blue and brown hues stand as a stark contrast to one another since there is no bluish brown or brownish blue in the way that red, orange, yellow, and green may mix with brown. When the brown of a blue-brown dyad has been derived from an underlying yellow (which produces a darkened yellow or chestnut brown) and, further, when it covers a larger area than the blue, the overall composition will appear slightly warm and so harmonious (as in 2 above). This is a fascinating pattern for Hollywood films

in that it combines in a single dyad both continuity and discontinuity, low contrast and high contrast. This pattern also mimics a common type of color blindness due to the absence of red retinal photoreceptors.

One way to remember this important blue-brown dyad is to keep in mind the four basic elements of Empedocles—fire, air, water, and earth, in that order.[2] The mediating elements, air and water, are used to separate, dilute, and give proportion to the extreme of fire (fiery heavens, rays of bright light, glowing *blue* sky), which makes the world visible, and the extreme of earth (dirt *brown*), which makes the world tangible (as theorized in the cosmology of Plato in *Timaeus* 31b–32a). Thus blue and brown mark the sky above and the ground below.

2. HIGH CONTRAST HARMONIES ("BALANCE")

High contrast harmonies are hues selected from a color circle or other geometry that are dispersed and discontinuous (dissimilar, "schematic"), but nevertheless appear *balanced* and equally opposed to one another in some fashion and so may act to simplify a composition by dividing it into two parts.

1. **Complementary** colors are a pair of hues—a dyad—that are directly opposite one another on a color circle and produce an achromatic gray when mixed in equal amounts or mixed in one's imagination. A hue that is matched with the two hues that are adjacent to its complementary forms a triadic scheme called a "split complementary." A four-hue scheme that matches each of a pair of adjacent hues (i.e., analogous, low contrast hues; see 1 above) with their complementaries is called a "double complementary" scheme.
2. **Warm versus cool** dyads were discussed in Chapter 5.
3. **Harmonious triadic** schemes are three hues that are related by simple and unambiguous intervals on a color circle. The intervals are orderly, often identical, and often also maximal. For example, the painter's primaries (red, yellow, blue) form a triad equidistant from one another on a typical color circle as do the secondaries (green, purple, orange), which result from combining the two painter's primaries that are not the complement of the given secondary; for instance, green is the complement of red and is formed through a subtractive combination of the remaining two painter's primaries, yellow and blue. More complex are the six tertiary colors that involve certain mixtures of primary and secondary hues. There are also quaternary and quinary hues. A rare but fascinating Technicolor triad is a variation on the secondaries: green, violet, gold.

   Other triadic harmonies may be selected by rotating an equilateral triangle inside an appropriately constructed color circle marked with equal

divisions—either mathematically or psychologically equal divisions. More complex two- and three-dimensional representations of an abstract color space may be employed in conjunction with the rotation of simple geometric shapes to determine apparently harmonious intervals; for instance, the points of a square or else a rectangle produce a tetrad. The assumption motivating these schemes is that spectators will unconsciously sense the regularity of mathematical or musical-like intervals among a set of colors selected in an orderly fashion from a color circle or other geometric shape.

4. **Saturated versus pale** colors are hues that are vivid and strong in a design placed against grayish, de-saturated hues.
5. **Light versus dark** colors are hues that have been lightened in a design to place them against hues that have been darkened. As a reference, here are some (very) approximate overall *contrast* ratios or dynamic ranges together with a measure of resolutions in pixels:

eye	25,000,000 to 1 (24 f-stops; contrast ratio/dynamic range)
eye nighttime	1,000,000 to 1
eye daytime	10,000 to 1
eye	500+ Mpixels
CMOS image sensor	11,000 to 1 (13.5 f-stops)
film projector	4,000 to 1
LCD/Professional CRT	1,000 to 1
Eastmancolor 35mm	128 to 1
digital camera	64 to 1
8K	33.2 Mpixels
6K cinema camera	19 Mpixels (16.5 f-stops)
35mm film	12+ Mpixels
UHD TV (4K)	8.3 Mpixels
digital camera	8 Mpixels
HD TV (2K)	2.07 Mpixels
photographic paper	120 to 1
paper reflectance	100 to 1
digital camera	64 to 1
color slide film	32 to 1
video	32 to 1
Plus X b&w film	43 shades of gray
NTSC/PAL video	13 shades of gray
NTSC	350,000 pixels

3. SOME POSSIBLE DISHARMONIES ("CONFLICT")

Disharmonies are combinations in which colors somehow fail either to connect with one another or to balance one another. The hues work to disconcert the viewer in some way or, at least, to offer a form of counterpoint or messy connection that upsets a sense of clear continuity or symmetry. (For other possibilities of conflict, see Chapter 5 above, section "Prominence: How a Color Becomes a 'Key Color' ")

1. **Unnatural luminance** results from hues that violate the natural order of lightness (see previous chapter). For example, murky green and hot pink placed together, or dark green and pale violet, or brilliant yellow and light pink, or dark yellow and turquoise.
2. **Failed complementaries** are hues that just fail to be complementary, e.g., a failed split complementary scheme.
3. **Intermediate colors** are hues that are neither adjacent (analogous) nor equidistant, e.g., neither complementary nor triadic and balanced. An intermediate color scheme offers neither passage among the hues nor sturdy contrast.
4. **Red-green alone or else blue-yellow alone**. These pairs of hues are harsh opposites that do not combine or mix with each other, i.e., there are no reddish greens or greenish reds, and no bluish yellows or yellowish blues.[3] No continuous transition or passage is possible between these pairs of hues.

   This category of disharmony would include certain variations on the red-green and blue-yellow pairs. For example, there are no reddish limes, yellowish violets, or bluish oranges; thus blue and orange areas stand apart from one another as do yellow and violet (reddish blue), and green and violet (bluish red).

   Also, importantly, no bluish browns exist. Thus a blue-brown dyad that strongly emphasizes the distinctness of each hue and their inability to combine might create a feeling of disharmony. For example, a large expanse of baby blue opposed to a murky olive-brown (cf. above, "unnatural luminance") or a medium blue paired with a yellowish brown (blue versus yellow).
5. **Naked primaries** are one or two primary hues that appear alone with no other hues to function as accents or additional hues to provide a transition between, or background for, the primaries.
6. **Conflicting secondary properties of hues**. For example, Kress and Leeuwen's notion of purity set against hybridity, e.g., a scarlet opposed to red-mixtures of orange or violet (Chapter 5, section 7, subsection 4). Another possibility for conflict among secondary properties might lie in mixtures among opaque, semi-transparent, and transparent hues; or, in violating Cornwell-Clyne's strictures on saturation (Chapter 2, section 2; Chapter 5, section 7, subsection 1).

7. **Brown-black and brown-gray dyads**. Traditionally it has been said that brown does not go together with black or gray, and maybe not with white. Brown does perhaps work with gray as a low contrast harmony, especially if brown-gray were a monochrome and the brown had a reddish tinge (sepia). As it is, I think of brown-black as "bad sepia."

   One might speculate about reasons for this disharmony. Brown is an indeterminate hue mixture—a hybrid of imprecise hues and proportions. Recall that brown is produced by surrounding orange light by anything lighter to make a very darkish, indefinite orange-yellow or muddy olive green. In one sense brown seems to be a mixture of hues while in Kress and Leeuwen's sense it may also seem "pure." This sort of brown is being paired with black or gray, each of which has its own indeterminate status, i.e., sometimes appearing as a hue while other times appearing as simply a dimension or quality of lightness within other hues. That is, there is something unstable about the nature of both brown and black-gray. The instability differs for each of the colors in the dyad, perhaps making them even more strongly incompatible. Furthermore, if black or gray as qualities of lightness were seen to combine somehow with brown to either blacken it entirely or else to lighten it in one's imagination, brown would vanish! To imagine brown as becoming black or else being lighter and so becoming orange-ish might create a tension within brown itself, making it seem both present and, if somehow darkened or lightened, also absent.

   At any rate, brown and black(-gray-white) intersect in strange ways that seem to tangle a clear logic of color; or, at least, to complicate our idea that there can only be a single logic of color. What is at stake here is not physics, but varieties of experience. And our experiences appear not on a blank slate, but result from embodiment in a world and in one or more cultures. We experience colors with our bodies and expectations, not just with innocent or blank eyes.

   An example of the use of a brown-black dyad is the evil uncle Scar, who is shown with a brown body and black mane in the final battle scene with Simba in Disney's *The Lion King* (1994). Note that Scar's colors are different earlier in the film. Similarly, the villainous Doberman Pinscher, Alpha, in Pixar's *Up* (2009) is brown and black, and is edged continuously in steel blue! The blue provides a sense of power because it is unnatural—dogs do not have blue hair—and, moreover, it is the only hue that will *not* mix with brown, producing a strong contrast (see 4 immediately above). There is also the villainous "Seduce and Destroy" showman, Frank T. J. Mackey (Tom Cruise) in Paul Thomas Anderson's *Magnolia* (1999), who first appears in brown and black onstage bathed in blue light, though by the end his clothes and lighting will have changed in time for a measured redemption. Another example is the

Louise Bourgeois painting, *Regrettable Incident in the Louvre Palace* (1947), which employs both of the dyads, brown-black and brown-gray along with several shades of blue accent, both as a pure blue and as smeared into gray.

8. **Unnatural transparency** is an arrangement of four hues that suggests a transparency effect, but a transparency that is impossible to achieve with light in the natural world. Four distinct hues are used in order to represent a background, background object, foreground object, and that portion of the background object that is supposedly being seen through part of the foreground object. Painters can easily create "impossible" transparency effects since they are free to choose the colors; filmmakers would need to employ special techniques. In general, filmmakers have not exploited transparency effects, except occasionally in animation.

   To elaborate: in *natural* transparency the hue that represents what is seen through the foreground object must be a subtractive color that lies in-between the hues of the background object and the foreground object; also the luminance of what is seen through the foreground object must not be darker nor its saturation higher than either the background object or foreground object. Thus there are strict limits on natural transparency. Subtle, disturbing effects might be possible by undermining the logic that determines the qualities of the hue, lightness, and saturation of natural transparency. Note that there is no transparent black or transparent white; clear milk does not exist.

   A transparency effect in painting occurs when a color is imagined as having its hue shifted because it lies behind, and is being seen through, a foreground color. Rudolf Arnheim asserts that there is a structural analogy between transparency and audition. He argues that sensing distinct hues by separating them into different spatial planes is analogous to hearing distinct melodic lines within polyphonic music. In all cases, what arrives physically at the eye and ear is a single, complex vibration that usually yields opaque color and harmonic music; only in special cases does the brain divide the stimulus into discrete components that result in perceiving superimposed areas of color and multiple lines of music.[4]

9. **Weak blends** is a term that I have invented to talk about certain combinations of blue-green and yellow-green. The first type of weak blend is a turquoise, which is a light bluish-green or light greenish-blue.[5] The second type of weak blend is a chartreuse, a brilliant yellow-green. It seems that there is a conspicuous cross-cultural dislike of these hues along with greenish-brown and a few other hues.[6] Variations of chartreuse, for example, are prominent in the coloring of the hands of the Wicked Witch of the West in *The Wizard of Oz* (Figure 6.8); a polluted pool in *Red Desert* (Antonioni, 1964); the strange garage-lab of

Primer (Carruth, 2004); the topless bar in the opening of *To Live and Die in L.A.* (Friedkin, 1985); the hospital corridor after the death of Shen Tao's father in *Mountains May Depart* (Zhangke Jia, 2015); the cannibal story of *Delicatessen* (Jeunet, 1991), which also utilizes brown-white dyads; the re-animating fluid in *Re-Animator* (Gordon, 1985) used to bring back the dead to a deadly kind of life; and the play of yellow and green on skin near the end of *Total Recall* (Wiseman, 2012) when "memory fluid" is about to be injected.

Based on psychological research, Paul Coates describes a three-part "arc of pleasure" that extends from pleasurable, blended blackened hues to neutral reds to less pleasurable, blended whitened hues. The former category includes green, green-blue, blue, blue-purple, purple, and purple-red; the latter category includes red-yellow, yellow, and yellow-green.[7] Presumably, an artwork may remodel predispositions about pleasure and unpleasure. There is evidence that the ancient Greeks favored a four-part palette: black, red, yellow (pale), and white. Blue was thought to be a form of black. If this is difficult to imagine, look at gray clouds in and against a blue sky. In early coloring of film, blue was used to denote night.

The four hues of the ancient Greeks are the hues of the Four Horsemen of the Apocalypse from the *Book of Revelation*. Yellow, however, was thought to be a simple form of white. If this is difficult to imagine, look at a bright yellow shirt in dim light. The Koine Greek word for "pale" that describes the color of the horse of the Fourth Horseman—Death—may also be translated as ashen, pallid, pale green, or yellowish-green, i.e., as a sickly or deathly pallor to a person or corpse. Such a four-color scheme would then reduce to three fundamental hues: black, red, and white, which are the focal hues of Coates's "arc of pleasure" and the first three landmark hues.[8]

4. DOES THE EYE SEARCH UNTIL IT *CREATES* HARMONY? (SOME EXAMPLES)

The low contrast harmony of an analogous color scheme may amount to a *mental principle* guiding what we claim to find in a pattern of colors, and not just a stray objective fact about an arrangement of colored surfaces. The same idea may apply to other sorts of color harmonies and disharmonies that we claim to see within a framework of what we take to be the basic hues. A person may adopt as landmarks those hues specified as fundamental by his or her culture; for example, the three colors of the "arc of pleasure," the four-color palette of the ancient Greeks, Newton's seven focal hues of the rainbow, or the eleven focal hues of the English language. The hues chosen would provide prominent features around which to orient other hues and culturally prominent relationships.

Consider the following comment by I.A. Richards about the search for color harmony:

> The fact that roses, sunsets, and so forth are so often found to present harmonious combinations of colour may appear a little puzzling. . . . But the vast range of close gradations, which a rose petal, for example, presents, supplies the explanation. Out of all these the eye picks that gradation which best accords with the other colours chosen. There is usually some set of colours in some harmonious relation to one another to be selected out of the multitudinous gradations which natural objects in most lightings present; and there are evident reasons why the eye of a sensitive person should, when it can, pick out those gradations which best accord. The great range of different possible selections is, however, of importance. It explains the fact that we see such different colours for instance when gloomy and when gay, and thus how the actual selection made by an artist may reveal the kind and direction of the impulses which are active in him at the moment of selection.[9]

Richards seems to suggest that a perceiver first encounters or absorbs a "vast range of close gradations" among the colors. This may be an implicit justification for the aesthetic strategy of 'delicacy,' discussed in Chapter 2, which depends on discriminating a range of carefully graded perceptual 'thresholds.' For Richards the many gradations may then be simplified and organized by a perceiver to create a distinctive mental image. The implication, I believe, is that there is a set of mental laws that governs the patterns we find in colors analogous to the gestalt laws governing our organization of lines and figures into shapes as well as a set of mental laws governing our ability to perhaps sense a larger, 'overall shape' to a series of local hues in a composition. Seven gestalt laws which may have analogues in the mental creation of color harmonies are as follows: color scission (transparency), color similarity, absolute contrast, similarity of area, good continuation (recall above the notion of "passage" in low contrast harmonies), proximity, and closure. In this approach, color harmonies would arise from our recognition of a few basic ordering principles of sensory variables.

The basic principle underlying gestalt perception is that perceptual processes tend toward the realization of the most regular, ordered, stable, balanced, and lowest energy state possible in a given situation. A perceiver's discovery that sensory fragments may be seen, or be made, to cohere quickly into the form of a simple 'object' is one such *ordered state*. For example, John McCracken's narrow, colored planks spaced equally apart against a white wall in "Rhythm" (2008) may be seen either as a series of separate, scattered colors or else as one hue that slowly evolves across the planks, or two hues that evolve and converge, to form a single gestalt object. Further, imagine that each plank is a distinct shot of several hues in a film, but that these compositions are evolving toward a larger design.

What seems important for Richards is that the final gestalt state reached by a perceiver will point toward the specific mental state of the artist (!) who initially created the color combinations. This requires that gestalt states are to be connected to basic emotional states and also connected variously to each other to create either a flow or an interruption among patterns at still 'higher' perceptual levels. Presumably for Richards, an artist's or beholder's specific selection of colors out of elaborate possibilities in a given display would reveal (and/or prompt/reinforce?) his or her state of mind, "gloomy" or "gay," he says.

If such color laws underlie harmonies (and their absence underlie disharmonies), then an artist may employ or subvert or extend these laws as the occasion demands, assuming that the artist has otherwise contrived to limit a viewer's choices. The artist must strive to put a viewer into the requisite mood to fully experience the designated combinations of hues in conjunction with the narrativized objects, emotions, and events being represented.

Let us briefly examine two examples of how a viewer may create a tight organization of hues amid a set of interlocking patterns that combine to yield a net impression. Figure 6.1 is a photograph by Melinda Szaloky of painted and actual lilacs. The high contrast harmony of the painter's primaries (red, blue, and yellow) provides a framework for identifying a series of subtle gradations among the reds, violets, and purples of the flowers, which hues and lines are reworked in the vase, as well as exhibiting a second progression among the yellows, golds, and browns of the background and picture frame. These two sequences that splinter the painter's primaries amount to low contrast harmonies of analogous color schemes. A common luminance ties together parts of the flowers. A blue-brown dyad with browns derived from yellows adds a slight degree of appealing warmth and a harmonious frame for what otherwise might be a too cool and oppositional set of blues. Some shadows are black, but many more reflect a range of hues creating the sense of a dynamic, multi-faceted volume heightened by a light foreground shifting gradually toward a dark background in the bottom half of the photograph that is balanced through reversal (in the manner of a high contrast harmony) in the top half of the photograph where a light background is shifting toward a dark foreground. These synchronized patterns and the carefully measured spaces that help define the patterns result in a vast harmony of parts that brings together in a single impression the compact idea expressed by the title of the photograph: *Art Imitating Life Imitating Art*.

Figure 6.2 is a photograph by Melinda Szaloky of palm fronds. The framework for the display of hues is a harsh series of red-green contrasts. The disharmony of red-green, however, is overcome by the aesthetic effects of a multitude of thin bands of hues producing an experience of delicacy, though a delicacy of intense, rather than subdued, hues (Chapter 2). Areas of splashed, soft cool blues accent the red-greens

and act to reduce the hotness of the many reds by making them more compatible with a profusion of cool yellow-greens and neutral-temperature greens. The overlapping palm leaves create a sense of depth that is negated and balanced by the lack of a horizon line and the generally common illuminance (and luminosity!). The overall effect is to reinforce a simple flatness that increases the prominence of hue qualities. The feeling of a multitude of precise, distinct hue oppositions is preserved while heightening the many delicate transitions between hues, which include an indistinct wash of hues in the leaf stems along with a number of leaky, wavy stripes of leaf color. The tense contrasts emphasize the clarity of lines and borders while the array of crowded and numerous hue transitions along edges make for ever-emerging new patterns of parts upon patterns of parts—a bewildering number of possible arrangements. The countless parts suggested by the many areas of distinct hues seem to make up a textured surface. The result is a state of large-scale unity in local diversity that promotes a tactile experience for a viewer, as if he or she might gently run a hand across these leaves to admire nature's variety. The many contrasts are thus displaced and unified through a sense of touch. Although the contour of each leaf is built by gestalt good continuation, the wholeness of a leaf moves in and out of a viewer's focus due to the anxious movements of its competing hues. Thus neither *disegno* nor *colore* predominates; each seems to arise from the other.

5. WHAT IS A *RULE* ABOUT COLOR HARMONY REALLY: IS IT A CAUSAL RELATION OF MIND OR WORLD (I.E., A SUBJECTIVE OR OBJECTIVE FACT), A HEURISTIC, A CONVENIENT DESCRIPTION, AN ARBITRARY INCLINATION, OR SIMPLY A MYTH?

Is color harmony a biological or emotional fact stirred by culture? Adrian Cornwell-Clyne:

> Colour harmony is, when it becomes significant, a product of creative imagination; and as such it is not subject to mathematical analysis. The colour harmonies of a great painter seem to possess the power of inducing responses from chords, so to speak, which lie deep within our natures, and which are but rarely stirred. How does this occur? Presumably deep-lying cultural association response-chains are set in motion. By what process of education does one learn to respond emotionally to the colouring of sunsets, and in the colouring of paintings to the light that never was on land or sea? It is all very mysterious![10]

Cornwell-Clyne pursues a number of analogies between color and music. In this passage he says that the human nervous system is musical, too: color harmonies

"possess the power of inducing responses from chords. . . which lie deep within our natures." By way of contrast, consider Ludwig Wittgenstein's comment:

> If there were a theory of colour harmony, perhaps it would begin by dividing the colours into different groups and forbidding certain mixtures or combinations and allowing others; and, as in harmony, its rules would be given no justification.[11]

If 'harmony' is merely a convenient, objective description (analytical, rhetorical, cultural) that exists in the eye of the beholder, or in the eye (or heuristic) of a filmmaker who wishes to achieve a 'consistent' look for a film, then a spectator will still require the ability to distinguish between color that has a purpose and color that is irrelevant.

Perhaps there is a middle position between Wittgenstein's hypothesized objectivity rooted in imposed conventions that are arbitrary and must be learned in order to facilitate communication and profit, and Richards's and Cornwell-Clyne's subjective and inter-subjective approach of authorial/artistic visions and response ("chords. . . which lie deep within our natures" together with "deep-lying cultural. . . response-chains"). Might we say, for example, that the high contrast harmony between red and blue is 'partially motivated'? That is, the opposition between warm and cool may be a useful way of discerning or creating *patterns*—a convenient description of (some of the) present objective facts—yet, at the same time, may be *partially* motivated—implemented—from our embodiment and subjective experiences in the particular world and social environment that we frequent and that we find replicated and examined in texts from our world.[12] In this approach, color is not determined mechanically by objective causal law nor by cultural consensus, nor does it exist in a vacuum of arbitrary possibility (rules). Rather, perceiving color is marked by an ability to shift among several contexts, permitting an artist to rely on both bottom-up and top-down mental processes in a viewer—on both the physicality of our bodies in an environment and on a person's expectations and background knowledge—when crafting a novel form of "light that never was on land or sea," even though light, land, and sea are quite real.

Rules that govern color combinations would seem to exist in a partial vacuum where there do exist objective starting points that can anchor an evaluation, yet a rule cannot always predict a color's significance for there are no necessary end points—only evolving and provisional end points prescribed/proscribed at a historical juncture by a disposition to see and speak—a desire to use established descriptions to promote our concerns. The process is not dissimilar perhaps to the evolving nature of metaphor in language.

COLOR HARMONIES IN FILM

I would like to conclude by illustrating some of the standard color categories and relationships discussed in this and the previous chapter. I will analyze in detail eight images from five films. The aim is to present a few basic norms of color design in classical Hollywood cinema and to demonstrate typical interactions with narrative structure.

1. *GENTLEMEN PREFER BLONDES*; AND, PASSAGE THROUGH TIME

Dorothy Shaw (Jane Russell) on the right is talking with close friend Lorelei Lee (Marilyn Monroe) aboard a cruise ship about a subject on which they have strongly opposing views: the pursuit of men (Figure 6.3). In this early scene from *Gentlemen prefer Blondes* Dorothy is shown to be an aggressive, forthright woman who makes herself the equal of men while Lorelei is a wily, golden-haired gold digger. The contrast can almost be seen in their poses—standing and confident versus seated and casual. Lorelei is examining the ship's passenger list to identify rich men and to plot her moves.

The hue design is triadic: yellow, red, violet. The red is quite prominent—though occupying only two tiny areas of the shot—because it is located in a notable place (the lips of two glamorous women) and is highly saturated (consistent with Cornwell-Clyne's first rule for color areas in film). Dorothy's yellow and Lorelei's violet are highly contrasting, near complementaries, which set warm against cool, purity against hybridity (Kress and Leeuwen), and, to the extent that Lorelei's violet contains "blue," the violet refuses to mix with Dorothy's yellow, i.e., there are no colors that are bluish yellow. At this point in the story, the most salient aspect of the women's relationship is that, though friends, they are antagonists and rivals.

We are drawn to Dorothy because she is dressed in a brilliant yellow coupled with a deep black outfit and black hair. This yellow-black dyad represents the strongest possible contrast in *lightness* between a hue and black, i.e., the two stripes of yellow offered by the scarf versus Dorothy's jacket, dress, and hair. By contrast, Lorelei is surrounded by a much larger area (consistent with Cornwell-Clyne's second rule for color areas) that contains two sequences of analogous hues (low contrast harmonies) offering two easy *passages* for the eye. (Recall I.A. Richards' notion that a person's attention is drawn toward a series of "close gradations" among colors.) Specifically, the eye can move from Lorelei's blond hair and matching complexion to her gold earrings to the gold-brown chairs, and even to the harmonizing yellow of the pencil she holds. (If the pencil were red, orange, or emerald green, for example, it would become a distracting magnet or pinprick and perhaps require separate justification as a thematic countercurrent.) It is as if we are being encouraged to

focus sharply on Dorothy while the point of view on Lorelei wanders and is more diffuse, or more positively, is expansive and generous. What is important is that the point of view on the two women as expressed in the color design divides into two different points of view.

Lorelei's violet outfit anchors a second passage for the eye. A spectator's attention is drawn along a series of close gradations of color since Lorelei's outfit seems to be echoed by a colored lighting effect on the background wall behind and to screen left of her. The lighting creates an analogous bluish or blue-violet glow or aura that is designed to mimic the look of light bouncing or reflecting off Lorelei's violet blouse onto the gray wall. The gray wall further de-saturates the light violet into a faint (traditionally feminine) pastel that *hints* through a common lightness—a type of color harmony—with Dorothy's yellow (together with the blue accent supplied by Dorothy's earrings) that the two friends will, much later in the story, find common ground. The light violet is conspicuous for a viewer and is able to suggest this connection between the women because it reverses the natural scale of luminance for a violet hue. In addition, Lorelei's "golden" glow is amplified by Dorothy's yellow scarf. The film narration here is working on two levels: most immediately, it is presenting a conflict/competition between the two women, but at the same time, it suggests in a subtle manner that the two women share an important bond that will later lead toward reconciliation.

An inspection of the background confirms what may be coming later in the story. The background, first of all, acts as a neutral in order to highlight the dominant, contrasting play of hues, and points of view, of the two women. Notice the rim light on Dorothy to separate her head from the dark wall. In character-driven classical Hollywood narrative, important characters must not fade into the woodwork. Notice also how the yellow scarf "crosses" over and into the light gold of the door hinges and into the brown door. This color movement ameliorates what otherwise might have been a more violent clash between the discordant hues of the brown door and Dorothy's black clothes and hair, though it would not amount to a clash of lightness values.

In addition, a strategic use of neutral grays in the background is designed to highlight a carefully modulated series of low contrast gradations connecting the two women. Dorothy's black-and-white checkered jacket is "projected" into a series of larger rectangular background shapes that contrast with the human shapes of the women, although an abstracted, gray oval window in the background echoes a head-shape, and so provides a transition between human and environment. Immediately behind Dorothy the neutral-gray wall is darkened at the right but lightened at the left (yielding a bright, light-gray) in order to frame Dorothy and to match her black-and-white outfit. The left edge of the bright, light-gray rectangle divides the shot into two equal areas, placing a woman in the center of each area.

Moving further toward screen left, there are shadow areas (yielding a middle gray in the areas around the darker windows) that rhyme with Dorothy's clothes and "hover" over Lorelei. The muted background colors that are opposed to Dorothy's brilliant yellow scarf follow Cornwell-Clyne's third rule for color areas. Finally, to the far left we encounter a large area of "bluish" or "blue-violet" glow from Lorelei's outfit that is superimposed on the wall, mingling with gray. This series of steps through several shades of gray and blue-gray creates a circuitous route from Dorothy's black-and-white outfit through and into the background behind Dorothy, then moving from right to left across the background, and finally bending back into the foreground from the gray chair cushions and bluish-gray wall to the light bluish-violet of Lorelei's outfit. This amounts to a use of Kress and Leeuwen's color "hybridity" to create continuity, passage, and a link between the women.

Thus the narration of the film employs subtle color overtones together with a series of color connections to join the two women. The link between the women will become progressively more evident as the narrative advances. If Lorelei's violet were a darker purple, then one could say that during the film she dresses in the three secondary colors, since later she will appear in outfits that are orange and then green (with a light violet sash to echo her earlier outfit). Still later, Lorelei will come to wear black outfits (foreshadowed by the "sensible" and "forthright" black-and-white clothes of Dorothy) and, finally, at the end of the film both women will wear white in a double wedding (where innocence and purity are reclaimed and rewarded). The color design of this early shot in the film states the explicit tension between the two women, but quietly predicts its resolution: they will reconnect as close friends in the moment when each finds her proper man in a proper manner.

The image from *Gentlemen prefer Blondes* illustrates one of the ways in which color relationships may enter Arnheim's "field of polyphony" (Chapter 5): an image full of color is fully narrativized both in the present of the plot and across time into the already-determined, but as-yet-unseen, future of the story.

2. *PICNIC*; AND, SPACE

Whereas the shot from *Gentlemen prefer Blondes* uses high contrast harmonies to depict a present conflict between Dorothy and Lorelei juxtaposed with low contrast harmonies to foreshadow their future reconciliation, a shot from *Picnic* emphasizes the power of low contrast harmonies to shape a space for character action, i.e., to both reveal character and define a space in which to realize character interaction (Figure 6.4). The low contrast harmonies in *Picnic* work as follows.[13] A few lightened hues have been set against a wide range of subtle gradations of *neutrals* with careful shadows and highlights, including the lighting of a minor object—the steering wheel (recall the discussion of *color-as-delicacy* in Chapter 2). On the right half of the image

strong blacks are found in several places: the molding around the side window; in Alan Benson's (Cliff Robertson's) dark glasses, shirt, and hair; and, in Mrs. Helen Potts's (Verna Felton's) dress and black-streaked hair. A series of variegated whites and grays spreads out from Alan's light gray suit, which is uninterrupted by fabric texture or pattern and is a perfect extension of the car's light gray seat cushions. On the left half of the image are streaked whites and grays plus Marjorie "Madge" Owens's (Kim Novak's) light pink dress and hair ribbon coupled with her mother Flo Owens's (Betty Field's) orange dress spotted with light blue ovals.

There is a slight discord between the pink and orange since these hues are "intermediate"—neither quite adjacent (analogous) nor far enough apart to act either as a form of continuity ("passage") or as a balanced contrast. The two hues thus state an ongoing tension between daughter and mother (specifically, concerning men and marriage) while their auburn red hair, lipsticks, eyebrows, and complexions are a match. If the orange is considered light enough to be more like a salmon pink, then daughter and mother would have been shown to be still "closer" to one another. Overall, the flesh tones of the four characters are rendered in the distinctive and attractive "Technicolor tan" (Chapter 2), bringing hardy "browns" into the mix of other neutrals. Madge's diaphanous pink dress adds the ideal feminine tint to the Technicolor tan of her body.

The mother's dress of orange and blue (two colors that do not mix, i.e., there are no bluish-oranges) makes her the center of attention and, in fact, it is she who directs the conversation in the car as she angrily interrogates Alan about his unemployed ne'er-do-well college buddy, Hal Carter (William Holden), who means well, but may have a hankering for Alan's girlfriend, and her daughter, Madge. Madge is not involved in the conversation and is mostly distracted and looking off left (lost in her thoughts about. . . Hal?). Though not speaking up for herself, the conversation is obliquely about Madge and her choices in life, and so she must not be forgotten by the spectator in this space. She and Alan are both given special side lighting for their heads. The large distance between her and Alan in the front seat reveals a growing distance between the two lovers (accentuated by the passengers in the back seat, where close inspection would seem to indicate that the mother is virtually sitting on the lap of Mrs. Potts). Madge's pink dress and hair ribbon is joined to Alan by only the very faintest of a pink flush to his gray tie. He is a straight-ahead sort of guy who can't be bothered looking at his tie, inspecting his surroundings for symptoms or nuances, or weighing other people's words for warning signs.

There is no editing or camera movement in the scene. Instead, color and composition interact to clarify and solidify the spatial articulation of the scene. A number of spatial conventions in the image are reinforced by the color design. The space is divided into thirds according to the rule of one-thirds: one-third from the right side of the image is Alan's right shoulder; one-third from the left side is Madge's

left shoulder. If one imagines two vertical lines extending from the edges of these shoulders, one has the beginnings of the sides of a rectangle. To form the top and bottom sides of the rectangle, one notices that approximately one-third from the top of the image is the horizontal line formed by the top of the back seat while one-third from the bottom of the image is the horizontal line of the top of the front seat. Within the "inner rectangle" formed by these two verticals and two horizontals is what is known as "dead" center—a place in the composition that is usually avoided in portraying action. The mother's head is located on top of the top horizontal, which is approximately the horizon line (in accordance with the rule of horizon line isocephaly), thus making her the dominant person in the composition as, in fact, she is in terms of color, dialogue, and action.

Spatial perspective in the shot is also conventional. If one were to draw imaginary diagonal lines through the two persons on the right and the two persons on the left, the lines would recede and converge at a point where the road vanishes in the far distance, creating a central, linear perspective. The daughter's and mother's pink and orange are warm, advancing hues that stretch out the space, which is especially helpful since the background is actually flat and rear-projected. The stable, linear perspective of the shot is further reinforced by the following color principles that relate to spatial articulation:

1. *Hues* should shift, becoming bluer in the distance (atmospheric perspective).
2. The *lightness* of hues should shift so that there is a progression from a light foreground or middle ground toward a darker background; or else the reverse, from a dark foreground to a lighter background. What is important is that a progressive change in lightness is matched to a perceived change in distance.
3. The *saturation* of hues should shift, becoming grayer in the distance.
4. The *contrast* among hues should shift, becoming less in the distance.
5. The *focus* on hues should shift, becoming less sharp in the distance.

"Classical shot space" may thus be summarized roughly as follows:

> The center of narrative interest is center-framed in central perspective in a single, usually middle plane of action with only one action occurring at a time (while other actions are temporarily "frozen"). The center of narrative interest is highlighted by a scheme of motivated three-point lighting along with such conventions as the rule of one-thirds, horizon line isocephaly, and appropriate color shifts of hue, lightness, saturation, contrast, and focus.

When this approach is applied to the space of indoor scenes, a horizon line must be imputed to the locale. All movements by the camera, characters, objects, colors,

and editing in classical narrative must be carefully monitored and, when necessary, the color combinations must be readjusted to maintain balance and to strengthen the sense of volumetric space following any momentary imbalances signaled by conflicts or revelations in the plot.

It is evident that great care has been taken in constructing this image from *Picnic* in order to integrate the choice of colors with space, character placement, shapes, and narrative action. There is nothing simple about classical Hollywood cinema. Enormous and painstaking work is required by a production team in order to create and then manage a composition that will make a spectator's task of comprehension seem as simple and easy as traveling a road.

3. *VERTIGO*; AND, THE RHETORIC OF COLOR CRITICISM

The first three shots of a scene in *Vertigo* introduce a spectator to a color zone that is composed neither of high nor low contrast harmonies, but rather stands as something intermediate, unbalanced, and perhaps vaguely disquieting (see shots 1–3 in Figures 6.5–6.7). Disharmony roils these images. The elaborate seduction of Scottie (James Stewart) from previous scenes by Madeleine/Judy/Carlotta (Kim Novak) continues in this scene. Scottie has walked from the street toward Madeleine, who lingers by his apartment door. He is surprised to see her at his apartment and has stopped on a lower step looking up at Madeleine, keeping a distance between them, as he cautiously begins to talk with her (shot 1). A possible brown-black color disharmony is stated through the characters' clothes: Scottie wears a deeply chocolate-brown suit and brown hat with his face a brownish Technicolor tan as opposed to the solid black outfit, gloves, and purse of Madeleine together with her gray scarf and prominent dark shadow on the wall. The large, brilliant red door with a slight orange cast stands between them. The startling prominence of the glowing red provides no continuity among the other hues in the shot and violates all three of Cornwell-Clyne's rules for color areas in film.

As an object, the door has no narrative significance. However, the *movement* of its color in the three shots is significant. The two positions on screen of Scottie and Madeleine do not change from shot 1 to shot 2; that is, if the shots were superimposed, we would see each character's body remain in place. The rectangle of red, however, has moved from near Madeleine (shot 1) to strike downwards through Scottie (shot 2). In shot 3, the vertical red reappears in the form of Scottie's tie. Moreover, at the very start of shot 2, Scottie is leaning toward the right with his hand on the metal railing. In the first two seconds of the shot, he removes his hand and straightens up, bringing the red of the door straight down through his body.

He remains unmoving in this position through the six instances of this set-up in the scene.

Scottie's movement in the first two seconds of shot 2 that allows the red to shift and appear to fall through him has been deliberately contrived. The continuity between shots 1 and 2 has been boldly violated in order to emphasize the red striking Scottie: Scottie had already taken his hand from the railing and straightened up at the end of shot 1. In fact, he is standing perfectly upright for about three seconds at the end of shot 1.[14] Nevertheless, the entire action is repeated at the start of shot 2. In this particular case, however, the obvious mismatch in action between shots 1 and 2 will almost certainly not be noticed by a spectator because of the limited resources of working memory, since the spectator is concentrating on dialogue between the characters and attempting to resolve puzzles in the plot. It would be a different matter entirely if Scottie, for example, were holding a gun in his right hand on the railing. Attention would be drawn to his hand, the railing, and his movement. In that case, the mismatch between shots 1 and 2 would no doubt be instantly noticed by a spectator.

Hitchcock must have calculated that it was far more important to make the "red" be seen to be slowly moving over to fall down through Scottie in shot 2 than to observe the strict continuity established by shot 1. These four movements of red that began with the red of the door near Madeleine in shot 1, followed by the red appearing to move nearer to Scottie, initially dropping through his left shoulder in shot 2, then slowly shifting on screen to move through Scottie's head in the first seconds of shot 2 (where it remains during six shots), and lastly, jumping to Scottie's tie in shot 3 are a simple illustration of employing a methodology derived from a "tracking theory" of color (discussed in Chapter 3). Red thus becomes prominent.

What might be said about a red hue in the context of a plot?[15] Hitchcock believed red to have special value:

> Color should start with the nearest equivalent to black and white. This sounds like a most peculiar statement, but color should be no different from the voice which starts muted and finally arrives at a scream. In other words, the muted color is black and white, and the screams are every psychedelic color you can think of, starting, of course, with red.[16]

Hitchcock's example of a person's "scream" will become in his next sentence, multiple "screams," and thus it seems that he believes many (psycho-delic?) colors will be available to match the many impending, climatic shrieks, starting, of course, with a primal red. Consistent with Hitchcock's dictum, Richard Allen locates "four main color systems" or "color patterns that exist across the entire range of Hitchcock's work," including a notable use of red. Allen asserts, further, that "These

patterns of color meanings are unique to Hitchcock's work."[17] The four groupings, each of which is linked to a distinct set of meanings, are as follows:

1. Hot colors: the warning series.

   > The "warning series" consists of bright, saturated, and solid yellow, orange, and red either individually or in sequence or combined in the same image to indicate progressively greater degrees of danger. Often Hitchcock's use of these colors, especially red, is highlighted in the image via a contrast with white and sometimes blue.[18]

2. Black versus white.
3. Cool colors versus warm colors or earth tones.
4. The presence of color versus the absence of color (black, white, gray) or else colorfulness (e.g., bright, highly saturated hues) versus colorlessness (muted hues, e.g., pale beiges).

Surely Allen's red "warning series" can be seen in Figures 6.5–6.7. These set-ups even include the prescribed contrasts of red—appropriately "bright, saturated, and solid"—against white and blue; in fact, several whites and four shades of blue. Allen's second category, black versus white, appears in set-ups 1 and 2. One might argue that the third and fourth categories are evident as well in these three set-ups. Indeed, the suspicion arises that there may be *too many* possibilities with these four groupings leading to too many possible interpretations of the ongoing narrative. Which is to say, if the four color patterns can be seen in too many ways and in too many places, they will no longer reliably predict or connect to narrative events.

For example, speaking of the black versus white grouping, Allen makes the following claim about the Kim Novak character in *Vertigo*:

> Before Judy Barton as Madeleine stages Madeleine's death as a suicide, she wears a black dress that is initially concealed under a white coat, suggesting the manner in which her threat to Scottie is concealed beneath her apparent innocence.

Later in the film Allen again finds a black-white dyad at work:

> When Judy Barton as Madeleine leads Scottie Ferguson to the bell tower in *Vertigo*, she wears a wool speckled black-and-white coat. The mingling of black and white here suggests the blurring of moral boundaries, as in the expression that something is not black and white.[19]

I would like to use this pair of claims about black and white to draw attention to two aspects of Allen's argument about color and its relation to narrative. Allen's method for thinking about color is typical of much color criticism. First, his rhetoric.

He is employing what I call a "stylistic metaphor."[20] He extracts an element of style from the film (here color) and projects it onto something abstract and invisible, onto a theme or general idea. Madeleine's black dress (in set-ups 1 and 2) is interpreted as being "concealed" (?!) under a white coat and then this notion of "concealment" is given a new, metaphorical sense whereby something—a "threat"—is "concealed" beneath Madeleine's innocence. However, we do not normally speak of one piece of clothing being "concealed" by another. We would not say, for example, that Scottie's white shirt and red tie (in set-up 3) are partially "concealed" under his suit jacket. Instead, the use of the word "concealed" betrays advance knowledge of future plot events that are being projected backwards onto the colors. This knowledge comes from already having seen the film, i.e., from long-term memory (as in a "standing," semiotic theory, discussed in Chapter 3). A spectator seeing the film initially would not use the word "concealed" to characterize the colors or clothes of any of the characters. This may even be true of a spectator who has seen the film many times.

I hasten to say that I have used (concealed) stylistic metaphors in my own descriptions of *Vertigo* above when declaring, for example, that red appears "to fall down through Scottie" in shot 2. Although a spectator does know at this point in the film that Scottie has a fear of falling due to vertigo, the spectator does not yet know all the implications of "falling" that will be exploited by the film, literally and figuratively, e.g., falling in love and free-falling in both literal and figurative nightmares. I have also discussed color patterns that imperceptibly foreshadow the end of a story in *Gentlemen prefer Blondes*.

I am drawing attention to the fact that neither of Allen's stylistic metaphors about black versus white—involving a "concealed threat" nor, later in the film, a "blurring of moral boundaries"—can be discovered by a spectator while watching the film for the first time. In particular, notice the subtle, but complex progression from literal to figurative in Allen's three sentences above that are employed to state this pair of claims using stylistic metaphors:

1. Black versus white (literally an absence of light versus the presence of light) → she wears a black dress under a white coat → black is "concealed" under the coat → she conceals her threat to him → her threat is "beneath" (under) innocence → threat is real, innocence is only apparent → thus her innocence is false → white coat covers (conceals) the "truth" → whiteness is deceptive (it is only a surface, blocks the truth) → the black dress is closer to her body than the white coat → black is her heart → black is true, white is false → hence black opposes white (now figuratively as well as literally).
2. Black versus white (literally) → she wears a black-and-white coat → not black *and* white, but "speckled" → her coat is really a "mingling" of black

> and white (though not becoming grayish) → a mingling is really a "blurring" of color boundaries (relative to an area marked on a scale that has been chosen) → a blurring of color boundaries is like a blurring of moral boundaries → thus morally her situation is *not* black-and-white (but in fact grayish?) → morality is judged on a relative scale by balancing factors (black weighed against white; one thing weighed against another) → color, too, is relative not absolute → therefore, *it is not*, after all, black versus white in absolute terms, either figuratively or literally, but instead the idea of black *relative to* the idea of white along a spectrum or else black and white commingled into shades of gray, either of which interpretations yields a negative metaphor that contradicts the starting point! → black versus white (negatively figurative since her situation is neither black nor white).

Both of Allen's narrative metaphors—signaling a "concealed threat" and a "blurring of moral boundaries"—become possible as interpretations only later in the film after a spectator has learned a number of secrets. These stylistic metaphors are *retrospective* projections based upon a spectator's memory of the film. This raises the second aspect of Allen's argument that I wish to highlight. Allen's "four main color systems" derive their utility from—and are resident in—a critic's long-term semantic memory. They do not derive from working memory or a tracking theory of color. Allen's categories function instead as *general signs* (postulated semiotic codes?) within a "standing theory" (Chapter 3). His categories cue a spectator's beliefs based on having already seen *Vertigo* and/or having seen many other Hitchcock films and then extrapolating broad tendencies from the films. These tendencies, in turn, derive still more generally, and vaguely since culture speaks with many voices, from our knowledge of what Allen calls, "deep-rooted cultural associations and iconography."[21]

Indeed, once one is committed to this sort of "standing" theory of color, why not look even further into the future of the plot of *Vertigo*? Why should one believe that the meaning of Madeleine's two coats is so clear? On the basis of the second stylistic metaphor above—the speckled black-and-white coat leading to a "blurring of moral boundaries"—why not say that the whiteness appearing earlier in the film, as interpreted by the first stylistic metaphor—the white coat over the black dress, set-up 2 of Figure 6.6—does function, after all, to *accurately predict* this woman's contrition and redemption from her initial "fall from grace" into villainy—i.e., seducing Scottie to unknowingly participate in a murder plot—when at the end of the film she falls to her death? That is, instead of saying that the whiteness of the coat denotes a false innocence, the whiteness could be said to show instead—to accurately predict—the eventual attainment of a true moral purity, or else to show the inherent moral purity of Judy from the beginning, which is proven true at the

end. The problem with this pair of claims about the meaning of whiteness is that they depend on a *concealed segmentation* of the film that seems arbitrary. Whiteness has the meaning it has, but only within a particular span of the film. How much of the future of the plot ought to be included in the meaning of a particular white? That is, for which period of time does white have one meaning rather than another? These questions arise because the method being used to assign meaning is based on resident semiotic codes retrieved from long-term memory together with an arbitrary selection of some portion of the plot. By contrast, in a tracking theory, one is describing a spectator's *expectations* about white that take the form of a spectrum of possibilities from likely to plausible to potential to improbable to nearly true, and so forth, including perhaps entertaining several possibilities at once adjusted to different perspectives on the plot.

My argument so far has focused on the second stylistic metaphor and has set aside the profound and disorienting ambiguity that surrounds Judy's death at the end, which would lead to further problems about the meaning of color. If, for a moment, you do wish to entertain the ambiguity that surrounds Judy's death (perhaps because you value ambiguity in a text and enjoy thinking of *Vertigo* as a kind of art film), then you will relish questions about the cause of Judy falling to her death and its implications for the color design of the film, including its use of black versus white. Here are some possibilities.

> Did Judy accidentally slip? Was she startled by the nun (who is dressed in black-and-white) or frightened by Scottie on the stairs? Was she frightened *of* Scottie? Did Scottie harbor an unconscious wish for her death that somehow materialized? Was she pushed? Was it suicide? Was it nervousness or guilt? Was it Fate? Justice? Was it a fateful coincidence? Was it the ghost of, or her thought of, Carlotta? Was it the ghostly jealousy of Midge eliminating a rival? Was it perhaps the return of the repressed Madeleine? . . . and so on.

The fact that the circumstances of Judy's death remain uncertain does not mean that a spectator cannot entertain several *alternatives* that, in turn, may be mapped back onto, say, the whiteness of her coats that were worn earlier in the film—the white coat and the speckled white coat emphasized in Allen's analysis—in order to see therein, for example, merely an ideal or else a relativity of moral value (and then, of course, one would need to ask, relative to what?). Even if a specific question posed by the plot were explicitly answered at the end of *Vertigo*, a spectator is not prohibited from entertaining "nearly true" alternatives raised by the plot—perhaps, in fact, encouraged to seriously reflect upon alternatives raised by the plot, both earlier and at the end.

In general, it could be the case that a color scheme early in a film refers to implicit or suppressed alternatives, rather than to an explicit moral proposed at the end of a film. A "semiotic, standing theory" of color is often too rigid about interpretation as well as too programmatic about the operations of the many types of human memory when it assigns a definite significance to color or else color to significance. This is especially true concerning a standing theory's use of long-term semantic and episodic memory.

The point I'm trying to make by mentioning Hitchcock's ambiguous ending to *Vertigo* is that there are numerous events and reversals in any given story, not to mention complications about the meaning of "whiteness" (discussed in Chapter 4). The analyst must decide what criteria are appropriate for selecting a specific point within the narrative that is to be interpreted and, in turn, used to interpret earlier color designs that were initially unrecognized or misrecognized.

Once one engages the future of a plot along with its interpretations and the matrix of available cultural associations, there would seem to be no limit to the number of meanings that may be played out against a specific color pattern in inconsistent ways. This may be a fact of life for texts, but it represents a very different orientation for critics who seek to explain why color appears and why it appears to have a concrete meaning in relation to a plot. A standing theory *assigns* meaning. A tracking theory *follows* meaning as it develops in fits and starts and multiplicities.

Let's briefly recall from Chapter 3 some of the advantages of a tracking theory. In a tracking theory, meaning is being pursued moment by moment while a spectator's responses and expectations are unfolding, that is, during the times when a spectator is evaluating and making inferences amid a flurry of alternatives and uncertainties. Some, or many, of these possibilities and experiences should be retained as important parts of the story, even after a plot has finally answered its enigmas, settled on its truths, and denied all else. Some of the unrealized possibilities in a story that are *nearly true* should not be discarded, misplaced, or forgotten by a spectator in favor of a solidified, final meaning.[22] The power of a text's style, e.g., its use of color, is vested in its connections to multiple ongoing processes of meaning. Accordingly, part of our understanding comes from tracking the progress of a text's many strands and weighing—fearing, celebrating, envisioning—what *might have happened* moment by moment. A standing theory reproaches a spectator with "you should have known the truth earlier." A tracking theory reminds a spectator: "you should not forget what was nearly true."

The different temporal orientations of standing and tracking theories do not rule out a productive use of both methodologies. My present goal is to underline the methodological differences and to suggest that, despite openness to a cultural semiotic, a

standing theory often settles quickly on an absolute meaning for a color design without sufficiently appreciating the contingency and relativity of interpretation.

Thus I am not saying that Allen's color categories are not perceptive and productive. I am only saying that these categories are different from—perhaps complementary to—a tracking approach to color. Allen's four categories would seem to function "after the fact"; that is, they are not aimed at an online, constructive moment-by-moment process of comprehending and judging a flow of color. As Allen concedes, in some cases a color pattern fitting one of the four systems may actually have had its meaning *inverted*—may have been used ironically by Hitchcock—or else the pattern may simply turn out to mean nothing. Alternatively, the color pattern may have been extended into quite *new* territory, as in Hitchcock's *The Birds* (1963), where the commingling of black and white goes beyond *Vertigo*'s "blurring of moral boundaries" to now signify

> the breakdown of category distinctions altogether that breeds an incipient chaos or monstrosity. The commingling of black and white is announced in the credit sequence as black birds that peck away as the credits flutter against a white background.[23]

How and when is one to know whether a color design appears according to some standard format or simply recalls the format, ironically or otherwise, as opposed to functioning in an entirely new way or else bearing no connection to the plot? Which context or segmentation of the film should be chosen to evaluate the meaning of a color or color design? In the case of *The Birds*, pecking birds onscreen are merged with fluttering (?) credits while the blackness of the birds is "commingling" (?!) with the whiteness of a background. . . all in order to suggest "monstrosity." And all of this before the story has begun. Can "monstrosity" be known at this time?

Finally, I wish to mention a few other aspects of these three set-ups from *Vertigo* in order to illustrate disharmonies at work, i.e., combinations of colors that somehow fail either to "connect" with one another (low contrast harmonies) or to create some form of "balance" (high contrast harmonies). Perhaps it would be better to say that low and high contrast harmonies reflect familiar ordering schemes and a measure of simplicity while disharmonies represent more complicated color relations that require additional processing time in working memory in order to discover internal correlations and possible functions in the plot.

Consider again the interaction of space and color in the first three set-ups from *Vertigo*, which account for 14 of the first 17 shots in the scene (Figures 6.5–6.7). The lighting does little to define volume and depth. The scene occurs in bright sunlight, yet only Madeleine casts a striking shadow, creating a "double" of her figure in shots 1 and 2, whereas Scottie casts no shadow in shots 2 and 3, and in shot 1 casts an amorphous, pitiful shadow near the base of the patio entrance, which might almost be one of the

bushes ("pitiful shadow" being a stylistic metaphor). In shot 1 Madeleine's head rises significantly above the artificial horizon line formed by the top metal bar of the railing and the bottom sill of the window while Scottie's head rests neatly atop this line in conformity with standard practice. Moreover, a skewed spatial perspective is created in shot 1 by the metal bar of the railing being inclined downwards to the right coupled with the shadow of the railing slanting down to the left and the dark V-shaped shadow on the wall at the top of the image angled in a third confusing direction.

Nor is there a progression of color changes from foreground to background that would reinforce depth in shot 1: the pink foreground railing is as pastel as the blue background molding around the door and the weirdly bluish gray-green bushes in the right foreground are close to the hue of the darker blue molding around the background windows. One might even say that the foreground bushes, including the greenish gray bushes, appear more desaturated than the colors in the background, which would reverse our expectation about the relation of saturation to distance. All of these strategies work to produce the impression of a space in shot 1 that has been severely *flattened* so as to appear to be cut up by the metal lattice motif of the railing into irregular and differently colored pieces bordered by a sort of dingy pink, like a Mondrian color field gone awry in a nightmare. If not unsettling, this space at least is a cause to pause.

In conclusion, one may ask whether the mélange of four shades of red (door, bricks, Scottie's tie, and the railing and pillar) set against four shades of blue (drapes, bushes, two shades of window and door moldings, plus variations in the lighting of these blues, as at the base of the long window) are enough to state a red-blue spatial dichotomy in the scene. Can these scattered areas be aggregated to form a warm-cool high contrast harmony? One should also consider Madeleine. Her pale silvery-blonde hair would seem to connect to the tiny gold clasp on her purse, the gold doorbell, and the gold mail slot (shot 2). But, at the same time, there seems to be neither passage nor contrast between the colors of her hair and the large areas of off-white on the wall behind her, including the areas of shadows and the pink pillar. Madeleine's color scheme thus seems to be both connected and disconnected.

If there is color coordination in the scene, it might lie in shot 1 in the links among Madeleine's pinkish facial coloring, the off-white wall behind her, the pink metal railing, and pink pillar, thus serving to magnify her spatial presence through the apartment's façade. But is Madeleine thereby being linked to the inanimate? The opening of the scene does possess a strangely inert quality since neither character moves during the first seventeen shots.[24] Only the color is alive—or dead—depending on one's suspicions about the characters. Perhaps in the end a spectator will conclude that each character is equally alive *and* dead, both red and blue, seeking and shunning, seeing and neglecting, each in different ways or in different spheres.

To summarize, it would seem that the color patterns in the first seventeen shots of the scene fall short of establishing either high-contrast or low-contrast harmonies. Alternatively, if one does detect the presence of a harmony, then one will find within or around that pattern significant disorder and uncertainty. I believe the overall effect of the design of this scene remains disconcerting, which fits with a spectator's anxious thoughts about plot actions.

4. *THE WIZARD OF OZ*; AND, TECHNICOLOR TECHNOLOGY

Represented color must be understood against the backdrop of the specific materials and processes that produce its qualities. The important relationship between color technology and its narrativization may be illustrated with an image from *The Wizard of Oz* (Figure 6.8).[25] First of all, the film's technology is able to create a range of striking hues and combinations. The crooked hands of the Wicked Witch of the West are shown as a smeared, light yellowish-green as she reaches for the sparkling red ruby slippers of Dorothy. Studies have indicated that there exists a cross-cultural dislike of the weak blend of chartreuse (see section "Some Possible Disharmonies ['Conflict']"). The strong contrast between the spotted, yellowish-green chartreuse color and the ruby red is also a contrast between hybridity and purity (Chapter 5). In addition, one knows, of course, that human hands are not this color. The film continues to make use of this yellowish-green dyad in a blackened, near monochrome of greenish-yellow as Dorothy and her friends in a later scene walk toward the camera in an ominous corridor of the Emerald City for their first meeting with the Great Oz, who, when met, is initially seen as a large disembodied and floating, angry greenish yellow–tinged face. The yellowish-green dyad has been revived because at this moment in the story, the magic of Oz is being depicted as identical, or very similar, to the evil magic of the yellowish green-and-black Wicked Witch of the West.

The shot of the Witch grasping for the slippers illustrates quite a few important issues regarding classic Technicolor design. Only three will be mentioned. First, Technicolor was exceedingly careful about its backgrounds in order to highlight the hues directly connected to the action. Here soft focus, muted neutrals, and geometrical shapes in the background were used to set off the foreground drama of color contrast, conflict, and organic shapes (cf. Figure 6.3). Amazingly, this geometrical background is *not* visible in the preceding master shot as the Witch begins to reach for the slippers! Instead, the background for the closer shot has been specially constructed to highlight the conflict between the Witch and Dorothy. The dark shadow behind Dorothy's legs, feet, and shoes serves to spatially separate these elements from the background while the "reverse-L" shape of the background shadow mimics both the "reverse-L" formed

by Dorothy's legs and feet as well as suggesting a kind of symbolic "boundary" that is about to be transgressed by the menacing hands of the Witch.

The Witch's yellowish-green hands are similarly set against a contrasting background that consists of another box-like shape of a lighter neutral gray transected by a faded, muddy-red horizontal stripe. The horizontal orientation of the contrasting red stripe suggests the horizontal movement of the Witch's hands as they inch toward the bright clarity of the ruby red slippers on the right side of the image. The brilliant, highly saturated yellow bolts of electricity that will next emerge from the slippers to thwart the Witch's grasp (evidence of a good magic protecting Dorothy), along with the highly saturated red of the slippers themselves, are in stark contrast to the pasty, pastel colors of the Witch's hands (bad magic) and the light blue socks of Dorothy, who is a mere human confronting something hideous. The brilliant magic of the slippers—i.e., brilliant red with flashing gold bolts of power—is shown to be something astonishing and unique. The range and subtlety of Technicolor's technology allows for these careful compositions.

The lighting on Dorothy's ruby slippers illustrates a second issue about classic Technicolor design. The slippers have red sequins sewn onto their exterior that serve to reflect light directly into the lens of Technicolor's specially designed three-strip camera. The prism optics of the camera did not do well with so-called specular highlights in that the highlighted area was completely "blown," i.e., overexposed to the point of losing all detail in a wash of absolute white. In addition, the affected area was surrounded by a kind of "reflective glow," which helped to form and enhance the glowing, saturated red areas immediately surrounding the white highlights. More specifically, one can discern a red reflective scatter from the slippers onto the stone step on which Dorothy stands, accompanied by refracted light within the massive cube of glass that was the prism itself within the camera. The result, in other words, were circular areas of a highly saturated red punctuated with blurry white splotches, i.e., in general, a lack of fine photographic detail. As a general rule, Technicolor went out of its way to avoid shiny surfaces or lighting that would cause these kinds of highlights unless, of course, the intent was to take advantage of this "technical defect" of the three-strip camera in order to create a kind of spectacular color effect, as was the case here. Incidentally, how should one interpret the whiteness of the whitened splotches (Chapter 4)?

Third, the image from *The Wizard of Oz* demonstrates the relative insensitivity of the Technicolor process—technically referred to as "latitude"—which is the ability of the color process to adequately register photographic detail at lighting levels both above and below that of ideal exposure. The flesh tones of Dorothy's legs, for example, quickly drop off into darkness. The Technicolor three-strip process was

characterized by a very narrow latitude, which was the reason why Technicolor, especially in the early days of the three-strip process using the first generation of negative film stocks, preferred bright, evenly lit scenes in which the *separation of spatial planes* was created through the use of color contrasts and *not* by a carefully modulated or modeled use of differing light levels, as was the case with black-and-white cinematography. Technicolor would, however, employ "dramatic" lighting schemes to create special color effects as in this D.W. Griffith-like close-up detail of desire and danger in the form of ruby slippers.

We may conclude as follows: in all four examples from *Gentlemen prefer Blondes*, *Picnic*, *Vertigo*, and *The Wizard of Oz*, color has been registered by a spectator according to its narrativization, and more broadly, one might say, according to its quality and adjustment through a process of careful manufacture within material limitations. If a spectator is to fully appreciate a "color-full" display, he or she must be sensitive to the relationships established within evolving color patterns onscreen as well as be aware of the commercial practices through which these patterns were created, put to use, limited, sold, and made to be resident in imagination.

5. *TWO OR THREE THINGS I KNOW ABOUT HER*; NATALIE KALMUS AND TECHNICOLOR STYLE

Technicolor introduced the first successful three-color motion picture process in 1932. Its president was Herbert T. Kalmus. By 1950 over five hundred feature films and countless shorts, both animated and live-action, had been photographed using Technicolor's three-strip process. Thus the Technicolor way of doing business became, by default, the Hollywood way of doing business. The subsequent history of color in the cinema, that is, the post-1952 era of multi-layer or integral tri-pack color film stocks, can be described in some cases as a continuation of things as usual and in other cases as a formidable revolt against the peculiar design style pioneered by Technicolor and its powerful "color consultants," who dictated comprehensive color schemes for the sets, costumes, lighting, and movements of the actors.

The classic Technicolor scheme is still useful today when analyzing color designs that are either revisions or radical departures from the standard. I will summarize this scheme as described by Natalie Kalmus, wife of Herbert, and then compare its strictures with two shots from a Jean-Luc Godard film, *Two or Three Things I Know about Her*.

Natalie Kalmus earned 381 film credits for color design from 1928 to 1950. She worked as a "color consultant" on virtually every live-action Technicolor feature released from 1933 to 1949. Her film credits probably exceed five hundred because she was involved with many of the Technicolor Process Number Four Technicolor

films and live-action shorts and inserts produced from 1932–1955.[26] Kalmus was director of the Technicolor Color Control Department, also known as the Color Advisory Service, from 1923 to 1948. Its function was to work with a studio during pre-production and production making color plans for every scene. By the late 1940s, she had up to sixteen assistants.

Historians and critics have generally treated Natalie Kalmus quite unfairly while celebrating her husband Herbert.[27] For example, when she is mentioned as a frightening, demanding, and even ludicrous figure in conjunction with her work on *Gone with the Wind* (Victor Fleming and George Cukor, 1939) in the documentary, *Glorious Technicolor*, the camera pans across a photograph of her. The panning movement exists precisely so that there will be sufficient time for us to hear the judgment being passed on her by the accompanying theme music of the Wicked Witch of the West from the film *The Wizard of Oz* on which, ironically, she was Technicolor Color Director (see Figure 6.8; also Figures 8.1–8.6).[28]

According to Angela Dalle Vacche and Brian Price:

> Natalie M. Kalmus is a woefully underrepresented figure in the history of American cinema. . . . One could say without risk of overstatement that Kalmus was a genuine auteur, a figure whose signature is as evident, if not more so, as the more celebrated directors with whom she worked.[29]

It would be difficult to overestimate the historical and aesthetic importance of Technicolor and of Natalie Kalmus's role in describing a specific collection of fine art principles that maximized the capabilities of Technicolor's unique technology. She did not invent these principles, but rather found effective methods by which to systematically implement specific kinds of color designs. What follows is a summary of these principles based on her 1935 essay, "Color Consciousness."[30]

> 1. Kalmus argues that like other elements of style, *color should be made subordinate to the story*—character, dialogue, and action—while at the same time color should work to reinforce ("harmonizes with") the dominant emotion or tone of a scene so as to "subtly convey dramatic moods and impressions to the audience" (26). Colors that are used in costumes should enhance already-existing character traits (28). Discordant color combinations must have a dramatic purpose (28).
>
> Kalmus was keenly aware of the decisive importance of technology: "The art director. . . in handling a color motion picture, must be forever mindful that the human eye is many times more sensitive than the photographic emulsion and many times greater in scope than any process of reproduction. Therefore, he must be able to translate his colors in terms of the process" (28).

2. *Color should be unobtrusive, psychological, and realistic*. Kalmus asserts that "the complete absence of color is unnatural" and yet "a super-abundance of color is unnatural, and has a most unpleasant effect not only upon the eye itself, but upon the mind as well" (25). "We must constantly practice color restraint" (29). Kalmus recommends that (i) excessive use of bright, saturated hues be avoided in favor of more "natural," harmonious and less intense, color schemes and that (ii) "the judicious use of neutrals" should be a "foil for color" (26). Kalmus admires the "subtle tones of a field of grain" and notes that "even when Nature indulges in a riot of beautiful colors, there are subtle harmonies which *justify* those colors" (25; emphasis added).

For Kalmus, the justification for the use of "subtle harmonies" (25) is firmly rooted in emotion: "Just as every scene has some definite dramatic mood. . . so, too, has each scene, each type of action, its definitely indicated color which harmonizes with that emotion" (26). "Each hue has its particular associations" and dramatic uses, including colors mixed with white, gray, and black (26, 27). For example, since "love gently warms the [redness of] blood," different shades of red "will suggest the type of love," and by introducing additional colors, "it will be possible to classify the type of love portrayed with considerable accuracy" (26). In general, Kalmus draws strongly on color patterns, harmonies, and associations that appear in nature (26, 27), though perhaps to some extent, on what we also believe or imagine appears in nature (24, 25).

3. *Color should be used to direct attention only to information important for the narrative (the "law of emphasis")*. Kalmus says, "the law of emphasis states in part that nothing of relative unimportance in a picture shall be emphasized. If, for example, a bright red ornament were shown behind an actor's head, the bright color would detract from the character and action" (28–29). The maximum point of color *contrast* should be associated with the principal actors in a scene while those actors who play "relatively unimportant parts" should blend into the background (28). Colors used in costumes should help separate important characters from their background because a classical narrative is about characters, not their backgrounds. [This provides a rationale for the use of color dyads and triads: simplicity and contrast.] Kalmus notes, for example, the lack of "color separation" of "a polar bear in the snow" (28).

Furthermore, because flesh-tones are typically warm, Kalmus explains, "we usually introduce the cooler[31] tones into the backgrounds; but, if we find it advantageous to use warmer tones in the set, we handle the lighting so that the particular section in

back of the actor is left in shadow" (28). Color contrasts (warm/cool, color/neutral, light/dark, high key/low key lighting) are used to separate spatial planes. Depth in the mise-en-scène is important for it allows the use of "interesting shadows and colored lights" in different spatial planes (28). A careful use of warm-advancing and cool-"retiring" hues (26) creates a pseudo-3-D effect (28). [This effect is a fundamental principle in Technicolor designs.]

4. Kalmus points out "how exceedingly important it is to consider *the movement in the scene in determining its color composition* because the juxtaposition of colors is constantly changing due to this movement" (29, emphasis added). By way of illustration, she says that orange would "appear more red than it really is" next to a blue-green because "each color tends 'to throw' the other toward its complement" (29). Thus one must carefully weigh the effects of character and camera movements, and especially of juxtapositions through editing (29). For example, editing may move a background hue to the foreground creating new hue combinations that will interact with each other, with hues in the previous shot, and with the developing story (see 1, 2, and 3 above).

"The same principles" that "make painting a fine art" will "guide" the new art of film and its "motion in color" (24). Kalmus advocates preparing a "color chart"—a color plan—for every scene and each character in a film. She compares the color chart to a "musical score" (28). [This would seem to suggest, I believe, that color itself should be conceived as being in motion in a film—as being a flow within a shot and from shot to shot.]

Let us now consider two shots from an opening scene in Jean-Luc Godard's *Two or Three Things I Know about Her* (Figures 6.9 and 6.10). These shots illustrate the principles of Natalie Kalmus by refusing to employ them. Notice in the first of these compositions the failure to distinguish spatial planes through the use of hues or gradations of lightness and the consequent flatness of scenic space (see 3 above). Weeds seem to sprout out of Robert's head and one of Roger's eyes seems to have come from a teddy bear (cf. the law of emphasis above). Does something also sprout from Roger's head? Hues seem to have been distributed at random and to fall into no discernable pattern despite, for example, the hint from the blues of two plates on the wall being connected to Robert's shirt, the reds of the teddy bear's ear and Robert's tie, and the greens of two plants. These connections, however, have no connection to the story (see 1 above). Instead, there seems to be a studied dispersal and disorder of hues that overpowers a sense of the painter's primaries, which might have formed out of the blues and reds together with the

slivers of a yellow tablecloth and chair. In fact, the slight presence of the reds leaves the greater expanse of Robert's blue shirt to conflict with the yellow tablecloth since blue and yellow are discordant (see section "Some Possible Disharmonies ['Conflict']").

Overall, the hues of Figure 6.9 do not clarify a center of dramatic interest—a focal point for a color plan or two focal points. Instead they create a "wounded" sort of stasis by placing the characters' heads along a dead-center horizon line in conflict with an askew symmetry. The twisted symmetry is formed by the tension between the horizontal symmetry of the blue plates and an angled asymmetry created by a small green plant mostly obscured by Roger's chair at the bottom of the left frame line that is set in opposition to a series of large green leaves lying unobscured along the right frame line from bottom to top.

Similar comments apply to the hue fragmentation of Figure 6.10. Godard has manifestly failed to avoid a background that is a clutter of distracting color and detail (see 3 above). Has he simply been inattentive to color design? Not at all. Other shots in the same scene employ startling and intrusive color combinations, which continue to develop throughout the film; and in previous films, Godard has shown an acute awareness of Technicolor style.

I believe that *Two or Three Things* is a meticulous exploration of experimental narrative as realized through an alternative color style.[32] Godard's comment on the film suggests some of its complexity:

> While the Americans pursue an immoral and unjust war in Vietnam, the French government, whose connections with big capital are common knowledge, sees to the construction in and around Paris of apartment complexes whose inhabitants, either through boredom or through an anguish which this architecture cultivates or through economic necessity, are led to prostitute themselves, notably (incidentally) to Americans returned from Vietnam.[33]

Godard also says, "During the course of the film—in its discourse, its discontinuous discourse, that is—I want to include everything, sport, politics, even groceries."[34] Furthermore, the "her" named in the film's title refers to as many as twelve different sorts of things, fourteen counting the main character Juliette Janson and the actress who plays her, Marina Vlady, who sometimes speaks on her own account.[35] At certain points in the film, the attitudes of Juliette and Marina are split from one another as if Godard were pursuing an analogy with "color separation" (see 3 above).

Natalie Kalmus recommends that color be subordinate to the story and illustrative of a mood or emotion (see 1 and 2 above). By contrast, for Godard, color helps to generate arguments in an intellectual cinema. His ambition—which is nothing less

than to create a strongly discursive, philosophical and dramatic, new poetic form—requires the invention of an entirely distinct aesthetics of color.[36] He begins *Two or Three Things* by returning to zero (Figures 6.9 and 6.10)—as demanded in another context by the film's final words—in order to wholly transform the principles of Technicolor style. The result reimagines what color can do within narrative form.

NOTES

1. Hermann Hesse, excerpt from "A Dream," in The Glass Bead Game (Magister Ludi), trans. by Richard and Clara Winston (London: Picador Classics, 1987), p. 437 (Hesse's emphases).
2. One might remember the four basic elements of Empedocles in their proper order by summoning a mental image of the *fire* of the sun warming *air* above the *water* of a sea above an *earth*en sea floor.
3. For an example of blue-yellow discord, see Mary Beth Haralovich, "'All That Heaven Allows': Color, Narrative Space, and Melodrama" in *Close Viewings: An Anthology of New Film Criticism*, ed. by Peter Lehman (Tallahassee: Florida State University Press, 1990), pp. 68–69; reprinted in *Color, The Film Reader*, ed. by Angela Dalle Vacche and Brian Price (New York: Routledge, 2006), pp. 150–151. Van Gogh's "Wheat Field with Crows" (1890) is a striking instance of strong blue-yellow and red-green contrasts. A red-green dyad/double is a structuring principle throughout the film, *The Double Life of Veronique* (Kieslowski, 1991). This dyad would seem to mimic a type of deficit in color vision. See Livingstone, *Vision and Art*, pp. 34–35.
4. On the relation of color transparency and polyphonic music, see Rudolf Arnheim, *Art and Visual Perception: A Psychology of the Creative Eye–The New Version* (Berkeley and Los Angeles: University of California Press, 1974), pp. 255–256.
5. A Louise Bourgeois painting, "Red Night" (1946–1948), illustrates the disturbing collision of a pure hue with a weak blend. The painting is executed in a red-turquoise dyad. There are a dozen or so close shades of red swirling about a bedroom with strange eddies, e.g., a barely discernible, dirty pinkish oval on the floor. Obscure shapes and lines are overwhelmed by the colors. The many slight variations in redness hint at an aesthetic approach of "delicacy" and thresholds (Chapter 2). The reds seem to be part of the background of the bedroom and expressive of psyche and action as well as a distinct physical substance because they obscure part of the bed, most of the room, and defy the darkness of night. A woman on the bed, and a wildly blowing window curtain pointing at her, are rendered in turquoise with accents of orange and blue, streaks of white, black, and gray. Another figure is obscured on the bed and the heads of three children appear at the woman's breasts and genital area. Although red is a pure hue, its many variations in this painting make it seem to be as mixed and indefinite as the turquoise. A kind of turmoil is seen to surmount the placid woman. Drops of turquoise are falling from the painted window curtain, unaffected by the wind storm of red. Some of the paint used on the canvas seems to possess its own material fragility in the midst of scenic propulsion.
6. On the cross-cultural dislike of chartreuse, see the intriguing book, *Painting by Numbers: Komar and Melamid's Scientific Guide to Art*, ed. by JoAnn Wypijewski (Berkeley: University of California Press, 1997), pp. 15–16, 92, cf. 59; and see <www.diacenter.org/km/index.html> (accessed Sept. 5, 2015). See also Chloe Taylor and Anna Franklin, "The Relationship Between Color-Object Associations and Color Preference: Further Investigation of Ecological Valence Theory," *Psychonomic Bulletin & Review* 19, 2 (April 2012), pp. 190–197; C.L. Hardin, *Color for Philosophers:*

Unweaving the Rainbow (Indianapolis: Hackett, expanded ed. 1993), p. 163. Is the dislike of chartreuse partially due to the fact that a black and blue bruise on the body will turn yellow and green?

Fine examples of deliberate discordant color combinations may be found in HBO's *Animals*, e.g., season one, episode six, "Pigeons"; FX's "Legion," season one, episode six; and Tero Juuti's illustrations of scenes from Stanley Kubrick's *The Shining* (1980) in Juhani Pallasmaa, *The Architecture of Image: Existential Space in Cinema* (Helsinki: Rakennustieto, 2nd ed. 2007), pp. 93–114.

7. Paul Coates, *Cinema and Colour: The Saturated Image* (New York: Palgrave Macmillan, 2010), p. 4; cf. 7–8.

8. In relation to the triadic scheme, black, red, and white, see Chapter 5 above, as these are the first three universal focal colors; and cf. Coates, p. 23, and his chapter 4, which is devoted to patterns of black, red, and white, pp. 69–86. See John Gage, "'The Fortunes of Apelles,' section 'The Four-Colour Theory'" in *Color and Culture: Practice and Meaning from Antiquity to Abstraction* (Berkeley: University of California Press, 1993), chap. 2, pp. 29–30.

9. I.A. Richards, *Principles of Literary Criticism* (New York: Harcourt Brace Jovanovich, 1985), pp. 155–156 (orig. 1925).

10. Adrian Cornwell-Clyne, *Colour Cinematography* (London: Chapman & Hall, 3rd ed. 1951), p. 645.

11. Ludwig Wittgenstein, *Remarks on Colour*, ed. by G.E.M. Anscombe, trans. by Linda L. McAlister and Margarete Schättle (Berkeley and Los Angeles: University of California Press, 1977), Part III, § 91.

12. On embodiment in film, see, e.g., Embodied Cognition and Cinema, ed. by Maarten Coëgnarts and Peter Kravanja (Leuven, Belgium: Leuven University Press, 2015). Mark Johnson in a foreword states a key tenet of embodiment: "the nature of our bodies (in interaction with their environments) determines both what can be meaningful to us and also how it is meaningful and how we reason about it" (p. 9; my emphasis of the first four words).

    On embodiment generally, see, e.g., George Lakoff and Mark Johnson, *Philosophy in the Flesh: The Embodied Mind and Its Challenge to Western Thought* (New York: Basic Books, 1999); Francisco J. Varela, Evan Thompson, and Eleanor Rosch, *The Embodied Mind: Cognitive Science and the Human Experience* (Cambridge, MA: MIT Press, 1991); George Lakoff, *Women, Fire, and Dangerous Things: What Categories Reveal about the Mind* (Chicago: University of Chicago Press, 1987); Mark Johnson, *The Body in the Mind: The Bodily Basis of Meaning, Imagination, and Reason* (Chicago: University of Chicago Press, 1987). See also Chapter 2 above, note 9, and Chapter 7 below, note 24.

13. Figure 6.4 from *Picnic* illustrates subtle flow lines created through low contrast harmonies with lightened hues. Fine examples of flow through low contrast harmonies with darkened hues may be found in *The Shepherd of the Hills* (Hathaway, 1941).

14. This mistaken continuity has eluded even the Internet Movie Database, which lists forty-two continuity "Goofs" for *Vertigo* <www.imdb.com/title/tt0052357/trivia?tab=gf&ref_=tt_trv_gf> (accessed May 24, 2016).

15. Space suddenly tightens as the red door appears to have moved much closer to Scottie in shot 2. This set-up appears 6 times in the scene. It cheats the position of Scottie's head relative to Madeleine and of his body relative to the pole and railing from the positions shown in shot 1. This may be the result of changing the focal length of the lens between shots 1 and 2. Also, the

shadows on the wall and door are different between shots 1 and 2. One of the reasons for these discrepancies, I believe, is to intensify the effect in shot 2 of the presence and proximity of the redness to Scottie.

To me, the red seems to advance in a threatening manner toward and onto Scottie. Paul Coates has a different interpretation. "If anything, red is thus associated rather with Scottie [than with Madeleine], and his potential passion may be declared by the red door of his apartment." Allen suggests a more general interpretation. "From a woman's point of view, *Marnie* [Hitchcock, 1964] can be understood, in part, as a response to *Vertigo*": the "phallic sublime" of the former as opposed to the phallic submission of the latter. See Coates, *Cinema and Colour*, p. 144; Richard Allen, *Hitchcock's Romantic Irony* (New York: Columbia University Press, 2007), pp. 244–245.

16. Alfred Hitchcock, quoted in Richard Allen, "Hitchcock's Color Designs" in *Color, The Film Reader*, p. 131.
17. Allen, "Hitchcock's Color Designs," p. 142.
18. Allen, "Hitchcock's Color Designs," p. 137.
19. Allen, "Hitchcock's Color Designs," p. 141. In Allen's book, he adds that Madeleine's speckled black-and-white coat is worn "over her black dress as her *dark* motives come to the surface." *Hitchcock's Romantic Irony*, p. 246 (my emphasis); see pp. 246–250.
20. On stylistic metaphors, see Edward Branigan, "'Story World and Screen,' section 'Screen Space and Stylistic Metaphors'" in *Narrative Comprehension and Film* (New York: Routledge, 1992), chap. 2, pp. 56–62. The mapping of plot and theme onto adjectives that modify style and into adjectives that describe a spectator's response to style is a venerable technique of critics. For example, in a single-paragraph capsule review, the experience of seeing *Vertigo* is summoned in terms of an unruly return of repressed passions and contradictions leading to punishment, which is said to apply to several characters as well as the spectator. This complex theme is mapped onto a dissolution of color from baroque to arid that reverberates with a spectator's secret fascination with states of frightful dissipation:

> The irrepressible allure of Hitchcock's visual extravagance—his baroque swirl of caustic greens, voluptuous purples, acidic yellows, and fiery reds, and the indecent glare of daylight—conjures a torrent of unconscious desires beyond the realm of dramatic machinations; his happy ending, of health restored and crime punished, resembles an aridly monastic renunciation.

New Yorker 88, 28 (September 17, 2012), p. 23.

Here's an example from Richard Allen:

> In *Torn Curtain* [Hitchcock, 1966], when Sarah Sherman, who elsewhere in the film wears redemptive earth tones, begins to threaten Armstrong's mission by her presence, she appears in a red robe, though it is worn over a white nightgown that demonstrates the underlying purity of her motive.

The adjective "redemptive" appears fifteen times attached to hues and other qualities in twenty-five pages that describe the four major color groupings in Hitchcock films (pp. 226–250). Also appearing are such words as redemption, redeeming, redeemed, and renewal. *Hitchcock's Romantic Irony*, p. 241.

Here's an example based on Eric Rohmer's *The Green Ray* (1986):

> She does the same with her own neckline, her hand dabbing around the broad collar of her pale blue sweater. Then, as she *loosely* entertains the idea of holidaying, Delphine idly twirls a canary-yellow feather, brushing it under her chin. Thoughts of going on holiday tickle her, irritating and diverting in their appeals. At the same time, for this character, such plans are only enjoyable when they are as *loose* and light as a pastel pullover and a feather.

Steven Peacock, *Colour* (New York: Manchester University Press, 2010), p. 1 (my emphases); see also pp. 4–5, 109. This example shows the close relationship between a stylistic metaphor and the schema, *being-colored-by*; see Figure 3.1. In both cases, a blending is asserted between style and story or between color and an abstraction or between color and something different.

21. Allen, "Hitchcock's Color Designs," p. 139.
22. On the importance of the "nearly true" to the comprehension of texts, see Chapter 7 below on "The Otherwise of Texts," and Edward Branigan, "Nearly True: Forking Plots, Forking Interpretations—A Response to David Bordwell's 'Film Futures,'" *SubStance* 97, v. 31, n. 1 (2002), pp. 105–114.
23. Allen, "Hitchcock's Color Designs," pp. 141–142.
24. Scottie enters the scene in shot 1 (set-up 1; Figure 6.5) and leaves this space in shot 17 (set-up 3; Figure 6.7). The first three shots and fourteen of the first seventeen shots in the scene are from set-ups 1, 2, and 3; two shots show Madeleine from a closer set-up than set-up 2; one shot shows Scottie from a closer set-up than set-up 3.
25. The image in Figure 6.8 is taken directly from *The Wizard of Oz*. It is reproduced in cropped form on the cover of *Color, The Film Reader*.
26. Some of Natalie Kalmus's film credits may be found at <www.imdb.com/name/nm0436124/?ref_=fn_al_nm_1> (accessed Mar. 2, 2014). It is surprising that there is no Academy Award to recognize the art of color design. Her husband, Herbert, one of the co-founders of Technicolor, received a star on the Hollywood Walk of Fame in 1960.
27. The autobiography of Herbert Kalmus—who was co-founder, president, and plant general manager of the Technicolor Motion Picture Corporation from 1915–1960 and the inventor of the company's unique trade name, and known even to his wife as "The Doctor"—omits all mention of his first wife, Natalie Kalmus, who played a central role in the color designs and success of Technicolor films. Herbert's second wife has added two chapters on Natalie to his autobiography that make for an entertaining, though perhaps not pretty, story, and which offer a different view than that of John K. Newman. See Herbert T. Kalmus with Eleanore King Kalmus, Mr. Technicolor (Absecon, NJ: MagicImage Filmbooks, 1993); and H.T. Kalmus, "Technicolor Adventures in Cinemaland," Journal of the SMPE (December 1938), pp. 1–18. Newman, "Profile of Natalie Kalmus: The Woman Behind Technicolor" in British Technicolor Films, ed. by John Huntley (London: Skelton Robinson, 1949), pp. 146–150.
28. The documentary *Glorious Technicolor* (Peter Jones, 1998) may be found on the 2003 restored, special edition DVD and Blu-ray versions of *The Adventures of Robin Hood* (Michael Curtiz and William Keightley, 1938; Technicolor Process Number Four; Natalie Kalmus, Technicolor Color Director). Both the documentary and the feature film provide an excellent glimpse of classic Technicolor style.

29. Dalle Vacche and Price, *Color, The Film Reader*, pp. 11–12.

30. Edward Branigan and John Kurten, "Natalie Kalmus's Principles of Classical Color Design," unpublished manuscript (2010). Page references in the text are to Kalmus's "Color Consciousness," reprinted in Dalle Vacche and Price, *Color, The Film Reader*, pp. 24–29. The reprint omits an introductory "Summary" from the original, which was given as a talk on May 21, 1935, and first appeared in the *Journal of the Society of Motion Picture Engineers* 25, 2 (August 1935), pp. 139–147. This volume is reprinted by Forgotten Books (London, 2015). Kalmus wrote an essay that closely tracks "Color Consciousness." Many sentences are identical; some material has been dropped while comments about Technicolor cameras have been added. See "Colour" in *Behind the Screen: How Films Are Made*, ed. by Stephen Watts (New York: Dodge, 1938), pp. 116–127.

    Kalmus's principles are adopted and elaborated in William R. Holm, et. al., *Elements of Color in Professional Motion Pictures* (New York: Society of Motion Picture and Television Engineers, 1957). On Kalmus's ideas, see Steve Neale, "Colour and Film Aesthetics" in *Cinema and Technology: Image, Sound, Colour* (Bloomington: Indiana University Press, 1985), chap. 9, pp. 145–158; Richard Neupert, "Technicolor and Hollywood: Exercising Color Restraint," *Post Script* 10, 1 (Fall 1990), pp. 21–29. Some of the problems Kalmus considers were mentioned in 1930: *Béla Balázs: Early Film Theory—Visible Man and The Spirit of Film*, ed. by Erica Carter, trans. by Rodney Livingstone (New York: Berghahn Books, 2010), pp. 178–181. It would be well to keep in mind that the look of a color design is only partly due to aesthetics; Dudley Andrew, "The Postwar Struggle for Color," *Cinema Journal* 18, 2 (Spring 1979), pp. 41–52.

    Kalmus is credited with writing "Basic Tips on Color," *Popular Photography* 12, 1 (January 1943), pp. 30–31, 83–85. This article has been severely edited from the original manuscript, which was ghostwritten by Ray Dannenbaum. The original manuscript is titled "The Black and White of Color Photography." It is fifteen pages of official typescript headed "Technicolor Motion Picture Corporation" along with two pages of typescript of "Suggested Captions for Stills." It may be found in the Natalie Kalmus Collection of the Margaret Herrick Library, Academy of Motion Picture Arts and Sciences, Beverly Hills, California, File #7, Ray Dannenbaum (file date 10/1/1942). The manuscript contains a detailed exposition of the physiology of color perception as understood in the early 1940s and offers ideas about the emotional values of particular hues, the effects of different lighting on fabrics, the creation of spatial planes of interest, and the use of harmonious combinations of hues appearing in various juxtapositions and separations. Although the essay is ghostwritten, it is possible to imagine Kalmus's voice, particularly in the descriptions of certain film examples and in twelve suggested captions for stills to accompany the published article. Unfortunately, only one of Kalmus's stills appears in the article. The editors of the magazine have added four insipid photos and all five color photos for "Basic Tips on Color" are reproduced in black and white.

    Kalmus is also the author of "Doorway to Another World," *Coronet* 25, 6 (April 1949), pp. 29–31 (about the death of her sister, Eleanor), reprinted from *Guideposts*, ed. by Dr. Norman Vincent Peale (Pawling, NY: Guideposts Associates, 1947). For those interested in her private life: *Natalle* [*sic*] *M. Kalmus, Petitioner, v. Herbert T. Kalmus and Technicolor Motion Picture Corporation: U.S. Supreme Court Transcript of Record with Supporting Pleadings* [1951] (Gale MOML Print Editions, n.d.).

31. The phrase "cooler tones" mistakenly appears as "color tones" in the reprinted version of Kalmus's "Color Consciousness" in *Color, The Film Reader*, p. 28.

32. On Godard's approach to color and narrative, see Edward Branigan, "The Articulation of Color in a Filmic System: '*Deux ou trois choses que je sais d'elle*' " (expanded version) in *Color, The Film Reader*, pp. 170–182. See also Alfred Guzzetti, *Two or Three Things I Know about Her: Analysis of a Film by Godard* (Cambridge, MA: Harvard University Press, 1981). Guzzetti's excellent book does not have an index. Some of his comments on color in the film may be found on the following pages: 15, 17, 27, 47–49, 51, 53, 99–107, 161, 225, 241–245, 343, 347, 349, 360–361, 363. Also in the film, there are many discussions of color and other phenomenal qualities in the dialogue and spoken commentary.

    For more on Godard's use of color in *Two or Three Things*, see the fine analysis by Paul Coates of the many social and political possibilities for "ensemble" (i.e., patterns!) to be found in the film and the implications of Godard's aesthetic "separation" into parts (i.e., fragmentation, dislocation, dissociation); *Cinema and Colour: The Saturated Image* (New York: Palgrave Macmillan, 2010), pp. 33–40. On Coates's aesthetic tenet of "separation," see Chapter 3 above, section "Color Splitting and Flow." There exists a large literature on Godard's *Two or Three Things*.

33. Godard quoted by Guzzetti, *Two or Three Things*, p. 47.

34. Godard quoted by James Monaco, *The New Wave: Truffaut, Godard, Chabrol, Rohmer, Rivette* (New York: Oxford University Press, 1976), p. 179.

35. Godard designed both the advertising poster and the trailer for *Two or Three Things*, both of which announce twelve references for the "her" of the title. On the poster, see Monaco, *The New Wave*, p. 180. The trailer appears on the Criterion DVD of the film.

36. I do not mean to imply that there are only two possibilities for color film aesthetics, Technicolor and Godard. To list merely a few unique styles not otherwise mentioned in this book: *Written on the Wind* (Douglas Sirk, 1956); *Equinox Flower* (Yasujiro Ozu, 1958); *The Red Desert* (Michelangelo Antonioni, 1964); *Ali—Fear Eats the Soul* (Rainer Werner Fassbinder, 1974); *Lancelot of the Lake* (Robert Bresson, 1974); *Stalker* (Andrei Tarkovsky, 1979); *One from the Heart* (Francis Ford Coppola, 1981); *Time Stands Still* (Péter Gothár, 1981); *The Element of Crime* (Lars von Trier, 1984); *In the Mood for Love* (Wong Kar-wai, 2000); *Platform* (Jia Zhangke, 2000); *Amélie* (Jean-Pierre Jeunet, 2001); *Bright Future* (Kiyoshi Kurosawa, 2002); *Far from Heaven* (Todd Haynes, 2002); *Speed Racer* (Lilly and Lana Wachowski, 2008); *Wild Grass* (Alain Resnais, 2009); *World of Tomorrow* (Don Hertzfeldt, 2015).

Chapter 7 **Track This in Place**

ABSTRACT

Chapter 7 examines the mental process of "tracking" and the creation of a flow of attention by a person's hard-working working memory that functions both in the moment and beyond and before. The concept of tracking derives from P.F. Strawson's notion of "reidentification," identified as a set of mental structures that Strawson employs in establishing a "descriptive metaphysics" of the world. I believe that Strawson's ideas apply equally to a spectator's mental construction of a descriptive framework for his or her comprehension of a film world. Reidentification permits a continuous recognition of things in spite of inevitable and persistent *ellipses* and fragmentation present both in the world and in a film world.

Adjustments and expansions of Strawson's notion of reidentification have been made by Evans, Millikan, and, I believe, for film narration by Wilson. I identify four areas where reidentification and tracking are important for understanding color consciousness. I argue that there are tight connections between color patterns and patterns of thinking and also between narrative structures and what we believe to be *nearly true* about a narrative, i.e., patterns of near possibility, both while watching a film and through reassessments after it ends.

There are three main aspects to an analysis of color tracking involving the selection of a scale of analysis, a type of color pattern to follow, and a type of linkage to follow that connects a color pattern either to narrative structure or to narration and point of view. I illustrate these aspects in the case where a color appears to reappear as the same color by analyzing a painting and two films.

> Of all internal functions that create the meaning of the world around us, the most central is the attention. The chaos of the surrounding impressions is organized into a real cosmos of experience by our selection of that which is significant and of consequence. This is true for life and stage alike. Our attention must be drawn now here, now there, if we want to *bind together* that which is scattered in the space before us. Everything must be shaded by attention and inattention. Whatever is focused by our attention wins emphasis and *irradiates meaning* over the course of events.[1]
>
> —Hugo Münsterberg

> I've been thinking a lot about how seldom we're in the moment, how we're always thinking about something else, making plans, remembering. But occasionally we get flashes of being right where we are.[2]
>
> —Richard McGuire

THE REALITY OF ILLUSIONS AND THE ILLUSIONS OF REALITY

Perceptual illusions in which a pair of lines, for example, are seen to intersect but are actually parallel, are quite numerous, and are studied by psychologists for what they reveal about the mechanisms of perception. Illusions may involve color and movement as well, such as the Hermann grid. Many people believe that shadows are simply black or dark, which is a first approximation that works in most situations. Is it correct, then, to say that shadows are *really* colored, to say that it is merely an illusion or misconception that they are black? What can be meant by calling certain effects "illusions"? In which contexts for which purposes is a percept an illusion? These phenomena are perfectly real when experienced. Indeed many of them cannot *not* be seen, even when it is explained that a perception does not match reality. This is strong evidence for the modularity of mind, cognitive impenetrability, and informational encapsulation.

On the other hand, why do we think that concrete reality is so real? When we see things, we are typically thinking about them along variegated flows of feelings, backgrounds, meanings, ideas, preconceptions, memories, projections, introjections, possibilities, actions, and intentions. We may also remember immediately prior thoughts, whatever they may have been and however connected. All such thinking, though subject to high-level distortions or dismissal, acts to place a thing in real contexts for a person and gives a thing a tangible, pragmatic quality. Nonetheless meanings are judged to be insubstantial as well as immaterial compared to rocks and trees. However, if we were to move beyond a first approximation of illusion versus reality, and meaning versus concreteness, then connections among both matters of mind and existential objects would become more complex. Is the real divided exactly in half between illusion and reality, or does illusion possess a measured degree of reality while reality holds an element of the illusory immaterial?

Let us not reverse the terms and say it is reality that is the illusion, since strong idealism and skepticism lie in that direction. Instead, let us consider whether illusion and reality interpenetrate to some extent. If so, then drawing a line that is sharp becomes arbitrary; or rather, a sharp line works only to support a chosen methodology and the delineation of a greater (or lesser) point of view concerning issues of the philosophy of mind and its objects. Pointing out the presence of an illusion requires an act of crossing between two frameworks, two contexts, two philosophies, two actions.[3] In the transitional space between illusion and reality, the two may mingle.

When assessing how color designs figure in our lives, the simple binary of illusion/reality is not helpful. A second approximation focuses on language and how it is used to express color by accentuating what we address, discuss, share, and say about a flow of thought about colored objects. We are speaking to and about ourselves as much as to others. In this context, language stands intermediate between embodying the meanings and memories we live by and pointing toward an everyday, lived world. Meaning and world are realized in talk about the actions we intend and favor. A starting point for constructing an architecture of a world—for fashioning a continuous memory of a world for our actions—is the *recurrence* in mind of previously sensed objects, persons, and places. The idea that not everything is *new* provides a basis (a precondition?) for thought and calculation. This process has been termed "reidentification." Color has a number of roles to play in this activity and, like the ongoing interplay between meanings and concrete things, color shows itself to be a fluid blend of illusory and real components.

THEORETICAL EXCURSUS (III): REIDENTIFICATION (STRAWSON)

P.F. Strawson draws attention to an important aspect of memory with the concept of "reidentification." Strawson employs the concept as part of an extensive "descriptive metaphysics" that is "content to describe the actual structure of our thought about the world."[4] Strawson's concern is to describe how a perceiver *constructs* a unified cognitive framework for perceiving the world. He does not speculate whether the world itself is somehow unified in a fundamental way. In what follows, I will stress the importance of reidentification in describing mental activities when a spectator is assessing how various color designs intersect with elements of a film world. I will not be concerned with supposed absolute qualities that may combine to make of color a fundamental constituent of the world or make of it a unity or a harmony of similars or opposites. Instead, I will concentrate on a spectator's evaluations of color moment by moment, especially when it is judged as having reappeared and thus seen as having moved through time and/or space.

Strawson's interest is in our continuous recognition of physical objects and persons that constitute fundamental, "objective particulars" (including the systematic *places* being occupied by the particulars) around which other sorts of qualities and the use of language are orientated and placed.[5] In this sort of physical framework, color is likely to be conceived as a material, e.g., paint or cosmetic. My interest is with a person's flow of thought about a narrative flow that is located inside a larger, everyday world where conceptual entities and stories—whether fictional or nonfictional—are being devised about material bodies and objective particulars. As I will demonstrate, narrative texts are not limited to material forms of color like

paint or metaphors based on paint, but are able to explore other, more fluid and abstract conceptions of color and its movements.[6]

According to Strawson, who no doubt has Kant in mind, we live within a conceptual scheme of a single, unified spatial-temporal structure. This scheme allows us to conceive of the world as "a system of elements every one of which can be both spatially and temporally related to every other."[7] We recognize that element B is now in the place formerly occupied by element A. This creates a "framework of particulars," for which "we must have criteria or methods of identifying a particular encountered on one occasion. . . as *the same individual* as a particular encountered on another occasion."[8] That is, we make use of specific criteria and methods in order to be able to continuously reidentify and reassert the unity of our world.

Strawson insists that "it is the essence of the matter that we use the same framework on different occasions." He explains the reason for this requirement:

> Our methods, or criteria, of reidentification must allow for such facts as these: that the field of our observation is limited; that we go to sleep; that we move. That is to say, they must allow for the facts that we cannot at any moment observe the whole of the spatial framework we use, that there is no part of it that we can observe continuously, and that we ourselves do not occupy a fixed position within it. These facts have, among other consequences, this one: that there can be no question of continuous and comprehensive attention to the preservation or change of spatial boundaries and the preservation or continuous change of spatial relations on the part of things mostly undergoing no, or only gradual, qualitative change.[9]

In life, as in film, there is no escape from ellipses and gaps. Space and time appear as a series of fragments: attention is contingent, memories and expectations are fluid, one falls asleep, and so forth. Strawson emphasizes that a limited vantage point in space and time makes ellipses inevitable and requires a continual reassertion of a personal orientation, including one's identity when waking in the morning—his or her own (reidentified) being being among the particulars. In order to describe our experience of watching a film unfold, Strawson's notion of a single, though interrupted, framework must be expanded into a framework of interconnected, embedded levels. The represented particulars of narrative film are located within a hierarchy of "levels of narration" and anchored relative to a "diegetic level," which is the represented spatial-temporal, physical world of the characters.[10] Levels above and below the diegesis provide a film with the power to speculate abstractly about properties of the diegetic world and to entertain non-material forms of color. For a spectator, watching a film becomes a mental performance that reconfigures textual materials on the screen into

mental particulars and event-sequences being seen under various epistemological constraints, i.e., under descriptions that are fit to relevant narrational levels.

Strawson's above list of the ways in which "the field of our observation" must of necessity be severely limited applies also to our observation of the embedded levels presented within a film. Aesthetic devices routinely create explicit or implicit discontinuities within and among narrational levels—distraction, ambiguity, fragmentation, confusion, concealment, unreliable narration—that act to both challenge and develop our ability to relate particulars across levels; examples include editing, image composition (camera angle, off-screen space, etc.), plot ellipses, non-diegetic sound, voice-over, and changing points of view. Even a continuous camera movement may violate a given framework of continuous space and time by invoking a different narrational level within a larger framework of levels. Modes and genres of filmmaking differ in the timing and degree to which a spectator is made aware of these shifts among levels when particulars are reidentified and situated.

One might further speculate about a viewer's general degree of awareness when immersed in a film. Strawson offers several fascinating comments on the nature and possibilities of a person's "disembodiment" relative to the world that may be applicable to varying states of awareness in watching a film. Though Strawson is not concerned with film viewing, the possibility arises that being subjected to represented events within levels of narration may result in a state of mind that Strawson refers to as "seeing and hearing in a disembodied state," thereby transforming the viewer into a certain kind of "mute and invisible witness."[11] For Strawson, our embodiment may become 'other,' i.e., displaced and embedded within a new spatial-temporal framework or new level within an overall framework.

In the case of film, theorists have proposed many sorts of alternative and disembodied states of mind, which are personified as types of 'invisible observers' for seeing the depicted events of film. The most productive formulation, I believe, is that of George Wilson, who argues for a conceptualization based on a viewer's experiencing of film from an impersonal, "unoccupied visual perspective." Further, he says, it is "indeterminate where, if anywhere, I am" when viewing a scene in a film and "indeterminate" how the viewing came about. Wilson's approach leads to far-reaching and complex issues.[12] Note that in his formulation, the 'camera' we imagine to be providing information and moving inside a film to view characters is no longer a physical camera with a weight and serial number.

Wilson's analysis suggests that a viewer adopts a sort of alienated point of view where the self is being imagined as viewing events in the third person in a measured, disembodied way. Strawson remarks that a person in a disembodied state would need to live in the *memories* of his or her previous life in order to retain a sense of individuality. As these memories of a previous embodiment and personhood

faded, so, too, would a sense of self. Thus, when applied to film viewing, it would seem fortunate that we wake to find our bodies.

Following a return from any temporary absence from the spatial-temporal framework of the world, one may find that things have changed. Strawson (quoted earlier) allows for the possibility that material particulars that have undergone "gradual, qualitative change" will need to be reidentified. Presumably, the recognition of a "gradual, qualitative change" means that a reidentification has now been slightly *re*framed. This raises a question about how much change an object or person can undergo while remaining 'the same'; that is, a slight reframing is itself subject to being reframed.[13] When does a series of reframings finally lose its object to become disconnected, meaningless, or connected to something new? One may also wonder whether entities other than physical objects and persons may qualify as being reidentified, e.g., elements of narrative structure. For example, do the following qualify as being 'the same'?

> The same color but darker; the same object under different lighting conditions; the same object but from an unexpected side; the same object but now used as a metaphor; a new object with the same shape as a previous object; the same place, though changed; the same effect but different causes; the same cause but different effects; the same meaning but through different words spoken; the same words spoken but with different meanings.

> Are there any limits on the reidentification of immaterial objects? For example, a character's unspoken thought; a character trait; an action; an emotion; an idea; a misinterpretation; an event functioning as another *obstacle* for a protagonist tracking a goal; a theme.

The matter of change is crucial to our comprehension of narrative structure since storytelling involves producing linked sequences of events that both alter and partially restore states of particulars. Though Strawson did not have narrative discourse in mind, his general principle still applies: "We cannot attach one occasion to another unless, from occasion to occasion, we can reidentify elements common to different occasions."[14]

For Ruth Millikan, the linking of a series of occasions into a continuity through reidentification is "the most central cognitive ability that we possess."[15] Millikan draws extensively on the work of Gareth Evans, who expanded Strawson's concept to cover a much broader set of mental processes involving "*re*-identification" as it functions within a "social informational system," which presumably would include the system of an artwork and its consumers.[16] For Evans, reidentification is a process by which diverse information is collected and formed into "bundles" about a particular object, no matter the roles of transmission, perception, inference, ideas, and memory. Evans emphasizes that reidentification is vital to the formation of "evolving clusters of information generated in a pattern of encounters."[17] For Millikan

reidentification is itself recognized as such when it becomes a "middle term" in making an *inference* (i.e., in overcoming an ellipsis) or else functions within some "other amplificatory information-processing, and/or for guiding action."[18]

The idea being pursued by Millikan and Evans is that reidentification is not a static stop in perception ('I just saw *that* and here it is again'), but has "information-using purposes" that are incorporated in a train of thought and in "tracking" (!) activities and/or becomes ingredient to response and action.[19] An object's *being* is revealed as temporal, not fixed. Thus, for example, the mere repetition of the color red may not be 'noticed' by a perceiver until it is identified as recurring (i.e., reidentified) and, importantly, as functioning in a train of thought along with *other elements* of color that may be combined with references to material bodies, actions, conditions, and abstracta. Reidentification for Millikan "is necessary in order to collect together over time knowledge of a thing's properties" and "is also necessary in order to apply one's knowledge of things."[20] In short, what counts as being 'the same,' is not absolute, but relative to shifting contexts and the tracking of developing information.

How, then, might reidentification bear upon a spectator's perception and use of color when comprehending a film? There are at least four areas to examine.

First, I believe that not just individual colors may be reidentified to create a singular, very basic pattern across time, but more importantly a certain *group* of colors in one or several shots may be reidentified as a recurring, complex pattern. In general, I believe that patterns represent abstract clusters of information that may mesh with other thematic elements of film to create what Evans characterizes as "evolving clusters of information generated in a pattern of encounters. . . which serves as the means to identify opportunities for using old, and gaining new, information."[21] The crucial importance of patterns and of training oneself to find or create them has been a major theme of the present book. Strawson's "methods" and "criteria" of reidentification (quoted earlier) would seem to require that a spectator learn to recognize important color patterns and relations, i.e., acquire an array of viewing protocols for assessing color.[22]

Even more important is the particular *language* we have chosen to express our felt logic of experiencing and using color in our lives. We act according to how we see and we see connections that are relevant to our lives through attitudes, frameworks, norms, and concepts. The notion of language-games or language-fields has been a major theme of the present book. From this perspective, a particular language-game is an interlocking set of *patterns of thinking* about patterns of color.

Second, since color may be a strong marker, its reidentification and placement within a *new context* may be used to assemble or accumulate properties of a particular thing or group of things in order to refocus significance. This suggests that color may prompt or activate the formation of a metaphorical connection between things

that, in turn, may act to summon, inflect, or adjust a thing previously seen. A reidentified thing may appear in a new light or else it may be employed to project a new light onto some other object taken to be similar in some respect. The 'same' always means 'the same in some respect,' since context always exerts an influence. Recall Strawson's notion that reidentified things may undergo "gradual, qualitative change."

Third, repeated reidentification may act to accrue a rich body of information that pushes toward a propositional structure and the possibility that a text may embody a developing argument.[23] Shifting among levels of narration may also move a color pattern through a series of informational states.

Finally, reidentification together with movements of conscious attention are key components of working memory, which is the form of memory that drives a spectator's ability to track a flow of color around and through the elements of narrative film.[24] The crucial importance of tracking, of course, has been a major theme of the present book.

Conscious attention has a unique role to play in mental life. Consciousness is neither a kind of froth or surface disturbance—the result of material having passed through the censor from a deep, unconscious sea—nor is it a passive filter for external stimuli tagged by emotions on their way to distant memory stores or the unconscious. Instead, working memory plays an active role in mapping, reforming, and navigating an environment that is present or present only in mind, including a film that is present on the screen or only in memory. Reidentification, tracking, and bundling information that is undergoing change is a vital function of conscious attention.

1. THE OTHERWISE OF TEXTS: WORKING MEMORY, REIDENTIFICATION, AND THE "NEARLY TRUE"

> Yet the spectator cannot hope to apprehend, however incompletely, the being of any object that draws him into its orbit unless he meanders, dreamingly, through the maze of its multiple meanings and psychological correspondences. Material existence, as it manifests itself in film, launches the moviegoer into unending pursuits. . . . Does the spectator ever succeed in exhausting the objects he contemplates? There is no end to his wanderings.[25]
>
> —Siegfried Kracauer

Part of the power of working memory is that it works on the future. Since reidentification is an aspect of working memory and attention, I believe that reidentification has a role to play in a spectator's feelings toward a film's future. Thinking

about the future involves predictions under uncertainty of *possible* scenarios that may be identified as identical in certain respects—that is, similar in form, effect, or efficacy—to something present or something past that has now been made present. Working memory is able to simulate effects that create future similar, possible situations through reidentification. This includes a hue or pattern of hues that are felt possible but did not, in fact, appear when expected or else a near hue or pattern appeared instead. Also, there are "nearly true" *contexts* for a hue pattern that drift in thought—an important effect that will be discussed in the next chapter in conjunction with the theories of Jacques Derrida.

I wish to briefly elaborate on the nature of these future forms of reidentification since they are crucial facets of comprehending a film. The fact of reidentification—something past returned to the present—may also become tied to something not seen but projected as "nearly true," which may linger in the film as a continuing possibility (i.e., linger in a viewer's memory as an expectation) or be discovered as an actual outcome in the future.[26] This conjunction of present and past that is projected into the future is perhaps a special case of the historical present, which is a narrative tense that allows something past to be described with the present tense (past-appearing-as-if-in-the-present).[27] In reidentification, the past is being described in light of its return in the present as something significant or else the past is being described as a feature of the present, even though the present of a text exists only as implicit narration from an indefinite narrative "future"; that is, the end of a story is always present, though disguised, at each point of its telling in a plot. When something is deemed to be "nearly true," for example, the present is being mentally described with a future tense (present-appearing-as-if-in-the-future). This occurs because working memory is not confined to immediate sensation, but is able to juxtapose and project other time frames to contextualize a past thing in its old, new, or potential place. A special case of the "nearly true" is the feeling that something will become a future reidentification.

The notion of a "future reidentification" reminds us that, as Strawson argues above, persistent ellipses and gaps are inescapable in life. Gaps must be managed and prepared for. The same may be said about the stream of separate parts in a narrative film. A film's plot unwinds with unavoidable gaps and deferrals. What haunts and inspires a spectator are these imagined gaps that may appear within one's comprehension of a text, namely, the unknowns that elsewhere in life may become actual by coming into presence. Perhaps those plots in our world or in a film that strike us as the truest are the ones less tight—the ones most respectful of mind, of a world of possibility, of what is desired or feared but lying somewhere else or possibly next. Those plots that stress the nearly true act less in a unidirectional way and more as a shifting cloud surrounding a depicted action.

The importance of the nearly true is that such thinking opens toward a broader recognition of the stochastic nature of many fundamental aspects of the world and thus also of our representations of the world. Our comprehension of a world and of a text often displays a probability distribution that can only be analyzed in statistical terms of degrees of improbability, possibility, and likelihood. Thus also interpretations of a film by critics often fall across a broad spectrum as a series of responses to the probabilities engendered by a narrative. A critic or spectator who confronts representations is navigating a maze of hypotheticals every moment, evaluating what was nearly true, including what in the future may be reidentified or, though possible, not reidentified as expected. The import of a story thus outruns its end, its middle, and its beginning.

Philosophical texts, like film texts, harbor interior gaps as well as gaps in application to a world. Both philosophy and film enact generality and possibility in the effort of overcoming ellipses in descriptions of a world. In this performance, lies the potential for film to act philosophically.

TRACKING IN PLACE

> In all movies, we take it for granted that clothes and décor—purple, yellow, and purple again—have been chosen. . . because they fit, and life turns into meaning.[28]
>
> —"Visualize This, Bitches," chap. 4 of *Breaking Bad: The Official Book*

I would like to illustrate in outline fashion how color patterns may be reidentified and tracked in films. It will be seen how reidentification may play a role in generating narrative situation, metaphor, theme, and argument—the four areas mentioned at the end of section 2 above—as well as generating a sense of the nearly true by fostering thoughts of possible (future) reidentifications.

Descriptions of tracking are relative to three choices that must be made by an analyst, who must select:

1. A scale of analysis from micro construction to large-scale structure; the choice of a scale influences the selection and fashioning of parts with which to divide (segment) an object or text in a "meaningful" or "appropriate" manner in order to reveal its functioning and relevant patterns;
2. A type of color pattern to follow; for example, from low contrast continuity, coordination, passage, or blockage to high contrast balance or conflict; and,
3. A specific focus for linkages to follow that extend from a color pattern either to narrative structure and theme—i.e., connecting to *what* is happening in a story and its moral—or else to narration/telling—i.e., connecting to issues of

when and *how* narrative facts are being disclosed, which, broadly speaking, are problems involving point of view.

In addition, it may be beneficial to consider whether color elements are being arranged as a hierarchy or as a heterarchy.[29] The former organizes elements from constitutive and important to subsidiary and less important; for example, a prominent hue appearing with accents. Alternatively, elements may be arrayed in a heterarchy; for example, as a richly networked dispersal of hues on some basis of "equality" or simply as scattered hues. Scattered hues may possess a narrative function. Elements of a heterarchy may be classified, sorted, and related in several different ways or not at all.

1. *WOMAN IN BLUE READING A LETTER*: COLOR AS DICHOTOMY—HERE AND THERE, NOW AND THEN

Let us consider Johannes Vermeer's painting, *Woman in Blue Reading a Letter* (1663–1664). Its large-scale structure displays a series of high-contrast color dichotomies: light/dark, warm/cool, neutral hues/rainbow hues, and brown/blue. (Recall that neutral hues are usually found within a range of whites, blacks, grays, and browns.) One could perhaps imagine an even larger dichotomy that merges—i.e., stresses the connections among—the four elements of the preceding first terms of the four dichotomies, namely, light, warm, neutral, and brown. This half of the larger dichotomy might be set against a combination of all of the second terms of the four dichotomies. One would then need to invent a name in a new aesthetic system for the larger dichotomy congruent with the direction of a critical argument; for example, energetic colors/passive colors, striving/patience, duration/stasis, down/up, being/nothingness, or else something quite formal and arbitrary, say, type r-3/type s-8. In my analysis, I will stick with the four separate dichotomies.

One notices that the four dichotomies employed to display objects in the room are brought together to center on the depiction of a woman in the center of the painting. Her figure collects the four dichotomies. The title of the painting also draws attention to the woman and to the color blue ("woman in blue"), the only figure and the only non-neutral hue in the painting. Her face, parts of her arm, and portions of her (cool) blue top and (somewhat warm) brown skirt are lightened in patches to represent the dispersed effects of light from a window. Some of the browns in the painting tend toward a warm chestnut hue. One way that Vermeer draws attention to the display of the four dichotomies within the room that condense on the woman is through omitting to show a window that she is facing while reading the letter. This means that a viewer must search for a way to motivate or

naturalize the disparate lightened areas throughout the painting. The viewer discovers that the areas may be pulled together to form a simple pattern across the painting as a whole and on the woman in particular when it is assumed that a strong light source emanates from a window off left. By not seeing the window, a viewer may first discover a simple pattern of dichotomies and realize that the colors lead in abstract directions and not simply toward light from a window.

A second tactic by which a viewer is encouraged to order the painting's gradations of color into larger principles is through noticing the indirect effects of light. There is a very subtle, speckled reflection of blue from the woman's smock onto the light yellowish-brown wall in front of her belly as well as a darkening of the wall immediately behind her (because her body has cut off the light), creating a distinctly yellowish-brown hue that lies between the rich brown of the wall map above (more about the map shortly) and the rich blue of the seat of the woman's chair below. In both cases, the viewer sees how blue and brown fail to mix or connect to one another, hence forming a dichotomy. (Recall that blue is the only hue that cannot form a hybrid color with brown, i.e., there are no bluish-browns or brownish-blues.) This keeps in place a *conflict* in the room and prompts a continued search for a kind of narrative *resolution* of the dichotomies, which will be found, if at all, in the representation of the woman herself—the place where the four dichotomies meet.

The dichotomies produce additional principles that structure the painting. The woman's face and letter are lit to a bright white—indeed it appears that the brightness of the letter lights her face. Her hair and neck, however, remain quite dark and posed against a similarly dark, indistinct map on the wall. The dark brown cast of her head and of the map results in very little visual separation between head and map, as if, in some sense, she is partly part of the map and the map ingredient to her thought. Further, her clothes may reveal that she is pregnant. Her rounded belly contrasts with the dominant and sterile angularity of the room and its objects, although in counterpoint, there exist small circular metal fasteners on the chairs, pearls on the desk, and a small globe anchoring the map, which is composed of especially dim and disconcerting, chaotic broken lines. It is not clear what the map is mapping.

The letter the woman is reading may have come from the father-to-be, who may be traveling in some distant locale depicted on the wall map. If this is the case, the father is present only in the letter, map, and unborn child. Vermeer's painting may thus be seen to use objects in the room along with the shape and color of the woman's clothes to measure the distance between father and mother while the coalescence of the color dichotomies across the mother's body and clothes, maps her hope to overcome separation. Lost in reading and thought, she stands at the window

rather than sitting in one of the chairs in the room. Colors render the position of her head—face lit by the letter, the rest in dark—as *both* distant from, and attached to, the murky map on the wall. Evidently, color patterns in the painting have been organized and interlocked hierarchically in concert with a developing, nascent story of a woman wearing blue reading words on a page.

Isn't the painting as a whole a kind of letter? Don't we *apply* words at least silently to interpret and frame our experience of the painting as if reading? If not, we know that appropriate words would appear, as if summoned, when needed. We hear our native language at work in our heads as an aspect of working memory. Words may also embody emotions. In this way, language acts to mediate between the painted spots and lines conjuring an illusory space on a two-dimensional painted surface while testifying to the fact of a world of objects, sunlight, persons, and events familiar to us.

There is certainly a distinction to be made between images and words—our experience of them is different. But how far does *phenomenal experience* go toward defining an experience of art? Take color, for instance. There are animals that are blind, see only in black and white, or perceive colors humans cannot. Does an experience, then, of color alone—new colors, no colors—make for *art*? The answer one gives in the case of color will apply to other sensuous qualities, e.g., line, shape, texture, depth, movement. All animals sense; only humans make and see art. The difference is that humans possess languages that are able to form and make an awareness of a world *into art* whether starting from images or words. One should at least ponder what counts as art within each of the three main systems of the brain: the neocortex (rationality, planning), the limbic system (emotion, reaction), and the brain stem (breathing, appetite, mating, survival). Art does not reduce to sensuous form even though some art is nonfigurative. Because language mediates sensuous form and representation, i.e., significance, there is no pure spectacle capable of seizing and erasing thought.

Consider the following remarks by Wittgenstein about the connection between images and words:

> What makes my mental image of him into an image of *him*?
>
> Not any pictorial likeness.
>
> The same question applies to the utterance "I see him now vividly before me" as to the image. What makes this utterance into an utterance about *him*?—Nothing in it or simultaneous with it ('behind it'). If you want to know whom he meant, ask him!

> (But it is also possible for me to visualize a face, and even to draw it, without my knowing whose it is or where I have seen it.)
>
> Suppose that while imagining, or instead of imagining, someone were to draw, even if only in the air with his finger. (This might be called "motor imagery.") Here one might ask: "Whom does that represent?" And his answer would be decisive.—It is just as if he had given a verbal description, which, after all, can also *take the place* of the image.[30]

For Wittgenstein, verbal descriptions and images are interchangeable in an important place in the human mind, a place firmly occupied by art (cf. Plato's "images of speech," *Sophist* 234c6). Note especially Wittgenstein's fourth paragraph, where he connects verbal descriptions and imagery to human memory. (His example is an instance of a type of memory known as "recognition memory.") Don't we sometimes visualize in mind—think of—an investigation or interpretation as a process of discovering pieces of a puzzle, like fitting together words and evidence into a jigsaw picture? I believe that descriptions, whether in the form of words-become-images or images-become-words, are open to nonspecific references and to fictions. In the case of the Vermeer painting, this would be a declaration that Vermeer need not have known an actual, specific "woman in blue," but, rather, that the viewer of the painting need understand only certain things in general, such as what reading is, what pregnancy may mean, what a wall map is, etc., so that he or she may thereby be placed in a position *to narrate* what the painting shows by pulling together sensuous form and human behavior into a descriptive language that speaks generally of conduct and value: *a* woman, *a* letter, *a* situation.

In the present instance, Vermeer has split apart various sensuous qualities of color to fashion a language that captures a meaningful set of pairings in a viewer's world.

2. *THE COOK, THE THIEF, HIS WIFE & HER LOVER*: COLOR AS THEMATIC LOCALE

> Color as an independent and constitutive structure [in *The COOK, the THIEF, his WIFE & her LOVER*] organizes bodies and rooms, maps out zones of happenings; released from objects, color floats to potentially redirect and land elsewhere.[31]
>
> —Eugenie Brinkema

Next, let us examine a film that sets up its own series of color dichotomies in six distinct locales and impels characters to move back and forth through these areas. In fact,

FIGURE 7.1 Scheming and Plotting in Color: Peter Greenaway's Menu for a Spectator of *The COOK, the THIEF, his WIFE & her LOVER* (1989)

	Six Color Schemes					
Five Sources For Themes	**black blue**	**black green**	**black red**	**white**	**black gold/brown**	**grayish-yellow brown**
locale	outdoors	kitchen	dining room	bathroom	book depository	hospital
activity	arrive from, depart into, the night: netherworld maggots and meat wild dogs	food: fertility virility aria	power: consumption consumed abuse humiliation	purification sterility	study	infirmity
act	VIOLATE	(PRO)CREATE	ATE	EVACUATE	CONTEMPLATE	DETERIORATE
event	eruption	concoction erection penetration intuition sublimation	introjection (enforced masochism)	projection	digestion interpretation reflection	cessation (no more *meat*)
state of being	irreality	ethereality surreality	reality	corporeality	realization	nonreality

a character's clothes sometimes miraculously change color as he or she enters a new locale.[32] Specific color dyads are associated with distinct actions and themes that occur in these places. Character movements through the spaces define chains of actions and connections—a kind of narration driven by color—that build into arguments being made by the film's overall structure. When the six locales are interpreted as portraying certain functions of the human body, color becomes enacted physiology. Characters must accommodate to the hues of each room to await the black smack of corporeal potpourri: violate, (pro)create, ate, evacuate, contemplate, deteriorate, and eliminate. Color, thus depicted as the subject of rapacious pursuit, becomes a delicacy, like food, to be consumed and purged (cf. Figure 3.1, *color-as-delicacy* and *color-as-spectacle*). The film is Peter Greenaway's *The COOK, the THIEF, his WIFE & her LOVER*.

Figure 7.1 is a summary of Greenaway's color schemes and themes. The id dominates the first four locales, with the superego and ego squeezed into the book depository. The remainder of human aims lie in decline in the hospital. The action is based on animal urges versus a striving for the creation of art with little between, though all of it is presented by Greenaway with exquisite attention to a highly refined filmic art of spiritual music, graceful camera movements, meticulous choreography, and precise and rapturous color both in the image and in the image of paintings and painting-like tableaux in the mise-en-scène.

The unresolved paradox of the film is that humankind's basest impulses—represented as versions of eating/excreting—give rise to, or coincide with, its loftiest aspirations and sensibilities. Is evil in the film the dominant impulse while art is a peculiar byproduct, or instead, is everything joined in an elaborate and mysterious, dangerous dance? The insistent use of lightness—in the form of a black, white, or gray hue as one of the terms within each of the major dyads for the six locales—suggests the existence of an ominous linkage making for a close association among the depicted states of being. The colors of imagery and photographed paintings would seem to depict at once sublime reflection and ignoble physiology. Does the locale of the book *deposit*-ory, for example, hint that a book is also like a toilet? Do the film's images exceed their nominal objects to reveal base urges?

Perhaps Greenaway's film is a baroque rendering of Wittgenstein's idea (by way of Aristotle) that "the human body is the best picture of the human soul."[33]

3. *WINTER SLEEPERS*: COLOR AS CHARACTER FATE

Finally, let us consider a film in which it is the characters who are color-coded and whose interactions produce dichotomies and permutations that are governed by rules associated with both subtractive and additive primaries. Color patterns in the film exert a steady force that reveals a *fate* that is moving characters toward a final disposition of

their hopes and goals. Spectators of the film, but not the characters, begin to understand what the colors demand. In Tom Tykwer's *Winter Sleepers* (*Winterschläfer*, 1997), each of the six main characters is strongly associated with a particular hue:

Red woman—Rebecca/Becky (receptionist at ski lodge and book translator)
Green woman—Laura (nurse and theater actress)
Yellow woman—Nina (skier)
Blue man—Marco (ski instructor and philanderer)
Black man—Rene (film projectionist with short-term memory loss [so re-seeing a film he projects is always an exciting new experience. . .])
Brown man—Theo (farmer)

The film's plot works out the intricacies of a crime that is interconnected with various romantic entanglements among the characters. The plot may be summarized as follows:

> Red woman, who has Yellow-blond hair, meets her romantic (?) match, Yellow woman, on a train at the very end of the film as a Blue boy is born to Red woman's roommate, Green woman (while the father of the boy—admittedly taking a broad view of fatherhood on the part of Red woman's womanizing boyfriend, Blue man, who is also sexually involved with both Yellow woman and Alexa, his attorney—disappears skiing helplessly into a dark crevasse on a mountainside: the final vagina of his unfaithful ways?), which leaves Green woman with a new career as a mother and Black man with no memory of his role as a possible father of Green woman's Blue baby and no memory of his role in an auto accident when driving Blue man's stolen blue sports car that leads to the death of Brown man's 10-year-old daughter (whose nurse is Green woman), but who remains secure in his latest false or forgotten memory that he is the father of Blue baby. Or, perhaps, Black man's sight of Blue baby—who is seen in four close shots dressed in blue but with a blackish cloth on the bars of a crib—in the final scene brings about a true and present memory of being the father. A thoughtful expression conceals his passing thought. If a memory of some sort has, in fact, dawned for Black man at the end, it is fated to soon fade into nothingness—like the death of the other possible father, Blue man—due indirectly to Black man's memory disability.
>
> There are several other connections among the six main characters that are not mentioned in the above plot summary. One idea motivating the story is that life is a matrix of combinations and re-combinations against a snowy White landscape become a chilly blank canvas ruled by the fate associated with the coincidence of one's color. The matrix is too large, and the combinations too vast, for any one character to fathom.

> To the extent that the characters intermingle on an equal basis, and certain moral issues settle evenly among them, the hues associated with the characters enact a heterarchy, not a hierarchy.

The plot of *Winter Sleepers* seemingly rejects the notion that Blue man is the father of Blue baby and instead provides evidence that Black man is the father. But, one may ask, do there exist thoughts that lie alongside the plot or in spite of the plot?[34] I would like to suggest that when a spectator entertains what might be "nearly true" about characters and events, or finds possibilities within interstices and ellipses, the prospect arises and persists that Blue man is, or might or could have been, in the depicted world or an alternate world, the father of Blue baby. The color scheme—to be discussed next—points in the direction of Blue fatherhood, as do a few subtle stylistic events.[35] This may result in a feeling that Blue fatherhood is true or at least nearly true, i.e., as a possible (future) reidentification of Blue man, now dead, in terms of his offspring, Blue baby.[36]

Let us carefully analyze a number of the color patterns that are at work generating the action of *Winter Sleepers*. Blue and yellow don't mix (i.e., there are no bluish-yellow hues) and blue and red are opposites (i.e., cold and hot). Thus Blue man is not destined to be with either Yellow woman or Red woman. Nevertheless, the color blue, as one of the subtractive painter's primaries (blue, yellow, red), functions to mediate the threesome through a sexual relationship with each woman—though neither woman knows of the other—and thus at the end of the film Blue man acts to unknowingly "bring together" Yellow woman and Red woman, who face each other as strangers in a train compartment. Green woman and Red woman do not go together (i.e., there are no greenish-reds). Yellow and red, however, do go together (i.e., there exist yellowish-red hues). The two women, Yellow and Red, are thus destined to meet by chance through the mediation of the man, Blue.

The pattern among the characters that we've just considered (blue, yellow, red) makes up the subtractive painter's primaries. Let's now analyze the other triad of hues involving Blue man, namely, the additive light primaries, blue, red, and green. As indicated earlier, blue and red are opposites (i.e., cold and hot). However, blue and green do go together (i.e., there exist bluish-green hues). Thus Blue man does not belong with Red woman, but does properly belong with her roommate, Green woman, with whom he may or may not have a son, Blue baby. Furthermore, the three light primaries (blue, red, green) *add* to white, which suggests a positive outcome for Blue man's death in the birth of a possible son, Blue baby, with Green mother. By contrast, neither of Blue man's relationships with Yellow woman and Red woman are possible in terms of the color relationships (i.e., no bluish yellow exists and cold is opposed to hot). Moreover, Blue man's relationships with the two of them in the painter's triad of blue, yellow, and red, *subtract* to black, which is suggestive of loss and emptiness (Blue man's death), and in the case of Black man, emblematic of his inability to remember.

Even when a character is not present, his or her distinctive hue appearing elsewhere in the mise-en-scène reminds us of him or her, allowing us to sometimes think of the absent person as actually playing a part in the events of the scene. Color seems to function musically with sequences, pauses, overtones, and climaxes. The distinctive character-hues may also bridge scenes and generate new action. In this way, color functions like a soundtrack that bridges two scenes, i.e., carrying over a sound from the previous scene or anticipating a sound from the next scene. Blue man, for instance, wears a red robe after sex with Red woman; a yellow hue on Blue man's ski jacket and on his skis accurately predicts that he will meet and seduce Yellow woman.

In all of this, the color white acquires a special prominence in the palette of *Winter Sleepers*. It is exploited simultaneously for its conventional positive and negative attributes as a blank slate of active possibility and also as lifeless void. Early on white is tied to the *narration* of the film and hence becomes a primary vehicle of the overarching sense of fate that governs the action and storytelling. The most powerful narrations in a film always come from the future since the end of a story is known at its beginning and hence the telling of a story must be continually adjusted moment by moment in order to maintain secrets and build suspense.

I would briefly like to elaborate on the potency of white as a form of narration in *Winter Sleepers*. The film opens with white titles on a black background. Four main characters—Red woman, Blue man, Black man, and Green woman—are introduced with their names superimposed in white. Interspersed with the four introductions, plus a phone conversation between Red woman and Green woman, are dissolves to and from five long-held shots of quickly moving, bright white snowscapes and mountain trees as if from the point-of-view of someone skiing, as happens at the end of the film—i.e., happens after events yet to occur bring about a future that sends Blue man on a fateful ski trip down the mountain. In fact, Blue man's introduction shows him looking through his fingers at a window that shows only a blank, blinding white sky. When we next see Blue man, he is introduced by a dissolve to the fast-moving white roof of his blue sports car as he drives into a black tunnel. At the end of the film, he will die flying on his skis into a black mountain crevasse.

There is still more to white. Each of three shots of Green woman on the phone with Red woman at the beginning of the film feature an arcing camera movement left to right (in the same direction as the five white snow shots) that causes a series of vertical white bars to move across Green woman's face (analogous to the close-up of Blue man's fingers that act like bars as he looks through them toward the window of white sky). The four characters, and especially Green woman, are thus tied to Blue man, to the color White, and to Blue man's ill-fated ski run at the end. What joins and directs the characters, and sets their destinies, is an emblematic and enigmatic White. The White is the best trace we have of the on-rush of the film's invisible, omniscient narration.

No single character understands the full story; in fact, each of the six labors under a private, serious misconception at the end. A spectator is in the best position to reidentify and track the intricate circulation of events,[37] though he or she may continue to wonder about Blue baby's paternity. The moral issues involving the death of Brown man's daughter are similarly complicated and far from clear: Red woman, Green woman, Blue man, Black man, and Brown man all bear degrees of responsibility for the death, though none understand nor acknowledge complicity. Black man, because of an accident in the army, suffers from anterograde amnesia, a short-term memory deficit (the same damage to working memory suffered by the main character in Christopher Nolan's film, *Memento* [2000]). Black man must continually take photographs in order to remind himself of what he has done during the day. His ability to track and reidentify is severely impaired and consequently he has lost a critical aspect of personhood. His study of a key photograph that he made on the day of Brown man's daughter's death fails to bring back an important memory (black and brown are traditional disharmonies). This is perhaps a reminder to the spectator of *Winter Sleepers* that while photographs (and films) may not lie, even though they may be fictions, they may still fail to disclose important truth.

All the characters of *Winter Sleepers* suffer from a kind of blindness and forgetfulness associated with partial and misleading points of view on their lives. No one is fully in control of their personal identity or knows completely who they have unwittingly become in *light* of events. The characters do accept with reluctance what has happened to them, but with no sense that the mobility and permutations of color patterns—powered by an opaque White—have conspired to bring about a final pattern, their final state. Only the spectator is able assemble the pattern of patterns by carefully tracking the colors. *Winter Sleepers* demonstrates that narrative—where characters and colors stand for states and statements—is fully an argumentative form.

NOTES

1. *Hugo Münsterberg on Film*—The Photoplay: A Psychological Study *and Other Writings*, ed. by Allan Langdale (New York: Routledge, 2002; orig. 1916), "Attention," *The Photoplay*, chap. 4, pp. 79–80 (emphases added).
2. Richard McGuire quoted by Jennifer Schuessler in an article about McGuire's graphic novel, *Here*, "Sharing a Sofa with Dinosaurs," *New York Times* (September 26, 2014).
3. The interpenetration of illusion and reality was first raised in note 12 and the accompanying text of Chapter 2 above. Another reason that the binary of illusion and reality fails is that reality may be conceived as a series of levels, some governed by determinism and some by indeterminism, making illusion and reality relative to a series of steps or degrees, not an absolute binary. See Chapter 1 above, note 6, and Chapter 8 below, note 5. From a Kantian perspective: Melinda Szalóky, "The Reality of Illusion: A Transcendental Reevaluation of the Problem of Cinematic Reality," *Acta*

Univ. Sapientiae, Film and Media Studies 1 (2009), pp. 7–22. From another perspective, see Timothy R. Holland's review of Richard Rushton's book, *The Reality of Film: Theories of Filmic Reality*, in *Film-Philosophy* 16, 1 (2012), pp. 299–302.

4. P.F. Strawson, *Individuals: An Essay in Descriptive Metaphysics* (New York: Routledge, 1959), p. 9.
5. Strawson, *Individuals*, p. 15.
6. For color conceived as paint or cosmetic as well as some of its other material and non-material forms in thought, see Figure 3.1.
7. Strawson, *Individuals*, pp. 15, 31, 35–36, 55.
8. Strawson, *Individuals*, p. 31 (Strawson's emphases).
9. Strawson, *Individuals*, p. 32; cf. p. 36.
10. See, e.g., the entry "Diegesis" by Eleftheria Thanouli in *The Routledge Encyclopedia of Film Theory*, ed. by Edward Branigan and Warren Buckland (London: Routledge, 2014), pp. 133–137; and Branigan, "Levels of Narration" in *Narrative Comprehension and Film* (New York: Routledge, 1992), chap. 4, pp. 86–124. Note that in the text above, I am describing narration as a concrete feature of a text. Equally important, however, is a description of narration from the standpoint of a spectator who seeks the future of a story. The two descriptions differ. See Branigan, *Projecting a Camera: Language-Games in Film Theory* (New York: Routledge, 2006), p. 31 and pp. 237–238, n. 31. On levels of narration as a form of "emergence," see Chapter 1 above, note 6.

    "Mind-game" films are especially adept at complicating the levels of narration in a story and making evident their operation; see, e.g., *Hollywood Puzzle Films*, ed. by Warren Buckland (New York: Routledge, 2014), and my essay in this book, "Butterfly Effects upon a Spectator," pp. 233–264. On the many types of interrelated levels and frames relevant to perceiving and thinking generally about film, see Branigan, "How Frame Lines (and Film Theory) Figure" in *Projecting a Camera*, chap. 4, pp. 97–149.
11. Strawson, *Individuals*, chap. 3, "Persons," section 7, "Disembodiment," pp. 115, 116. Strawson explains how a feeling of disembodiment may arise (p. 115):

    > From within our actual conceptual scheme [of the world], each of us can quite intelligibly conceive of his or her individual survival of bodily death. The effort of imagination is not even great. One has simply to think of oneself as having thoughts and memories as at present, visual and auditory experiences largely as at present. . . whilst *(a)* having no perceptions of a body related to one's experience as one's own body is, and *(b)* having no power of initiating changes in the physical condition of the world, such as one at present does with one's hands, shoulders, feet and vocal chords.

    Strawson compares a hypothetical "disembodied" state of a person to that of a "spectator at a play" (p. 116). See the following note. Do the prisoners in Plato's cave *begin* in a disembodied state? See my entry, "Apparatus Theory (Plato)" in *The Routledge Encyclopedia of Film Theory*, pp. 21–33.
12. George M. Wilson, "'Le Grand Imagier' Steps Out: On the Primitive Basis of Film Narration" in *Seeing Fictions in Film: The Epistemology of Movies* (Oxford: Oxford University Press, 2011), chap. 2, p. 41; see also pp. 37–42, 45–51, 109–112, 138–139.

I construe Wilson to be arguing that there exists a visual equivalent of the subjunctive conditional mood—a counterfactual conditional. There may even exist a nested series of subjunctive conditionals to include, say, a character's point-of-view shot. This series would amount to a sequence of visual dispositional "sentences" that track a spectator's evolving interpretive stance about things already fictional, e.g., about a character's view in a room. Where are these "sentences" to be found? One could ask a spectator what he or she is seeing. Do different spectators find different sentences? What, then, is the *potential* of an image?

By contrast, if a theorist were to mistakenly think of screen images as "present" or in the present tense (or the past tense for some theorists and there are other variations), the theorist would be converting a subjunctive conditional into a present (or past) indicative and then wondering about the need for some observer for a view on the screen on the model of an observational sentence where the essential basis of cinema is believed to be photographicity, i.e., a sort of visual, propositional or quasi-propositional statement produced mechanically and chemically. Instead, a theorist must recognize that despite present sensation, screen images have *no* tense other than what is fictionally (conditionally) ascribed to images by a narrative discourse.

Is "seeing without being," i.e., having a point of view without being at the point, a contradiction? To hold that a disembodied view is, in fact, a contradiction, one needs to define what counts as "being there" (see previous note). How is one to be in a fiction? This may mean defining multiple kinds of "seeing from there" (imagining, dreaming, guessing, drawing a picture, seeing a picture, telling a story, hearing a story, remembering. . . .) associated with multiple kinds of "present being" or "being present." The result is that ontological entities may greatly proliferate in spite of the attempt to hold "being" together as a unity with a view from a distinct point separate from an objective, fictional reality whose view is not from a distinct point. The cost of maintaining anthropomorphism as merely an illusion—one can't really be there—and hence the failure of narrational theories that insist on the centrality of the human is apparently to create a vast proliferation of ontological entities, even several different entities for each occasion. An alternative approach based on the subjunctive mood is more appropriate for theorizing narration because in a narrative at any one moment we don't know everything: the narration works to ensure that certain things are left open along with encouraging curiosity and suspense, concealing secrets and potentialities, and employing a judicious dollop of misdirection and misrepresentation. That is, we only seem to be present when judging the film's "present." Thus our seeing something in a film and evaluating it in relation to our lives may remain anthropomorphic but without a body being in a definite place.

What a sentence displays are words, but that's not usually what we see in a sentence. Similarly, what an image shows are graphic shapes and two-dimensional spaces (etc.), but that's not usually what we see in or through a narrativized image. And for both the sentence and the image, there is a variety (hierarchy? heterarchy?) of modes of seeing and selves that may be invoked by, for example, narrative context, e.g., the subjunctive conditional. It's not the *objects* in an image—or the points or persons from which they are seen—that have the last word, but how they are related in a spectator's thought. A close friend of mine and a sophisticated viewer was crying at the words and actions of a puppet in a live-action film. The movements of thought are not illusory.

The subjunctive mood represents something not as a fact but as contingent, merely possible, and marked by doubt, wish, or desire, as in the verbal forms "would," "seem," "could," "should," and "might." Here also is the notion of "missed opportunities," the "nearly true," "deferral," and "paths not taken," i.e., "paths not presently on." Is a spectator deluded in thinking about alternative paths? Is fiction a profound mysterium or, rather, does it simply require a different temporal

structure than nonfiction? It is important to note that a contrary-to-fact conditional does not necessarily mean contrary to *all* facts; indeed, it usually presumes many facts. See, e.g., Nelson Goodman, *Fact, Fiction, and Forecast* (Cambridge, MA: Harvard University Press, 4th ed. 1983). A film shot through with the subjunctive as well as unusual serpentine color is Wong Kar-Wai's *In the Mood for Love* (2000). For more on fiction, see Chapter 4 above, note 11.

13. The problem of deciding when something has remained essentially the same—retaining the same "form" or "identity" or important properties—in spite of change is a philosophical issue with an exceedingly long history and involves the nature and role of "causation." My approach to some aspects of the problem may be found in *Projecting a Camera*, chap. 5, "When Is a Camera?," section "Sustaining and Other Causes," pp. 178–191.

    A crucial part of the process of recalling something—reidentifying it—is that one is retrieving a memory *of* a memory, not retrieving an original copy firmly stored and stable in the brain. Hence, already in reidentification there exists the potential for qualitative changes to a given memory due to a range of contextual influences. On memory "reconsolidation" and other memory processes that are involved in qualitative changes to a memory (e.g., Freud's deferred revision) and are operative in comprehending and interpreting media, see Edward Branigan, "If-Then-Else: Memory and the Path Not Taken" in *Transmedia Frictions: The Digital, the Arts, and the Humanities*, ed. by Marsha Kinder and Tara McPherson (Berkeley: University of California Press, 2014), pp. 45–79; on "reconsolidation," a memory of a memory, see pp. 49, 73 n. 16, 79 n. 46. There may also exist memories of the future, i.e., when we expect to reidentify something in the future; narrative "foreshadowing" would be an example. See Branigan, "If-Then-Else," section "Implicit Memories of the Anticipated Future," pp. 65–68. See text below accompanying note 26 on the "nearly true." For a philosophically informed account of memory knowledge based on simulation rather than causation, see Kourken Michaelian, *Mental Time Travel: Episodic Memory and Our Knowledge of the Personal Past* (Cambridge, MA: MIT Press, 2016).

14. Strawson, *Individuals*, p. 32.

15. Ruth Garrett Millikan's complete statement: "I have tried to show that the ability to reidentify things that are objectively the same when we encounter them in perception is the most central cognitive ability that we possess." Note that what has been reidentified is relative to a specific framework. There is no necessary implication that what is perceived and reidentified in the framework of a film—i.e., within a specific level of the film's narration—is also an objective particular in the framework applied by a person to his or her world. *On Clear and Confused Ideas: An Essay about Substance Concepts* (Cambridge: Cambridge University Press, 2000), p. 109.

16. Gareth Evans, *The Varieties of Reference*, ed. by John McDowell (Oxford: Oxford University Press, 1982), pp. 125, 126. Although Evans distinguishes his general concept of "*re*-identification" from the more limited one of Strawson by using an italicized prefix and hyphen, I will pursue broad and expansive forms of reidentification, but retain Strawson's orthography. Strawson's example of reidentification seems deliberately reductive and in need of expansion. Strawson discusses a drawing of nine simple geometric figures where the square at the bottom right is reidentified as the square from the top left. Strawson, *Individuals*, p. 33.

17. Evans, "Recognition-Based Identification" in *The Varieties of Reference*, p. 126; chap. 8, pp. 276–277. A subtype of declarative long-term memory is "recognition memory," and includes the possibility of a merely vague feeling that a thing seems to be familiar, though it cannot be immediately reidentified. Related to reidentification, I believe, are effects of "primacy" (long-term, explicit

memory), "priming" (implicit memory), and "recency" (working memory) where a specific type of recall impacts the present. Also related are *déjà vu*, the uncanny, *jamais vu*, and misidentification. According to Evans there are a number of anomalous, but important variants of reidentification where information, encoding, retrieval, recognition, and familiarity are misaligned or else the variants are entertained but negated by degrees; pp. 277, 296–298. Presumably, such cases may be fashioned/refashioned and productively mined by artists. See Branigan, "If-Then-Else."

18. Millikan, *On Clear and Confused Ideas*, p. 144.
19. Millikan, *On Clear and Confused Ideas*, pp. 38, 144–146; cf. 80, 151, 188–192. Reidentification is explicitly tied to "tracking," pp. 76–80, 155–158, passim; and tied to "language," pp. 92–94.
20. Millikan, *On Clear and Confused Ideas*, p. 38.
21. Evans, *The Varieties of Reference*, p. 277.
22. On methods and criteria of reidentification, see Strawson, Individuals, pp. 31, 32, 55, 207. On the importance of patterns, Strawson states that our conceptual scheme "must allow for discontinuities and limits of observation. So it must lean heavily on what we may for the moment call 'qualitative recurrences'—that is to say, on the fact of repeated observational encounters with the same patterns or arrangements of objects" (p. 33).
23. On reidentification and propositional structure, cf. Evans, *The Varieties of Reference*, pp. 296–298.
24. On the connection of working memory with our ability to track, see Chapter 3 above, note 3, and accompanying text. On ellipses created through the ebb and flow of consciousness that are a part of working memory, see Chapter 3 above, note 10, and accompanying text. It is perhaps instructive that Hugo Münsterberg in the epigraph to the present chapter employs a color term in describing the movement of working memory, thus transferring movement to color: "Everything must be shaded by attention and inattention."

    Considering the presence of pervasive and unnoticed ellipses in our lives, perhaps philosophers have been too much concerned with the issue of obvious forms of perceptual illusion. The film medium itself, of course, is constructed upon ellipses starting with individual film frames and editing. In general, I believe that a broader approach to cognition, memory, and reality is required to adequately theorize film since there is often a component of "reality" in seemingly illusory and elliptical presentations. See generally, Daniel D. Hutto and Patrick McGivern, "How Embodied Is Cognition?," *Philosophers' Magazine* 68 (January 13, 2015) (online journal: <www.philosophersnet.com>). Hutto and McGivern mention that an "embodied" approach to cognition includes a series of closely related approaches: enactive, extended, embedded, ecological, engaged, emotional, expressive, and emergent. See the preface above, note 4; Chapter 2 above, notes 1 and 9; Chapter 5 above, note 37; Chapter 6 above, note 12.

    I will more or less use the terms "reidentification" and "tracking" interchangeably in describing the recurrent presence and variation of a textual feature. If necessary, a distinction can be made on the basis of whether the recurrent feature is continuously observed (tracking) or observed at different times (reidentification). Psychologists refer to sustained attention or concentration on particular features as "vigilance."
25. Siegfried Kracauer, *Theory of Film: The Redemption of Physical Reality* (Princeton, NJ: Princeton University Press, 1960), p. 165. Kracauer says that a spectator approaches "the murmur of existence," after having traversed the "maze" of an object's "multiple meanings" and "after having probed a thousand possibilities." I owe the phrase the "otherwise of texts" that appears in the subheading to Daniel Morgan (personal correspondence).

26. "Nearly true" may mean "about to happen" or "didn't happen, but might easily have happened" or "may yet happen in the future." On the importance of what is "nearly true" to the comprehension of texts, see Edward Branigan, "Nearly True: Forking Plots, Forking Interpretations—A Response to David Bordwell's 'Film Futures,'" *SubStance* 97, v. 31, n. 1 (2002), pp. 105–114; Branigan, "Butterfly Effects upon a Spectator," *Hollywood Puzzle Films*, pp. 233–264. See also notes 11 and 12 above; and Benedict Carey, "Study Shows Brain Stores Seemingly Trivial Memories, Just in Case," *New York Times* (January 21, 2015) (on "retroactive consolidation"). See generally Marie-Laure Ryan, "Diagramming Narrative," *Semiotica* 165 (2007), section 4, "Diagramming Time," pp. 22–29. Three examples of "possibility" as an explicit theme are Kate Atkinson's novel, *Life after Life* (New York: Little, Brown, 2013), Paul Auster's novel, *4 3 2 1* (New York: Henry Holt, 2017), and the film, *About Elly* (Asghar Farhadi, 2009).

    In watching a film, a spectator tracks the action while projecting hypotheses and employing memory. The momentum created by text and thought finds expression in acts of projection. On diverse types of memory in media, see Branigan, "If-Then-Else," pp. 45–79. See generally, Michael Spivey, *The Continuity of Mind* (Oxford: Oxford University Press, 2008).

    As an illustration of the "nearly true," consider these two counterfactuals:

    1. If I were to drop this automobile or this time machine [or whatever, but I do not have anything at hand] out the window, it would be accelerated by gravity at (about) 32 ft./sec.^{2}
    2. If I were to drop a pear out the window [which I am holding and plan to drop], it would be accelerated by gravity at (about) 16 ft./sec.^{2}

    The first fiction is true, since by physical law anything dropped out a window accelerates at that rate; the second is false, but nearly true, because 16 ft./sec.^{2} may be thought to be nearly 32 ft./sec.^{2} and, further, in a folk psychological sense that ignores strict logic and is prominent in film comprehension, a pear is more likely than an automobile or time machine to be dropped from a window. From this standpoint, then, "realism" in film is not an image that looks familiar; instead, cognitive and contextual factors of our thinking must be evaluated.

27. Here is an example of the historical present tense:

    > Scottie followed Madeleine in his car as she visited several places in San Francisco and was surprised to see her *arrive* at his apartment building, *emerge* from her car, and *walk* to the door of his apartment *carrying* a letter. Scottie *parks* his car, *gets out*, *starts walking* toward her, and *says*, "A letter for me?"

28. "'Visualize This, Bitches,' section, 'Color Usage and Meaning'" in *Breaking Bad: The Official Book*, ed. by David Thomson (New York: Sterling, 2015), chap. 4, p. 132 (ellipsis points in original).

29. Philosophers have created many types of hierarchies and heterarchies from Plato's method of "collection and division" to Gilles Deleuze and Félix Guattari's "rhizome." It would seem that a key question involves making a decision about what sort of logic will govern how the system connects and expands outward, for example, from the top down, bottom up, or laterally. Perhaps also, there may be several ways to visualize a "direction" of relevant movements in the system. For a glimpse at a diversity of possibilities, see Manuel Lima, *The Book of Trees: Visualizing Branches of Knowledge* (New York: Princeton Architectural Press, 2014); Caroline Levine, *Forms: Whole, Rhythm, Hierarchy, Network* (Princeton, NJ: Princeton University Press, 2015); Nikolaus

Gansterer, *Drawing a Hypothesis: Figures of Thought* (New York: Springer-Verlag/Wien, 2011). In addition, consider note 42 of Chapter 8 below.

30. Ludwig Wittgenstein, *Philosophical Investigations*, trans. by G.E.M. Anscombe, P.M.S. Hacker, and Joachim Schulte (Oxford: Wiley-Blackwell, 4th ed. 2009), Part II, §§ 17, 18, which is now known as *Philosophy of Psychology-A Fragment*, III (Wittgenstein's emphases; double and single quotes in the original); cf. § 23 (IV).
31. Eugenie Brinkema, *The Forms of the Affects* (Durham, NC: Duke University Press, 2014), p. 173.
32. Eugenie Brinkema notes that in Greenaway's film "the changing colors of the spaces and the metamorphosis of Georgina's fashion [as she moves from room to room] are surprisingly under-analyzed in criticism." Brinkema argues that "In place of colors belonging to the mise-en-scène. . ., color is from the outset claimed by the mise-*n'en*-scène, put into the film as what fails to be put into the scene except as its instability." (my emphasis). *The Forms of the Affects*, pp. 172, 173; see also pp. 87–89, 231–235.
33. Wittgenstein, *Philosophical Investigations*, Part II, § 25 (*Philosophy of Psychology—A Fragment*, IV). For Aristotle, the soul is a range or set of human capacities or potential abilities that are possible to acquire. "The soul must be a substance in the sense of the form of a natural body having life potentially within it. But substance is actuality, and thus soul is the actuality of a body" (*On the Soul*, 412a20). "It follows that the soul is analogous to the hand; for as the hand is a tool of tools, so thought is the form of forms and sense the form of sensible things" (432a1).
34. On a spectator's thoughts that lie apart from the plot, see note 21 above. The first shot of *Winter Sleepers*, for example, is a close-up of Red woman's fist with her upright. . . erect. . . thumb oozing a drop of blood. This image doesn't fit with the immediate action, so where does it—will it!—fit?
35. Among the subtle events that point toward Blue man, rather than Black man, as the father of Blue baby, are the following. In the pre-main-title "introductions" of the central characters, we see Blue man driving his blue sports car into a dark (symbolic?) tunnel (cf. his death in a crevasse at the end of the film) and then see emerging from the darkness of a tunnel a train in which Green woman is seen staring out of the train's window. Is the time frame between Green woman's sexual encounter with Black man sufficient for pregnancy? Consider also that in a major scene at the beginning of the film before she has met Black man, Green woman inexplicably faints and is not conscious for a considerable time; and she again faints in a later scene prior to the scene of having sex with Black man. The fainting is never explained; is she already pregnant? Further, are there suppressed ambiguities in several scenes between Green woman and Red woman involving the person of Blue man? On the other hand, when Green woman is pregnant and lying with Black man, she wears a green top and black bottom.
36. On the "nearly true" and possible (future) reidentifications, see text accompanying note 26 above.
37. Like *Winter Sleepers*, the characters in *Teknolust* (Hershman-Leeson, 2002) are color-coded. The dominant color for characters and locales is keyed to each of five levels of reality in *Hero* (Yimou Zhang, 2002). Characters in *Reservoir Dogs* (Tarantino, 1992) and *Clue* (Lynn, 1985) are named for colors. And let's not forget the Power Rangers.

    Here are a few films, like *Winter Sleepers*, that depend on a "circulation" of events with no outlet: *Man from Reno* (Boyle, 2014); *Birdman or (The Unexpected Virtue of Ignorance)* (Iñárritu, 2014); *Sex and Lucia* (Medem, 2001); *Lovers of the Arctic Circle* (Medem, 1998); *After Hours* (Scorsese, 1985); *The Earrings of Madam de* (Ophüls, 1953); *La Ronde* (Ophüls, 1950).

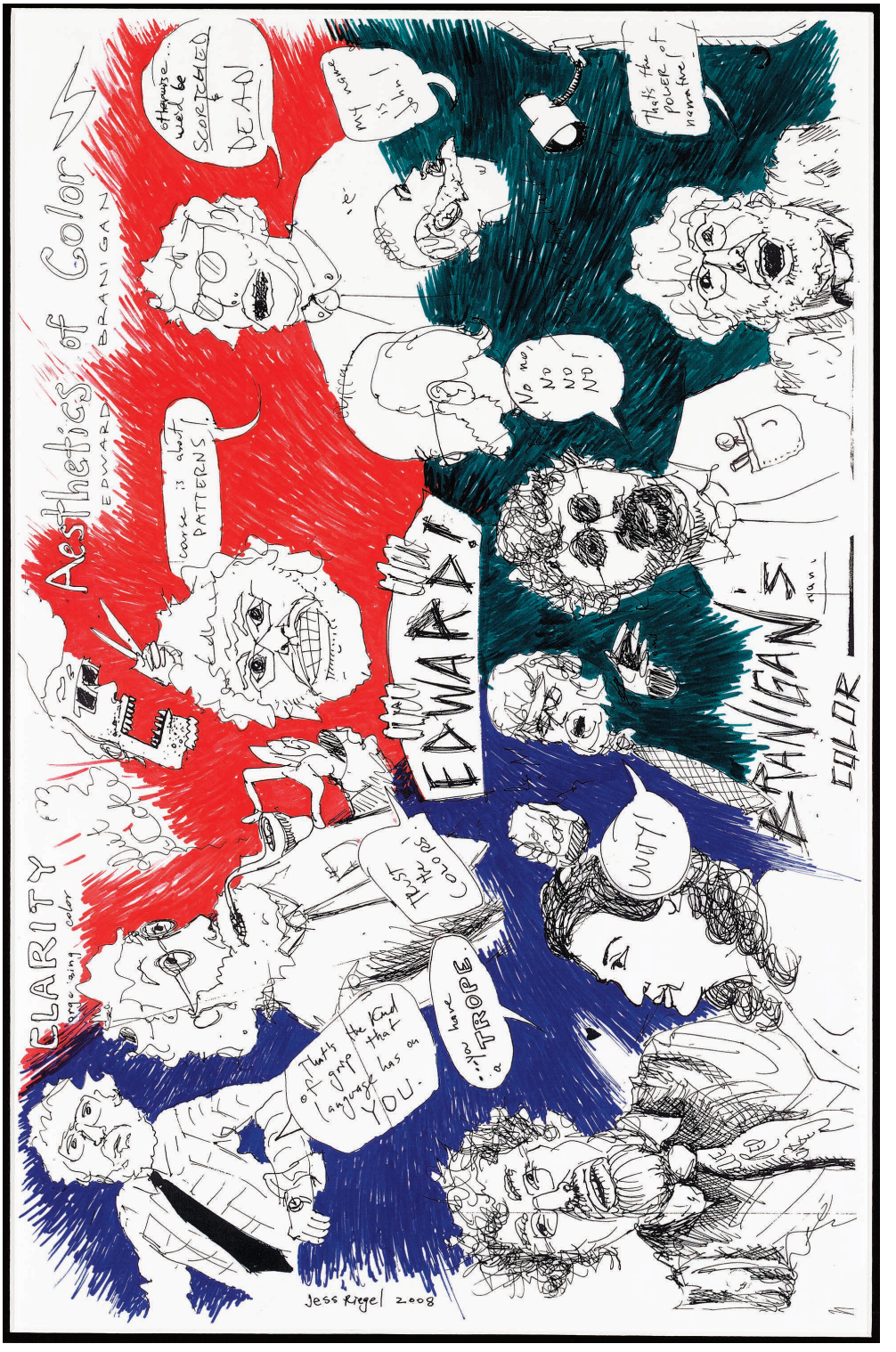


FIGURE 0.1 The Basics of My Aesthetics of Color Course

Jess Riegel, *Aesthetics of Color: Edward Branigan* (2008), pen and ink on paper, 11 × 17 in., from the collection of Edward Branigan. Photo by Ron Finley Studio—Westlake Village, California. Some of the dialogue from the drawing: "Course is about PATTERNS!" "Otherwise we'd be SCORTCHED & DEAD!" "That's the kind of grip that language has on YOU." ". . . you have a TROPE." "Trust the colors!" "UNITY!" "That's the power of narrative!" "My name is John [Kurten]!"



FIGURE 1.1 Roy Lichtenstein, *Two Paintings: Dagwood* (1983), oil and magna on canvas, 90 × 64 in. (228.6 × 162.6 cm) © Estate of Roy Lichtenstein



FIGURE 6.1 *Art Imitating Life Imitating Art*; photograph by Melinda Szaloky (2016) incorporating an untitled oil painting by Erzsébet Fazekas (1996), both from the collection of Edward Branigan.

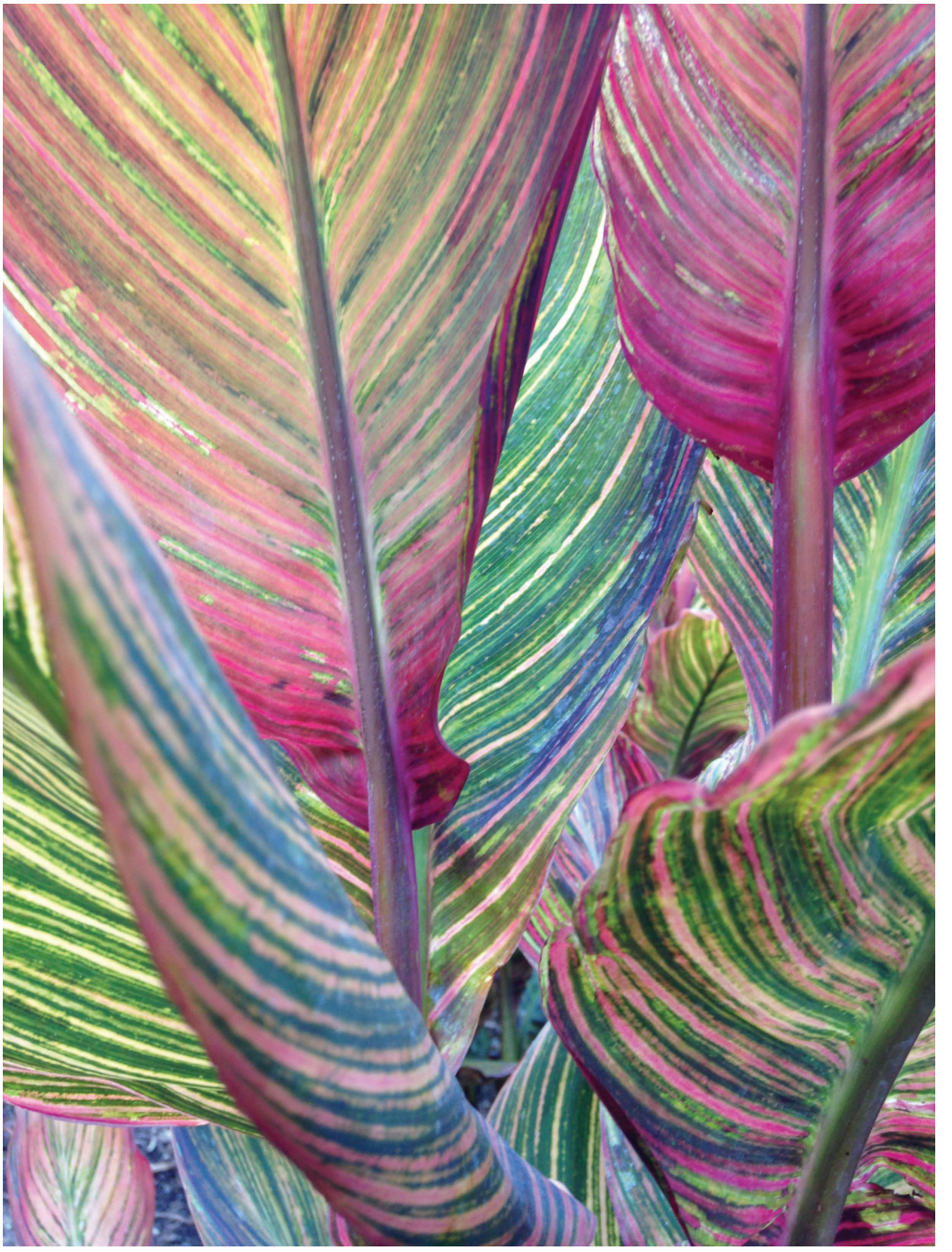


FIGURE 6.2 *Morning at Colorado Center, Santa Monica* [California]; photograph by Melinda Szaloky (2016) from the collection of Edward Branigan.



FIGURE 6.3 *Howard Hawks' Gentlemen prefer Blondes*
(Howard Hawks, 1953; Technicolor Process Number Four, third generation Eastman Kodak negative film stock, which represented a shift from a nitrate to acetate base, digitally restored; Leonard Doss, Technicolor Color Consultant)



FIGURE 6.4 *Picnic* (Joshua Logan, 1955; Eastmancolor CinemaScope released in Technicolor Process Number Five; Henri Jaffa, Technicolor Color Consultant)

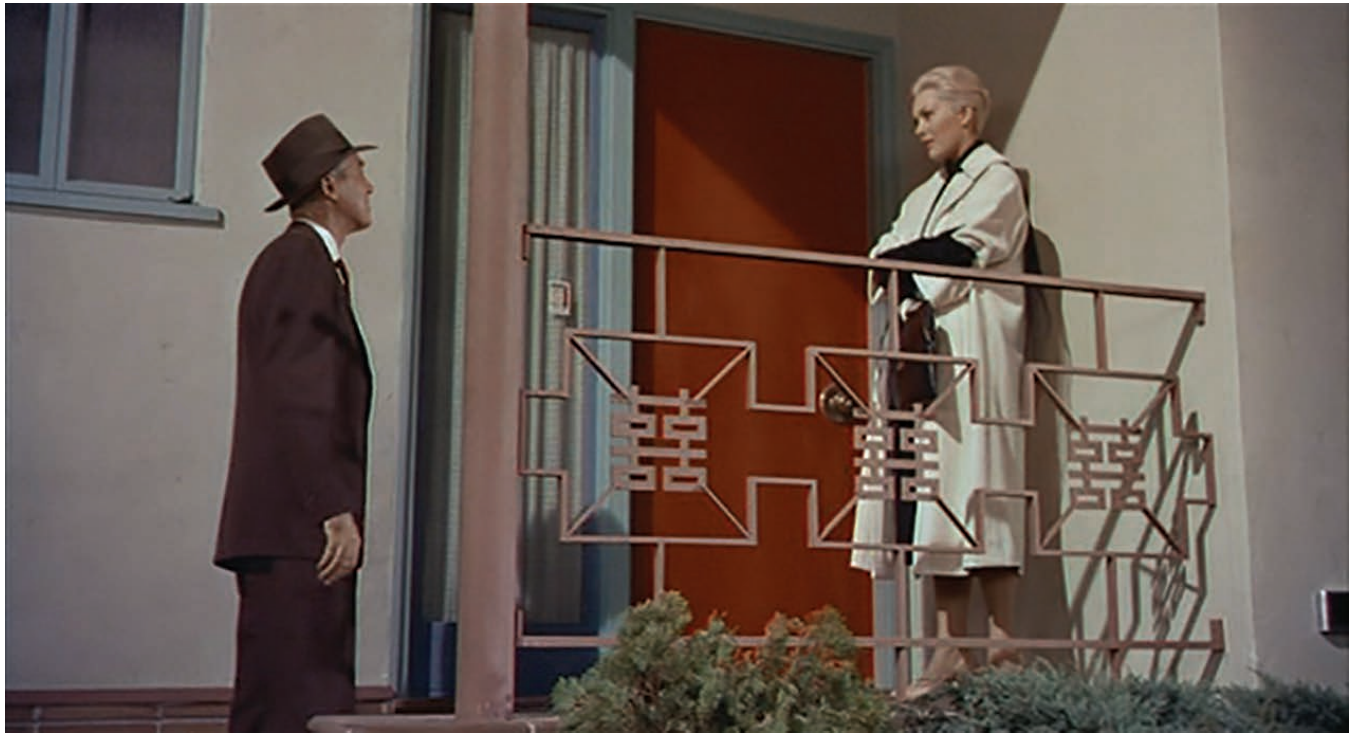


FIGURE 6.5 *Vertigo* (Alfred Hitchcock, 1958; Eastmancolor VistaVision Technicolor Reduction Dye Transfer, Technicolor Process Number Five, digitally restored; photographed in an aspect ratio of 2:1, but "ideally" composed for 1.85:1, or the equivalent for video of 16:9, as reproduced here; full-frame versions of the film can be found) (shot 1; this set-up appears once in the scene)



FIGURE 6.6 *Vertigo* (shot 2; this set-up appears six times in the scene)



FIGURE 6.7 *Vertigo* (shot 3; this set-up appears seven times in the scene)



FIGURE 6.8 *The Wizard of Oz* (Victor Fleming, 1939; Technicolor Process Number Four, first generation Eastman Kodak negative film stock, though some evidence points to second generation, digitally restored; Natalie Kalmus, Technicolor Color Director, Henri Jaffa, Associate)



FIGURE 6.9 Robert Janson (blue shirt), his friend Roger, and the shortwave radio in *2 ou 3 choses que je sais d'elle* (*Two or Three Things I Know about Her*, Jean-Luc Godard, 1967; Techniscope Eastmancolor)



FIGURE 6.10 Juliette Janson addressing the camera while Robert and Roger are heard off-screen with the shortwave radio in *2 ou 3 choses que je sais d'elle*

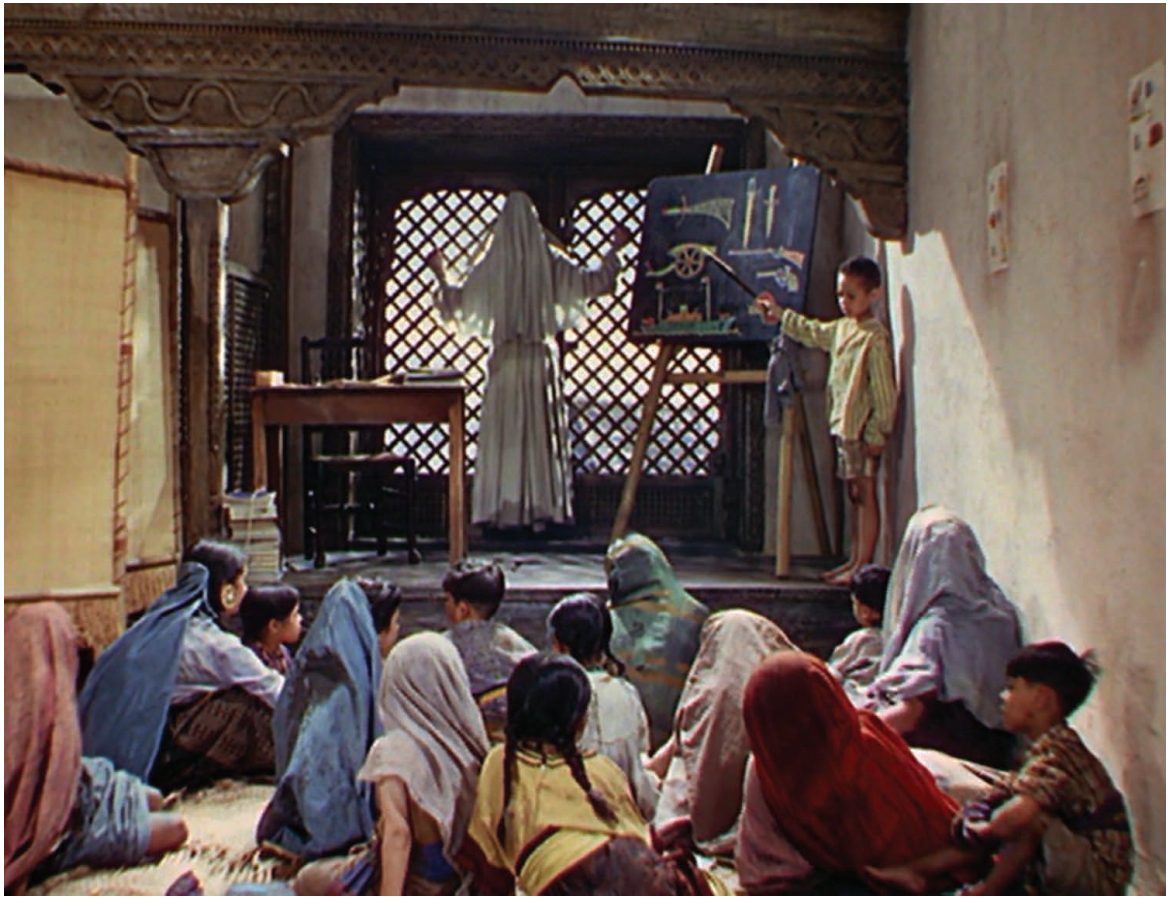


FIGURE 8.1 Sister Ruth's classroom in *Black Narcissus* (Michael Powell and Emeric Pressburger, 1947; photographed in Technicolor by Jack Cardiff utilizing Technicolor Process Number Four with second generation Eastman Kodak negative film stock; Natalie Kalmus [pro forma], Colour Control, Joan Bridge, Associate)



FIGURE 8.2 Sister Ruth's glass of milk (frame left) in her classroom in *Black Narcissus*



FIGURE 8.3 Sister Ruth spying on Sister Clodagh and Mr. Dean in *Black Narcissus*



FIGURE 8.4 Dissolve from Sister Ruth spying on Sister Clodagh and Mr. Dean to an outdoor scene of natives beating on drums around a fire in *Black Narcissus*



FIGURE 8.5 Presumably (and later shown to be) somewhere in the Palace/Convent of Mopu after a three-shot outdoor scene of natives beating on drums around a fire in *Black Narcissus*



FIGURE 8.6 Sister Ruth in a crimson dress confronts Sister Clodagh in *Black Narcissus*

Chapter 8 **Track That in Movement**

ABSTRACT

In this chapter, I examine the situation where a color reappears in a new guise, i.e., when a hue appears to shift and change its color. This leads to a definition of four basic types of reidentification and tracking that involve the cognitive notion of *radial association* and the philosophy of Derrida. A radial series is a series of contexts, each of varying strength that continues to inform an often arbitrary starting point. An advantage of this approach to film comprehension is that one can readily see how moving along certain radial lines of thought will bring parts of a metaphor together, literalizing it, and making real a provisional fiction, i.e., a counterfactual that has been proposed by a spectator or prompted by a text. Is the single plot of a fictional story really the only strand to find? Does a story stand only for itself? I think instead that a story is about a lived reality complete with contingencies. This augmented story is the interface between world and art.

In the previous chapter as well as the present one, I employ a binary of tracking color either in place, when it seems not to change, or in movement, when it seems to change hue. I conclude this chapter by discussing how the binary may be condensed into a single mental image of a spectator walking through an ornamental garden. Some formal techniques of film are mentioned that may drive color tracking. I also consider whether color "movement" is merely a somewhat specious metaphor. It is not.

> For [Jean] Epstein, [cinema generates] a new dimension of movement, where everything that appears to us as solid is shown to be in constant movement. Cinema transgresses the solid perception and renders objects into a liquid or even in a gaslike state.[1]
>
> —Trond Lundemo

THEORETICAL EXCURSUS (IV): WHAT MAKES COLOR MOVE

What animates movement? What makes things go and brings on a spectator's tracking? To start an answer, let us back up.

The Pre-Socratic philosophers began by worrying about the nature of material itself, the thing that moves: what stuff makes up the world and cosmos (for example,

Empedocles, Hippocrates)? They found four basic materials: fire, air, water, and earth. But, then, what causes these materials *to move*, including the creation of various intermixtures and transformations; in short, what makes for change? Assuming that movement and change are not simply fundamental illusions (as Parmenides, Zeno of Elea, and Melissus believed), one of these four materials may be taken to be a primary mover of the others: fire (Heraclitus; fire and night, i.e., light and dark for Parmenides), air (Anaximenes, Diogenes), water (Thales), or earth (Xenophanes). Or, a fifth material may be added to account for special heavenly movements, e.g., aether (Aristotle). Perhaps, instead, there is something still more primordial underlying these four or five materials, such as the unlimited action of the *apeiron*, which is an unseen, volatile mixture of the opposites hot and cold, wet and dry (the four combinations of which yield the four basic materials for Anaximander), or certain other invisible dynamic opposites (Anaxagoras, Heraclitus), or the limiting action of certain mathematical objects (Pythagoras), or else actions of the soul (Plato).

Alternatively, perhaps mobility is simply intrinsic to material so that movement arises from *within* each of the four basic elements due to indestructible "atoms" forever in motion (Leucippus, Democritus, and the startling ideas of the Hellenistic philosopher Epicurus). This last answer is very close to the answer of modern physics. However, modern physics does not directly address movements of immaterial objects, even though Democritus believed that mental objects, e.g., perceptions and thoughts, were equally composed of *physical atoms*. Today, of course, the nature of the connection between events of the mind and the physics of the brain remains a subject of vigorous debate.

Let us now jump ahead and work backward toward the material world from, say, the movement of color in mind. We might look for mobility in all of the same places—inside and out[2]—explored by the Pre-Socratics as they searched for the source of change in material and mental things. Do material things in the world move color, even if they are not directly seen to be at work? For example, one might attribute the movement of color in film to a film projector, or to a character or camera that actually moved when the film was being shot, or to a juxtaposition created in post-production by splicing shots, or to the ideas of an art director. Perhaps, instead, color moves intrinsically because of what it is or has become. For example, the sight of a green hue—or the thought of an archetypal greenness—may spread through parts of the mind, making us see green in a new light that collects material things literally and figuratively green along with feelings and meanings directed toward, i.e., moving toward, other things literally or figuratively joined and related to green.[3] Alternatively, there may be a primary mover of color, such as an imposing line (line, outline, drawing, *disegno*) or the actions of a light-and-dark design or key color that enforces a hierarchy upon hues.[4]

Furthermore, it may be that a color pattern that lacks one of its elements—making for a kind of negative space within the pattern—creates a "vacuum" or exerts a pressure that calls to mind the missing element (cf. the visual illusion of a subjective contour). If so, this suggests that still broader contexts, such as narrative structure and color theory, may push and pull hues. A pattern or a pattern-in-the-making is more than a collection or list of elements; one sees in a pattern, or in an emerging pattern, a mini-logic worth remembering, for it helps to simplify and explain still larger groupings of data. A pattern arises not from a single material source, but from *relationships* found in mind among specific dispersed sets of materials or nodes in a structure. This suggests that the fact of being unlocalized or distributed may produce the feeling that a pattern, though demonstratively real, also possesses an illusory, immaterial quality, which is neither quite here nor there but something fleeting and mental, something existent in working memory.[5]

To complicate matters, a color may continue to look the same when reidentified while its significance and use within larger patterns changes dramatically; that is, its position and function in thought may develop in new directions through time. In this approach to color movement, the language activity of working memory—the phonological rehearsal loop and visuo-spatial sketchpad—mediates between the memories we live by and the world and films with which we seek to interact. (Memory is not just about the past, not just about how we *have lived*, but equally about the future, about prospective action.) Language shifts color from place to place to be reidentified and sometimes relocated within new patterns. In general, our use of language echoes the fact that we think of objects as moving whether they are material, immaterial, or hypothetical. Our search is never-ending for regularities, patterns, and patterns among patterns through acts of dissection and assembly. The mobility of our bodies is a template for the felt movement of thoughts and language.

Consider the following common types of mobility of intangible objects: the movement of an argument from point to point; following a plot or quasi-plot; being moved by a sequence of feelings while reading a story; rearranging in a systematic way a set of possibilities. An example of the latter might be attempts to plot the source of physical movement—the movement of/in things—by the Pre-Socratic philosophers, who worked within a series of closely related ontological frameworks. Another example would be a person entertaining hypothetical scenarios within the ontological framework of a specific story world—the work of many Hollywood producers.[6] Gilles Deleuze sought in his pair of books on the movement-image and the time-image to bring the sensory-motor schema that delineates physical movements in the external world together with the movements of cinematic imagery and the movements of mental imagery in consciousness and memory.

One of the reasons Plato posited the existence of Forms was his worry that movements of all sorts would get out of control and that there would be no way

to measure enduring qualities, commonalities, generalities, universals, and laws, nor to establish linguistic conventions and agreements about premises to underpin communication. Plato notes in the *Sophist*: "Any discourse we can have owes its existence to the weaving together of forms [Forms]. . . . Discourse [is] one of the kinds of things that exist. To rob us of discourse would be to rob us of philosophy."[7]

Aristotle sought to dispense with Plato's Forms—those entities that were posited to be timeless and perfect prototypes and that were physically existent in another realm (thus "physical" and stable in a new sense)—while holding onto the effects of Forms, so as to establish regularities and laws in the world rather than ceaseless change and chaos. Incorporeal, heavenly realms were, in effect, pulled by Aristotle into the innards of objects making objects fully and exclusively secular. One of Aristotle's strategies in replacing the Forms was to stress the notion of *innate potentiality* as a fundamental limit for a thing, which amounted to a rule of measured mobility and possibility for the present and future actions and uses of a thing according to its "essence," which essence was, roughly, matter and form informed by the appropriate efficient and final causes. Almost by definition, such a rule could not be fully and completely known or else it would become a Rule or Form.

For Aristotle, what *can* happen or *can* be done is now part of what a thing *is*, as if one could in imagination follow an object into its past and future. This perhaps is a recognition that relationships to *nearby* things, even intangible things, as well as relationships to an observer, are formative. Therefore, what does it really mean to attempt to limit talk about an object to matters of direct contact with the object? Indeed, what can be meant by direct theories of an object's representation? Perhaps the question should be, "how *can* it make sense to talk about types of contact and mobility at a particular time under specified conditions?" The aim is to limit an object's intrinsic nature, its being, to an appropriate temporal range while not reducing the object to a point. Although "potentiality" sounds like an occult force—like its double, the psychological concept of "affordance"—it purports to refer to actual conditions and circumstances that *may* occur. In my view, the measure of these conditions is the *language game(s)* that presently embrace an object. Thus I believe it is misleading to speak of potentiality as being of/in an object rather than speaking of potentiality as referring to the *kind* of recognizable circumstances that will or may occur to/with/alongside an object. I also believe that relative to a given language framework some of these possible conditions may be called *nearly true* conditions.[8] What we call an "object" is thus an abbreviation for complexes.

To conclude, it would seem that perceiving color is merely to find an arbitrary starting point, which is at rest *only* in relative terms since reidentification is always a possibility within the ongoing reidentifications of many kinds of linked things. A color that has appeared may be discovered as already a reidentification from the past or as

an anticipated reidentification in the future. The motive force of color movement is a public language that serves as a descriptive backdrop for cognition, desire, and use.[9] Language itself, as a material network of changing neuronal states and activation patterns, suggests that there may be no single cause that drives mental movements, like one billiard ball striking another. Instead, movement may be the fact—the embodiment—of a restless, mental network whose dynamic architecture allows memory and thought to develop among successive states of equilibria, whether actual or potential.

1. *CHRIST'S ENTRY INTO BRUSSELS IN 1889*: THE GREEN RIVER

Color is busily flowing in James Ensor's large format painting, *Christ's Entry into Brussels in 1889*, in concert with a marching band and parade of Mardi Gras revelers (1888; 8 × 14 ft.). Against a vast, dense, and packed frame in the predominantly bold hues of vermilion red, carbon black, chrome yellow, and lead white, there flows a particular, pure emerald green at the left and right edges. Like a river, this green acts as a meandering "passage" through parts of the painting.[10] Tracking the green provides insight into some of the ideas woven into the scene; or, rather, the scene gives up its thoughts by its representations, even if not exactly, since, exactly as in the case of words, there remains a margin of indefiniteness and openness. There is always a limit to how far one can see clearly and distinctly with image and word. Some critics assign especial aesthetic value to what is unknown and vaguely indistinct about an artwork, as if what can be confidently said is akin to the monotony of the everyday that overlooks flights of fancy, forms of spirituality, or whatever else is believed beneath or beyond a world of surfaces.

Ensor's painting has been described as "an astonishing assault on any conventions of beauty, in which the harsh ugliness of the subject is matched by bilious color, compositional confusion, and the violent collapse of a one point perspective scheme."[11] Contrasts among the five basic colors are at a maximum since intermediate hues and mixtures have been avoided. Thus the "passage" and flow (like Ariadne's thread?) provided by the emerald green becomes a crucial feature linking elements together to make meanings cohere.

Ensor signed and dated the painting at the bottom right within a large block of green. Below the signature and toward the right corner, the head of a spectator appears and possibly part of his clothing—the Marquis de Sade, who may be dressed in green.[12] He is smiling at an outsized church dignitary who is leading a boisterous Mardi Gras parade. Nearby is a man whose nose seems to be a penis (cf. carnal/carnival). The same green appears at the far left side of the painting, beginning a winding course downwards from the upper left corner. In what follows, I would like to trace the movement of this stream of green as it touches nearby elements. Though omitting many important details about the painting as a whole, the movement of the green will bring to the fore several key themes, only some of which I will mention.

In the upper left corner, we see a dirty, smeared blackish bluish-green that may represent faraway clouds and buildings.[13] Dropping down slightly (but suddenly into a new, closer space), we see a smear of someone dressed in green on a balcony among many revelers. The long balcony that stretches toward the right is green. Moving along the balcony, we see a person defecating over the side and next to him or her, another person vomiting over the side. Dropping down from the right end of the green balcony we see a mysterious white smear with green splotches within it that seems to be part of a wall, but that eventually at its rightward extension turns into a pair of vaporous curving green lines (of what?!) that bisect some heads.[14] This perverse green that appears to color things both solid and not, along with a man in green, act as a framing device for a large sign further below painted in splattered variations of white and green that reads, "Dogmatic Fanfares Always Succeed." From the bottom part of the sign, a thin streak of green bordering the bottom edge of another banner carries the eye downward and back leftward toward someone who may be dressed as a clown or ogre with a green face and an anguished expression looking upward, possibly toward the defecating and vomiting pair in the far distance. What is the meaning of his open mouth. . . ?

Continuing to move downward and left, following a large green object hanging on a green cord around the clown's or ogre's neck but flying outward toward another reveler—now clearly seen—dressed in green with his face painted partly in blue with the remainder, plus a large cone-shaped hat, painted in a garish pinkish white flesh color. This person, who appears to be dressed as a court jester, is extending an arm toward the left and downward over the head of a person in a devil-like mask with green eyes. The court jester is apparently gripping the black handle of a knife that has been plunged into the tall top hat of another reveler who is wearing a skull mask. Or, perhaps, this is not a reveler, but Death himself appearing as a celebrant! The Death figure is in the immediate foreground; his body below the shoulders is cut off by the bottom frame of the painting. We have thus traveled from the far distance represented in the upper left corner in a twisting green loop toward the right and then back left and down to the immediate foreground near the lower left corner.

The Death figure is wearing a stark green tunic and black coat. The lower two-thirds of his top hat are green and the upper third is black. There are indistinct, dark Prussian blue smudges in the blackness of the coat and hat. Thin blue, dripping lines extend down from the black of the hat into the green of the hat, as if the hat were bleeding. Death looks no less alive for having a knife in the head. There are also a series of faint gray spatters in the hat that may trace the descent of the knife blade. This diffuse, lumpy gray color gestures toward the possibility of a line, bringing to mind what a line could be,[15] just as the overall progress of the green from the upper left draws a line through some of the ideas in the painting.

We are witness to a vast parade of people through mangled spatial perspectives, as if the enormity of the event were caught in the process of becoming gaseous

(cf. the Lundemo epigraph). A gigantic procession of bizarre "fanfares" is depicted where lives are lived only by vacuous dogmas and slogans that are as artificial and deceptive as the masks of the revelers. A number of inscriptions—e.g., "Long Live Jesus, King of Brussels"—are seen along with Death and Resurrection. The resurrected Christ, the ostensible center of the celebration, is riding on a mule in the center of the painting all but lost to view, together with a peculiar hovering between death and life in the bloated figure of the church prelate, or someone wearing a cleric's clothes, who appears to be either leading the procession at the bottom foreground—as seen impossibly upwards from his feet below the frame edge—or else lying flat and dead as others are carrying him all the while being watched by a smiling Marquis de Sade.

Small dislocations and gaps in the overall flow of green on the left side enact a sort of "broken" passage through a beholder's ideas and emotions, aligning various pieces of the painting slightly out of line.[16] Though fragmented and not spatially contiguous, the green serves to direct our attention through a chaotic scene. Ending with the Death's Head (or alternatively, flowing outward and upward from it), the world is shown to be a haphazard collection of frivolous activities and values that amount to mere vanity while the normally life-affirming connotations of a bright green hue are reversed from their usual association with springtime, renewal, and fertility. Death and disruption lie resolutely in life and near, not separate from it. Though determinedly modernist in style, Ensor's painting is a thorough rejection of pointillism and social realism since he may have seen those two styles as attempts to justify, reify, or become immersed in simple perception and the present state of the world. Ensor's painting of grotesquerie exhibits alarm over the irrationality of the masses along with deep skepticism about the possibilities of religion, state, and modern life. These crucial ideas in the painting emerge through a moving green line.

2. *BLACK NARCISSUS*: THE RED RIVER OF METAMORPHOSIS

Color flows in two different ways in Michael Powell and Emeric Pressburger's 1947 film, *Black Narcissus*. As in the Ensor painting, there is a tracking color; in this case a viewer tracks red. In addition, the red appears as a transformation or conversion of a white hue. I wish to briefly outline these two sorts of color movements. For this purpose, I will broadly segment the film's color schemes as follows.

I. [0 to 51–1/2 min.] Arrival of Anglican nuns to establish a convent, school, and infirmary at the Palace of Mopu located in a remote valley of the Himalayas.

Basic, pure midrange hues; bright, flat lighting with tight and defined shadows for depth; textured objects are highlighted with patterns, shapes, and colors to emphasize distinctness and to mark spatial planes.[17]

II. [51–1/2 to 72–1/2 min.] Christmas.

This section begins after a long fade out as a black screen is held, then fade in on Sister Clodagh (Deborah Kerr), who is using black ink on a Christmas card to color a figure's clothes a solid black.

III. [72–1/2 to 95 min.] Sister Ruth's ordeal and struggle with Sister Clodagh.

Red appears strongly in most shots, gathers *momentum*, and transforms white into red.

IV. [95 to 100–1/2 min.] Coda: Departure of all of the nuns from the Palace/Convent of Mopu.

Green bamboo and blue skies; Spring-like colors and neutrals; rain and renewal. Mr. Dean (David Farrar) is asked by Sister Clodagh to tend to the grave of Sister Ruth (Kathleen Byron).

Black Narcissus is divided into four main sections, each with its own color strategy corresponding to a classical narrative pattern (Figures 8.1–8.6).[18] A beginning plot kernel (I) expands globe-like sweeping up characters toward a point of maximum volume (II) followed by a rapid contraction (III) toward a moral lesson implicit in the events (IV). Figure 8.1 is illustrative of the color schemes of section I, which schemes are essentially at rest, despite tracking some short-term plot disruptions. Figure 8.1 reveals a scattered, potentially discordant, but limited palette (a "bad rainbow" of hues?) against neutrals of browns, blacks, grays, and whites for the three walls, ceiling, and floor of Sister Ruth's classroom.[19] In other shots of section I, the palette will be reduced further to satisfy a need for expressive objects to populate the story or to coalesce hues into basic contrasts that are keyed to a present mood or event. Schemes in section II are darker and less stable as plot complications and conflicts accumulate. The first scene of section III (Figure 8.2) initiates a sustained and powerful trajectory of intense reds that ends with the death of Sister Ruth in a fight with Sister Clodagh at the edge of a cliff, after which a new color scheme is inaugurated in section IV as a prolonged epilogue offering closure with section I.

A spectator can hardly avoid tracking the rhythm of reds in a series of objects that appear and reappear in section III (Figures 8.2–8.6). The reds appear in myriad guises from shot to shot in the form of highlights, accents, overtones, mixtures, details, and dominants. The opening scene of section III (Figure 8.2) installs an astonishing red tone throughout the space of the scene while containing very little dialogue and action. Joseph, a servant boy, enters Sister Ruth's classroom to bring her a glass of milk. A stern Sister Ruth, alone in the room, is staring fixedly into space without initially looking at him. Frightened by Sister Ruth's frame of mind, Joseph flees the room. Sister Ruth then pours the milk out of the window. While pouring it out, she spots Sister Clodagh and Mr. Dean in the courtyard. Sister Ruth

immediately rushes down through the monastery to get closer to them in order to eavesdrop on their conversation. There have been sixteen shots in one hundred seconds inside the classroom, all featuring a prominent red. The use of red continues in each of the next twenty-three shots until the shot of Figure 8.5.

In Figure 8.2, one notices pronounced streaks of red light on the habit and hands of Sister Ruth in conjunction with her look of suppressed resentment, bitterness, frustration, and rage. This description of red and the range of its expressiveness in subsequent shots does not depend on how the red came to be during production of the film, such as through the use of a red fill or background light, but rather depends on how red is functioning in the plot, even if (as in Figure 8.2) it is functioning as a twice-true metaphor; namely, being the literal result of a sunrise (itself simulated) along with a figurative indication of Sister Ruth's mental state. As a spectator will soon come to appreciate, the meanings of red will expand in subsequent scenes to become associated with Sister Ruth's jealousy of Sister Clodagh, Sister Ruth's sexual passion for Mr. Dean, Sister Ruth's frustration with Mr. Dean's rejection of her, and, finally, her determination to kill her rival, Sister Clodagh, which is expressed at the conclusion of section III by her glaring eyes shown outlined in red.

The white of the milk in Figure 8.2 is also twice-true. During this first scene in section III, we learn that the milk has been sent by Sister Clodagh. Whether or not it was Sister Clodagh's explicit intention to prod Sister Ruth, the white color reminds a spectator that Sister Clodagh stands for the idea that in her opinion Sister Ruth should continue to wear the white habit of the Order of Saint Faith and not give up her vow of chastity, which also is conventionally denoted by white. The clash between the women is made evident by the composition. In the far background, we see the vague shape of a snow-covered mountain in the dawn turning a rosy, light pinkish color. In the near background, there are vague, murky shapes of wall surfaces tending toward an abstractness that erases the possibility of identifiable objects that might pull our attention away from the milk and Sister Ruth, who remains staring at nothing, which is to say, looking inward at objects not visible to us.

Figure 8.3 shows Sister Ruth secretly spying on the conversation between Sister Clodagh and Mr. Dean in the courtyard. The image has become darker and more abstract than Figure 8.2 and includes strange, unmotivated blobs of hot, dark red light in the upper left corner that contrast strongly with the cold, bluish cast of Sister Ruth's habit in half-light.

When the conversation ends, the image of Figure 8.3 is slowly transformed through a dissolve to a group of natives seated around a fire insistently beating on drums. For a moment during the dissolve, the fire seems to be burning inside Sister Ruth (Figure 8.4). In a striking metamorphosis, part of her habit, along with the architecture at right, takes on a new color, transforming into scarlet.

After three shots of the natives beating on their drums, there is a cut to an abstract and indeterminate space (Figure 8.5). There is no way to judge where we are or why. There is no way to identify the space or how one might navigate it. It is as if we had suddenly reached an alien terrain in the film and must start afresh to understand the troubles of the characters. This marks a pause in the stream of red and a crescendo of abstraction, but a momentary one and only so that red can reappear in new, more virulent forms to shock the viewer.

Sister Clodagh finally appears in this space carrying a lantern. She walks down what we now see to be a flight of stairs toward the camera and then moves along a hallway. Eventually she enters Sister Ruth's sleeping chamber and is horrified to find that Sister Ruth has put aside her habit to wear a deeply crimson dress (Figure 8.6). Her discovery is framed with disorienting architectural elements heightened by a profound schism in the image between red and blue (cf. Figure 8.6 with Figure 8.3).

Shots in the remainder of the scenes of section III continue to track red and various red objects, including a startling shot that symbolizes Sister Ruth's fainting spell after Mr. Dean tells her that he has no love for her. Mr. Dean appears from Sister Ruth's point of view as a transparent red slowly spreads across the screen; the image then tilts wildly as she faints and the screen goes to black, which is held before fading in on a collapsed Sister Ruth.

We have previously tracked and seen color metamorphose: the snow of a mountain become rose-colored (Figure 8.2); and the changes in Sister Ruth's habit from white (like the glass of milk) to grayish to grayish blue (Figures 8.1–8.2) to streaked with red (Figure 8.2) to blue (Figure 8.3) to scarlet (Figure 8.4) to a crimson dress (Figure 8.6). Finally, at the climax of section III, the red will reach the skin itself, circling Sister Ruth's murderous eyes as she advances behind Sister Clodagh at the edge of a cliff. Was the red indicative of a drive, always present but initially unseen? Or, was the red the result of something that appeared when the English nuns attempted to establish a convent in British India? It may be that we are to interpret the red both psychologically as a return of repressed sexuality and culturally as overlaid from an environment proving to be resolutely foreign. Soon after the release of *Black Narcissus*, India achieved independence from Britain.

FOUR TYPES OF COLOR REIDENTIFICATION

Two films discussed previously—Peter Greenaway's *The COOK, the THIEF, His WIFE & Her LOVER* and Tom Tykwer's *Winter Sleepers*—illustrate a prototypical type of color reidentification, namely, the same hue reappears in, or for, the same object, person, or locale. The repetition of the same hue acts as a sort of "landmark" and may be the basis for a developing figure of speech. There are over thirty figures

of speech based on repetition. We have also seen a second type of color reidentification wherein the same hue is tracked as it moves among, and accrues associations in conjunction with, different objects at different times or places. This possibility was illustrated in the James Ensor painting as a green hue appeared to move like a "river" among certain objects and in *Black Narcissus* as red appeared to flow throughout section III of the film. Common rhetorical figures based on "flow" are metonymy and synecdoche.

A third, more complex type of color reidentification was illustrated in *Black Narcissus* when a single object—Sister Ruth's white habit—was systematically altered step-by-step from white to grayish blues to reds to scarlet, and finally to a dress of crimson (Figures 8.1–8.6). Similarly, in Greenaway's film there are a few instances when the hues of a character's clothing instantly and inexplicably change to fit a new locale, i.e., to fit a new metaphysics chameleon-like, whether the change is voluntary or involuntary (see Figure 7.1).[20] A realistically motivated version of this third type of reidentification would be the situation where an object is shown only partly in light or an object or place suddenly acquires new colors when a light switch is snapped on. A common rhetorical figure based on a selective comparison or "blending" of a thing that is both the same and newly different is metaphor, including negative metaphor and irony. The next section will consider this type of reidentification in more detail.

Particularly intriguing are fleeting remarks by Paul Coates. At times, he seems to be pointing toward examples of this third type of color reidentification. In his book, a caption to an image from Kieslowski's *Three Colours: Red* (1994) states that in a character's room there is a "central multicoloured fruit [of yellowish-orange and green brownish-black] masking with realism its status as a two-tone, red-black."[21] This seems to say that the fruit and its immediate environs is actually the color of another color (red-black), which is seen in the same image. Such a color metamorphosis would then be a type that has been condensed into a single moment, not spread out in time as is my example from *Black Narcissus*. Coates's analysis of *Three Colours: Red* provides no elaboration on this claim.[22] His claim amounts to an assertion that a visual representation of a red-black hue in Kieslowski's film *has-been-colored-by* another hue (see Figure 3.1, pt. 6). Elsewhere Coates says of Kieslowski's *Three Colours: Blue* (1993) that the "inherent multiplicity" of blue has a "tendency to shade or break up into other colours, to summon them dialectically." Coates says of the film's title that "the 'mismatch' of plural [*Colours*] and singular [*Blue*] expresses the implicit multiplicity of a singularity that is only apparent."[23]

Coates seems to be suggesting that a film may set up its own set of "primary" hues that act narratively to generate the other hues in the film as well as slide into and out of each other.[24] The morphing of hues is a much more complex effect than assigning a fixed meaning to a hue, e.g., assigning death to black. In general, a color

becomes significant whenever a relationship it has with itself, with another hue, or with a potential hue is matched to a relationship in a developing narrative, which relations may be internal to a film or external. One should keep in mind that sometimes a sequence of hues may simply mark phases – within a single, emerging functionality – rather than each hue standing in stolidly and separately for something else, e.g., green connoting fertility, red for envy.

A fourth, more problematic type of reidentification occurs when both hues and the objects of hues are changing together. The reason this case is especially difficult to discern through mental tracking is that it must be distinguished from the situation where hues and objects are simply changing with *no set of discursively relevant associations* among them. There is, however, one common version of the fourth type of reidentification that was discussed earlier in terms of Vermeer's painting, *Woman in Blue Reading a Letter*. This version involves a progression of systematic changes in the coloring of painted hues making for a multi-hued "river" that flows over or near a series of different objects in order to create an association with a fictional source of light—an unseen window in Vermeer's case. In film, of course, the fictional source of light in a scene need not be identical to the actual source of light employed to obtain exposure for the film in the camera. A fictional source of light may be made important to a story in a variety of ways in addition to the distances that are being revealed across space by the apparent light. In the Vermeer painting, for example, a general notion of such distance, signaled by the letter writer's absence and the woman's possible pregnancy—which, if true, would be distant in time—may be the artwork's true subject.

The four types of color reidentification represent methods or criteria that a spectator may employ to recognize patterns, causal or non-causal, across space and time. I will name the types for easy reference.

First type: *repetition* or *reappearance*, in which the same hue—or the same hue with slight variation—returns in, or for, the same object, person, or locale. What counts as being the "same," of course, may be a matter of interpretation.

Second type: *river* or *passage*, in which the same hue—or the same hue with slight variation—strings together a series of objects, as in the green river of the Ensor painting, the red river of *Black Narcissus*, and the white river of *Little Dutch Mill* (to be discussed later). Again, what counts as being the "same" as well as which linkages have been activated by a river are matters for interpretation. For example, in Corneliu Porumboiu's *Police, Adjective* (2009), a striking blue river flowing among ordinary objects throughout the film would appear to make blue an adjective for daily life parallel to the way that omnipresent blue-uniformed police enforcing strict laws are shown to surveil and modify daily Romanian life.

Third type: *morph* or *metamorphosis*,[25] in which different hues appear to characterize the same object, person, or locale, as in the systematic changes to Sister Ruth's habit in *Black Narcissus*—which enact a kind of shifting morphology of the character of Sister Ruth—as well as the cleansing of what turns out to be a dirt-stained rug in *Little Dutch Mill*. What counts as being a stable "object" whose color evolves may be a matter of interpretation. Another example: when visiting a website, one may be prompted to make a decision by the appearance of colored rectangles or circles on a screen. Clicking one of them gives the shape's hue a new hue; we do not say that an entirely new rectangle or circle has appeared.

Fourth type: *library* or *archive*, in which hues and objects are both changing under a rationale of associations or according to a theme that is being expressed through these changes, which amounts to a sort of multi-colored river linking multiple objects of different colors, thus distinguishing this case from a situation in which hues and objects are simply randomly changing. An example is the systematic variations in color across a series of different objects associated with an apparent or real source of light.

More broadly, reidentification through the means of a *library* offers a spectator the possibility of "illumination" through an emerging idea or general feeling, much as a collection of books of different sizes and colors may, or may not, *share* or develop a topic in their pages; for example, a collection of books about auto mechanics or a collection focusing on ancient Greek philosophy and mythology or simply a box of random books that are all to be given away. On second thought, the books in the box and their colors may be random, but at least they do relate to the fact of the box and its destination. This latter fact suggests a further analogy: if a given film text seems not to be a collection of designed, colliding texts connected to an overall rationale, the film is at least building a bookcase for a spectator and acquiring "books" for a collection in a box. . . The name for this type of reidentification was inspired by Jorge Luis Borges's short story, "The Library of Babel."[26]

Complex types: More complex reidentifications are situations involving the four types above where a "single hue" is replaced by a "pattern of hues" or "several hues" are replaced by "several patterns of hues."

REIDENTIFICATION, RADIAL ASSOCIATION, AND DERRIDA

What does it mean to entertain "relevant associations" among hues and objects in the four types of color reidentification? Human knowledge, i.e., memory, is highly organized, intricately interrelated, and formed into bundles—clouds of associations

and links. Knowledge is not a random collection of facts, nor a series of experiences from first to more recent, nor copies of perceptible things, nor a catalogue, nor an inventory, nor a set of memorized lists. Long-term memory in humans, i.e., knowledge, is exceedingly powerful because it is distributed and content-addressable—hence, we usually know at once when we do *not* know something. Human memory is based on principles of gist, pattern recognition, matching, approximation under constraint (incomplete resemblance!), prototypes (core and periphery), linking, and blending, rather than based on explicit criteria, searchable "indexes," algorithms, and location-addressable information, which are the methods used by computers.[27] Human knowledge is densely interconnected, structured, and bundled. When dealing with the artificial memoria of an aesthetic text, a spectator or reader will anticipate making associations and following lines of thought in constructing models that are meant to simulate the story productivity of the text. Thus a key component of comprehension and interpretation is assembling out of a text relevant bundles of associations.

I have elsewhere offered two diagrams in which the associative process is depicted as a series of permeable "boxes" that overlap, touch, or else are near one another, creating a temporary gap or interruption.[28] Each box stands for a single meaning of a word or other signifying item, while the set of boxes taken as a whole represents a collection of associated meanings, e.g., polysemy. These distinct, though linked, meanings, I believe, would be termed "radial" by George Lakoff and exhibit relationships that Wittgenstein would call "family resemblances." A radial series provides a series of contexts, each of varying strength that continues to inform a starting point.

I believe that the radial nature of concepts applies also to percepts, such as color, and that a radial process illuminates some aspects of Jacques Derrida's theories. Consider these seven steps:

1. Let a box in one of the diagrams stand for a "concept." Every word we use is a concept of sorts standing also for something more general, perhaps excepting proper nouns, but let us focus on those words that act as long levers to move large groups of other words, e.g., the concept of "good" or of filmic "suture."[29]
2. From this box, there will emerge a stream of radial extensions of the concept or word into the present and past. The flow of these words (additional boxes) is a *movement* providing greater definition and context to the starting point.
3. Some of these extensions (new boxes) may sprout their own radial extensions.
4. With respect to the "concepts of colours," this radial process accords, I believe, with Wittgenstein's injunction that one "connect what is experienced with what is experienced," rather than connecting "what is experienced

with something physical."[30] This advancing flow (of boxes) thus involves a person's memory—i.e., knowledge and use of a grammar—rather than being driven by initial causation, e.g., for film, the physics of photography.

5. Gilles Deleuze would seem to advance a claim similar to the preceding argument by Wittgenstein about connecting experience to experience. Deleuze, however, is operating from within a different theoretical framework when he argues for the crucial importance of a spectator's stream of "virtual conjunctions" of experiences that move separately from external causation and hence are not explained completely by an initiating physical cause or situation. This is especially true for experiences of narrative concatenations of actions and events. Deleuze contends:

> As [Béla] Balázs says, however much the precipice [in a film] may be the cause of vertigo, it does not explain the expression it produces on a face. Or, if you like, it explains it, but it does not make it *comprehensible*: "The precipice above which someone leans perhaps explains his expression of fright, but it does not create it. For the expression exists even without justification, it does not become expression because a situation is associated with it in thought." And of course, power-qualities [which are affects that enter into "virtual conjunctions"] have an anticipatory role, since they prepare for the event which will be actualised in the state of things and will modify it (the slash [power] of the knife [whose blade is first seen under a quality—the *brightness* of the light], the fall over the precipice). But in themselves, or as expresseds, they are already the event in its eternal aspect, in what [Maurice] Blanchot calls "the aspect of the event that its accomplishment cannot realise."[31]

6. To return to my analogy of a succession of related boxes: if one follows along these radial pathways, one will likely find a conceptual box that questions, subverts, undercuts, opposes, negates, contradicts, transgresses, has been suppressed, has been repressed, is contrary to, reverses, etc., the content of the starting box, which was the original word (meaning or experience) being examined for its conceptual faithfulness, solidity, fullness, and consistency. Thus the starting point will have been shown to be subject to some sort of inner contradiction and heterogeneity.
7. This newly found negative-box that disrupts and displaces a conceptual starting point will also raise significant *doubts* about the starting point itself and is the box Derrida searches for in the approach of *deconstruction* and is why the "de-" of deconstruction signals, not a mapping of a semantic field, but a tearing down and dissection that leads to a re-viewing and re-situating of the original concept and context both semantically and pragmatically. The result is that there is no certain center, i.e., no center free of all occasions, no

> metaphysics apart from history, no logic apart from grammars, and no privileged starting point or origin.

In the preceding view, deconstruction is a special sort of etymology that exposes a set of differences that emerge from a word to indefinitely defer its final meaning. (Embodied cognition is another special sort of etymology that traces the meaning of an expression back into a pattern of sensorimotor and bodily experiences.) Derrida calls this process of deconstruction one of "différance" or the felt presence of a "trace."[32] Nonetheless, one might see it more holistically and from the other end as less destructive and more an acknowledgment of interwoven semantic and pragmatic networking. Meaning is closely entwined with the processes of individual and social memory. This fact of language need not always be as fatal in all cases as Derrida may imagine, since writing means making choices, setting limits, assigning weight, and choosing to communicate to a specific audience. Writing also means (usually) holding at bay the most virulent forms of skepticism about the act of writing and its relationship to a word and world. Not *every* connection to a concept need to count equally in order to use the concept to *do* something; and doing something implicates a person's plans and set of values. Language use is about taking your chances with measured contingencies.

Still, the last word—and "last word" with its sense of closure is definitely not in the spirit of Derridean theory—may belong to Derrida since the goal of communication is a description or account—an argument about a world—and in thus employing a public language there are always other, near values at stake—implicit, assumed, underlying, hidden, suppressed, repressed, once active. Whatever is Other lurks as the ground of intelligibility for—as the cost of—communication. A ground has emerged from *somewhere* and will eventually shift. Language moves along with persons, projects, and circumstances.

Derrida concentrates on words and their status as "concepts," but aren't images and percepts similarly to be placed and displaced, i.e., aren't they, too, conceptual and subject to elaboration, subversion, and reversal? In particular, doesn't a color percept depend on the possibility of its being other or other in another pattern? For example, the Ensor painting discussed earlier works to reverse the usual associations of a bright green hue[33] from fecundity toward collapse. Furthermore, isn't there a measured incertitude or contingency about images and percepts?[34] When encountering an image, we may jump toward visualizing something directly relevant—an instant mental image of the image—but in doing so we may miss seeing other streams of images condensed in the initial view. As an analogy, consider how we instantly imagine an unseen window in the Vermeer painting while more

careful deliberation is required to see how light from the window acts to pattern objects and how that pattern connects to other color patterns that connect to unseen issues.

It would seem that no matter how definite a concept or percept appears, there remains an underlying schematic or diagrammatic quality that is generalizable and generative to other concepts, percepts, or scenarios. That is, there exist *potentials* within a word or image since various contexts may be made to fit and revise a given starting point, especially in fictional discourses where feelings linger and evolve after an initial presentation and point toward new properties and events. As feelings linger, new connections are produced that defeat any purported final meaning. Though a word may fasten on a specific thing, it still has come from, and moves toward, other places bearing/baring Derridean "traces."

Color conventions and color spectacles, though apparently static and resistant to movement, may nonetheless be fit to the idea of radial change. For example, imagine that the first box in a series appears as if drawn in mental bold type to signify a convention or spectacle. Moving outside of the first box and forward along a progression of boxes would then involve a greater than usual mental effort on the part of a viewer. Overcoming such inertia, however, is not impossible and would open the starting point to reflection, nuance, and context.

For Derrida, a given context for a word or image is always merely provisional:

> Is there a rigorous and scientific concept of *context*? Or does the notion of context not conceal, behind a certain confusion, philosophical presuppositions of a very determinate nature? Stating it in the most summary manner possible, I shall try to demonstrate why a context is never absolutely determinable, or rather, why its determination can never be entirely certain or saturated [saturée].[35]

This is perhaps an appropriate moment to mention that one should not think that finding a pattern—much less *the* pattern, *the* context—is simply a fact and an objective enterprise. There is no *certain* context and thus there are many possible patterns in a display depending on a person's starting points, assumptions, and radial associations, even though not all discovered patterns will be equally fruitful. To mention only two factors, a person may be influenced by a confirmation bias and the well-known hindsight bias—a knew-it-all-along effect—where a creeping determinism works to integrate an outcome with previous information (i.e., by finding similar instances and assigning them causal or predictive influence) in order to create an appropriate whole that nonetheless may greatly underestimate the importance and even existence of other patterns. Memory and thinking are practices of verbal-and-image making as we remodel what is seen on screen. This process of transformation of screen information cannot be clearly separated from how we

personally and collectively remember, pattern, and think about all of what we have seen or expect to see; indeed, how we decide if we see at all.[36]

I have been seeking in this book to describe various starting points and assumptions through which a person may orient the discovery of patterns. Chapters 5 and 6 outlined various kinds of contextual and schematic qualities that may become appropriate in thinking about a particular color in situ—i.e., when a color enters consciousness attached to an object. I have attempted to elucidate some of the philosophical presuppositions about color when animating it in mind. The third ("morph") and fourth ("library") types of reidentification are near Derrida's notion of a 'structural non-saturation of context.' Recall the example of Sister Ruth's habit changing color for the third type and the example of color and object mutually changing in response to streaming light from an unseen window in the Vermeer painting for the fourth type. These two examples are conservative since they illustrate only a portion of the possible range of these two types of reidentification. Both examples track a controlled teleology, namely, Sister Ruth's sexual repression verging on psychosis and a window that brings to light a literal and figurative illumination on/of a woman with a letter. The arc of a character's emotional disturbance and the action of light are familiar events in classical filmmaking. Presumably, there exist more difficult and less visible forms of the third and fourth types of reidentification.

Though perhaps accidental, it is fitting that Derrida links concepts and contexts to a term that also refers to a dimension of color in both French and English, "saturation" (see preceding quote). The term "saturation" brings forth a mental image of words and images as being figuratively colored in multiply hidden gray tones (i.e., the gray beneath the hues), or else of words and images having absorbent and leakage qualities (i.e., being filled and poured out) where a color may drain away to be replaced by something new taken in from an environment. Meaning itself thus acquires fluid hue qualities. Hence the ubiquitous metaphor of *being-colored-by* (see Figure 3.1).

As Derrida would probably appreciate, there are also semantic accidents from inside or outside a text that nonetheless inflect or deflect a key word or image, changing its 'color' and fit with other words. Derrida insists that there is no certain end to a circulation of association, nor presumably a sharp edge to a flow of color from beyond, beneath, or around that temporarily "saturates" a meaning. Language brings a world to mind, making accidents merely signs of *management from a distance* like a meaning or emphasis that appears too soon in an argument or too late.

1. *LITTLE DUTCH MILL*: THE WHITE RIVER AND THE COLOR OF A COLOR

The third and fourth types of color reidentification—morph and library—involve a color that acquires a new color. This is consistent with how we speak about any

entity potentially *being-colored-by* (Figure 3.1). Let us examine more closely radial association as inflected by Derrida's approach to language and the notion of a color being colored.

The Max Fleischer cartoon, *Little Dutch Mill* (Dave Fleischer, 1934), features a villain in discordant browns and blacks, who announces at the start that "I hate [the] neat and clean: I'm dirty and I'm mean."[37] He also stomps on three tulip beds. Events ensue in accordance with the villain's self-assessment. Nearby townspeople arrive just in time to prevent him from cutting out the tongues of two children so that they would be unable to tell anyone about discovering the villain's hoard of gold. The townspeople decide to rehabilitate him by scrubbing clean his house and person.

As part of the rehabilitation process, we see a woman in the house using two sticks to beat on a rug of a solid, reddish brown color. Before our eyes, the rug is transformed into a sparkling and fancy geometric design edged in light blue containing two new brown hues along with a dark blue. The reddish brown is gone; only a dark brown fringe at the rug's ends lingers as a reminder of the original. The villain is similarly transformed by a forced cleansing in a tub of water, though certain fringe details of his original person remain, now manicured. At the end, we see that more than his appearance has been altered, for he decides to give away his hoard of gold: "I never knew that cleanliness should bring one happiness. All my gold has brought me naught. Joy like this cannot be bought."

The metamorphosis of the rug stands as an emblem for the movement of a narrative that has labored to reverse a character trait of the villain in order to reestablish communal order by converting him into a benevolent member of society. Though the idyllic state of Dutch life that was shown at the beginning of the story was disrupted by the entrance and actions of the villain, it is shown by the end to have been repaired and remedied. The true colors of the rug have likewise been reinstated. A spectator realizes that the rug's genuine colors had a false veneer and unseemly hue caused by excessive dirt. But which hue is which? Which receives priority? The fact that dirt is represented in the story as being merely *dirty* biases a spectator toward a colorful *original*, namely, the rug restored to its beginning state in bright multiple hues.

Immediate context seems to rule, i.e., a near radial association. The reason is that *Little Dutch Mill* is not a story, say, about the farming of tulips where dirt is fertile, valuable, and clean. Instead, a good rug is one that makes a house (and owner) attractive and is emptied fully of dirt. That is, a true rug displays the presence of an absence (being empty of dirt) as testified by its newfound brightness. But if we are to benefit from Derrida's theory about the relativity of context and concept, we should remain open to additional contexts and reversals, however seemingly minor. The idea here is that objects and ideas in, and of, narrative discourse possess radial

qualities that, given circumstances, require degrees of suppression in order not to destabilize the drift of a narrative toward its moral conclusion.

When we are thinking of dirt as simply an undesirable condition imposed by a local context, then perhaps we also see that *in general* there need be no metaphysical presumption that the original substance or state of a thing is to be preferred over its latest manifestation or manufacture. In *Little Dutch Mill*, we are being influenced by narrative context, i.e., a specific argument is being advanced that dirt is bad and deficient with respect to rugs and persons. And also, that dirt figuratively, but then actually, affects a person's inner person and virtue. However, in other contexts, the most recent color of a color might be the preferred one when, for example, a hue is viewed in a soft haze; or through smoke, transparency, or a stained glass window; or acquires a halo, aura, or rainbow; or is stained in new and more attractive hues; or is made pleasing through the action of some new and better framework. In these situations, we might proclaim that something intrinsic to the thing has finally been realized—forced to become actual—or that the thing's sorry initial state has at last been transcended so that it has finally "come *into its own*." Indeed, how many contexts might bear on a color or thing in realizing itself? Which particular contexts are being foregrounded by a given narrative, which ones suppressed?

In the case of *Little Dutch Mill* our thought has been constrained because we are thinking of dirt-brown as a color that has been simply *painted* on the rug or applied as a (negative) *cosmetic* (Figure 3.1). By contrast, conceiving color as radial, would be to adopt the master metaphors *color-as-camera* or *color-as-causative* and to embrace future reidentifications, possible patterns, and the "nearly true." The general problem of the color of a color has also appeared as the "archive dilemma" in which a narrative justification is offered about restoring a film to its "original" state while leaving aside cultural viewing protocols and the contexts that an original audience had supplied to the perceived qualities of color (Chapter 3). To be clear, the archive dilemma is not an argument against restoring films, but only an effort to clarify the limits of what can be seen in a restoration and to disclose the philosophical weight being given to, and associated with, the word "original."

The idea of searching for a true and original state is connected to a formidable perceptual metaphor: the idea of cleansing and purifying. Whatever is inessential (dirt) must be removed to achieve a purified state. This yields such mental images as a white cloud, blank screen, snow, and a clear liquid, all seen as pure, even if contaminated. The reason is that a hue appears homogenous when no other hue is discernible, no matter if it were created by mixing hues. Blue and yellow paints, for example, simply make green, while blue and yellow lights create a white purity. The goal of cleansing to achieve a look of purity may be extended to justify unsavory political practices and be projected onto philosophical endeavors. A person's

thoughts about abstractions and doctrines are often projected from tangible aspects of perception. Theories and thoughts come from sensation and the world (where else?) and return to the world.

The color morph of the rug in *Little Dutch Mill* is part of a developing associative pattern whereby cleanliness of rug and person is demonstrated to be next to brightness and goodness, which is next to, or leads to. . . what? . . . godliness? If not godliness, then goodness is next to another sacred value—a neat and clean, 'good orderly community' in which the townspeople are depicted as wearing clean white aprons, white collars, and white caps, and where everyone helps everyone else, where everyone is in their place, and the whole community thrives.[38] There is an innate goodness to each individual, even if sometimes obscured by grime from the world. As the villain recognizes at the end, gold brings a person "naught," a kind of emptiness or dirtiness, i.e., nothing good, no good dirty gold. He stores his gold in darkness inside a wall where the gold takes on the darkened hue of dirt. Ironically, the gold is stored directly behind a burning candle (radiating light) on the wall—but then, what is irony? a prediction? where the only choice is the opposite (the need to lighten the gold)?

The story argues that the real color of gold should be exposed to the light—made to be free of dark and dirt—like the white clothes of the townspeople. Gold and greed are naught(y)—and threaten children—while community is good. This is a timely message considering that the cartoon was made during the worldwide Great Depression of the 1930s. Still, giving away gold at the end, as does the villain, is giving the gold to individuals for whom it may again bring naught. It is the accumulation of gold and the encouragement of greed—as in capitalism—that is assuredly bad. A rug needs to be emptied of dirt just as the villain's body needs scrubbing. Like a rug, gold needs fair distribution to be free of its filthiness. That is, a gold cache is empty of true value, but in a new and different sense of *empty*; indeed, in an opposite sense of empty. Gold becomes its own dirt as it piles up, becoming empty; its true value increases only as it is spent for the good of the community becoming meaning(full) for all. The rug moves from full of dirt to empty; the gold from empty to full. In a further irony, the fullness of gold lies in its being spent, i.e., lies in its disappearance.

And what shall we conclude about the villain's use of the word "naught," which is akin to empty? Presumably the narrative argument of *Little Dutch Mill* is not itself about "nothing." It is gold that brings naught. However, even here there remains an ambiguity, for the villain distributes his gold by bringing it to the people, throwing it to them in handfuls, but can one distinguish between a hand that holds something but brings nothing (gold is nothing, naught) and a hand that is merely open—a hand that may be holding back and hoarding or a hand holding itself out

in friendship or a hand not signifying anything? What does one need to know in order to know whether a hand is truly empty or full? Is there a (negative) image that goes with "naught," "empty," or "doing nothing," or is it instead a matter of applying one's *expectations* to a hand, i.e., one's prior beliefs about what is or is not, or should be or should not be, in hand? Thus the duality of naught as both absence and presence—gold as naught yet a windfall to the townspeople combined with the abiding presence of "no" (as in, "one shall not. . . abide greed") coupled with "yes, here is a present"—survives and clings to the gifts of gold to the townspeople. The implicit pairing of such ethical imperatives as "naughty and nice" is unextinguished because undistinguished.

What survives is an assertion of the value of community as seen in the opening and closing shots that frame the story. The opening shot shows a display cabinet that keeps white cups, white containers, and decorative plates free of dirt. The glass doors slowly open by themselves and then hands of an unknown person hangs cups and a woman sings, "There's a story nearly everyone knows, and I'll tell you how it goes." The hands have put a plate into the cabinet facing the camera. The camera moves in (to check for dirt?) toward the plate, which shows an engraving of a little Dutch mill on a little Dutch hill that then transforms into a little Dutch mill on a little Dutch hill with its windmill turning to introduce the first scene of the story. The woman sings her final words, "There's a little Dutch mill on a little Dutch hill." The two children protagonists with their white duck enter the scene.

At the end of the story, we see the little Dutch mill—now transformed into a glowing, golden brown hue at the center of a joyous circle dance by the townspeople (celebrating their new wealth?). The scene is noticeably *brighter* than the opening scene of the mill and the sky is flush with light blue and sparkling white clouds. There is a dissolve back into the little Dutch mill depicted on the decorative plate. The camera pulls back from the plate (all is now clean in the community). The glass doors of the cabinet then slowly close on their own as the lesson of the story is declared to be undeniable and to no longer require human agency in the form of hands bringing things (plate, story, moral) to a spectator as if distributing a kind of gold (golden rule). The woman from the beginning shot is heard singing about "the little Dutch mill on the little Dutch hill," and about the two children: "now they're happy and safe." The hanging cups and plates in the cabinet are all securely in their place like an orderly society or logical proof. The doors have closed by themselves. The world has spoken. The gift is complete. The spectator has witnessed for himself or herself. The camera (like gold) is properly invisible, its gift complete. A title card appears atop a horizontal wooden frame for the glass panes of the cabinet, "The End."

It is almost possible to reduce both the lesson and the Derridean countercurrent of *Little Dutch Mill* to a "white river" that assumes (morphs into) new hues

while streaming amid colored objects and intangible things, themselves forming a "library" of white reidentifications coalescing into a major message containing faint internal contradictions. For example:

1. The whitish, flashing sparkle of gold coins that materializes in single, separated frames of a scene in which the villain sings and plays with his hoard of gold coins. The glints of white become identified with the golden hue of the gold. "Gold's like blood—it's life itself," he sings. But don't we already mind the color difference between gold and blood? Does gold flow and move? Will it?! What "body" is being nourished or not by gold?
2. The lily white duck, faithful friend of the imperiled children (victims of the sight of gold), who runs squealing and squawking, shedding white tears, to summon the townspeople to the rescue.
3. The white soap bubbles—like the white-clad townspeople—at work cleansing and cutting the unruly, spiky black hair of the villain during his enforced bath. Two large white soap bubbles are burped out by the villain to indicate that metaphorically his innards are also being cleaned.
4. The white highlights in the villain's hair cast by a newly created light beamed from his cleaned and brightened home as he appears transformed—reclaimed by the community—in his trimmed black beard, combed hair, and fancy clothes. White has worked its wonders on the villain, renewing him, and in the process has been reversed into its opposite—the shiny black of his hair.
5. The *lack* of whitish glints to the gold (naught glints, naught allure) as the villain at the end of the story opens his bag of hoarded gold and throws handfuls of coins to the townspeople. White become gold finally does move like blood through the body politic (as predicted, white becomes a mental red!), but paradoxically the loss of gold's physical sparkle reduces a sense of its materiality. The material of gold no longer lacks emptiness[39] because it has become substantial in a new sense to offset its loss of material glint. At the same time, however, the gold is moving closer to again becoming naught and invisible since the loss of the coins' sparkling quality and their flight to new hands foretells a loss from sight and the renewed possibility of empty hoarding (accumulation of capital) far from sight. White disappears into the golden brown of a little Dutch mill in a little Dutch land.
6. Step-by-step, we've seen the "white river" become multi-colored: white sparkles become golden yellow coins, white tears and soap bubbles for a villain, black hair highlights, red blood; when the work of white is finished, it transforms into white clouds over a mill newly minted as golden brown. White has joined ideas together and morphed into other hues until it finally vaporizes like the gold at the end to become all the colors of a land returned to harmony. Of

> these key colors, only the black is a cool hue. But by the end, like the figure of the black-haired villain, it has been refigured into the cool blue of fair skies with dancing white clouds (watch them move) over a white- and blue-streaked little hill upon which sits the freshly brightened (it shines), golden brown home of the villain, which has become simply *a* little Dutch mill.

With the sixth step, the river's white—its meaning and hue transfigured from one moment to the next—has been fully absorbed and morphed into the lesson of the story. The white has disappeared into a spectator's evolving thought, visible only in altered forms of feeling and idea.

What has been said so far about types of color movement through reidentification and the links to cognition through radial associations should apply to any percept including perceptions of shape, size, texture, lighting, depth, movement, sound, and so forth. Thus a spectacle of sensations may well not stand as simply a festival of inchoate feelings, but be linked to thought through reidentifications and movements through successive states. Spectacle is not absolute, but relative and radial.

FORMAL TYPES OF COLOR MOVEMENT

> I made lists and lists of these kinds of things [familiar everyday paradigms, such as people telling jokes or looking for things they can't find or filling the awkward silences of everyday life]. In the e-book version [of my graphic novel, *Here*] when some of them get reshuffled, I see things I didn't realize were there, or a theme emerges I hadn't thought about. I definitely think of the whole structure as being musical. I had it all up on my wall in my studio. I was shuffling around the pages constantly trying to get flows of things so they feel right. I wanted crescendos. That's what's exciting about the e-book version, in particular, the way it's shaking me up—to discover something new—every time I see it.[40]
>
> —Richard McGuire

Reidentification as theorized by Strawson, Evans, and Millikan is an important mental ability because it fosters tracking and lends movement and flow to whatever has been actually, approximately, or *seemingly* reidentified since spectators, sometimes quite willingly, make mistakes while many texts create important effects by encouraging spectator missteps. Can a mistaken movement that produces associations later be canceled in mind as if it were an accident and had never appeared? A spectator's interpretive process includes testing ideas by *rearrangements* and by applying new frameworks to find meanings that stick.

The present chapter and the previous one consider the issue of tracking colored things that appear either "in place" or "in movement." Each chapter employs illustrations from painting and film. Is there a way to combine these two modes of tracking into a single mental image?

John Kurten has proposed an analogy based on the design principles of ornamental gardens. This approach seeks to merge the moving image arts with the fine arts of painting, sculpture, drawing, watercolor, graphics, set design, and architecture. It also attempts to merge color in its dynamic forms of energy, action, and development—which summon such metaphors as flow, circulation, current, and fluidity—with various frozen forms characterized by proportion, balance, equilibrium, and harmony, even if there may be disharmony, tension, or rhythm (periodicity) among material elements that are fixed in place. In a painting, for example, reidentification may work to create a temporal pattern of movement and resolution.

Kurten elaborates on a model of color design based on ornamental gardens as follows:

> The ornamental garden works as a kind of analogy to the color motion picture as most gardens are designed to be viewed from pre-defined or idealized points of view (e.g., patios, outlooks, benches, pavilions, etc.) from which one is invited to pause and to contemplate the compositions of the surrounding greenery [a "color node"]. However, most gardens are also designed to be viewed from some sort of path or system of pathways, which, in essence, introduces the element of motion and all of the various compositional complications that goes with it [a "color morph"]. Note too that the analogy of the garden also includes the element of lighting, both in terms of the changes in the angle of the sun during the course of the day as well as the changes in lighting brought about by one's continually shifting sense of perspective while strolling down some garden path.[41]

Kurten's emphasis on "the color character of the next 'node' " will produce effects on a spectator's *chromatic memory*. Kurten's description of color movement through a series of nodes embraces numerous types of movements in films. Consider these:

1. *Editing* that creates immediate juxtapositions and sometimes creates more complex patterns within a single shot or involves patterns based on much earlier or later shots, e.g., nonlocal editing, distance montage, and hyperdiegetic narration[42] that shift our attention within working memory while retrieving selected information from long-term memory (see the discussion of *Vertigo* in Chapter 6). Similar movements may be based on changes in the

position of characters, lighting, optical effects, and a moving camera, all of which may constitute "montage within the shot."
2. *Narrative and narration*; for example, the "arc" of a character, the "flow" of events, and changing points of view.
3. *Speculations* moment by moment by a viewer concerning what might happen in a plot; for example, a viewer's sense of "forking paths," radial meanings, alternatives, and what turns out in retrospect to be only "nearly true."
4. *Off-screen forces* that impinge upon a "garden" from places outside—that is, influences that lie beyond the present and visible: degrees of uncertainty, future and nearly true space, disharmony, transcendence, endlessness, chance, nature, commerce, production circumstances, cultural pressures, the imagined intentions of a garden's author, other films, and the matter of "found" color, "weed" color, color schemes beyond the border of the garden, and nonstandard color schemes.
5. *Reveries* and half-conscious repressions about a film while watching, analogous to a person who is distracted, lost in thought, or wandering partly dazed through a garden. (Recall from Chapter 7 Strawson's comments on the nature and possibilities of a person's "disembodiment.")
6. *Writing and talking* about a film—the film in and of one's memory—the figurative refigured.[43]

To understand what a film means to a viewer, one cannot limit movements of mind to the movements of graphics across a screen. Color spots on a screen are firmly tied to mental movements, which may be designated as the field of memory-chromatics. Moreover, mobility and fixedness are relative. The modes interpenetrate since movement can be described as "balanced" or "repetitive" (going nowhere) or shown to be slowing to a stop while that which is deemed stationary may express a tension that is set to tear apart its elements toward something new. What is incipient or potential—the power to be(come)—may be expressed within either movement or fixity. Thus the analogy of an ornamental, color-full garden encompasses both dynamism and stasis lying within each other awaiting only a viewer's measured reidentifications, associations, trackings, and expectations, whether the viewer happens to be walking or standing.

Is the movement of color fundamentally real or merely an impression that is intangible and illusory? We speak of the literal movement of a person and of an object like a train as well as their literal potential to move when at rest. But we also speak of a literal *train* of thought, of being *hit* by an idea, and about a *door* that opens in mind. Which is literal, which figurative? What counts as being literal

depends on one's choice of a place to start. If truth is linked with the literal, then it lies equally with the literal-as-figurative.

Is color literally static and figuratively in motion, or the reverse?[44] The answer depends on where and when one starts to track. For a color theorist, the answer depends on where he or she begins to tell a story by setting the "point of attack" and "initiating event" in a philosophical account of color. If one begins by saying that color moves only in a figurative sense—by, for example, treating color perception like a photograph and working memory like pages of photographs—then the truth of this metaphor about mobility will lie in a future effort to literalize it. Alternatively, one can observe patterns fill out and change, and then imagine them having once been frozen in place merely ready to move. Untangling movement and stasis to start with one or the other is arbitrary.

NOTES

1. Trond Lundemo, "The Colors of Haptic Space: Black, Blue and White in Moving Images" in *Color, the Film Reader*, ed. by Angela Dalle Vacche and Brian Price (New York: Routledge, 2006), p. 90.
2. The locution that I've selected in the text—"inside and out"—to characterize a set of permutations explored by the Pre-Socratic philosophers seems to suggest that all possibilities have thus been subsumed and exhausted, or may be subsumed with further diligence. One should, however, weigh words carefully, especially those that steer quietly like "inside and out." Nonetheless, I will speculate that philosophers who share a certain framework, or several closely related frameworks, will distribute their views across the possibilities allowed by the framework as if filling in the squares of a tabular array.
3. On the intrinsic, usually affirmative qualities of the color green, see, e.g., Michel Pastoureau, *Green: The History of a Color*, trans. by Jody Gladding (Princeton, NJ: Princeton University Press, 2014); and Natalie Angier, "Ever Green," *New York Times* (April 21, 2015). Angier asserts, "Evolution has attuned humans to a color that pervades the natural world, a combination of pigments that prompts feelings of comfort, well-being—and occasionally, deep distrust." The usual connotations of green receive an ironic and bitter reversal in a James Ensor painting to be discussed later. There is a small library of books on the supposed intrinsic qualities of individual hues.
4. On the battle between line (drawing) and color, *disegno* versus *colore*, see chapter 1 above, section 2; chapter 5, section 1, subsection 1; and note 15 below. On key color, see chapter 5, section 7.
5. Perhaps like the invisible relationships of visible points in a pattern, space itself is neither quite here nor there. Furthermore, we encounter the space around us in two dimensions, not three; the brain constructs the third dimension as an abstraction out of a 2–1/2 dimensional mental sketch. Thus there should be nothing unusual about a feeling that there is an aura of the abstract and immaterial surrounding our sense of the relationships in a color pattern. Some mental abstractions arise strongly from material and are caused by material conditions. Again, as argued in the first section of Chapter 7, there is an overlap between illusion and reality. How much of an overlap? On

spatial construction by the mind, see, e.g., Steven Pinker, "'The Mind's Eye,' section 'Seeing in Two and a Half Dimensions'" in *How the Mind Works* (New York: W.W. Norton, 1997), chap. 4, pp. 256–261.

6. On the movements of immaterial objects and figurative objects, see Edward Branigan, *Projecting a Camera: Language-Games in Film Theory* (New York: Routledge, 2006), pp. 22, 153–157, 201–208, and index entry, "motion/movement, abstract/nonphysical kinds of." Consider also the very different kinds of movements in the world associated with "causation"; George Lakoff and Mark Johnson, "Events and Causes" in *Philosophy in the Flesh: The Embodied Mind and Its Challenge to Western Thought* (New York: Basic Books, 1999), chap. 11, pp. 170–234.
7. Plato, *Sophist*, 260a (trans. Cornford).
8. Cf. chapter 7, section 2, subsection 1. On the notion of potentiality, cf. the concepts of gravitational potential energy and electrical voltage. A book contract mentions a few aspects of the potential of a manuscript.

   Aristotle explores the actual and potential movements of material and immaterial objects in a variety of contexts in *Physics, Metaphysics, Nicomachean Ethics*, and *On the Soul*. Both Plato and Aristotle should be better appreciated by film theorists, including those searching for the material (or other) specificity of film, so far not found, not even close. The search is a fool's errand. Consider generally, Brian Price, "The Steady Unsteadiness of Theory: On D.N. Rodowick's 'Elegy for Theory'," *New Review of Film and Television Studies* 12, 4 (December 2014), pp. 463–480.
9. On language coloring a specific color, see Chapter 4.
10. On "passage," see Chapter 6, section 1, subsection 1 and section 2, subsection 1.
11. Mark Leonard and Louise Lippincott quote Robert Rosenblum and H.W. Janson in "James Ensor's *Christ's Entry into Brussels in 1889*: Technical Analysis, Restoration, and Reinterpretation," *Art Journal* 54, 2 (Summer 1995), p. 18. See the excellent book by Patricia G. Berman, *James Ensor:* Christ's Entry into Brussels in 1889 (Los Angeles: J. Paul Getty Museum, 2002). Berman takes note of "the thick trails, masses, and dots of unblended colors" in the painting and Ensor's "use of intense, scalding colors and dissonant contrasts" (p. 11). The painting is owned by the Getty Museum.
12. Leonard and Lippincott, "James Ensor's *Christ's Entry into Brussels in 1889*," p. 20.
13. If one instead follows the blue hue down from the upper left corner of Ensor's painting, one finds a saintly figure mysteriously entombed.
14. See figure 4 of Leonard and Lippincott, "James Ensor's *Christ's Entry into Brussels in 1889*," p. 21.
15. In the battle between drawing (line) and color (*disegno* versus *colore*—see note 4 above), it is also possible that each may dissolve and blur into the other; for example, J.M.W. Turner's stubbed and staccato effects in his painting, *Rough Sea with Wreckage* (1840–1845). Further examples: Jasper Johns, *Numbers*, ed. by Roberta Bernstein and Carter E. Foster (Cleveland, OH: Cleveland Museum of Art, 2003). Cf. the Renaissance modes of painting known as *sfumato* and *unione*.
16. Another example of "broken" passage may be found in Vincent van Gogh's *Corridor in the Asylum* (1889), whether one chooses to follow the green, yellow, or red hues. This shows that "continuity," its ellipses, and reidentification, is not strict, but relative to a norm of judgment.
17. John Kurten (private correspondence) points out that British Technicolor films, by and large, retain the tradition of black-and-white filmmaking that utilizes finely modeled gradations of studio lighting. *Black Narcissus* may thus be considered a Technicolor "art film" made within the traditions

that typify the studio era. For Sarah Street, "*Black Narcissus* can. . . be read as an extremely 'open' text, which explains its suitability for appropriation. . . . It provides a rich source of images, sensibilities and associations that can be transformed into new meanings." Street also finds two instances of an "*hommage* to Vermeer's paintings" where "the most dominant light source is strong beams streaming in through a window." As discussed in Chapter 7, Vermeer's *Woman in Blue Reading a Letter* makes light from an unseen window a compositional principle with important thematic consequences. Street, *Black Narcissus* (London: I.B. Tauris, 2005), pp. 87, 32.

18. Figures 8.1–8.6 derive from the first of two digital restorations, both supervised by cinematographer Cardiff. Though not certain, it would seem that the first restoration appeared in 2000 and the second, supervised by Cardiff and Thelma Schoonmaker Powell, an editor and widow of Michael Powell, appeared in 2010. Both restorations were done by Criterion. DVD, laserdisc, and videotape versions of the film published prior to 2000 are likely based on Eastmancolor copies of original Technicolor materials.

19. For Michael Powell, director of *Black Narcissus*, "Giving the nuns off-white robes, or rather the colour of oatmeal, was an inspiration. Their robes gave a key to the picture to which all other colours had to conform" (p. 584). The entirety of the action and dialogue of section III was shot and edited in strict conformity to an existing musical score that, I believe, provides the color design a vital pulse (see pp. 581–584). *A Life in Movies: An Autobiography* (London: Faber and Faber, 1986).

    Note in Figure 8.1, a relative softness in relation to fine detail that is characteristic of Technicolor IB (imbibition-blank) printing onto nitrate-based blank IB print stocks. (Camera negative and matrix film stocks were also nitrate.) The softness has to do with "image definition" as opposed to a tonal softness of the hues. With imbibition printing, the applied dyes had a tendency to spread out in a lateral fashion that effectively blurred or softened the otherwise hard edges of a photographic image in much the same way that a fountain pen, when its tip is placed on a piece of blotter paper, will create a spreading ink blot as opposed to maintaining the hard-edged contour of the pen's sharply defined point. According to John Kurten (private correspondence), what was lost during the shift from nitrate to acetate (circa 1950) was nitrate's specific transparent base, which produced a peculiar, though subtle, luster or sheen to the hues.

    Figures 8.1–8.6 were derived from a Criterion DVD from the first of two digital restorations, both supervised by Jack Cardiff, who was trained by Technicolor and was the director of photography for *Black Narcissus*. The first restoration (2000) is probably the more accurate of the two in terms of the appearance of an original Technicolor IB release print. The second restoration (2010) no doubt was the basis of the Criterion Blu-ray edition of the film. The second restoration—stunning in its own right—is likely an "optimized" version of the film in terms of the image information—including sharpness—present in the original three-strip negatives. These inherent qualities of the original negatives make the second restoration perhaps more "relevant" in terms of contemporary viewing tastes. It is not, however, what an audience of 1947 would have seen. As in all matters Technicolor, the nonpareil authority to whom I am indebted is John Kurten.

20. Additional well-known examples of the third type of color reidentification, "morph" or "metamorphosis," created through lighting are the changes to Vladimir's face in Sergei Eisenstein's *Ivan the Terrible, Part II* (1958, Agfa Bi-Color) and *La Cucaracha* (Lloyd Corrigan, 1934, Three-Strip Technicolor). One might include Rainer Werner Fassbinder's *Lola* (1981, Fujicolor). On Eisenstein's notion of color "flow," see discussion in Chapter 3, section 3. For a stronger version of "morph"

that includes both seen and unseen colors of colors, see Apichatpong Weerasethakul's *Cemetery of Splendour* (2015).

21. Paul Coates, *Cinema and Colour: The Saturated Image* (New York: Palgrave Macmillan, 2010), p. 71.
22. Coates, *Cinema and Colour*, pp. 70–74.
23. Coates, *Cinema and Colour*, pp. 104, 105.
24. Coates, *Cinema and Colour*, pp. 79, 84. On Coates's hints about color metamorphosis or some similar effect, see pp. 74, 82, 93, 94, 95, 99, 101, 102, 103, 104, 105, 140.
25. On the third type of color reidentification, "morph" or "metamorphosis," cf. *Meta-morphing: Visual Transformation and the Culture of Quick-Change*, ed. by Vivian Sobchack (Minneapolis: University of Minnesota Press, 2000).
26. For visualizations of Jorge Luis Borges's library in his short story, "The Library of Babel," see "Fairy Tale Architecture: The Library of Babel," &https://placesjournal.org/article/fairy-tale-architecture-the-library-of-babel/& (accessed July 16, 2015): "The library's impenetrability becomes clear, and the illusion that all knowledge is somehow close at hand slips away." For more, see <www.smithsonianmag.com/innovation/digital-library-contains-every-phrase-that-could-ever-be-uttered-180956623/?utm_source=smithsoniantopic&no-ist> (accessed Sept. 27, 2015).
27. This paragraph derives from my essay on memory and media, "If-Then-Else," note 8.
28. On assembling relevant associations among concepts and percepts, see my "Introduction (II): Concept and Theory" in *The Routledge Encyclopedia of Film Theory*, ed. by Edward Branigan and Warren Buckland (London: Routledge, 2014), pp. xxviii–xxix and two sorts of semantic box diagrams in figure 1.
29. On the radial nature of the concepts "good" and "suture," see my *Projecting a Camera*, sections "World, Language, Ambiguity," pp. 98–102, and "Caesura and Suture," pp. 133–145. For more on Lakoff and Wittgenstein, see chapter 4 of *Projecting*, "How Frame Lines (and Film Theory) Figure," pp. 97–149. On the radial nature of the color "white," see Chapter 4 above. For more on suture, see Chapters 3 and 5 above, notes 3 and 55, respectively.

    The epigraph of chapter 4 in *Projecting a Camera* is a quotation from Wittgenstein asserting that the word "good" is a family resemblance concept in which "there is a continuous transition from one group of things called good to another," but "they may not have anything in common." The example that I offer of the polysemy of the adjective, "good," with a few additions, is as follows (p. 102). The word may describe:

> a film, a day, a breeze, a concept, someone's conscience (!), a feeling, a friend, a car, a road, a memory, a meal, a conversation, a scolding, a color, a generalization, a time, a voice, a pencil, a thing, a hand, a reason, a possibility, a beginning, as well as the real goods and what is good for nothing.

    Furthermore, a (good) question like "how far is the airport?" might elicit the response, "a good hour." Is truth a thing that "holds good" of something? One may also wonder what would count as a "good" film or painting. Keep in mind that an imperative statement ("Here is a painting that you *should* believe is good [and should approve of].") cannot be derived as a deduction from a series of declarative statements no matter how many, e.g., "the central area of the painting is balanced in red and blue; the figure standing in the water bucket is scowling." An "ought" cannot be derived from an "is." On the word "good," cf. Aristotle's *Nicomachean Ethics*, I.6.

30. See epigraph to Chapter 5.

31. Gilles Deleuze, *Cinema 1: The Movement-Image*, trans. by Hugh Tomlinson and Barbara Habberjam (Minneapolis: University of Minnesota Press, 1986), p. 102 (my emphases and two footnotes omitted).

32. Derrida's notions of "différance" and "trace" refer to a person's felt sense of a path or paths of closely related meanings spreading out from a word with the energy of simulation (similitude) perhaps in a manner similar to the fact that modern physics discovers in the vacuum, not profound emptiness—emptiness fully contained—but virtual particles. On a larger scale, a novel or film—*Black Narcissus* is both—stands as a transformation of a series of other texts much as a radial series of semantic "boxes" flowing from a word manage to partially enclose one another, touch, or are separated by small gaps; see notes 28 and 29 above.

    "Under erasure" is another of Derrida's concepts relevant to the skein of a word's associations. "Under erasure" refers to a concept that is written, then withdrawn by crossing it out, but nevertheless shown as crossed-out like a phase of the word's meaning or a fictional truth or perhaps quasi-fictional truth. (Does this mean that "under erasure" applies at least to certain fictional images: what is being seen is invisibly crossed out?) The notion of being "under erasure" also provides a way to think about *negation*, that is, to evaluate an assertion of *the* non-presence or non-being of something, in a way that does *not* presume in advance presence (Presence) or being (Being) as the starting point (Origin) of meaning or existence. Note in the preceding sentence the misleading use of the word, "the," which already suggests definiteness and would thus need to be itself crossed out, put under erasure. The problem about negation is that in an Origin lies also its End, hence closure and completeness, presence and being. These are not new issues in philosophy. See, e.g., Robert B. Pippin, "Negation and Not-Being in Wittgenstein's Tractatus and Plato's Sophist," *Kant-Studien* 70, 1–4 (1979), pp. 179–196.

    Some of Derrida's other concepts, e.g., supplement, archi-writing, and pharmakos/pharmakon fit the general picture of radial meaning that I have been outlining and also fit the flow of contexts emphasized by Derrida that revise and undermine a word. In a similar spirit: Michel Foucault, "Nietzsche, Genealogy, History" in *The Foucault Reader*, ed. by Paul Rabinow (New York: Pantheon Books, 1984), pp. 76–100.

33. On the usual associations of green with fecundity, see note 3 above.

34. On the incertitude or contingency of color percepts, recall an example inspired by Foucault where red was transformed into its opposite, blue, under a particular theory discussed in Chapter 5, section 4, subsection 3.

35. Jacques Derrida, "Signature Event Context" in *Limited Inc*, trans. by Samuel Weber and Jeffrey Mehlmanin (Evanston, IL: Northwestern University Press, 1988), p. 3 (Derrida's emphasis); see also p. 21.

    In this essay, Husserl's example of an agrammatical sentence, "the green is either," is repositioned by Derrida into an argumentative context positing endless contexts for words while employing the color term *saturation* as a meta-contextual term to describe a particular state that may be reached by any context, which in the present instance delineates a context for an anomalous, but somewhere possible impossible green. The French word Derrida uses to describe this general contextual state is the same in English, "saturable." Derrida asserts that "there are only contexts without any center or absolute anchoring" (p. 12). Implicitly he has shifted the problem of meta-level contexts for contexts onto other sorts of abstract things and contexts that possess concrete *color qualities* (saturation, lightness, hue). For Derrida, then, one may wonder what sort

of abstract light brings such a saturable hue to view. In this situation, would there not also be possibilities for the arrangements of colored contexts to form harmonious, disharmonious, and other sorts of patterns? What will this mean for cognition? Similar problems ensue if the context for the meta-concept of the *saturation* of a given context shifts from the notion of "light" and hue saturation to an occupying liquid that makes things "sponge-like" (i.e., saturable) and fluid to circumstance.

Note that Husserl's sentence could be made grammatical if the *punctuation* were erased and the sentence allowed to continue: "The green is either emerald green or nearly so." On the flow of concept and context in new and old media, see especially Jeff Scheible, *Digital Shift: The Cultural Logic of Punctuation* (Minneapolis: University of Minnesota Press, 2015).

36. This paragraph on the relativity of relational patterns owes a debt to comments by Brian Price (personal correspondence). One might start to think about this issue with Richard Rorty, "The Pragmatist's Progress" in *Umberto Eco, Interpretation and Overinterpretation*, ed. by Stefan Collini (Cambridge: Cambridge University Press, 1992), pp. 89–108.

37. On the discordant dyad brown-black, see Chapter 6, section 1, subsection 3. *Little Dutch Mill* (two-color Technicolor, which is red and green, and their combinations) may be found on the 2014 Blu-ray disc by Thunderbean Animation, "Fleischer Classics, Featuring *Gulliver's Travels*."

38. On whiteness and cleanliness generally, see Chapter 4. *Little Dutch Mill* enacts a metonymic skid from whiteness, sparkling brightness, cleanliness, neatness, goodness, and clean living to orderly community values. One of the examples of this line of thinking discussed in the text above is the villain's *dirty* rug, house, person, and gold, all of which need to be cleansed before he can be reintegrated into the community.

On the other hand, consider the following general observation by Roland Barthes:

> The fact, in its purity, is best defined by not being clean. Take an ordinary object: it is not its new, virgin state which best accounts for its essence, but its worn, lopsided, soiled, somewhat forsaken condition: the truth of things is best read in the castoff. The truth of red is in the smear; the pencil's truth is in the wobbly line. Ideas (in the Platonic sense) are not shiny, metallic Figures in conceptual corsets, but somewhat shaky maculations, tenuous blemishes on a vague background.

Barthes's claim may be elaborated in several directions. For present purposes, I would stress that the natural state and color of the rug in *Little Dutch Mill* is not necessarily that of its clean version. Also, Barthes's claim would seem to be congruent with Wittgenstein's assertion that the essence and truth of a word or object lies in its use, i.e., in its *dirtying*. With use and the traces of labor, a thing has had time to declare its function in a particular world. Barthes, "The Wisdom of Art" in *The Responsibility of Forms: Critical Essays on Music, Art, and Representation*, trans. by Richard Howard (Berkeley and Los Angeles: University of California Press, 1985), p. 180. On Barthes, see also Chapter 5, note 5.

39. I believe that the syntactic ambiguity of my phrase in the text, "no longer lacks emptiness"—that is, referring equally to "now lacks emptiness/now has presence, is present" and "no longer is lacking emptiness/now has emptiness, is empty"—leads by different narrative paths in *Little Dutch Mill* to the same place of ambivalence about gold.

40. Interview of Richard McGuire by J.C. Gabel, "Expanding on His Corner of Space," *Los Angeles Times* (December 7, 2014).

41. John Kurten, private correspondence. For more ideas about the temporal dimension of color, see William Johnson, "Coming to Terms with Color" in *The Movies as Medium*, ed. by Lewis Jacobs (New York: Farrar, Straus & Giroux, 1970), esp. pp. 224–229.

42. On nonlocal editing, distance montage, and hyperdiegetic narration, see Branigan, *Projecting a Camera*, chapter 4, note 63. On hyperdiegetic narration and heterarchy in the present book, see Chapter 3, note 9. The problem of thinking about editing as one juxtaposition after another is that there arises a tendency to think of editing as simply being built up one piece at a time. Such a mental image leaves out the possibility of thinking further ahead or further in the past—engaging prospective and retrospective memories—to construct larger patterns in which something quite new *emerges* from the separate pieces that is not a sum of the pieces. In addition, remote associations and reidentifications are still associations, still placed by editing, and still work to map paths through one's conscious and nonconscious memories even though not contiguous. In such nonlocalized editing, ellipses are, in effect, replaced by interpolated elaborations and distractions.

43. On reveries and repressions, writing and talking, about film within a "garden" of words (including concealed metaphors that point out paths through the garden), compare James Elkins, "Writing as Reverie" in *Our Beautiful, Dry, and Distant Texts: Art History as Writing* (New York: Routledge, 2000), chap. 10, pp. 254–271. From a different standpoint than Elkins, David Bordwell examines the conventions and dominant schemata of writing about film in *Making Meaning: Inference and Rhetoric in the Interpretation of Cinema* (Cambridge, MA: Harvard University Press, 1989).

44. Time, like movement and color, is both literally in motion and literally static depending on one's perspective and purpose. The double nature of time is said to be true for film, language, and physics. See, e.g., Gilles Deleuze's two books on the "movement-image" (flowing time) and the "time-image" (crystallized time); Lakoff and Johnson, "Time" in *Philosophy in the Flesh*, chap. 10, pp. 137–169 (moving time versus moving observer; time as orientation versus time as [frozen] resource); Michael Lockwood, *The Labyrinth of Time: Introducing the Universe* (Oxford: Oxford University Press, 2005), chaps. 1–3, pp. 1–70 (clock time and block time).

    Is there also a double nature to color?! Compare the problem of "what it is like" to experience redness; see Chapter 2, section 5.


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Chapter 9 **Summary**

ABBREVIATIONS

What follows is a summary of the key ideas and arguments in this book, sometimes from new angles and perspectives. Important concepts and themes appear in **bold**. Numbers in parentheses refer to the part of the book, then the section, and, if applicable, the subsection or endnote number of the relevant preface, chapter, or appendix. For example, (5, 7, 1) refers to Chapter 5, section 7, subsection 1.

I have added sub-subheads to the summary as an aid to the reader.

The summary is followed by general conclusions in Chapter 10 that arise from the arguments of the book and then the appendix. Given the length of this summary and some of its reframings, it would seem to be almost a condensed book within the book.

SUMMARY

PREFACE

Color Consciousness

The preface outlines key arguments of the book: color acquires meaning in language; images create words and words images (2, 3; 7, 3, 1). Color is relational, not a substantive, not simply a gleam and name, but emerges from descriptive networks, values, and goals in a human context (P, 1). Think of color as a temporal pattern driven by expectation and working memory, not as an a priori static sign arrived from long-term memory (P, 2–3). Color tends to vanish when seen in film (P, 4). It can be better brought to mind through situation models and the methods of embodied cognition (P, n. 4).

Four Themes: Norms, Languages, Memories, Sensations/Spectacles

Theorists, critics, and historians are busy telling competing stories about the logic of color and its comprehension. Explaining color is a matter of selecting facts and

arranging them under a set of invented concepts to tell a specific story. The **color story** in this book weaves together four themes (P, 3; 2, 1):

1. Conventions and **norms** of color (chapters 5 and 6);
2. **Language systems** available for expressing a felt consciousness of color (chapter 4); color will shift in mind as we talk about, remember, connect, and narrativize what we have sensed and expect from one moment to the next;
3. Mental processes for memorializing and reidentifying color that may be collectively termed the realm of **memory-chromatics** (e.g., the use of **root metaphors** and folk theories; Figure 3.1; 5, 4; chapters 7 and 8), which realm is much broader than textual inferences, signs, or cues from a screen, i.e., broader than adherence to explicit instructions (signs, cues) and sensitive instead to the implicit, the suppressed, the background, and the interpretive; and, finally,
4. The possible roles assigned to sensation and **spectacle** in the life of the mind, which, in turn, requires a closer look at the interlacing of mental processes (chapters 2 and 3). Is spectacle simply a retreat from rationality and language or is it a description that connects with thought (7, 3, 1)?

CHAPTER 1: INTRODUCTION AND OVERVIEW THROUGH *TWO PAINTINGS: DAGWOOD*

Remembered Languages

A close analysis of the color design of Lichtenstein's painting, *Two Paintings: Dagwood* (Figure 1.1), provides an occasion to introduce major ideas of this book. Despite the title, Lichtenstein's painting may be seen as a three-part painting, just as this book uses a third part to explain the interaction between two aspects of color aesthetics: theory and practice. The third part of the argument in this book acts as a catalyst between the philosophy of color design and its figurative and practical manifestations in media. This facilitating part is the **remembered languages** and insistent mental images that a perceiver utilizes to frame his or her understanding of a colored display (1, 3; 5, 4).

Lichtenstein's painting brings out a fundamental concept of analysis: the existence of **patterns** as an essential aspect of a person's interpretive practice (1, 1; 1, n. 2). A pattern has **emergent** properties not present in its elements and these emergent properties are causal in ways the elements are not (1, 2; 1, n. 6; A, 2, 1; A, n. 7). Transitions among patterns are especially important since the resources of our mental imagery work to produce a sequence of parallel transitions in our thinking in the form of **expectations** (P, n. 3; 1, n. 11), hypotheses, changes, and developments

that propel comprehension. The present book focuses strongly on this intermediate stage of transitions between patterns wherein a perceiver is **tracking** the (re)appearance of patterns of hues (1, 3; 3, 1; chapters 7 and 8) and projecting future patterns. This middle stage of transition generated by remembered languages and schemata concerns what is *incipient* and points toward mental movements that are neither exclusively sensation nor yet fact or idea but that prefigure what may come to be (1, 1): the true, **nearly true** (7, n. 26; 7, 2, 1; 7, n. 25; 8, 1), and the fully figurative (1, 2). This book investigates aspects of mind that are materialized in language and involved in tracking color.

Color versus Drawing Lines

Traditionally, color has been seen as static and unmoving—glued into place—and opposed to a line or drawing (**colore vs. disegno**; 5, 1, 1; 5, n. 9; 8, n. 15), which, though also unmoving, is still said to be inherently dynamic (secretly dynamic?) because a line encircles an object that has moved or will soon move. A viewer's eye stops on a spot of color but is said to follow a line. A color spot thus seems to be defined/confined by a line or border. But shouldn't one say instead that a line needs color in order to be seen and a color spot has a shape that defines a line?

Philosophical views about the nature of color and line, and their relative priority with respect to each other, lead to differing aesthetic theories. There is also a deep connection here between philosophical rhetoric and works of art when one realizes that a philosophy is said to draw lines to distinguish its concepts that, like color, fill out a picture of a world (1, 2; 1, 5). Recall Wittgenstein's insistent question in *Philosophical Investigations*: how sharp must lines be for concepts to function in a situation? Often not very, he says, plus concepts have an inner motility, as with family resemblance and radial association (8, 2).

The connection between philosophical rhetoric and art highlights the emphasis that this book places on details of **ordinary language** when analyzing philosophical abstractions directed at the theory and practice of color (1, 3; 1, 5; Figure 3.1). What choices do we make among assumptions, methods, and schemata in talking about our experiences with an artwork? How are words found? Which **vocabularies** are selected to out*line* our perceptions and adopt or revise mental images (chapters 5 and 6)? Color perception, I believe, has an essential, top-down cognitive component and thus color may become conceptual and hence a medium for thought. Moreover, color in the medium of film often allies itself with, fuses into, the medium of narrative argumentation (1, 4). The color we experience in a film has frequently already been **narrativized.**

CHAPTER 2: LIVING WITH CHROMOPHILIA

The Insides of Color: Black and White All Over?

What is in mind when color is in mind? The word "in" in this sentence has a profusion of possible meanings. The encoding of visual experience—the significance of sight—does not lie "in there" in biology but *out* in socially anchored linguistic systems. Each human is defined by his or her use of a variety of languages (**language-games**) fit to changing goals and local beliefs. Humans make and remake themselves for specific interactions. In particular, to borrow the title of a book by Wittgenstein, aesthetic languages are fit to "culture and value." Color that appears *in* an aesthetic text is dual: ordinary and real while at the same time fictive because a participant in fictive possibilities of culture and value. Color may also characterize the authorship of words, even potentially exposing an author's insincerity (2, 3 and 4; see Figure 3.1, **colored-as-intended**). Hues need not be confined to sensuous spots, but perform *in* a rhetoric about actions taken *within* culture and value. One does not see a hue and then *add* a context, one sees by way of context. In looking, one never forgets one is human and has had experiences.

Eisenstein distinguishes films that are merely colored or colorized from ones that are genuinely *in* color. One tactic, though not the only one, for elaborating what counts as being truly *in* is to imagine plunging deeply *inside* color. What will one find inside? For some critics, though not Eisenstein (3, 2), one finds an essential black and whiteness. Cornwell-Clyne, Herbert and Natalie Kalmus, Coates, and Aristotle provide variations on this theme (2, 1; 4, 3; 2, n. 13). In Newton's version, a pure white *contains* all the other hues (5, 2; 5, 3). These approaches install color as a simple addition to a black-and-white image, a supplement that must be specially justified (5, 1, 1) and then perhaps made to flow as painted objects move and connect (3, 1; 3, 2; 8, 1). In this way of thinking, black and white are not really colors—they stand for an unseen, colorless essence of color—or perhaps not really hues or, if they are hues, black and white are seen as more fundamental, deeper and different (4, 3), located at the very base of color, or at the base for black and the pinnacle for white. Black and white are viewed as somehow closer to the essence of the solidity of objects. Sometimes such talk envisions "white" as being a *symbol* for light itself (chapter 4), i.e., the lightness of a hue is confused with light. A certain pure, uncontaminated form of light is assumed to be the essence that makes the other hues appear as mere colorful, flighty surface phenomena—fragments of white that cover a hidden metaphysical scheme.

G-words

Talk of the status of black and white has consequences. For example, the binary of Eisenstein's color*ed* versus *in* color (in-color?) as well as the binary of coloration

versus black-and-white lead to critical evaluations that are similarly divided into competing yin and yang halves: gaudy and garish versus glorious and gorgeous, or merely colorized versus purely radiant. I call these evaluations of disapproval or approbation the **g-words** and there are eleven of each (2, 1). Disapproval that employs the gaudy-type-g-words may incorporate virulent political and social rationales (Goethe and Blanc: 2, 1; 5, 1, 1).

Another consequence of installing black and white as the source of all hues is that a spectator's mind, impressions, and memory are similarly conceived as being split in two to be serviced by either an improper or proper use of color techniques. This mirrors the fear stated by Aristotle and repeated to the present day that sensation threatens always to overwhelm rational thought—that spectacle diverts us from contemplation of the beauteous and sublime (2, 1; 7, 3, 1; see Figure 3.1, **color-as-spectacle**). Mind and memory, however, are enormously more complex, modular, and multiple than a pair of yin and yang halves (chapters 7 and 8). Instead, color aesthetics will need to be fit to the diversity of mental functions with its vast array of schemata and types of active memory. Accordingly, one must examine cases where color melts into various memory modules (2, 1), becoming invisible, but active and efficacious (5, 8; 8, 2, 1). In short, we must look elsewhere for the specification of what is *inside* color: we must look toward a broader conception of mind and toward color's instantiation in *language-actions* (3, 3; chapter 4).

Color Restraint and Delicacy

For Cornwell-Clyne, the antidote to spectacle, excessive sensation, gaudy color, and/or an overabundance of glorious color is to seek restraint in subtle color designs (2, 2). He maps out a sophisticated six-part "cycle" of color aesthetics in which the fourth, "mature" phase achieves hue combinations of "utmost subtlety." His thinking may be further extended toward designs that explore delicate transitions and perceptual thresholds (Figure 3.1, **color-as-delicacy**). The general mandate is to employ color to carefully assess the resolving power of sensoria *and* cognitive systems—to fine tune them, as it were—in two ways: to provoke a sense of the emerging presence of something and then to edge toward the discovery of a tentative recognition/identification of the thing that is emerging into presence or else advancing incrementally toward an *ambiguity* that will make several things be present. The emphasis thus falls on probing interstices and liminal states, prolonging attention and contemplation. Think of two nearly identical colors or of very slight changes within a hue or of small changes between successive patterns. Color is being seen, but in the act of vanishing quietly into thought. Cornwell-Clyne advises artists to "let your tones be difficult to describe and difficult to remember."

Note that Cornwell-Clyne is fashioning a *story of color* by winding together claims from all four major themes of the present book (P, 3; 2, 1; 5, 1), namely, he selects certain color norms/techniques (e.g., by endorsing subtlety and thresholds); employs a special language (in this case, psychological language); draws upon memory (in this case, a collective memory of the human life cycle from birth to death is projected onto color as a body); and addresses the problem of sensory extravagance. There are, however, many other ways to map these four themes onto qualities of color patterns in order to create alternative stories about the theory, criticism, analysis, aesthetics, and history of color.

The Human Face as Color Poem

Take the human face as an event for creating a four-part story about color involving norms, language, memory, and the place of sensation/spectacle (P, 3; 2, 1; 2, 2; 5, 1, 1). Certainly the face is a key norm and identification point for film spectators. In classical narrative film, the face leads directly to the soul, the emotional center that directs a flow of plot events. Different color technologies and their associated techniques may be distinguished from one another according to how the face is being represented through a subtlety (however defined), or not, of selected tints, shades, tones, temperatures, and transparencies of hues. A significant factor in Technicolor's success was its ability to render the face in a very specific way, known as the "Technicolor tan" (2, 3; Figures 6.4, 6.5, 6.8) as opposed to other technologies in which the face was deemed to have become a garish or distracting spectacle. Color is often the tie that *binds* face to a narrative language of flesh, sensation, feeling, desire, spirit, soul, intent, and action. Color thereby acquires a certain kind of metaphorical body and mind that is expressed by a spectator in psychological interpretations of hues whether a hue pattern is seen on a screen or through words.

A close analysis of six lines of a John Updike sonnet reveals color's intimacy with the human body through words and imagined color (2, 3). Making a thing *mean* something is about forging mental links through memory whether imagined from a screen or from words. In the sonnet, emotions collide with fragments of mental images of a face and body as phrases weave together numerous impressions that both attract and repel each other. The sonnet turns on a link between "tears" and "diamonds" that sparks a *remembrance* on the part of both reader and author of how tears and diamonds are conventionally said to "glisten" and "sparkle" that, in turn, brings about an expression of what it feels like to cry at as well as to (un)dress for, a special occasion. Color in the sonnet is employed to bring together face, flesh, and relevant feelings. The sonnet, though (semi-?)fictional aims at real emotions that can arise in actual situations with persons. Must a summoned color, then, be either fictional or real?

Fiction

When fiction is strongly opposed to nonfiction, or else mistakenly aligned against (or with) narrative, color is deemed too real, fit only for realism or else only for its opposite, exotica and fantasies, as it were, extreme fictions where the real has been resolutely discarded. The opposition between fiction and narrative, or nonfiction and nonnarrative, fails because built on a presumed, but specious relationship of alterity (or else similarity) that amounts to a kind of category mistake made prior to the attempt to relate the terms. Again, strong binaries fail to capture the fluidity of color. The problem is not with the nature of color, but with commonplace theories of fiction that set the fictive in opposition to the real and demand that color function in only one way.

Many theories about the nature of **fiction** are wholly misconceived. Logically, it is a counterfactual, a mere framing device or mood, for what is being said, which prescribes no advance or stable place for color (7, nn. 12, 26; 4, n. 11). As Christian Metz notes, fiction is characterized by an interlocking of mental states in an on-going oscillation, rather than somehow being on one side or the other of a bold line that divides reality from **illusion**, simulation, pretense, or lies (2, 4; 2, nn. 1, 10; 4, n. 11; 7, 1; 7, n. 24; A, 1). I believe that Metz also connects the activity of *theorizing film* to the construction of quasi-fictions (!) because both draw upon rhetorical structure, hypotheses, counterfactuals, and projections in an effort to express a theorist's or artist's felt experiences of a world—what there is—and so also a person's felt experiences of a filmed world (2, 4). Film theories are rooted in the languages and values of a community because the object film cannot be cut off from represented faces, bodies, and actions—that is, cannot be cut off from the lives of those who make and consume film. Thus to understand a theory of color in its textual manifestations, one must understand the "manners of speech" (Metz) of a community, its concerns, values, and folk theories that act to place color *in* a particular place and time (2, 4; 3, 3; 2, n. 35).

Issues involving painting, palette, and color return in the rhetoric of James Wood as he evaluates Updike's oeuvre (2, 4). Shall we say that Wood offers a *theory* of Updike's work by painting over some of the walls of rooms in an Updike building? (Compare the discussion of Metz's "poetico-didactic fictions" involving paint on the walls of a film theory; 2, 4.) Fictions of color are also painted by Gilles Deleuze in his film theory and in his comments on certain films by Jean-Luc Godard, which, he says, extend color from the human body into social and political contexts (2, 4). Deleuze finds in Godard's work a model practice whereby color functions top-down to exemplify *categories of thought*, i.e., political and social norms, stereotypes, and patterns.

Top-Down and Bottom-Up (Working Memory)

I believe that instead of asking the classic philosophical question, *what it is like* to actually experience experiencing a particular color, it would be more productive to ask what a color is made *to be like* in a situation of viewing, i.e., how memory and mind are making of color sensation a metaphor from the top-down—a link, a significance—to be fit to, onto, into, and for, an immediate occasion (2, 4). **Top-down perception** is any mental process utilizing task-oriented cognition and long-term memory. Top-down perception works to contextualize stimuli and, like a blueprint or a *pattern*, often utilizes spatial forms of time. By contrast, **bottom-up perception** is involuntary and driven by stimuli from an environment. In the case of film, a spectator's bottom-up perception is tied to the irreversible movement of shot-lengths through a projector that acts much like a clock. The conceit is that each moment of a film is unrepeatable and quickly lost, except when an image on the screen is repeated later in a new screening (but has it been repeated?).

CHAPTER 3: TO STAND IN PLACE OR TO TRACK?

Semiosis or Tracking?

One possible top-down theory is semiotics where a thing, A, *stands* for something else, which is absent, B. This theory draws on **long-term memory** and learned (a priori) connections. I call these theories **standing** or **stand in place** approaches. The approach that I explore in this book is somewhat different. It may be called **tracking** and is based on the operations of a middle-level memory system, **working memory** (3, 1; 7, 2; 7, 2, 1). Working memory is made up of a series of discrete components including a specialized language processor, phonological rehearsal loop (inner speech), visuo-spatial sketchpad, episodic buffer (which binds the phonological loop and sketchpad to episodic and semantic long-term memory systems), phonological and visual stores, sensory buffers, general workspace (i.e., consciousness and a specialized short-term memory store), along with executive functions, namely, the self-control, attention, saliency, and planning functions necessary to **track** and use time efficiently (3, n. 3; cf. the film theory concept of "suture," 3, n. 3). Working memory interfaces with textual events to track, follow, and anticipate emerging patterns in real time.

I offer two detailed visualizations of tracking at work: first, **color-as-camera** (Figure 3.1), which amounts to a new type of language-game, and, second, the interaction of two irregular wavy lines along with four modes of color tracking (3, 2; text accompanying 2, n. 19). I adapt a passage from Susan Blackmore to illustrate the flow of conscious attention that drives tracking. A key aspect of tracking—the **reidentification** of particulars—is examined in Chapters 7 and 8. One of the reasons

we track ceaselessly is to try to prepare for what will be next, to project **possibilities**, including what is "nearly true," on the basis of what has been and what is desired. The aim is to ensure that what there *is*, we will be *conscious of*, so as to better manage the future. Thus a narrative experience is more than what is immediately present and what eventuates. The nearly true remains potent as truth even if not occurring in a given narrative.

The Archive Dilemma

Tracking in the mode of *color-as-camera* (Figure 3.1) is especially adept at displaying the **movement and flow** of color. Writers as diverse as Natalie Kalmus, Cornwell-Clyne, and Eisenstein have urged that color be designed to move in a film (3, 3; chapters 7 and 8). Color may acquire a sound and move like music once it is "separated," says Eisenstein, from its object. Narrative discourse is among the invisibilia in film that color may bond with to achieve a ghostly flow. Hues that bond with, and fit into, patterns are able to partially offset the fact that a film may not display its *original* colors due to deterioration, lost technology, and especially due to the loss of the specific cultural categories and aesthetic preferences through which original audiences perceived color (cf. 8, 2; 8, 2, 1; 8, n. 19). This poses a problem for color restoration that I call the **archive dilemma** (3, 4; 2, 1; 3, nn. 18, 19; chapter 4).

Furthermore, there are philosophical problems that arise from a fundamental divide between theories of film built on ontology—the search for, and the attempt to recapture, an origin/originality—and those built on epistemology—the search for conditions of knowledge and belief that are contingent on time and place, not timeless, objective, and universal (3, 4; 2, 1; 3, nn. 18, 19; chapter 4). In theories that stress epistemology, approximate color *may* be enough to measure color within patterns. Of course, we should be pursuing both goals with color: seeking its restoration as well as its status in an invisibilia of **relational structures** and topological and semantic spaces.

If a color and filmic theory is selected that stresses epistemology, then the question of what's *in* a hue will have a different answer than an ontological theory that attempts to *specify* inherent or original qualities of the hue—its specificity. I believe that what is *in* a hue is better described as a look we bestow upon it, to put something *into* it: what expectations do we impose on a colored object? A tracking theory focuses not on what color is, but on how it functions, what it does for us, what we make of it. What a color does for us is part of what it *can do* and this aspect derives from a master schema or core metaphor that dictates inferences and extensions. One powerful, everyday schema we use *in* thinking about color is **color-as-paint** (3, 5). *Color-as-paint* fits especially well with a static, semiotic standing theory, as does **color-as-cosmetic**. I describe ten mental pictures for perceiving

color, each of which sets in place different expectations that are mapped onto color and its nature (Figure 3.1).

Warhol Weighs In

Andy Warhol's "Do It Yourself" paintings illustrate a way of seeing that may be fit to a tracking theory (3, 5). Warhol's paintings stand as a symbol of the inadequacy of a semiotic standing theory of **color-as-symbol**. The unpainted, numbered shapes in the paintings suggest not one-to-one correlations of a semiotic code, but an open multiplicity. Warhol's paintings are built upon a sense of uncertainty, chance, and noise. The unpainted shapes in their hollowness and emptiness also suggest the futility of ontological theories dedicated to object inspection with a fixation on supposed conditions of existence and originality. Of greater relevance are epistemological theories that track fluctuating appearances, such as *color-as-camera* and **color-as-context-sensitive/color-as-causative** (Figure 3.1). What lies in color resides not inside an inner sanctum, but in color talk, in the talk of viewers and critics, and therefore what animates chromophilia are the metaphors that we choose to see by (2, 1).

More generally, one may wonder about the novelty of a world—when all is said and done—that is always so *familiar*.

A final section of Chapter 3 is devoted to a detailed summary of Chapters 2 and 3.

CHAPTER 4: WHAT'S *IN* WHITE?

White and Whiteness

A test case for the above ideas is the color **white**. Is white defined by a simple essence, *whiteness* (cf. Appendix)? Our talk about colored objects that count for some sort of white reveals a surprisingly large number of hues called white, though they are not white, and sometimes are not visible. Still, saying that something is not really white does not mean that calling it white is not accurate. What counts as white would seem to depend on the **grammar** of its use as it is being fit to a particular situation. Does white possess a core definition specifying pure "whiteness" or must one instead inquire about the *need* or desire to describe something using the word "white"? I argue that white exists in an assortment of forms, but whiteness does not.

Film theorists have been busy "seeing the light" on screens, finding it white. Since white light seems to work the same way reflected by/in mirrors and filtered through curtains, the film screen can be said to "reflect" a world when not strategically veiling it. The metaphor of light allows one to use the white screen in two

opposite ways depending on the needs of a theorist's argument; that is, white is transparent or opaque according to whether the action of white light on a screen shows forth (screens, reveals) or screens out (hides, screens). Film theorists have also discovered several striking ways to argue that film does both at once—revealing and concealing—for example, with theories based on photographicity and psychoanalysis. Stephen Heath's theory of narrative film as a "**memory-spectacle**" may be extended into a productive way of conceiving the ability of a hue to shape-shift as it moves irresistibly across mental terrain bringing new ideas to light through successive links and leaps.

Hollis Frampton finds in the rectangle of white light projected on a white-colored screen something essential and also, he says, "eternal" that defines "all films." He speculates that perhaps the "sheer presence" of the feel of whiteness "has as much to tell us as any particular thing we might find inside it." White will no longer link or compare, be the subject of, stand as a metaphor for, or become a safe fiction acting as a convenient veil for defense and denial. White will simply be left to be. Without language, thought, or intent. A sense of whiteness reduced to its primal essence is Frampton's **starting point** while theorists employing phenomenological, formalist, and semiotic methods find other starting points for film, e.g., motion, two-dimensionality, shape, shadow, sound, frame/framing, style, cues, signs, and more. Yet: do film and white need a starting point or essence?

Other theories related to Frampton's approach are discussed, including Coates, Cubitt, and Deleuze. These are some of the "rhapsodic" color theorists as opposed to an alternative tradition, the "linearists."

Inside White Is Language

I believe that white does not possess whiteness. White is seen in a place and seen to have connections with its place. What else is necessary? White has no immutable quality, no certain essence, but is rather a summation of our present uses for the word and for whatever percepts that bring the word to mind in the world we live. There is no being to white, no stable core for its being white, no starting point—there are only a series of different occasions and connotations, loosely connected in **radial** fashion that define the various appearances of white in *descriptions* emerging from a form of life, i.e., emerging from present culture and value (cf. 8, 2).

Ordinary language—and the community life it supports and springs from—is a form of *manufacture for exchange and use* like other technologies in society. Film theory, too, is an object manufactured for exchange and use. Ordinary language colors color as it also colors the arguments of film theory and leads to the construction of norms.

CHAPTER 5: MAKING IT COLOR-FULL—RELATIONS AND PRACTICES

Grammatical Norms

The Wittgenstein epigraph of Chapter 5 states an important principal: simply looking at colors does not reveal the conceptual structures that have been used in seeing them and finding relationships. Instead, our talk about color is often a vague blend of **grammatical norms** about contexts and systems, coupled with particular experiences. A viewer finds order—i.e., constructs or borrows a nonconscious norm for finding order—out of several experiences, likely focusing, as Wittgenstein notes, on only a single instance of the use of a hue—whatever first comes to mind. An artist who deploys color patterns according to a new scheme may be aiming to expand the possibilities for experiencing **conceptual relations** among depicted objects while risking confusion and noise. In order to better understand how viewers and artists pursue aesthetic experiences through carefully selected patterns, one needs to understand some of the basic norms that are available to generate or recycle color practices.

What Is Color?

What mood should govern an investigation of color norms? A major tradition in art history views color as decisively subordinate to the *rationality* of line and drawing (*colore vs. disegno*; 5, 1, 1; 8, n. 15). For Roland Barthes, color is superficial and an artifice, like cosmetics applied to paint corpses (5, 1, 1; Figure 3.1, "color-as-cosmetic"). For Charles Blanc and Goethe, color is one of the lower forms of nature (5, 1, 1; 2, 2). By contrast, for David Batchelor, color harbors great complexity and remains indispensable (5, 1, 2). Batchelor argues forcefully that color has long been marginalized as either alien and dangerous or else trivial.

For many writers color seems to be paradoxically both vital and unimportant: at once concrete and insubstantial, profoundly objective and decisively subjective. This combination allows one to hold any number of contradictory views in assessing color's nature and aesthetic worth (real, too real, sublime, ephemeral, insignificant. . .). Wittgenstein argues that color—along with, one imagines, other kinds of percepts—has acquired a falsely ineffable quality because one is tempted to misconstrue it as a *private mental state*—a secret phenomenal seeing of just *that* redness, that feeling I now have, that I alone possess. The idea of a secret seeing of something real—including a private language of that seeing in order to tell oneself of the seeing—cuts off all thought and uniquely just *is* for each person. (On the incoherence of the notion of a "private language," see Wittgenstein's *Philosophical Investigations*, §§ 243–271.) The answer for Wittgenstein is not to say that color is simply physics, but that color exists as a form of action directed through a community language.

The result of the nature of color being both this and that for many writers is a peculiar sort of objective subjectivity or a subjective objectivity. But what happens when all such thought is cut off in favor of an incorrigible, interior appraisal? No longer is there the connection to a community from which arises the notion of intelligible communication in the first place. For Wittgenstein, color requires an implicit *vocabulary* that situates it in our present lives—a vocabulary that describes how it appears in concrete and public contexts, how it achieves and reworks relationships that *matter* to an individual (5, 1, 3).

The Prism (Prison?) of Physics

The vocabulary of physics is a prominent way we talk about and locate color within categories and relations. Engraved in our cultural imagination is the image of natural light streaming from a window being split into pieces by Isaac Newton's prism. The prism becomes a symbol of the objective principle of the unity of light. Equally engraved in culture is the name Roy G. Biv or his acronymous relatives: Red, Orange, Yellow, Green, Blue, Indigo, Violet (5, 2). A person can detect only about two hundred hues in the prism's light—a casual glance shows five, not seven—but ten million otherwise; paint can reproduce about 500,000 (5, 2, 1). Newton's first choice was to identify eleven basic hues, but decided on seven to match the seven notes on a musical scale (5, 3, 1).

The prism and its seven magical hues hold enormous sway in imagination. It seems perfectly natural, too, as we marvel at nature's occasional gift of a prism in the sky that produces a beguiling rainbow—or produces something extra-enticing on rare occasions, a double rainbow. The continuous blending in a rainbow and its *ordering* of hues makes for one sort of aesthetic standard. (See Figure 3.1, **Color-as-Jewel-or-as-Rainbow**.) We appreciate the apparent unity, simplicity, and completeness of a rainbow despite the fact that major colors and qualities are missing in the display (e.g., pink, brown, magenta, gray; 5, 3, 2). **Brown** is an especially important non-spectral hue (5, 3, 2; 5, 4, 1; 5, 7, 4; 5, n. 28; 6, 1, 1, 3; 6, 1, 1, 6; 6, 1, 3, 4; 6, 1, 3, 7; 7, 3, 1).

The tight organization of hues that spills from a prism or rainbow is an objective fact about natural light, though not other kinds of light. What logically *follows* from the fact of a prism, however, is only partially objective. A rainbow represents a highly selective set of facts that becomes still more arbitrary when fashioned into an aesthetic norm. The prism may be replaced by various mathematical contrivances, for example, by color circles and a wide range of other linear and nonlinear geometries (5, 3, 1; Figure 5.1). Similarly, many choices exist for selecting the number and shades of "primary hues," all of which are convenient abstractions strictly relative to purposes and materials, but only partially objective (4, 3; 5, 1, 3; 5, 2, 1; 5, 3).

Major systems have employed three, four, five, six, seven, eleven, and twelve primaries. The arbitrariness of aesthetic decisions concerning color is obscured by making successive connections to other (arbitrary) cultural decisions. One never quite reaches a state that is fully arbitrary, however, since there is always a measure of deferral in favor of the next link so that color remains **partially motivated** by objective criteria. (Cf. the partial motivation of "radial association" in 8, 2.)

As an analogy, think of the many locations you visit in a day and in a year. These places leave their imprint on your experience as you leave behind parts of yourself. You may be able to recall previous versions of yourself visiting these places. In a similar manner, the experience of color moves toward many places, not naturally, but according to purposes informed by chance and memory whether a textual motivation is relevant, partially relevant, or apparently irrelevant (chapter 4). As Wittgenstein notes in the epigraph, a theory of color is built by connecting what is experienced with something else that is, or has been, or is expected to be, experienced: this is the imperative of created pattern. Aesthetics is thus more than a theory of stinging private sensations: art works to bring to mind connected experiences that have come from one's movement from place to place in a culture guided by a set of values.

Many of the colors we see are *manufactured* and designed for profit (5, 2, 1). To understand color, one must understand the unique material properties of specific colors and how and why they have been fabricated to become meaningful commodities. Physics thus plays a role. But being meaningful comes with purposes: rhetorical, discursive, aesthetic, commercial. Thus to theorize color aesthetics, one must include the narrativized objects—tangible and intangible—that display luminous hues as well as non-spectral, complex, and impossible reflective hues. A color theory will require at least three separate, large-scale logics (5, 3, 2) and many sorts of language-games adjusted to many sorts of norms and situations.

Color Divided and Organized, Again and Again

For example, hues may be organized according to their supposed tactile qualities, e.g., **warm reds versus cool blues** so as to be extensively correlated with, mapped onto, non-tactile cultural qualities thus making up a method and "language" (5, 4). The warm-cool dichotomy, which is capable of multiple and complex references is cross-cultural and motivated by at least six facts from physiological to linguistic, though its full application to a specific text depends on cultural factors. Whether text and perceiver must be identically situated, and the degree to which this is an ideal match or even possible, is another issue. The mutual projection between the warm-cool dichotomy and culture leads toward profound issues: the nature of relationships, how a hue may become resemblant, textual analysis, the expression of thought, and the "language" of cinema.

Descriptions about hue relationships may also be built upon the natural scale of luminance values starting with white and yellow and descending to purple, brown, and black (5, 5). The *perceived* interrelations of hue, lightness, brightness, luminance, subjective and objective color temperatures, value, and saturation are quite intricate and nonlinear allowing artists to confound and snarl expectations when desirable. Only some of these relations are considered in this book.

Is there a "natural scale" by which the names for hues come into existence in language? A study of 110 languages discovered an extraordinary fact: there is a mostly inflexible sequence to the acquisition of as many as twelve fundamental color terms known as universal or landmark focal colors, i.e., prototypical hues (5, 6). All languages, for example, contain terms for black and white; in this situation, the perception of all cool hues are included in the term "black," all warm hues are called "white." If a language employs a third term for color, it will be "red." And so on. Thus language would seem to reflect a universal way in which diverse social groups assign significance to the presence of a hue so that, for instance, a red hue is the next most important color to mark out and differentiate after black and white. Or: is red the first hue while black and white are primary and something else?

Perhaps the universal mapping that creates color prototypes in a fixed sequence (in spite of slight variations in the quality of each prototype), may be extended still further into mental schemata that function to direct a perceiver's expectations (P, 3) and organize **semantic fields**. What is important is how the semantic fields of a given language apply color to cultural matters (cf. the mental pictures/schemata of Figure 3.1).

Organizing colors in text and mind often requires some colors to become more prominent than others. Myriad ways exist to forge a **key color**, including adjusting color areas (5, 7, 1: saturation and area, lightness and area, hue and area), creating magnets, pinpricks, blots, stains, punctures, and accents (5, 7, 2), among other strategies (5, 7, 3). Artists and writers have also created self-contained systems that specify rules and hierarchies for color usage (5, 7, 4). The idea behind organizing color into prominent patterns is to single out specific colors whose function is to move, link, and participate in important textual/conceptual structures.

Wittgenstein and Impossible Color

Must a key color always be visible? An analysis of Wittgenstein's discussion of the material color, reddish yellow, and the **impossible color**, bluish yellow, leads toward the conclusion that visible color, when deemed prominent and significant, need not be visualized in mind while certain non-spectrum colors may be visualized even though nonexistent in the real world (5, 8). Visible color is thus not always visual while color that can never be visible may become visual. For example, an impossible

luminous gray may still be imagined (5, 3, 2; 5, nn. 26, 27, 66). The reason is that major components of color consciousness in one's working memory come from top-down mental processes that are relatively independent of bottom-up perceptual processes.

Color consciousness depends crucially on *patterns* and *contexts* and, in turn, these depend on **relationships**, all of which are always conceptual and invisible, i.e., top-down (4, 1; 8, nn. 5, 6). Relationships among hues are invisible, even when two hues are visible and adjacent. For example, the invisible relationships of being "next to" or "to the left of" are identical to being "apart from" in that these three circumstances depend on a further relationship to an abstract conception of *space*. Relationships are moveable in mind and may combine to form mental patterns, some of which may take the form of images. In like manner, delineating a context for something is a philosophical challenge because a context is composed of a set of invisible relationships with visible elements (cf. Derrida's notion of the mutability of context; 8, 2). The nature of a context is thus similar to that of an impossible hue.

In this spirit, one may ask a still more general question: how much of our experience of a film is actually *visible* on the screen?

CHAPTER 6: MUSICAL HUES: COLOR HARMONIES

Harmonies

Conceptual structures that are based on relationships to music have been devised to generate color patterns. Traditionally, six low contrast **harmonies** are identified, which seek to coordinate colors to create a flow, continuity, or **passage** among them, along with five high contrast harmonies, which attempt to balance colors against one another (cf. 8, n. 16). In effect, these two types of harmony attempt to define adjacency (being appropriately "next to") and distance (being properly "apart from") by spatializing color into ordered, "melodious" pairings. In addition, nine possible **disharmonies** are described that may work to tangle and create conflict and discord among hues (chapter 6; the arc of pleasure and unpleasure, 6, 1, 3). Harmonies and disharmonies depend, like music, on an orchestration of *intervals*.

What might be the aesthetic basis for these three types of rules/patterns—low and high contrast harmonies and disharmonies? If beauty is in the eye of the beholder, does the eye search until it settles on one of these states of harmony or disharmony (6, 1, 4)? That is, these rules may act as gestalt principles that function to create unified or disunified pairings of colors. If so, a series of such gestalt-like states may create a new type of harmonious flow, blockage, balance, or conflict at a still *higher* perceptual level that may then produce further movements—continuous or staccato—among groups of large-scale patterns. This idea is illustrated by analyzing two photographs.

What is the philosophical status of rules about harmony and disharmony? For example, do the rules reflect causal relationships within mind or world (i.e., as subjective or objective facts), or are they instead either a heuristic, a convenient description, an arbitrary inclination, or simply a myth (6, 1, 5)? These six possibilities lead in various directions when constructing a theory of color. In general, the present book assumes that color is incompletely motivated—i.e., implemented in and for a context, but only partially—so as to keep in mind other possible contexts, expectations, background knowledge, and vocabularies that may come to bear upon a color pattern. (Some of these contexts are listed in the final section of this summary of Chapter 6.) My approach is to remain sensitive to—to track—the radial extensions of color as it appears in working memory and narrative discourse (chapters 7 and 8). This also means that there may exist types of color patterns that are significant, but neither harmonious nor disharmonious.

Five Film Examples of Relations and Practices

Chapter 6 concludes by closely examining portions from five films in order to illustrate some of the ways that the color relations discussed in this and the previous chapter may be activated and partially motivated through narrative context by mise-en-scène.

In *Gentlemen prefer Blondes*, high and low contrast harmonies in a particular shot are motivated by a *temporal* circulation—passage and blockage—that mimics the forward movement of the plot in accordance with effects from the future since a film's narration straddles both present and end (6, 2, 1; 6, 2, 3; 7, 3, 3; 7, n. 10). Color design in the shot moves ahead marked by the future, i.e., time will tell what has been foretold in the telling of the story. A spectator's narrative journey always takes him or her back to the future.

Picnic motivates its low contrast harmonies by the scenic *space* inhabited by the characters (6, 2, 2; 6, 2, 3). Scenic space for classical narrative generally comprises at least thirteen factors interwoven with principles of color design.

Vertigo employs neither high nor low contrast harmonies in creating an in-between color design that is vaguely disquieting in a particular sequence of seventeen shots (6, 2, 3). The ambiguous realism of *Vertigo*'s patterned color and the prominence of red provides an occasion to examine a typical *rhetoric of film criticism* that seeks to interpret color design in many sorts of films. This rhetoric relies on **stylistic metaphors** (6, 2, 3)—i.e., the mapping of an aspect of plot or theme onto an adjective describing style or a spectator's response to style. An example is the claim that Madeleine's black dress is concealed under a white coat that acts to mask a concealed threat (blackness) to Scottie beneath Madeleine's apparent innocence (whiteness) in the same way that in the plot Madeleine's false identity conceals Judy

and Judy's plot against Scottie, which, in turn, conceals Gavin Elster's plot against Scottie beneath the fictions of Madeleine and Carlotta. This type of rhetoric relies on long-term semantic memory characteristic of a semiotic "standing" theory that seeks to assign definite meanings rather than to follow ("track") a spectator's evolving expectations within working memory. Standing and tracking theories rely on different memory systems and so may, or may not, complement one another on a large scale, though each addresses quite different ways that a spectator engages film moment by moment (3, 1; 5, n. 55; 6, 2, 3; chapters 7 and 8).

Represented color depends on the exact properties of textual materials and techniques. An image from *The Wizard of Oz* illustrates the way in which a specific *technology*—Technicolor Process Number Four—frames a color design by enhancing certain properties, suppressing others (6, 2, 4).

Technicolor's technology required a design *style* to maximize its effects and hide its defects. This style is examined through a film, *Two or Three Things I Know about Her*, that rejects Technicolor's style in the process of creating an alternative to classical Hollywood cinema (6, 2, 5).

Color Contextualized

To sum up this summary: in this and previous chapters, we have seen color organized and described in a wide variety of **contexts**. It has been attached to the lower and least strata of being and nature—evicted, as it were, from both ontology and anthropology (5, 1, 1); subordinated to line (*disegno*) in the history of art (5, 1, 1); found in natural light (5, 2; 5, 3, 1; 5, 5) and not found (5, 3, 2; 5, 8; 5, n. 66); recruited to physics and material science (5, 2; 5, 2, 1); limited by its technology of reproduction (chapter 4; 5, 2, 1; 6, 2, 4); fit to geometry and mathematics (5, 3); subjected to the mathematics of music (6, 1; 6, 2, 1–2); made tactile—marking a return to the human and culture (5, 4); established in cultural politics (5, 1, 2); divided capriciously and broadly into glorious versus gaudy and into restraint/realism/nature versus spectacle/fantasy/distraction/pleasure (2, 1; 5, 1, 1–2; 5, 7, 1); submitted to critical rhetoric (6, 2, 3); named in a specific order in 110 languages (5, 6); akin to a gestalt law (6, 1, 4); placed within a variety of systems of speech, rhetoric, and semantic fields (Wittgenstein; 5, 1, 3; 6, 1, 5; 5, 4, 1; Figure 3.1); absorbed into film theories, which is to say, absorbed into theories about the preferred uses of ordinary language in describing film abstractly, i.e., generally (chapter 4); systematized in aesthetic systems (5, 7); narrativized in any number of ways in almost every case; and, twisted into interpretations of film (6, 2). These among other blendings. Should some of the blendings be broken into finer categories and then re-assembled into larger shapes making for new, impromptu theories of color?

Color is not merely a visible spot, but lives in contexts and moves among them. Hence the fascination of color for philosophers and spectators. The task for the color analyst is to determine which contextual elements drawn from community life have been selected to blend with color in order to fit color to an objective, which objective is to infuse physical and nonphysical objects with color in order to form a rhetoric of evolving patterns and arguments. The critic then analyzes and measures the fitness of the fit.

CHAPTER 7: TRACK THIS IN PLACE

From Ascetic Binaries to Aesthetics

Chapter 7 looks more precisely at the mental process of "tracking" and the creation of a flow of attention by a person's hard-working working memory that functions both in the moment and beyond and before (3, 1; 3, n. 3; 7, 2; 7, 2, 1). Patterns appear in consciousness, new properties emerge from patterns (1, 2; 1, n. 6), "melodies" are made and remade of perceptual events, and patterns of patterns may come into existence. In short, a text is not flat under working memory, and is not like a ribbon of celluloid running through a projector, but rather is composed of a series of levels and intervals where relations among patterns on a higher level may be significantly different than among elements (or patterns as elements) on a lower level. A notable example of these strata in a film text are **levels of narration** (7, 2; 7, nn. 10, 11, 12; 1, n. 6).

A color may literally or figuratively change color by acquiring new sets of relations and meanings when it appears on a new level while "moving" through a text (chapter 8). A color's information content is subject to change depending on changing contexts. When a text is conceived as broken into levels—even though certain aesthetic styles seek to make some levels relatively invisible—a margin of overlap, mediated by language, may come to exist between what a perceiver believes to be "reality" and what is taken to be merely "illusory." An overlap contains aspects of both reality and illusion. Thus some parts of reality may be seen as illusory and some illusions and fictions may harbor reality (7, 1). Along with other binaries encountered in this book, the binary of reality and illusion needs to be made more responsive to the actual processes of mind and language.

Some of the binaries that are being recast in this book by attending more closely to memory-chromatics and the operation of **language-fields** (i.e., semantic fields, mental images, schemata, and language-games) are the following: glorious/gaudy, rainbow/black-and-white, *colore/disegno*, impossible hue (e.g., bluish yellow) versus possible hue (e.g., reddish yellow), spectacle-emotion versus normality-reason, perception/cognition, practice/theory, motivated/unmotivated, illusion/reality, fiction/nonfiction, and figurative/literal.

Reidentification (P.F. Strawson)

Harsh binaries give way to gradations, transitions, levels, and frameworks. A person constructs in memory a unified framework for perceiving his or her world and for measuring changes to that world—a structure into which persons, places, and objects may be slotted and their characteristics known as they evolve. P.F. Strawson undertakes a "descriptive metaphysics" by examining mental structures that create a "framework of particulars" such that "a system of elements [exists] every one of which can be both spatially and temporally related to every other" (7, 2). A key process in a given relationship of elements is a person's ability to **reidentify** "objective particulars." I believe that Strawson's ideas apply equally to a spectator's mental construction of a framework for his or her comprehension of a film world.

A mental framework permits a continuous recognition of things in spite of inevitable and persistent *ellipses* and gaps from occasion to occasion that are the result of, for example, limited views, movements from place to place, fluctuating attention, distractions, changes, and sleep. Film techniques and narrative style (like the world) contain abundant ellipses through which a constructed framework must remain the same in order to accommodate and compare new and reidentified elements within a film's levels of reality, one of which, the diegesis, is taken to be the physical reality lived by the characters (7, n. 10).

A gap of varying distance exists between the framework a spectator deploys in a film and those frameworks he or she utilizes in the real world. Experiencing a screened world is not identical to embodiment in a lived world. A spectator knows that he or she is watching a film. Film theorists interpret this gap and distance in many different ways. George Wilson argues persuasively that relative to an actual world, a film's **narration** positions a spectator's state of awareness as if he or she were a disembodied witness, i.e., experiencing a film's world from an impersonal, "unoccupied visual perspective" (7, 2; 7, nn. 11, 12). I should add that the camera we imagine moving *in and around* such a film world would likewise be disembodied, i.e., the camera is no longer conceived by a spectator as being a physical entity with a weight and serial number. Instead, the camera has become invisible and all-powerful, though indisputably real, not illusory, in that it provides a spectator with highly pertinent information and views of the characters, despite flagrant omissions and misrepresentations in the service of telling a good story.

Reidentification of particulars within a framework must also accommodate gradual qualitative changes in the particulars. This opens up new sorts of problems for identifying the recurrence of a slightly changed physical or nonphysical element, e.g., a material object or an element of narrative structure, but also opens new possibilities and powers for a text, where a text is visualized not as a flat sheet of evenly spaced lines, as it were, but as a set of wrinkled valleys and mountains with

lines bent and sometimes torn and irregular, though still serving as a viable "map" for a spectator who has become a "disembodied witness" and traveler (cf. 3, 1). The analogy of irregular mapping aims to capture a measure of real world contingency, unpredictability, and changing expectations in the form of, for example, a spectator's feelings of narrative mysteries, surprises, and suspenseful moments.

Gareth Evans and Ruth Millikan have expanded Strawson's definition of reidentification. In effect, they reidentify his definition and slightly refashion it in order to fit a larger context, a "social informational system" that would seem to be a better fit for analyzing aesthetic texts and color patterns (7, 2). Though Strawson did not have narrative discourse nor color patterns in mind, his general principle still holds: "We cannot attach one occasion to another unless, from occasion to occasion [despite pervasive and inevitable ellipses], we can reidentify elements common to different occasions."

Reidentification and Color Consciousness

Acquiring the methods and skills needed for reidentifying elements impacts at least four areas of color consciousness: (1) sensing the recurrence of abstract information contained in color patterns and relationships; (2) marking (then remarking and remarking on) a hue as the basis of a metaphor that reframes present elements; (3) reidentifying informational states and following the movements of color patterns through levels of narration leading toward propositions that form narrative arguments; and (4) employing working memory to reidentify elements that are crucial in tracking and comprehending hue patterns (7, 2).

Critical to these four areas of color perception is the particular *language* we have chosen to express our felt logic of experiencing and using color. We act and see connections relevant to our lives through adopting—consciously or nonconsciously—frameworks, concepts, norms, and attitudes. The notion of language-games or language-fields has been a major theme of the present book. From this perspective, a particular language-game may be viewed as a framework that supports an interlocking set of *patterns of thinking* about patterns of color (7, 2).

Reidentification and the Nearly True

Reidentification is a function of working memory and conscious attention (7, 2, 1). Reidentification addresses the future in terms of a person's expectation, i.e., through an expectation of a future reidentification. One aspect of thinking about the future is a "nearly true" hope or fear. Event x has occurred, but event y nearly happened instead; or, event x has occurred and event y may be next and near. This amounts to thinking broadly about the composition of narrative events. More narrowly, the

presence of a hue may count as an "event" capable of triggering the thought of a nearly true hue and/or a nearly true, hue-related plot event.

A crucial purpose of narrative, whether intended or not, is to implant in a spectator's imagination a series of "lingering doubts" in the form of a set of nearly true events. In general, narrative discourse operates to bring about a spectator's awareness of the chances of life, some of which perhaps have been temporarily or permanently thwarted, or else advanced by the failure of the nearly true to become true. Nonetheless, the nearly true will not have been abolished from mind. What is *otherwise* in the world or a text remains caught in our lives.

Analyzing Color Tracking

To describe color tracking and the "nearly true" in a narrative context, an analyst must select (1) a scale for analysis from micro to macro; (2) a type of color pattern to follow; and (3) a type of linkage to follow between the color pattern and narrative (7, 3). Color patterns may appear in the form of (4) a hierarchy or heterarchy.

A color pattern may be tracked as it remains in place to become, for example, distinctive of locale or character; or, color may be tracked as it moves and changes. I discuss three examples of each of these kinds of tracking. The first three examine tracking in place where the color pattern persists from one time to the next.

Three Examples of Tracking Color in Place

Vermeer's painting, *Woman in Blue Reading a Letter*, illustrates how color may be divided into a series of four high-contrast dichotomies that serve to articulate a particular space, character, and action (7, 3, 1). Though the dichotomies are realistically motivated by light from an unseen window, a viewer may be prompted by his or her impressions of certain prominent color contrasts and similarities to entertain a number of ideas about unseen matters being depicted by the painting. Vermeer shows a woman reading a letter, though we are reading the painting itself as a kind of letter. Wittgenstein argues that images and words (and hence images and words in, and about, artworks) are interchangeable in an important place in the human mind (7, 3, 1). Only humans possess both language and art. Because language is always present to mediate sensation and its world, there can be no pure spectacle that, like a drug, expunges all thought.

Greenaway's film, *The COOK, the THIEF, his WIFE & her LOVER*, utilizes six distinctive color dyads to mark its six locales, each of which are connected to specific bodily and metaphysical states, actions, and themes (7, 3, 2). Sometimes a character's clothes will instantly and miraculously change hue to match a new locale that he or she has entered. Color in the film is placed as the principal sensuous

correlate and dire paradox of decadence and art, of aspirations both low and high, of body and soul.

Tykwer's film, *Winter Sleepers*, stamps each of six major characters with colors that determine his or her fate in accordance with the *logic* of both the subtractive painter's primaries and the additive light primaries (7, 3, 3). The perils and prospects of a future reidentification—i.e., a spectator's expectation—in the form of what is "nearly true" are raised as a spectator follows the character of Black man, who suffers from a deficit of working memory (anterograde amnesia) that is eerily similar to the one that afflicts a spectator, who, in watching a film, is trying to piece together the future of a plot while trying to recall details that will become significant and discarding what seems irrelevant. One of the issues raised about Black man is whether it is true or nearly true that he is the father of Blue baby at the end of the film.

Trapped within a network of desire, unintended consequences, and chance events, no character at the film's conclusion is able to recognize his or her true role or nearly true role in a child's death. Instead, each character is misled by a limited perspective on a world of concrete objects, local colors, and grand color schemes that seemingly dictate all. Only a spectator can see the whole picture and assign degrees of responsibility for the child's death. A spectator thus benefits mightily from the film's omniscient narration. Stepping further back from the film, a spectator steps into his or her own limited perspective on life.

Remarkably, the narration of *Winter Sleepers* is also assigned a color: white! Within the film's use of conventional color associations, white is a perfect expression of an ambiguity at the core of all storytelling (cf. 6, 2, 1; chapter 4). White is at once a blank starting point for the appearance of hues that will dictate the interactions among color-coded characters while also the symbol of an invisible, all-powerful force that withholds information and manipulates plot events toward a predetermined end. For a spectator, white at every moment is simultaneously beginning and termination.

CHAPTER 8: TRACK THAT IN MOVEMENT

Movement and Color

Chapter 7 illustrated how a color pattern may be tracked as it remains in place to become, for example, distinctive of locale or character. This chapter considers color undergoing movement and change.

Objects in the world, whether moving or not, may make color move in mind. Some perceptual illusions produce a sense of motion. Memory may move color. The Pre-Socratic philosophers sought to explain mobility in general by producing

a systematic inventory of answers as to its sources (8, 1). Today, there are still more answers, which depart from the methods of the ancient philosophers, though not always from their conclusions.

In the case of film, some theorists seize on the movement of images through a projector or on various formal properties of images, e.g., editing and camera movement, including elements *missing* in images that require an act of mental filling-in not unlike the operation of some perceptual illusions. When thinking about relationships, patterns-in-the-making, and patterns of relationships, the focus falls on the activity of working memory, reidentification, expectation, and language processes in making and seeing things change.

Often we are prevented from seeing colors move by the *mental image* we possess of the idea of "movement" that dominates our thought by restricting movement to percepts of motion, as when billiard balls, cars, and the like are taken as exemplars of movement. Even when motion percepts fail to appear after the wheels have fallen off, we continue to summon the same mental image, now augmented by the idea of a *potential* for motion in those billiard balls and cars, which amounts to a sort of moving non-movement or unmoving movement. Thus we miss important cases of other kinds of actual movements in the world and in texts that are not determined by motion percepts or their momentary lack. These movements include moving colors, color patterns, narrative and stylistic patterns, immaterial objects, relationships, and figurative entities (8, 1; 8, n. 6).

Two Examples of Tracking Color Movement

Ensor's painting, *Christ's Entry into Brussels in 1889*, demonstrates how a green hue may move, which I visualize as a "green river" (8, 1, 1). A particular shade of green weaves and winds together a set of ideas as a beholder's eyes follow it from the upper left of the painting down to the lower right (or in the opposite direction). Ideas emerge from the sorts of grotesque objects that the green moves through or touches. The cumulative effect promotes a sense of despair about certain aspects of modern life and modern painting. This pessimism in the painting reverses the usual connotation of green as signaling renewal and fertility, instead stirring—causing—ideas of depletion and death. Color moves and generates ideas.

Powell and Pressburger's film, *Black Narcissus*, uses red as a tracking color, a "red river" (8, 1, 2). The film also illustrates a different kind of color movement when we see a white hue transform/morph step by step into red. Color flows and also leaps from idea to idea in developing a theme of political and psychological oppression, repression, and their potent return.

Four Types of Color Reidentification, Radial Association, and Derrida

The five examples of color tracking discussed so far in Chapters 7 and 8 illustrate the first three of four basic types of color reidentification: (1) repetition/reappearance; (2) river/passage; (3) morph/ metamorphosis; and (4) library/archive. The third type suggests that a film may create *its own* set of "primary hues" that generate the other hues in the film that, in turn, may slide into and out of each other (8, 2). The fourth type will be illustrated below in conjunction with *Little Dutch Mill*. The fourth type is, in effect, a multi-colored river that links multiple objects of different colors (8, 2). This type will be difficult to distinguish from the case in which hues and objects are simply changing with no set of discursively relevant associations to bind them. The rarity of this fourth type of construction is probably due to the severe limitations imposed by working memory on both dynamic and static forms of conscious memory (two or three items and five to nine items, respectively). Artists seeking *impressiveness* practice within these limitations on immediate memory. More complex versions of the above four types of color reidentification involve the movement of patterns of hues instead of individual hues (7, n. 22).

Critical to all types of reidentification is the determination of *relevant* associations and the construction of a rationale for linkages among a cluster of hues and objects. The associations may be categorized and related to rhetorical figures and to the ways human memory organizes knowledge (8, 2; 7, nn. 13, 17, 24, 26). Distinct, though connected, meanings will form a set whose logic is based on radial extension or family resemblance (Lakoff, Wittgenstein). A radial series is a series of contexts or concepts, each of varying strength that continues to inform a starting point. This makes patterns less objective, less immediate, less stable, more mutable than what may first appear at a given **starting point** (4, 3; 4, 5).

I believe that the radial nature of associations applies also to images and percepts, such as color, and that it clarifies several aspects of Derrida's theories, including the *mutability of context* as well as his use of the term, "saturation," which, through radial association, is applied to color! (8, 3; 8, n. 32). Small gaps and interruptions among associations create differences while the stream of associations as a whole acts to postpone any final meaning (Derrida's *différance*) while continuing to encourage new semantic conjunctures and apparent "accidents" of meaning. Thus, in theory, a perceiver's thought remains active and contingent whether working with concepts or percepts.

The Example of Little Dutch Mill

Max Fleischer's cartoon, *Little Dutch Mill*, illustrates both river and morph forms of reidentification. It demonstrates how a color may acquire a color in spite of our

tendency to take material objects, and the lines that define their shapes, to be a *starting point* and more fundamental than their color (8, 3, 1; cf. *colore* versus *disegno*). More fundamental for what, one may ask? A line seems to outline an object by separating its edge from whatever is not it. However, the danger in focusing attention on a single material moment or quality like "line"—or on some other spectacular phenomenal or formal event—to the exclusion of, say, the color of a color is that one may lose track of the movements of radial narrativization, i.e., lose track of one or more states of affairs under causal or non-causal change (8, 1)—of what remains old and potential in an object, of the evocation of what is "nearly true," and of the possibility of oppositional (deconstructive) forces. Why should "line" or an initial hue receive priority over other features when assessing relevant associations of a radial concept or percept?

The river of color in *Little Dutch Mill* is white. Combining the "white river" in the cartoon with a countercurrent of other colored objects, suppressed associations, and ambiguities makes for a "library" or "archive" of reidentifications, which is the fourth basic type of color reidentification. Questions arise in the cartoon concerning the notions of "empty," "absence," and "naught"; concerning the nature of a hand bringing or holding onto gold; and, concerning the agency of hands that open the cartoon by opening a display cabinet containing a decorative plate of "a little Dutch mill" with an evident narrational will.

Is the plate the narrator of the story? Is a collection of plates in a cabinet a gift or an instance of hoarding, like the villain's hoarding of gold? The villain's hands are shown to be metaphorically dirty because of his actions, but the hands that open the film and introduce the story can't be so easily judged because the circumstances and person opening the cabinet are not shown and the storyteller's subsequent actions are indirect, disguised, and opaque. A "library" of reidentifications thus moves along both an explicit story path and an alternative, radial shadowy path.

The third type of color reidentification, a color morph, occurs when a rug is being cleaned and we watch as it transforms from a dirty reddish brown to a bright multi-hued pattern containing light and dark blues. But which of these two color states of the rug counts as genuine and original? Employing which assumptions? (Note the similarity here to the "archive dilemma," which involved the restoration of a film to its original color; 3, 4.) There is a sense whereby each of the two color states of the rug may color the other depending on the choice of a *starting point*. Which is the more natural state of the rug when each color may be realized out of the other? The cartoon's narrativization of action and *hue* pushes a spectator toward a particular orientation and conclusion whereby dirt is bad—seen as dirty brown in the rug—along with the hoarding of gold, which is

equally bad since the gold is darkened, made dirty-looking, inside a wall where it is kept hidden by the villain.

The cartoon argues that cleanliness and whiteness is the proper starting point and is next to a radial and radiant brightness, goodness, godliness, and prevailing community values. In discursive contexts not actualized in the cartoon, however, dirt and hoarding may be seen as a virtue, for example, in the fertile soil of farming and saving for the future, which also are community values. As a discourse, narrative steadily works to suppress certain associations and possibilities while judiciously marshalling concepts, connotations, and (new, old) contexts in bringing about an explicit conclusion. To adapt Wittgenstein's comment about the work of a (storytelling?) philosopher, the work of a storyteller "consists in marshalling recollections for a particular purpose." What, then, are a spectator's recollections about purposes? Which spectator at which time—an original spectator in 1934 or one today? (For more on Wittgenstein's comment, see Chapter 10.) *Little Dutch Mill* struggles to argue that colors clean and bright are the original and proper state of all things, including invisible community values (but see Barthes's celebration of "discoloration"; 4, 3). What assumptions are involved in this search for what a thing is *first* of all, its *starting point*?

The Garden Metaphor of Reidentification

Imagining color as a static material, as a matter of "quasi-paint" or as a "cosmetic," serves to limit meaning in the same manner as the controlling and censoring acts of narrative discourse. A more expansive, mindful approach to color and narrative conceives of them as opportunities for a person to actively track patterns and possibilities, as in the "camera" and "causative" schemata (Figure 3.1). Furthermore, color movement through reidentification and radial association should apply to the movement of many other kinds of percepts. Thus a spectacle of sensations does not stand as simply a chaotic carnival of feelings, but is linked to cognition generally. Spectacle is not absolute, but relative to the multiplicity of mental schemata deployed by a person.

The six illustrations of reidentification from two paintings and four films in Chapters 7 and 8 have divided the tracking of color into two parts in which color is seen to be changing while "in place" or while "in movement." These two modes of tracking may be combined into a single mental scenario to represent encounters with color whether in a moving image art or a fine art, such as painting (8, 4). Envisioning an ornamental garden is one way to create an integrated scenario. The garden has paths and benches that provide "passage," "blockage," and sustained viewing of a succession of color areas while walking or sitting. This analogy

embraces all four types of reidentification and may be expanded in several ways to include, for example, changes in lighting, point of view, and out-of-garden color. Movements from area to area of the garden simulate the perception of color in an artwork as a person's **chromatic memory** is continuously updated and altered.

In the case of film, there are many formal techniques capable of generating color movement; for example, editing, montage *within* the shot, and distance montage (3, n. 9; 8, n. 42); narrative and narration; speculations by a viewer; off-screen events; reveries and repressions; and, not least, how a viewer talks about a remembered film (8, 1; 8, 4).

Is Color Movement Literal or Figurative?

To comprehend film, one cannot limit movements of mind to the movements of graphics across a screen (8, 4). There exist literal movements and literal potential movements in the world like a train in motion and one waiting in a station. There also exist literal movements of invisible objects, such as a train of thought, and literal potential movements of invisible objects like a next train of thought. Is one of these types of movements more literal and the others less literal or merely figurative? The answer depends on the assumptions and norms adopted by a person when thinking about movement together with the selection of a reference or starting point and criteria for measuring the ensuing movement.

The foregoing raises a critical question: is the *mobility* of color literal or figurative? The answer depends on the choice of a starting point for tracking. For a color theorist, the answer depends on the choice of a final destination for his or her aesthetic prescriptions and thus *where* he or she begins to tell a particular aesthetic story by choosing and setting in place a narrative "point of attack" and "initiating event" in a theoretical account of color. If one begins by saying that color moves only in a figurative sense—by, for example, treating color perception as a sensory photograph and working memory as a page of photographs—then the truth of this metaphor about mobility will lie in a future effort to literalize it and marginalize change (cf. 2, 5).

CHAPTER 9: SUMMARY

CHAPTER 10: CONCLUSION: HOW IT FINALLY MATTERS

Diogenes and Wittgenstein

The conclusion to the present book explores connections among three searches: the search by the philosopher Diogenes for an honest man by using a lighted lamp in daylight; the search generally by philosophers for knowledge using Wittgenstein as a beacon (4, 3; 5, 1, 3; 5, 8); and the search by perceivers for knowledge of a colored

world. A search is always within a methodological context. All searches, including that of the present book, begin with tacit assumptions and criteria (preface and chapter 1), i.e., a search has reached there from *here*. There is always a here—a trace of here in there. The key to color consciousness lies in a present interaction between patterns in an artwork and patterns of new and old thought.

APPENDIX: WITTGENSTEIN/CONTEXT: TWO PHILOSOPHY LESSONS ABOUT COLOR AND SOUND

The Appendix examines two paradoxes posed by Wittgenstein plus a venerable philosophical puzzle in an attempt to clarify the influence of *context* on our thinking about both color and sound.

White Darkened

Wittgenstein observes that a rose in the dark may be seen either as quite black or as red. Elsewhere he wonders about the status of an impossible color, such as "bluish yellow" (A, 2, 1; 5, 8). If one were to put these two ideas together, then the rose, though quite black, may be seen as red through one's memory and perhaps also impossible colors might be created through imagination by joining incompatible parts to make a new whole (perhaps as a mythical griffin is a lion-eagle composite). In addition, Wittgenstein offers a color paradox, "If white turns into black some people say 'Essentially it is still the same.' And others, if the colour becomes one degree darker, say 'It has changed *completely*.' " This raises the question whether there is an essence to white (chapter 4) that can be called "whiteness," and for that matter, whether a few or many or all other hues, even apparently inconceivable ones, have essences.

Obviously one cannot say that each hue—there are about ten million—that has become one degree darker has a new essence since the idea of an "essence" would then have no meaning, i.e., each thing would be its own essence and essence would be dependent on a person's acuity of vision; nor can there be a new essence for every degree of saturation or of luster or of any other color quality that has changed by a single degree. How many qualities and degrees of color are there?! This argument is a variant of the sorites paradox which asks, for example, whether there can be an exact number of grains that make for a "heap," i.e., where exactly is the boundary between a heap and non-heap or white and a darker white or red and orange. This paradox, in turn, is related to the crucial problem of parts and wholes tackled by Plato in five of his late works since a heap of parts does not truly make a whole. It would seem that one must create a ranking system for all types of changes with fewer and fewer changes in appearance or behavior allowed until one reaches a fundamental changeless state, which is the essence of an entity. An essence is the

necessary and sufficient core that makes a thing exist exactly and only as such. Despite such problems, can an essence be discovered for a particular hue or for the status of "color" generally?

Some hues (but which ones?) are not "pure" and so may not have a distinct essence; orange and violet hues, for example, are (apparently) mixtures of red and yellow, red and blue (5, 7, 4). There is also the issue of whether the primaries may be the only colors with essences; but, if so, how many sets of primaries exist—many numbers of primaries have been offered—and *which* hues of the different primary reds, for example, are the real ones (5, 2, 1)? Once one sees how a hue changes in different contexts—within physical, perceptual, cognitive, linguistic, cultural, logical, and aesthetic systems—the problem becomes intractable. The numerous qualities of sound, too, spread out in many of these same directions and thus sound, too, defies an absolute, context-free definition.

A more promising approach for thinking about the nature of color is to inquire what color is *said (or judged) to be* through an acquired *grammar* and rule system fit to an occasion, a practice. A particular color grammar involves a succession of concepts that generate descriptions, categories, criteria, perceptible objects under descriptions, and projections (A, 1). The same approach works for sound.

Multiple Grammars

If multiple grammars are possible, then we can talk about a white hue without contradiction in both of the ways Wittgenstein depicts in his paradox, i.e., white is both unchanged when darkened and changed completely just as a rose in the dark may be seen as quite black and as red. Words are relative to a way—a grammar—of thinking about color in particular contexts. White as radiant "light" and white as "paint," for example, may form two quite different contexts for grammars. As C.L. Hardin indicates, color is relative to many sorts of contexts (A, 1, 1).

Furthermore, contexts have contexts. Analytical frameworks provide a general orientation and method of investigation when implementing various descriptions, i.e., such frameworks offer meta-descriptions of pertinent color relations and what will count as potential change within a system of rules fit to specific hue contexts. The following hoary philosophical paradox will serve to highlight four methodological contexts for theorizing hue contexts: "A tree falls in a forest, but no one is there to hear it. Does it make a sound?"

Four Contexts for Contexts: Objective, Subjective, Inter-objective, Inter-subjective

The puzzle is always stated in terms of sound, never color. Why isn't the puzzle stated in the following way: "A tree has leaves and bark in a forest, even though

no one is there to see them. Are the leaves green, the bark brown, when no one looks?" What sort of *essence* are we imagining for sound such that it does not exist without a perceiver, but color always does? We tend to think that sound essentially *comes and goes* and so apparently may or may not really *be there*—as we, too, come and go in the forest, and may or may not be there. By contrast, it just seems natural to think that color is always *out there* in a definite place in some permanent fashion, like the *shapes* of trees and leaves, with the result that no similar paradox arises with color even though the perception of both sound and color depend on the movement of physical waves (A, 2; cf. 5, 3, 1). Nonetheless, we believe that sound must be made and continuously remade in order to continue to exist—or else it must exist elsewhere in some mysterious, soundless irreal realm made of each thing reduced to its essence, so that hearing would need no vibrato.

How do we imagine the color of an enduring hue? Is it *paint-like* (Figure 3.1)? And if scratched off to show a new hue, is the new color simply a different literal or metaphorical paint? One way to escape the mental image of endless layers of paint giving color its color(s) is to think of hue as fluid and immaterial in terms of tracking, reidentification, and evolving change (and what might "evolving" mean here in light of the sorites paradox?). Mobile color under the description *color-as-camera* and *color-as-causative* (Figure 3.1) is color as an emanation like sound. In similar fashion, sound may be reconceived as permanent like the hue of leaves and bark in a forest; for example, when we hear a melody that repeats endlessly. This amounts to color and sound swapping mental schemata and thereby acquiring new properties and powers. Thus there are two distinct analytical frameworks at work, each of which may host color and sound on appropriate occasions.

Color and sound as solid, shaped, immobile, and paint-like appear in an *objective* frame; color and sound as emanations appear in a *subjective* frame. In addition, there are two other prominent frameworks in which color and sound may appear with new properties and powers: the *inter-objective* and the *inter-subjective* (A, 2, 1; A, nn. 7, 9). The inter-objective context includes self-organizing systems with emergent properties. The inter-subjective context includes the cultural skills we commonly summon in order to employ different language-games or language-fields in order to talk about color and sound. The methodology of the present book lies within the general framework of the inter-subjective.

Therefore the word "color" is at least four-way ambiguous, for it may refer to something both here, and in various degrees, not here, i.e., conceptual, depending on the mode of bringing it to mind and placing oneself in relation to its relations. The sense of color created by the four methodological frameworks need not

be considered as a defective realization of an exact but invisible and unattainable prototype—an ideal, perfected form of color—because color may function quite efficiently, though differently, in each situation (to be) of use. Being variously situated is color's being. There is no essence to color and sound, only an existence in a chosen language responsive to frameworks (Chapter 4). Frameworks exist because there are diverse things to do in the world.

Chapter 10 **Conclusion: How It Finally Matters**

> It is, for me, a source of real pleasure that colour. . . remains so resistant to analysis and, not least, to language. In one sense colour is here, now, around and in front of me, a part of objects and atmospheres, as real and commonplace a presence as anything; in another sense colour is nothing—certainly no thing—at all.[1]
>
> —David Batchelor

> Someone who speaks of the character of a colour is always thinking of just *one* particular way it is used.[2]
>
> —Ludwig Wittgenstein

DIOGENES'S LAMPLIGHT

Diogenes of Sinope supposedly carried a lighted lamp in daylight during the fourth century BCE searching for an honest man. Isn't this what we do when looking at color and wondering what sort of thing it really is, through and through, if it is something after all, if only it could be seen in an honest and authentic state? Color seems both pervasive and nothing, as if presence and absence were stamped into each other. Color is before us in daylight, perfectly in plain view, yet we can't seem to find its source along with its hidden affinities, as it were, down in the murky depths where being is said to lie. We continue to believe that color somehow lies between materiality and immateriality, and that each hue must exist somewhere in a secret, yet-to-be grasped, unconditional state. Still, there is real pleasure to be found in a cunning sensory elusiveness.

Diogenes may have been seeking to show the futility of lighting up the interior of a person's character or the futility of casting light on philosophical abstractions about honesty. For him, finding nothing—that is, demonstrating that nothing is to be found—was itself something gained. In ancient accounts, Diogenes was said to be merely searching for a human being. His lamp shining into dark corners of society may have been a sign of despair over a prevailing norm for humanity or a standard of honesty, i.e., the people he met with the lamp, including their reactions, were being revealed with a measure of scorn as having disqualified themselves as

being truly human or honest. Diogenes disliked Plato's search for "abstractions" and had a notorious dispute with him over the definition of being a "human," and accordingly, what living a life should be like.

With the lamp, Diogenes was surely doing philosophy while making productive use of a paradoxical fiction. His notion of philosophy was quite unlike the popular image (still popular today) of philosophizing envisioned by Rembrandt in the painting, *Philosopher in Meditation* (1632), where, having come down a winding staircase from a murky black, tunnel-like space above, the philosopher sits quietly in deep, dream-like thought in a lonely, poorly lit space that rubs up against metaphysical lights streaming from blazing window panes that scarcely light the room, but that nonetheless promise to create an ordered world like the intricately patterned stone floor the philosopher has turned toward.

WITTGENSTEIN'S LAMPLIGHT

> Philosophy just puts everything before us, and neither explains nor deduces anything.—Since everything lies open to view, there is nothing to explain. For whatever may be hidden is of no interest to us.
>
> The name "philosophy" might also be given to what is possible *before* all new discoveries and inventions.
>
> The work of the philosopher consists in marshalling recollections for a particular purpose.[3]
>
> —Ludwig Wittgenstein

The quest of Diogenes with a lamp may be slightly reframed by taking seriously Wittgenstein's view of the kind of search in which one is engaged when doing philosophy. Wittgenstein maintained that philosophy doesn't discover new, deep things, but only rearranges things to show the logic of what is in plain view before us in daylight, whether or not highlighted with a lighted lamp. One does philosophy not in a lonely place waiting for enlightenment through strength of will, as does Rembrandt's philosopher, but amid the commotion of a world. Wittgenstein does not assume that cases of deception, where the being or nature of something is deliberately hidden, is a cosmic principle:

> Being and seeming may, of course, be independent of one another in exceptional cases, but that doesn't make them logically independent; the language-game does not reside in the exception.[4]

Instead, for Wittgenstein, language and behavior, in fact, speak clearly. Careful attention to how we speak and act on any particular occasion—i.e., coming to understand the specific language-game in play—will show, as it were, the true, if provisional, colors of a thing. Wittgenstein notes that "the rule-governed nature of our languages permeates our life."[5] The aim is to discover the languages in play, not to drain them from life hoping to touch a purified datum or raw reality with a suitably blank mind or to find a single perfected language or logic as an essential residue free of all observers and actions or to confirm the real nothingness of color. Using language is to proceed to experience, form events, and act.

Imagine, for example, walking through an area from many different directions in order to create a map. Taking many directions across a given terrain is analogous to repeatedly "shining a light" on a series of areas that seem familiar. For Wittgenstein, the goal of philosophy is to discover "perspicuous (surveyable) representations" by reassembling what we already know into an "overview" that exposes the analogies, models, folk theories, grammars, and dissimilarities at work within the maze work of our preferred language-fields and action-games.[6] A language-game that has been selected is not about individual *words*—much less whether we have words, or can summon words, to name the ten million hues we can see—but about the rules for a *limited system* that generates and author(ize)s certain kinds of linked *descriptions*. Such models, architectures, overviews, and mappings are not assertions or mirrors of individual psychological states or poignant feelings, but merely careful clarifications or changes of point of view that reflect language behavior and engagement with a world. A better model than Rembrandt's *Philosopher in Meditation* for Wittgenstein's approach to philosophizing might be Cindy Sherman's photographic series, *Untitled Film Stills* (1977–1980). In each of these photographs, a character's moment of inexpressive attention instead brings into view—brings to mind—a sharply focused, formulaic situation through a rich description of locale, clothes, objects, and immanent (imminent) action.

I believe that in Wittgenstein's approach the many sorts of color circles and color geometries as well as the different numbers and shades of "primary" hues amount to first-order rearrangements of specific grammatical (normative) rules and are not meta-representations or deep abstractions. Instead, what counts are concrete situations and goals. For Wittgenstein, a color circle and primaries represent merely the rules/facts of one way of interacting with a specific material situation.

BEING COLORED IS BEING CONTEXTUALIZED

"Being colored" is part of what is meant by the notion of "material object," which object is defined as being extended in space and time and subject to light. But

color is not limited to physical objects whether solid, liquid, or atmospheric. Color moves from matter to what matters, i.e., it may appear in conjunction with immaterial objects as a category of thought. For example, there is no limit in language to what may be said to be, or to have been, *colored-by*. Anything at all, physical or abstract or even a hue may be said to be colored by a color in the course of thought or, indeed, to be *narrativized* and, hence re-colorized, by anything else not a color! (See 3, 5; 8, 3; Figure 3.1.)[7] What this demonstrates is that color depends not on names, but on descriptions and changing contexts, including relationships with perceivers whose desires, favored descriptions, and expectations drive color mobility.

All too often, it seems, theorists and critics search for color in accordance with some preconceived or privileged notion that there exists an a priori conception of color to be found apart from perceivers and their lives. I do not believe that color can be generalized into a single abstraction that embraces all color at all times, nor can a hue be singled out as somehow basic to all color, such as white (4, 1; 4, n. 9; A, 1) or black (4, 3) or black and white (2, 1; 5, 1, 1; 4, 3) or some other hue (4, 3). Nor can a few hues (e.g., the primaries; 5, 2, 1; 5, 3; 5, 1, 3) or a few functions (e.g., the harmonies; 6, 1) be made to be foundational beyond all contexts. Color is a radial concept (8, 3) embedded in ways of thinking that include our recognition and use of languages, conventions, memories, and sensations, where even a spectacle cannot defeat all contexts.

Universal statements about color are simply vacuous or else the result of an aesthetic prescription and do nothing to advance the analysis of specific functions of color or to understand a perceiver's reactions. Limited generalizations may be valid so long as the generalizations are tied to specific materials, relationships, patterns, rules, norms, contexts, and purposes. Like yards and meters, a claim about something is always measured against a *standard* for a circumstance.

Still less is it productive to block out one's mind through a phenomenological reduction and attempt to feel the feeling provoked by a hue hoping to find a hue's true nature in sensation or to move closer to finding an essence or else to find that true color, after all, is just nothing, as if it were a permanent mist in the midst of evaporating, or is nothing but a vivid custom (Democritus) because nothing exists but atoms and void. Statements announcing a specificity for a hue or some hues or all hues or color-ness are utterly specious. Such claims fail to invent intermediate cases and fail to pursue streams of *radial* sensory and conceptual connections, associations, variations, interactions, differences, and contingencies, all tied to specific ways of life (8, 3). One should look to practical life, not metaphysics, for the motivations that generate specific grammars.

PATTERNS RULE

It is distracting to hold that "color-as-spectacle" distracts or stands apart from mind (7, 3, 1; Figure 3.1). Too many theorists have isolated color (and film) from the mind by bracketing and reducing it or attempting to break it into a series of immutable parts or to simply feel its glow simultaneously from everywhere and nowhere. Color is thus cut off from types of reidentification and from a full sense of movement, tracking, and cognition. The key to color consciousness as well as awareness of other sensory qualities—e.g., depth, motion, form—lies in plotting sensory patterns. This means locating patterns in an artwork, reidentifying patterns, uncovering patterns of the narrativization of qualities, and employing *patterns of thought* through appropriate metaphors, criteria, models, schemata, and language-games. Otherwise, a quality like color is left to become static, immemorable, and a non-event (2, 5).

Color is more than an appearance or smudge, just as a written word is more than its stick-figure lines. The task for a philosophy of color is to discern which conceptual structure in a particular case has produced sight and significance—and to locate the remembered language or grammar that has been selected to push hues into patterns. Thus, restoring a hue to its "original" state in a medium does not guarantee restoring the significance it had for an original audience nor the memories it elicited for an original audience (2, n. 10; 3, 4; 3, nn. 18, 19; chapter 4; 8, 3, 1).

The approach outlined in this book amounts not to a relativism, but to a radial relationalism. The aim has been to raise doubts about a single method or single answer to all color problems and to provide some of the elements for recognizing and constructing grammars and patterns that memorialize what is being seen, thereby illuminating a perceiver's search to know the colorful more fully.

NOTES

1. David Batchelor, *The Luminous and the Grey* (London: Reaktion Books, 2014), p. 31.
2. Ludwig Wittgenstein, *Remarks on Colour*, ed. by G.E.M. Anscombe, trans. by Linda L. McAlister and Margarete Schättle (Berkeley and Los Angeles: University of California Press, 1977), Part I, § 73 (Wittgenstein's emphasis); cf. Part III, § 213.
3. Ludwig Wittgenstein, *Philosophical Investigations*, trans. by G.E.M. Anscombe, P.M.S. Hacker, and Joachim Schulte (Oxford: Wiley-Blackwell, 4th ed. 2009), §§ 126–127 (Wittgenstein's emphasis); see also §§ 89, 92, 122, 124, 125, 132, 436.
4. Wittgenstein, *Remarks on Colour*, Part III, § 99.
5. Wittgenstein, *Remarks on Colour*, Part III, § 303.
6. See note 3 above.
7. Citations in the text follow the abbreviations used in Chapter 9.


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Appendix **Wittgenstein/Context: Two Philosophy Lessons about Color and Sound**

ABSTRACT

The Appendix extends the discussion from chapter 8 of radial association and Derrida to examine two paradoxes posed by Wittgenstein plus a venerable philosophical puzzle in an attempt to clarify the influence of *context*, i.e., the search for a larger gestalt on our thinking about both color and sound. These philosophical starting points provide an occasion to offer a pair of philosophy lessons designed to elucidate the interconnection of color and sound through a series of contexts that promote different ways of talking about color and sound. In addition, four methodological contexts of contexts—objective, subjective, inter-objective, and inter-subjective—are examined as large-scale filters for analyzing color and sound. The four frameworks explain the differences between them while paradoxically revealing their intimate connection.

PHILOSOPHICAL PARADOX I: WHEN IS WHITE?

Ludwig Wittgenstein notices the following paradox: "If white turns into black some people say 'Essentially it is still the same.' And others, if the colour becomes one degree darker, say 'It has changed *completely*.' "[1]

How should one resolve this paradox about the nature of being called white? Do things in the world have an unshakable 'essence'—in this case, a hidden 'whiteness' quality—that resists or is simply indifferent to change, i.e., an underlying core substance that allows for modifications (e.g., becoming "one degree darker"), allowing even transformations (e.g., becoming black), while "essentially it is still the same"? Is the world filled with changeable and deceptive appearances that create illusions to trap unwary persons while unseen and underneath all is solid and certain? Well, is it? Do we feel that 'illusions' must be *penetrated* to find their essence in a search for what is 'real'?

In what follows, I will suggest that some cognitive illusions are the product of a kind of inertia set in place by the notion of essence and that some illusions may be dissolved, rather than pierced, by weighing contexts fostered through the circulation of language. I will argue that in these cases the inertia that may exist is due to the image or schema we choose as a 'starting point' in thinking about color.

As an example of the belief in an 'essential' quality, you may be convinced that a piece of melted wax in which many of its properties have been altered from its original solid state is still essentially *wax*, as was Descartes.[2] You may also be convinced that you are today the same *you* from a month ago despite changing your hair, clothes, and learning more (hard lessons) about life, even though presently filled with many new feelings and positive thoughts. Perhaps the verb "filled" in the preceding sentence helps you to think in this way by suggesting that you are a certain kind of (thick-walled) "container" that can be occupied with different sorts of transitory content while the container—the essential you—remains unchanged. Don't these situations involving melted wax and your personal identity support the analogous idea held by some people that black and gray are essentially versions or forms of whiteness and not something new—that black and gray are only *apparent* and are subject to something more permanent, such as changes in the intensity level of a pure (abstract!) *light*?

Indeed, what is to prevent us from asserting that one or more kinds of black hues—or, for that matter, gray hues—reveal the 'real' substance or quality here (i.e., an underlying blackness or else grayness) while white and gray (or white and black) are mere variations? Perhaps we are biased toward whiteness because we think of it as the color of Light and the source of sight and, more generally, as an incarnation of *presence* and an affirmation of the presence of the present, of being alive and aware, able to see and to be ourselves. Thus presence becomes something that we *carry* with us. (One should think carefully about the grammar of "presence" . . .) If, for example, we take white to be the sum of additive primaries (e.g., red, green, and blue), then shouldn't these primaries—or else simply "blackness"—be the true basis of white? Or, if we take white to be the sum of the rainbow hues, then shouldn't these particular hues, which are a tiny subset of all hues—or, again, simply "blackness"—be the true basis of white? What is it that we should assume to be fundamental for white, if anything—that is, some special thing that unites all appearances of white, which makes it *be* in the face of 'surface' changes, like becoming one degree darker but still being white? Are we under an illusion to think this way? Perhaps using the word 'illusion' here is itself a mistake and part of the problem.[3]

Are there concepts that might take the place of illusion or deceptive appearance and not lead to a simple dichotomy of illusion/reality, appearance/essence? Note that these strict dichotomies are mandated by the notion of "essence" where something simply is or it isn't. As one possibility, consider the following shifting phases of language use:

> A category → is elaborated into a description → allowing objects to appear (forming a group) *under* the description → then metaphorical and other rhetorical projections of that description → create a richer vocabulary → i.e., a grammar → which can be employed to solve specific problems involving the objects under that

> description (thus making up a "practice" with the objects) → constituting one part of a way of life (i.e., identifying and solving specific problems about living) → giving rise to criteria for applying that grammar (to solve other problems) → which generates a series of new categories → that create new descriptions (allowing additional objects to appear under the descriptions) . . . and so on, until there are objects and uses for objects that resist appearing under the descriptions (thus calling for a new concept to enable a new description and compliant categories).

What, admittedly, has just been *described* is one of the ways that language may circulate in defining a form of life and thought. Wittgenstein's emphasis on principles of language, description, grammar, and criteria that support life is not a two-step process where (1) a word and its definition is (2) applied to name an object. Instead, Wittgenstein's approach is closer to a one-step process whereby a word or description is acquired and learned through employment in a situation—through active engagement with/in a world. A formal definition is merely a convenience created by abstracting and simplifying a number of pragmatic situations into a kind of shorthand or ideal. In actuality, the various meanings of a word exist because of particular contexts of use or applications.

Well, then, does color have a true *essence* applicable for all contexts, all descriptions? What color would such a colored essence be? Very abstract and colorless? What is certain is that there are numerous ways we talk about things and choose to act in circumstances. Perhaps, therefore, we should seek to identify those sets of circumstances in which a change is *not* relevant—say, where we say that white becomes darker but is *still* the same—as opposed to other circumstances in which a tiny degree of change prompts us to state that the alteration has become final and has produced something (as Wittgenstein remarks) "completely" new. For example, we tend to ignore change when we think of white as a form of illumination (light/lightness) in which degrees of change—including the elimination of all illumination—do not alter its essential nature as being white like white light(ness). A white thing can then be said to still be white even in darkness (more on this later). The opposite conclusion may be reached when thinking of white as a specific hue of paint; for example, in the situation where an artist decides to change the hues of a vase in a painting from white and red to black and red. Here, white has changed completely; that is, we would not be inclined to say that the black-and-red vase is really, despite appearances, white-and-red or should be seen as white-and-red because it may still *be* white-and-red in the painter's head.

Suppose we dissolve the paradox by allowing a person to talk in both ways—where white stays the same despite changes and becomes something completely new after a change? This more liberal approach would set the stage for a person to act in different sorts of ways when perceiving and planning in accordance with different ways an object

may act or serve. This illustrates that we are often misled by language—here, language about color—because we attempt to reduce many different ways of talking and acting to a single language that speaks of a single "essence" of a thing for all cases.

Consider, for example, a photograph that shows a highlight on a surface painted in a light hue.[4] The photograph highlights the fact that the "white" of both the highlight on the surface and the "white" of the light hue of the painted surface look quite similar. Although physically, the two appearances are different, the tendency is to reduce or flatten the two situations into a single way of talking about light and white. Light has made the highlight white while white paint has lightened the hue painted on the surface. Thus light striking the surface is misleadingly analogized to a lightened hue of paint. A highlight and a light hue have come to have the same essence. The conflation of these two sorts of white will lead to many other inconsistencies and contradictions in the ways we talk—if and when we seek to create a single master mathematics of color appearances instead of seeking to adjust our language to a purpose and situation by choosing among language-games.

Matters become still more confusing when one seeks a comprehensive and unified theory of color by incorporating other types of appearances of a hue besides lightening, such as adjacency effects (simultaneous and successive contrast), afterimages, forms of luminosity and transparency, impossible transparencies, color constancy, non-spectral hues, the problem of brown, impossible hues (e.g., bluish yellow), impure hues (e.g., orange), groups of primaries, and cultural and linguistic preferences. Could there be a single language-game to rule them all? Rather than searching for a final essence, then, let's consider a few contexts that authorize different kinds of talk through the use of distinctive descriptions and grammars.

SOME CONTEXTS FOR COLOR TALK

According to C.L. Hardin, "A proper full-blooded account of color would include the respective roles of the phenomenology of color perception, its biological substrate, the evolution and ecology of color vision, the color-relevant properties of physical materials, the affective, aesthetic, and communicative dimensions of color, the relations that colors bear to each other, and the uses of color language."[5]

For example, with respect to the evolution and ecology of color vision, Hardin notes that human vision is designed for "forest floor" daylight and for high acuity; and that although chromatic vision is only slightly better than achromatic vision at detecting and recognizing objects, chromatic vision excels at "biological signaling" involving, one assumes, situations of fright, flight, fight, and. . . mating. These actions speak directly to the social existence of human beings. In addition, chromatic vision is important in distinguishing the chemical composition of objects through their surface color, such as ripe red fruit against green foliage. In this context, we seldom use color

to describe the properties of light sources per se apart from lit objects; some exceptions are rainbows and traffic lights. The lesson is that the nature of color is being relativized to a particular context—here, the evolution and ecology of color vision. Color assumes its nature according to a use. Or, as Wittgenstein asks, "What is the experience that teaches me that I differentiate between red and green?"[6]

These areas of investigation make for a range of pertinent contexts, as Hardin has noted, and also presumably, contexts within contexts. It may be that our talk about hues and their changes should be made relative to such frameworks rather than to a timeless substance fit to all occasions. A context, however, also requires a general orientation and method of investigation to implement its descriptions. Which analytical framework will best describe pertinent color relations and permit judgments about change? A second philosophical paradox will serve to highlight methodological contexts for color contexts.

PHILOSOPHICAL PARADOX II: COLOR IS IN, SOUND IS FROM

Colors seem to exist "outside of us," somewhere in the surroundings, because the brain operates by processing color information together with shape information. (Where exactly *is* the shape of an object? in edge contrasts?) Color thus appears to be an inherent property of a shaped material object—to be "in" the material, to be possessed by it far out there—even though from a physical standpoint light waves are being continuously reflected to our eyes in a manner very similar to the way sound waves are reflected to our ears. Sound, however, is heard as "coming from" a source rather than being embedded "in" the vibrating material object that produces the waves of sound. Thus, although both color and sound depend on waves—light waves reaching the eye, sound waves reaching the ear—color is perceived to reside firmly *in* its object, intrinsic to its object (like shape), while sound moves through intervening space and seems instead to emanate *from* its object. This is an interesting and peculiar difference that nevertheless makes for a difference in our perception and cognition. It also makes a difference for artists who are constructing objects for effect.

"*A tree falls in a forest, but no one is there to hear it. Does it make a sound?*" This is a hoary philosophical puzzle. The puzzle is always stated in terms of sound, never color. Why? Why isn't the puzzle stated in the following way: "*A tree has leaves and bark in a forest, even though no one is there to see them. Are the leaves green, the bark brown, when no one looks?*" Is there an answer in the preceding paragraph? Don't we tend to think that sound essentially *comes and goes* and so apparently may or may not really *be there*—as we, too, come and go, and may or may not be there in the forest? By contrast, it just seems natural to think that color is always *out there* in one place in some permanent fashion, like the shapes of trees

and leaves, with the result that no thought-provoking, similar paradox seems to arise with color. Color just always *is* while sound must be made and continuously remade in order to continue to exist. Alternatively, sound might exist elsewhere in an ideal Form within a new kind of abstract space, a mysterious soundless realm made of things reduced to their essence, needing no vibrato.

To put it another way: sound seems to depend on the intervening air, not on a vibrating object, and may at times seemingly evaporate into the air like steam and disappear into silence while color—though equally dependent on energy waves—seems always to be *right there*, locked tightly and permanently into or onto a material object as if it were, say, solid *paint* or shape. (Recall the ten mental images of color in Figure 3.1.) But, one may wonder, what accounts for the color of that paint? Would there be a second very thin coat of paint on top of the paint, or else under the paint should it be scrapped away to show a new hue? Is there an end to layers of "paint" as a hue changes?

What accounts for the asymmetry between our beliefs about seeing the solidity of color versus hearing immaterial sound? It may be that a hidden and seductive analogy is responsible for the asymmetry that underlies the philosophical puzzle about what resides in the forest. It appears that a person "not being there" is like a sound "not being there," and since both may or may not be in the forest, it would seem natural to ask, what really *is* there "there-then": could sonic sense exist in the forest without a sensibility, could sense exist and sensibility not? Thus one is encouraged to puzzle about the status of sound in the forest, but apparently not to puzzle similarly about color, despite both sound and color being contingent upon physical waves. It is harder to imagine that trees and leaves are colorless outlines or are blank shapes when no one is present.

By contrast, sound in its comings and goings is analogized to—and made dependent upon—*persons* in their "comings and goings." Sound is embodied while color remains disembodied and detached. Furthermore, it is a fact that the vibrations of sound in air can be felt as they strike our bodies, flesh, organs, and bone. Sound rubs against and within us; it does not merely arrive through the ears. Indeed, sound affects us even though often acting below conscious awareness. But not color, which needs no medium (e.g., air or water) for its transmission. We can make sounds ourselves as we breathe and speak, but we do not have the same control over color. Thus sound is contingent upon, and coincident with, humans and life-giving air in a way that color is not.

What is it that troubles us about the perceived asymmetry between sound and color? Do we worry that some sensory qualities may be inexistent in a forest, e.g., sound, while other sensory qualities in the forest refuse to go away and thus are more fundamental than sound, are more primal, e.g., color or the touch of tree bark? Sound would seem to be more personal and reassuring because it depends on us, it comes to/for us. Sound is measured by our (mortal) bodies. By contrast, color appears to be objective and distant—immune to time and apart from us, alien

even. But is this story about sound and color absolute or does it depend on the first image or schema that comes to mind? Haven't I so far merely described our default assumptions about sound and color? In fact, there are a number of contexts, metaphors, and ways of situating sound and color that produce different results.

FOUR ANALYTICAL CONTEXTS

Suppose one were to think of color, not as "paint," but as an emanation like sound in a forest by employing such mental schemata as *color-as-camera* or *color-as-causative* (see Figure 3.1). In similar fashion, sound might be reconceived as permanent like the color of leaves and bark in a forest; for example, the sounding of an alarm clock or a musical melody that remains "stuck in one place" and repeats endlessly. This would amount to color and sound swapping their usual mental images and thereby acquiring new properties and powers. Thus there would be two distinct analytical frameworks, each of which might host color and sound on appropriate occasions. Color and sound as an emanation would appear in a *subjective* framework, and color and sound as solid, shaped, immobile, and paint-like would appear in an *objective* frame. In addition, there are two other prominent frameworks in which color and sound may appear with new properties and powers: the *inter-subjective* and *inter-objective*.

Consider four contexts that may be employed in judging the nature of sensory qualities. Descriptive frameworks may be fashioned as (1) objective, e.g., the physics of wave motion; (2) subjective, e.g., mental qualia or cognitive response; (3) inter-objective, e.g., an interactive network in which material factors converge or collide producing emergent qualities; and (4) inter-subjective, e.g., an interactive network in which cultural factors converge or collide producing emergent qualities.[7] The latter inter-subjective context is the methodology employed in the present book to conceptualize color. This framework includes the skills we commonly summon in order to talk about sound and color; that is, the ways we create appropriate expressions in a particular language and vocabulary aimed at a situation, including writing philosophy, imagining a puzzle about a forest, and visualizing an impossible color like Wittgenstein's "bluish yellow" (Chapter 5). The inter-subjective plays a key role when we immerse ourselves in fictional scenarios so as to imagine what it would feel *like* to be in a forest alone with night drawing nigh and a tree about to fall, or not, and hearing sounds we wish we didn't.

Won't the character of sound and color be different in the four contexts? The status of sound and color is strictly relative to an orientation and method of investigation, i.e., to a way of knowing. Here is an experiment in order to aid in visualizing the four methodological contexts. Look around a room and (1) observe the distant color of an object (objective); then (2) close your eyes (or *pretend* to close your eyes—it doesn't make a difference, does it?) and remember the color (subjective); next (3) look again at the object situated in its complex environment and grasp

the nature of its surface interacting with other nearby colored surfaces and light sources, and wonder whether it would be this same color if the lights were dimmed or turned off? No and yes! For as Wittgenstein observes, a rose in the dark may be seen either as quite black or as red[8] (inter-objective); and finally (4) write a message to someone, or frame your thoughts in a speech, to describe the color you saw and felt. What will another person come to see through your communication that draws on shared conventions and imagery (inter-subjective)?

Do the colors in the four contexts all refer to a single, perfect (*perfectly unseen*) essential color? If not, the word "color" is at least four-way ambiguous, for it may refer to something both here, and in various degrees, not here, depending on the mode of bringing it to mind and placing oneself in relation to it, as if visualizing it from a selected node in a network. The sense of color created by the four contexts need not be considered as a defective realization of an exact but invisible and unattainable prototype—an ideal, perfected form of color—because color may function quite efficiently, though differently, in each situation (to be) of use. Being situated in a framework is color's being.

To put it another way and to conclude this discussion: color comes and goes, as does sound, but what makes for its "comings and goings" is the sudden appearance of a new kind of forest, i.e., a new, pertinent context of use. Color and sound can be experienced within any of the four methodological contexts. The difference between the two is that each has a different default or typical context. Color is usually conceived in an objective frame; sound in a subjective frame.[9] Contexts, however, may shift when we are prompted to do so by circumstances or an artwork. When color appears in a subjective or inter-subjective frame, it becomes mobile and may be tracked and reidentified. There are many choices to be made by artists and spectators, and hence there is an evolving ethics in creating and experiencing, but there is no essence to color and sound, only a given framework promoting existence.

NOTES

1. Ludwig Wittgenstein, *Culture and Value*, ed. by G.H. Von Wright, trans. by Peter Winch (Chicago: University of Chicago Press, 1980), p. 42 (Wittgenstein's emphasis). Further problems about "white" are discussed above in Chapter 4.
2. René Descartes, "Second Meditation" in *Meditations on First Philosophy (with Selections from the Objections and Replies)*, ed. and trans. by John Cottingham (Cambridge: Cambridge University Press, rev. ed. 1996), pp. 16–23.
3. For more on illusion, see Chapter 7, section 1.
4. My example and argument derives from Ludwig Wittgenstein, *Remarks on Colour*, ed. by G.E.M. Anscombe, trans. by Linda L. McAlister and Margarete Schättle (Berkeley and Los Angeles:

University of California Press, 1977), Part I, §§ 1–5; cf. Part I, § 49, and Part III, §§ 95, 131, 171; cf. knowing a hue with knowing a pain in Part III, § 304.

5. C.L. Hardin, *Color for Philosophers: Unweaving the Rainbow* (Indianapolis: Hackett, expanded ed. 1993), pp. xix–xx.
6. Wittgenstein, *Remarks on Colour*, Part III, § 167.
7. For a detailed discussion and application of the four contexts—objective, subjective, inter-objective, and inter-subjective—see my essay, "If-Then-Else: Memory and the Path Not Taken" in *Transmedia Frictions: The Digital, the Arts, and the Humanities*, ed. by Marsha Kinder and Tara McPherson (Berkeley: University of California Press, 2014), pp. 45–79. In notes 24 and 25 of "If-Then-Else," I situate color only in a subjective frame, not as it might exist in all four frames. My purpose in these two notes is merely to offer an illustration of the four frameworks and to show how they may be utilized to analyze the nature of a "material object." In the Appendix of the present book, my purpose is to apply each of the four frames to the analysis of our beliefs about "phenomenal perception" and about the "sensory qualities" of color and sound.
8. Ludwig Wittgenstein, *Philosophical Investigations*, trans. by G.E.M. Anscombe, P.M.S. Hacker, and Joachim Schulte (Oxford: Wiley-Blackwell, rev. 4th ed. 2009), §§ 515, 671.
9. We could close our eyes or stick fingers in our ears with the result that color or sound would disappear. More usually, however, the sun continues to shine or the light bulb in the ceiling remains glowingly bright giving us continuous color while the music and noise of vibrating objects tends to wear out and cease. The difference between color and sound here has to do with the way our lives are being lived—being in the sun and screwing in light bulbs versus using our muscles, say, which are subject to fatigue, to vibrate a guitar string—and this, in turn, is reflected in the way we talk and move in our lives, i.e., in our language, which is to say, in our expectations since language is a primal vibration of our various selves in motion through the world. Thus, while color and sound can be experienced in any of the four methodological quadrants—objective, subjective, inter-objective, inter-subjective—we tend to start with particular schemata for our thoughts and feelings about color and sound, i.e., there are different default imagery for color and sound based on life and language.

   On color and sound within objective and subjective frames, see my two essays, "Sound, Epistemology, Film" (revised version) in *Film Theory and Philosophy*, ed. by Richard Allen and Murray Smith (Oxford: Oxford University Press, 1997), pp. 95–125; and "Soundtrack in Mind," *Projections* 4, 1 (Summer 2010), pp. 41–67. For Stanley Cavell, the distinction between visual and auditory stimuli turns on visibility where the auditory is invisible: "what is heard comes *from* someplace [that is not seen], whereas what you can see you can look *at*." *The World Viewed: Reflections on the Ontology of Film* (Cambridge, MA: Harvard University Press, enlarged ed. 1979), p. 18 (Cavell's emphases).


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Works Cited and Further Reading

There are nearly 240,000 books on color listed on Amazon. Roy Osborne lists 3,200 major treatises on color (see below, A2). And that's just the beginning. Color seems so basic, we see it day and night. Yet it is a deeply tangled and controversial affair that has provoked intense interest across many disciplines since Empedocles, Plato, and Aristotle.

What follows are the works that are cited in the present book together with a highly selective list of further reading. For convenience, the citations are organized into approximate subject categories. I have also provided a few annotations and comments. Works marked with an asterisk (*) are especially important to the arguments of this book. Here are the categories:

A. Color as Phenomena
   - A1. Physics and Physiology of Color
   - A2. History, Field Observation, Practice, and Polemics
   - A3. Semiotics of Color
   - A4. Systematization of Color

B. Film
   - B1. Aesthetics of Film Color
   - B2. Analyses of Particular Color Films
   - B3. Anthologies of Writings on Film Color
   - B4. Film Theory
   - B5. Overview of the Aesthetics, History, and/or Theory of Film Color
   - B6. Sound
   - B7. Technicolor

C. Painting
   - C1. Color Composition for Painters
   - C2. Color Painting/Psychology/Criticism

D. Philosophy
   - D1. Cognitive Science
   - D2. Philosophy
   - D3. Philosophical Aesthetics
   - D4. Philosophy of Color

E. Residuum
   - E1. Knowledge Representation
   - E2. Literature
   - E3. Memory/Consciousness
   - E4. Narratology

A. COLOR AS PHENOMENA

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